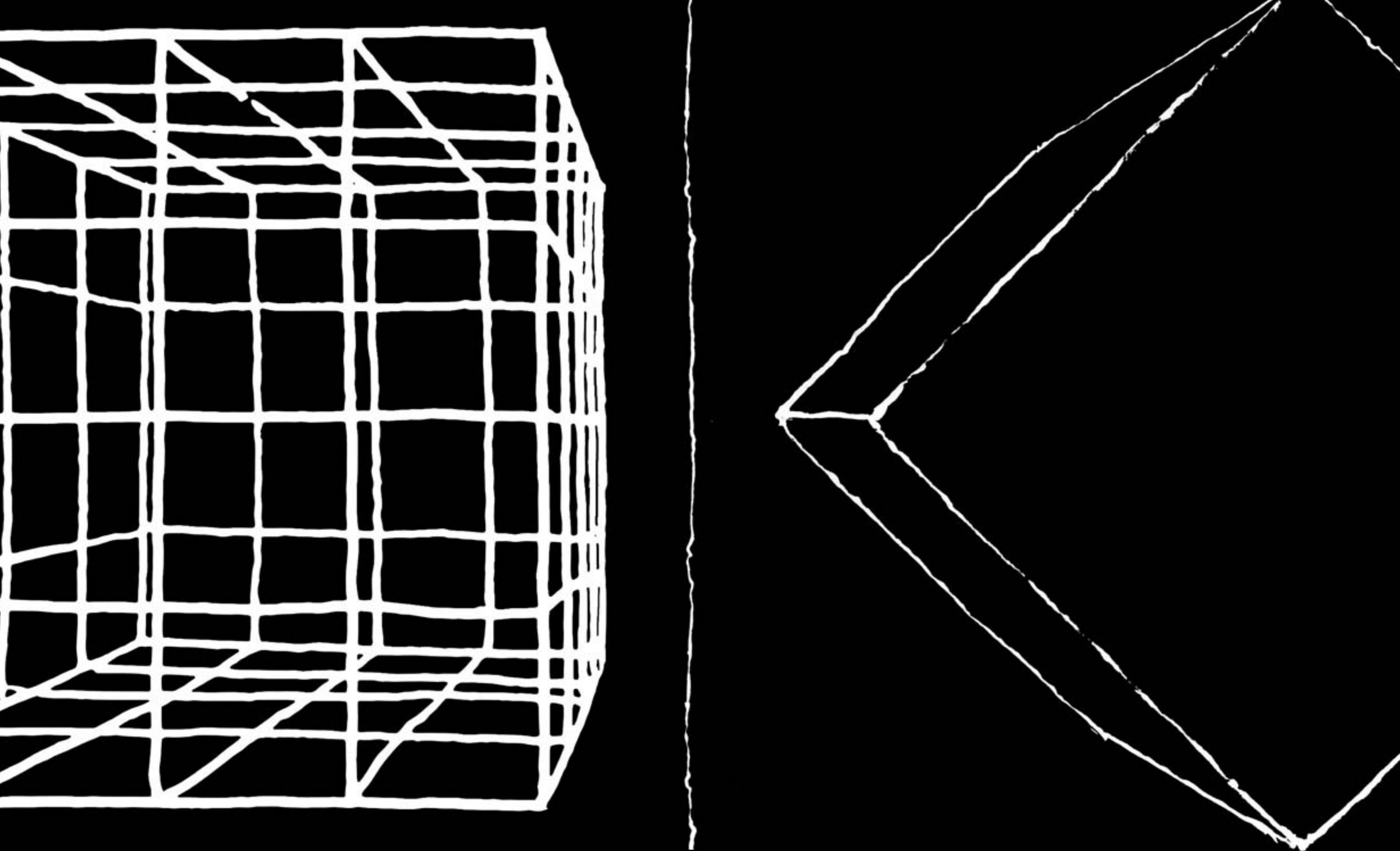

ARCHITECTURE OF CHALLENGES
EXPERIMENT IN SPACE — THE SPACE OF EXPERIMENT

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ARCHITECTURE OF CHALLENGES
EXPERIMENT IN SPACE — THE SPACE OF EXPERIMENT

collective work
edited by Anna Maria Wierzbicka

Warsaw 2026

The fourth edition of the Architecture of Challenges conference, held in 2026, brings together researchers in architecture and urbanism who are interested in discussing the theoretical foundations of creative practice. Continuing the tradition established in previous conferences, the event combines expert debate with design-based inquiry and hands-on engagement in real space, actively involving students in the process. Work conducted with physical models as well as real materials is not treated merely as a technical exercise, but as an integral component of reflection on how architectural ideas are generated, developed, and critically verified. The central notion emerging from the conference title, its theses, and many of the contributions included in this volume is that of experiment. In architectural discourse, this concept is interpreted in a broad and multifaceted way. It extends beyond activities that meet the strict criteria of scientific experimentation and encompasses a wide range of exploratory practices. What connects these approaches is a shared emphasis on searching for architectural solutions through iterative processes of trial and adjustment. These methods often involve users and unfold either in real spatial conditions or within increasingly sophisticated simulated environments.

From the perspective of a strict understanding of the empirical method – shaped by the intellectual achievements of the Enlightenment – it could be argued that the term “experience” might more accurately describe architectural practice than “experiment.” In its rigorous scientific sense, experimentation is difficult to achieve in architecture and, as a result, remains relatively rare. Historically, architects such as Antonio Gaudí and Frei Otto sought to create conditions that approximated laboratory environments, enabling more systematic observation and testing. Today, digital technologies have significantly expanded the possibilities of simulation, allowing designers to model complex spatial scenarios with increasing precision. However, even the most advanced tools cannot fully replicate the dynamic and often unpredictable interactions between users and space. Moreover, ethical considerations prevent architects from exposing users to risk or discomfort, which further limits the scope of experimentation. Despite these constraints, broadly understood “experimentation” plays a vital and enduring role in architecture. The accumulation of knowledge through direct experience – particularly through engagement with users – has been an essential aspect of architectural practice for centuries. Full-scale (1:1) prototyping, has become an important educational and design tool. It enriches the learning process, strengthens design skills, and allows architects to test ideas in conditions that closely approximate reality. Such practices increase confidence that proposed solutions will respond effectively to users’ needs while contributing positively to the urban and social context. At the same time, they often reveal subtle spatial, material, or behavioral conditions that remain inaccessible to purely analytical or theoretical approaches.

Thanks to the dedicated efforts of the conference organizers, this monograph provides an opportunity to pause and reflect on the evolving role of experimentation in architecture. It brings together a diverse set of perspectives on architectural and urban experiences, offering readers insight into contemporary approaches as well as inspiration for their own exploratory and experimental practices. In this sense, the publication not only documents the outcomes of the conference but also contributes to an ongoing dialogue about the methods, responsibilities, and future directions of architectural design.

Prof. DSc PhD Eng. Arch. Jan Słyk
Vice-Rector for Academic Affairs, Warsaw University of Technology

Contemporary architecture operates within a reality shaped by profound civilizational transformation. Changing patterns of social life, the accelerating development of technology, the climate crisis, new modes of mobility and communication, together with an increasing awareness of the psychological and cultural dimensions of space, mean that design can no longer rely on stable or permanently established models. Architecture today faces the ongoing necessity of redefining its own tools, methods, and ways of understanding the relationship between human beings and the built environment. Within this context, the concept of experiment acquires particular importance, understood not as a purely formal manifestation of novelty, but rather as a cognitive and exploratory process through which the boundaries of existing knowledge about space and its social consequences may be expanded.

Within scientific methodology, experimentation occupies a unique position as a means not only of confirming assumptions, but also of exposing their limitations through falsification. The significance of the research process lies not merely in achieving the anticipated outcome, but perhaps even more importantly in the capacity to submit accepted assumptions to critical examination. Negative or unexpected results frequently possess greater epistemological value than those that simply confirm a hypothesis, because they reveal the inadequacies of existing models and compel a reconsideration of established modes of thinking. The advancement of knowledge therefore occurs not solely through validating the effectiveness of particular solutions, but equally through recognizing their weaknesses, insufficiencies, and unforeseen consequences. From this perspective, experiment may be understood as a practice of intellectual humility and as evidence of a willingness to confront uncertainty and the complexity of reality.

Architecture has long evolved through a comparable process of searching for and testing new responses to changing conditions. This applies equally to technologies and materials, to the organization of space, to ways of inhabiting, and to the social relations embedded within the structure of the city itself. At the same time, the specific character of architecture means that the concept of experiment requires exceptional interpretative caution and ethical awareness. Unlike laboratory-based research, the consequences of architectural decisions unfold directly within the everyday lives of individuals and communities, affecting the environments they inhabit and co-create. Designed space is never neutral. It shapes social behavior, influences patterns of movement and interaction, affects communal relations, and frequently determines the sense of identity, security, belonging, and broadly understood quality of life. For this reason, experiment in architecture cannot be understood as the transfer of risk onto users. Rather, its role lies in the responsible pursuit of new forms of knowledge concerning the shaping of space, while remaining conscious that the ultimate subject of all design activity is human existence itself.

In this sense, architecture requires from the designer not only technological competence and creative imagination, but also empathy and the ability to understand the experiences, expectations, vulnerabilities, and limitations of other human beings. Empathy should not be regarded as a secondary addition to the design process, but rather as one of its fundamental conditions. It allows us to recognize that architecture is not simply an abstract composition of forms, but an environment of everyday practices, emotions, memories, and social relationships. Through such an approach, experimentation in design may become a tool of responsible inquiry rather than a mechanism through which uncertainty and risk are imposed upon users. Particularly significant in this regard are methods grounded in social dialogue, participatory processes, prototyping, and the gradual implementation of solutions, all of which make it possible to confront architectural ideas with lived human experience at both individual and collective scales.

It is also important to emphasize that contemporary design processes increasingly assume a research-oriented character, drawing upon scenarios, prototypes, simulations, data analysis, and temporary interventions. The rapid development of digital tools has opened entirely new possibilities for conducting architectural experimentation. Parametric modeling, environmental simulations, digital twins, augmented and virtual reality, as well as systems employing artificial intelligence, make it possible to analyze spatial processes long before any physical transformation takes place. Increasingly advanced technologies enable the prediction of user flows, environmental performance, the influence of light and microclimate, energy efficiency, and even potential perceptual or social responses associated with the configuration and use of designed space. Such tools do not eliminate, of course, the uncertainty inherent in architectural practice. They do, however, make it possible to shift a significant part of the experimental process into the realm of simulation and predictive analysis. As a result, numerous variants and scenarios may be tested without directly exposing users to the consequences of potentially flawed decisions. Digital models and algorithmic methods therefore create the possibility of reducing the ethical burden traditionally associated with experimentation conducted within real social environments. This does not imply the replacement of human sensitivity or responsibility by automated systems, but rather an expansion of the possibilities for critical reflection, informed anticipation, and more responsible decision-making within the design process.

The conference “Architecture of Challenges” opens a space for reflection on the contemporary meaning of experiment in architecture, urbanism, and education. It encourages us to perceive design as a form of research while simultaneously reminding us that architecture remains deeply rooted in the world of human experience and values. In an era of increasing complexity, the capacity to undertake responsible exploration becomes especially significant - exploration capable of combining innovation with critical reflection, discovery with awareness of consequences, and technological advancement with an empathetic understanding of human needs. Perhaps it is precisely within this balance - between the courage to experiment, the possibilities offered by emerging technologies, and responsibility toward the user - that one of the most important challenges of contemporary architecture reveals itself today.

DSc PhD Eng. Arch., Assoc. Prof. Krzysztof Koszewski,
Dean of the Faculty of Architecture, Warsaw University of Technology

Architecture is one of the most complex fields of applied art. It cannot be reduced to shaping of form and arrangement of function; it involves constant exploration of relationships between space, activity, and people. In design practice, people remain central, and every project – to some extent – is a deeply experimental process in which theory meets real life. When treated as a field of experimentation, architecture ceases to be a collection of static objects and becomes a dynamic space of events. From this perspective, experimentation reveals tensions between what is designed and how space is experienced and used in daily life. It also becomes a tool for disciplinary boundaries, provoking new ways of experiencing and sharing space, and deepening understanding of a reality extending far beyond material and pragmatic design assumptions. In the face of contemporary climate, social, and cultural challenges, reinterpretation of the concept of the experiment gains crucial significance. It becomes a conscious search for balance between technology, ecology, and ethics. In this context, the Vitruvian triad – firmitas, utilitas, venustas – regains relevance, structuring the design process from testing structural limits, through redefining functions, to the development of aesthetics and symbolism, which together form a coherent and responsible whole.

This publication is the fourth edition of the series. The first was dedicated to the reconstruction of Ukraine, the second to new materials in the service of sustainable human and environmental development, and the third to the idea of the New European Bauhaus and community building. The fourth edition focuses entirely on experimentation as a key method for the development of architecture and urban planning, presenting it not only as a tool for innovation but, above all, as a design mindset. Of particular significance is the work of an interdisciplinary team – researchers, designers, educators, and practitioners – whose collaboration and exchange of perspectives gave the monograph its multidimensional character. The team became a laboratory of ideas, where experimentation was both the subject and the method of creation.

This monograph, "Architecture of Challenges: Experiment in Space and the Space of Experiment", was born out of a deep conviction that prototypes, installations, pavilions, and temporary interventions serve as true laboratories of architectural thought. They are places where the "experiment in space" meets the "space of experiment." Each such gesture engages in mindful dialogue with its context – landscape, history, culture, and local community – through critical reinterpretation of tradition. The book consists of four main parts that explore experimentation on various scales and in different dimensions: in architecture – as micro-worlds of innovation, in urban planning – as a city laboratory, in education – as a space for learning through action, in the service of society – as a tool for dialogue and building shared responsibility.

It is our deep conviction that the ultimate task of architecture remains the creation of experiments that truly serve humanity – spaces that are meaningful, functional, ethical, sustainable, and deeply rooted in the real, complex world. I hope that reading this monograph will become for readers not only a valuable source of knowledge, but also a powerful inspiration for their own bold and responsible experimentation with space – in the spirit of care, creativity, and deep sensitivity to the challenges of the present day.

DSc PhD Eng. Arch., Assoc. Prof. Anna Maria Wierzbicka
Chairwomen of Conference Committies, Warsaw University of Technology

Faculty of Architecture
Warsaw University of Technology

This collection of articles showcases the latest research from scholars in the architecture and urban planning discipline, alongside insights from practitioners and educators engaged with the built environment. It vividly embodies the theme of the upcoming fourth conference organized by the Faculty of Architecture at Warsaw University of Technology, as part of the "Architecture of Challenges" series. The central motif of this edition is broadly conceived experimentation, reflected in the compelling and expansive title: "Experiment in Space – Space of Experiment."

For architects, researchers, and tutors operating within the fields of architecture and urban design, it should be evident that, inspired by the ideas of Aldo Rossi and his reflections published in "A Scientific Autobiography," architecture should be regarded as a domain of continuous experimentation – an ongoing synthesis of memory, imagination, and the pursuit of innovative solutions. Architecture and a city are not static objects or places but rather spaces of phenomena, where relationships between form, function, and users are subject to relentless testing and reinterpretation.

In the face of global challenges such as climate change and social inequality, architectural and urban experiments take on particular significance as tools for seeking a balance among technology, ecology, and ethics. Increasing importance is also placed on experimentation in architectural education – not only through creative interventions in urban space but also through innovative pedagogical methods, a fresh approach to engaging with students and outreach beyond university walls.

Within the scientific community, experimental research, experimental tools, and simulations are highly valued. In the context of the interests represented by the authors of the included texts, the "research by design" method is also important. This approach, which intertwines design and research processes, can serve as a means to evaluate experiments and acknowledge that not all experiments are successful.

The scholarly texts and experiential descriptions compiled in this publication focus on various aspects of experimentation in space – from microcosms of innovation and urban laboratories to educational settings and social architectural initiatives. Organized into four thematic blocks – ranging from innovative micro-worlds and urban laboratories to education and service to society – these contributions demonstrate how experimentation in architecture and urban design can create beautiful, functional environments crafted ethically and responsibly, addressing the challenges of the contemporary world.

I recommend engaging with the articles gathered here, which constitute not only reports on research and projects but also sources of inspiration for further exploration and creative dialogue within the disciplines of architecture and urban planning – as spaces of ongoing experimentation.

Prof. DSc PhD Eng. Arch. Krystyna Solarek
Chairwoman of the Scientific Council for Architecture and Urban Planning, Warsaw University of Technology

Faculty of Architecture
Warsaw University of Technology

Experiment in architecture is not the exclusive domain of the avant-garde. It is an inherent part of design practice, since every project develops under conditions that cannot be fully anticipated. Needs may change during the course of realization; materials may behave differently than expected; users may appropriate space in ways not foreseen by the designer. Such outcomes should not be understood as failures of the process, but rather as intrinsic to its nature.

The theses of this conference draw attention to an issue that can easily be overlooked in everyday practice: architecture has always functioned as a field of experimentation. The key question, therefore, is not whether to experiment, but how this is done, and with what degree of awareness and responsibility. The Vitruvian triad – durability, utility, and beauty – reappears here not as a prescriptive formula, but as a framework for inquiry. Where are the limits of construction? How might function be redefined as modes of living evolve? What constitutes an aesthetic rooted in specific spatial and temporal contexts? In the face of contemporary climatic and social challenges, these questions acquire particular urgency. Sustainable architecture, especially when conceived within circular systems, may be regarded as a significant ethical experiment, insofar as it redefines the scope of the designer's responsibility.

Warsaw provides a particularly relevant context for such a discussion. The city was rebuilt from near-total destruction within a single generation, often under conditions of incomplete information, institutional pressure, and uncertain outcomes. Some of the decisions taken during this period proved effective, while others have been subject to critical reassessment. What characterizes them collectively is that they were made without the possibility of fully anticipating their consequences. These consequences continue to shape the city today.

The Royal Łazienki Park, where the pavilion will open on June 28, introduces another dimension to this context. As a site with a strong landscape and cultural identity, it requires that any contemporary intervention engage with existing conditions. This engagement does not necessarily entail the rejection of tradition, but may instead involve its reinterpretation. The conference theses articulate this point explicitly: architectural experimentation operates in dialogue with landscape, history, and community.

The four thematic sections – architecture, urban planning, education, and society – should not be understood as discrete domains. Experimentation at the level of a student studio and at the scale of an exhibition district share a common methodological basis: both require a willingness to reassess initial assumptions and a commitment to critically evaluating outcomes. The contributions collected in this monograph demonstrate the diversity of approaches to such experimentation, shaped by differences in scale, tools, and context. At the same time, they share a common premise: that architecture is not only a physical artifact, but also a medium for dialogue and the co-creation of shared space.

The Academy of Fine Arts in Warsaw is pleased to participate in this discussion, together with the Faculty of Architecture of the Warsaw University of Technology, the Royal Łazienki Museum, and the National Institute of Architecture and Urban Planning.

I invite you to read this volume.

Prof. DSc PhD Błażej Ostoja Lniski
Rector of the Academy of Fine Arts in Warsaw

ACADEMY OF FINE ARTS IN WARSAW

Ladies and Gentlemen,

The fourth edition of the Architecture of Challenges conference series is taking place in the unique setting of the Royal Łazienki Museum. This latest event continues our reflection on contemporary architectural trends and the possibilities for its development. I am proud that this year's conference places experimentation at its centre – both as a creative tool and as a research method – enabling us to move beyond established design conventions and rethink the relationships between space, the user, and the environment. I am confident that the conversations sparked here will resonate far beyond our immediate community on an international level, and I am deeply grateful for that. Central to this edition is the unique context of the former royal residence itself. Historically, this site was associated with exclusivity, representation, and a strict hierarchy of access. Today, the museum's role as a public space opens up rich new possibilities for interpretation. In many respects, the museum is evolving into ever more inclusive, accessible, relational, and participatory space. This year's conference explores the creative tension between these two identities: the legacy of elitism and the modern drive towards openness in design.

The theme of experimentation finds its most vivid expression in the concept of the summer pavilion, which serves as a testing ground for new forms, materials, and modes of use. Situated within the historic garden complex, the pavilion acts as a bridge between tradition and the future. The conceptual and creative processes behind it extend the boundaries of architectural research, inviting us to examine how architectural form can mediate between a site's heritage and its contemporary, democratic purpose. The excitement surrounding experimentation extends beyond the current conference to the workshops and discussions that will take place at the Royal Łazienki Museum in September 2026, bringing together representatives of Europe's most significant royal residences as part of a conference organized in collaboration with the Network of European Royal Residences (ARRE), entitled Royal Residences as Spaces for Contemporary Artistic and Social Experimentation.

The publication accompanying this conference brings together texts, projects, and reflections by conference participants that showcase the diverse ways we can approach experimentation in architecture. The work presented here grapples with the limits of innovation, the designer's responsibility, and the role of architecture in a rapidly changing world, and considers what a royal residence might become when it is treated, simultaneously, as a museum and a laboratory. I would like to express my sincere gratitude to Professor Krzysztof Zaremba, Rector of the Warsaw University of Technology, and to Warsaw University of Technology Professor Krzysztof Koszewski, Dean of the Faculty of Architecture, for their institutional and organizational support that made this fourth edition of the conference possible. My warm thanks also go to Warsaw University of Technology Professor Anna Maria Wierzbicka for her creativity, commitment, and generosity of spirit. The inspiration, knowledge, and dedication of her and the entire team at the Faculty of Architecture have been indispensable. I would equally like to thank the whole team at the Royal Łazienki Museum who contributed to bringing it to life. I look forward to our future collaboration, and wish you all enriching and inspiring experiences – both here at the Royal Łazienki Museum and at royal residences across the globe.

PhD Marianna Otmianowska
Director of the Royal Łazienki Museum in Warsaw

ROYAL ŁAZIENKI MUSEUM

The National Institute of Architecture and Urban Planning is pleased to present the fourth monograph in the Architecture of Challenges series – a multi-author volume emerging from the 4th International Scientific Conference "Experiment in Space – Space of Experiment". The conference was organized in collaboration with the Faculty of Architecture at the Warsaw University of Technology, the Łazienki Królewskie Museum, and the Academy of Fine Arts in Warsaw.

For many years, the Institute has pursued its core mission of fostering public awareness and promoting high standards in architectural and urban culture, actively supporting research, educational, and outreach initiatives. Architecture is approached not as a static object but as a dynamic field of experimentation – a continuous testing ground for ideas, relationships, and possibilities. The regular organization of this conference is a key component of this mission.

Following the ideas of Aldo Rossi, who emphasized that architecture arises from memory, experimentation, and imagination, this volume illustrates how deliberate transgression of disciplinary boundaries, interdisciplinary dialogue, and ethically responsible practice can address the pressing spatial, climatic, social, cultural, and technological challenges of our time. Experiment, understood in a multidimensional sense – as a design method, pedagogical tool, temporary social intervention, and laboratory of ethical responsibility – constitutes the central and unifying theme of this monograph. The contributing authors juxtapose classical urban theories with recent advances in cognitive psychology, cognitive science, neurobiology, digital technologies, algorithmics, and the circular economy, showing that architecture can simultaneously function as a space of events, a platform for authentic social dialogue, a catalyst for innovation, and a practical instrument for fostering sustainable futures.

Now in its fourth edition, the international conference has established itself as a recognized forum for exchanging ideas among researchers, designers, educators, and practitioners worldwide. Each edition broadens perspectives, deepens reflection, and opens new avenues of inquiry – this year with particular emphasis on the social, ethical, educational, and environmental dimensions of architectural experimentation, as reflected in the volume's rich content. The National Institute of Architecture and Urban Planning regards this publication as a significant contribution to national and international discourse on the role of experimentation in 21st-century architecture and urbanism. We firmly believe that only through bold, ethically grounded testing of new solutions – alongside respect for cultural heritage, local context, and human needs – can the quality of public space in Poland be systematically enhanced. This is a long-term process, a task for present and future generations.

It is our hope that "Architecture of Challenges. Experiment in Space – Space of Experiment" will serve as a valuable source of inspiration and practical insight for architects, urban planners, researchers, doctoral students, educators, policymakers, and all those for whom the public realm represents a shared cultural and social asset.

Prof. DSc PhD Eng. Arch. Bolesław Stelmach
Director of National Institute of Architecture and Urban Planning

NARODOWY INSTYTUT
ARCHITEKTURY
I URBANISTYKI

The National Union of Architects of Ukraine welcomes the organizers and participants of the International Scientific Conference "Architecture of Challenges. Experiment in Space – Space of Experiment." and expresses its respect to them.

The events of recent years have clearly shown that for its survival, humanity must experiment, seek new models of cooperation between people in creating its living environment. Architects, by the very essence of their profession, play a leading role in this search. That is why it is so important for architects from different countries to experiment in the context of forming the human habitat. The current experience of Ukraine can be very useful for architects and other people who are professionally concerned with the problems of forming an environment for the safe functioning of civil society.

The experimental space was created during the implementation of the international project UREHERIT with the participation of representatives of architectural communities from 8 countries, in which the Union of Architects of Ukraine is one of the main participants. During the implementation of the project, an opportunity arose to intensify scientific and practical work in 12 thematic areas related to spatial solutions of both individual monuments-buildings and urban planning solutions of large urban formations, such as Dnipro or Mykolaiv.

We are confident that the creation of spaces for joint experiments, implemented by architects from different countries in different fields, will help Ukraine overcome the consequences of the aggressive war and lay a solid foundation for the future reconstruction and development of the country.

PhD Eng. Arch. Oleksandr Chyzhevskyi
President of the National Union of Architects of Ukraine

PhD Eng. Arch. Oleksandr Baranovskyi
Project Coordinator of the National Union of Architects of Ukraine

Open Letter from the Association of Polish Architects (SARP)
on the occasion of the opening of the scientific conference
“Architecture of Challenges.
Experiment in Space – Space of Experiment”

Ladies and Gentlemen,

Twenty-first-century architecture has reached a turning point. On the one hand, it is confronted with the growing complexity of climatic, demographic, technological, economic, and cultural challenges. On the other hand, it now has access to tools that make it possible to transcend the existing boundaries of design, planning, and thinking about space. Within this tension emerges the need for a new language, new methods, and new forms of experimentation.

The conference “Architecture of Challenges. Experiment in Space – Space of Experiment” raises the question of the role of experiment as a cognitive, design, and social tool. We are interested both in experiment in space – that is, the courage to test new formal, material, and technological solutions – and in space of experiment, understood as a system of institutions, procedures, regulations, and practices that may either foster or constrain innovation.
At the center of our reflection lies the conviction that contemporary architecture can no longer remain merely reactive. It must become a laboratory of the future – a place where models of dwelling, mobility, resource sharing, adaptation to climate change, and the relationships between human beings, the environment, and technology are explored. Here, experiment is not an aesthetic gesture, but a method of responsible design under conditions of uncertainty.

Referring directly to the conference theses, Architecture – if viewed through the spirit of Aldo Rossi – is not merely a profession, nor even solely an art form. It is a way of thinking about the world that constantly intertwines the personal with the collective, and the remembered with the yet possible. In “A Scientific Autobiography” Rossi guides us through his own inner city, built from fragments, memories, and afterimages of places that shaped him. And in this very gesture he reveals the most important truth: that architecture is born not from abstract theories, but from the life that exists within us.

In Rossi’s understanding, memory is not merely a record of history. Rather, it is a slow, organic process in which images, forms, and experiences overlap, creating an inner map from which the creator draws inspiration. Each of us carries within us such a city, composed of the streets of childhood, our first fascinations with space, and the buildings that formed us – often without even realizing it. Architecture thus emerges from memory not because it repeats the past, but because the past is the only material we truly possess. Yet memory alone is not enough. Rossi reminds us that architecture requires experiment – not as a spectacular gesture, but as a necessity.

Experiment is an attempt to answer the question of how to transform what we remember into something that may serve the future. It is a risk undertaken by the architect, knowing that every form is a hypothesis and every city a continuously written essay. The Experiment is therefore not the opposite of tradition; it is its continuation in another language.

And finally, imagination – the third element of this triad – which in Rossi’s thought is not fantasy, but the most precise instrument of cognition. Imagination allows us to see what does not yet exist, but what may come into being if we give it form. It is what makes architecture not merely a response to needs, but also a proposition of meaning. Imagination enables us to design not only buildings, but also relationships, communities, and future memories.

Today’s conference is an opportunity to look at architecture in precisely this essayistic and reflective light. To consider how memory – both personal and collective – shapes our design decisions. How experiment may become a tool of responsibility rather than merely innovation. And how imagination may become a space of dialogue between what is permanent and what is changing.

At the same time, today's conference carries a special significance. It is not only a space for reflection on architecture, but also a continuation of the celebrations marking the bicentenary of the Warsaw University of Technology, which falls in 2026. This jubilee honors an institution that for two centuries has shaped successive generations of architects, engineers, and thinkers. In this sense, the conference becomes an important voice in the dialogue on how architecture can build community, solidarity, and social responsibility. It is a moment in which academic tradition meets the challenges of the contemporary world, and institutional memory meets the experiment of thinking about the future.

As the Association of Polish Architects, we believe that architecture is one of the most important ways in which society tells its own story. It is a form of thinking that requires both sensitivity and courage, both rootedness and openness. In the spirit of Rossi, we encourage you to treat architecture not as a collection of technical solutions, but as a continuous essay about the world – one that we write together, each through our own experience, memory, experiment, and imagination.

We wish for this conference to become a space of free reflection, creative exchange, and intellectual attentiveness. May it be a place where architecture, as in Rossi's thought, reveals itself as a story that never ends, because it is written in the rhythm of our lives.

With sincere respect,

The Association of Polish Architects (SARP)

MSc Eng. Arch. Marek Chrobak
President of the Association of Polish Architects (SARP)

ODDZIAŁ WARSZAWSKI
STOWARZYSZENIA ARCHITEKTÓW POLSKICH

Architectural creation is inherently connected with experimentation. Design activities that, under controlled conditions, generate specific phenomena concern both material elements and the spaces between them.

Experimentation with materials enables the application of knowledge of the physical properties of individual elements and entire structural systems through their creative recombination. New solutions create opportunities for novel applications, whether by offering greater cost efficiency or by exploiting their unique characteristics.

Experimentation with space involves the use of spatial arrangements developed by culture and their reconfiguration in unexpected ways. Beyond creating new functional properties, such creative work enables the discovery of entirely new sequences of meaning assigned to parts or objects, allowing them to be reinterpreted and understood anew through creative synergy.

Finally, experimentation also concerns the very reception of an architectural work. When beginning a project, the creator does not know the true result. They can only infer, based on their own experience, what it might look like. Likewise, they cannot know how their work will be perceived by other users, who interpret it through their own experiences.

Thus, the Architects bring together experimenters who shape space – the condition of human existence – through matter, achieving a designed form with a specific semantic message.

The Warsaw Branch of the Association of Polish Architects (OW SARP)

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ODDZIAŁ WARSZAWSKI
STOWARZYSZENIA ARCHITEKTÓW POLSKICH

The fourth edition of the Architecture of Challenges conference creates a platform for dialogue between architects, urbanists, researchers, and practitioners seeking new ways of understanding and shaping the contemporary city. Rooted in the idea of experimentation, the conference presents architecture not as a fixed result, but as a continuous process of exploration - developed through testing, observation, collaboration, and direct engagement with space and human experience.

At Flora Development, this approach closely reflects the philosophy that defines our work. As a “craft developer,” we believe that meaningful architecture is created through attentiveness, responsibility, and the continuous refinement of every stage of the design process. Each project becomes a form of conscious experimentation - an attempt to balance human needs, urban context, material quality, and environmental responsibility. Like the conference itself, we see design as an open dialogue between theory and practice, vision and everyday experience.

The experimental dimension of architecture reflects the challenges of contemporary cities. Creating valuable spaces today requires not only technical precision, but also empathy, interdisciplinary thinking, and the courage to seek new solutions while respecting local identity and community life. Full-scale prototyping, collaborative design processes, and direct interaction with materials and space have become essential tools for understanding how architecture truly performs in reality.

Experimenting in architecture means not only creating new forms, but also boldly reinterpreting what already exists. Without demolition. With respect for the existing structure and the urban context. Flora Development is shaping a new direction in thinking about reuse, demonstrating that projects of this kind can become a new standard for responsible urban design. Reuse can be understood as a form of architectural experimentation - where every building, context, and structure requires a different approach, and no two transformations are ever the same. In this sense, reuse is not a fixed method, but a process of adaptation. It is based on the idea of “user-friendly” architecture, focused on people and their everyday needs. For Flora Development, reuse is not only a conscious and responsible approach to architecture, but also about the courage to seek new solutions and to give existing places entirely new value.

This publication is more than a documentation of the conference. It is an invitation to reflect on the future of architecture as a discipline that evolves through experimentation, collaboration, and conscious engagement with real space. It also reminds us that the most valuable solutions emerge at the intersection of knowledge, experience, and the courage to explore new possibilities

MSc Dominik Różański
President of Flora Development

FLORA

Architecture emerges from memory, experiment, and imaginaion.

In the spirit of Aldo Rossi's thought, inspired by *A Scientific Autobiography* (1981)

Architecture is one of the most complex fields of applied arts. It cannot be reduced merely to shaping form and arranging functions; it also involves testing the relationships between space, action, and the user. In design practice, the human being is always at the center, and every project – to some extent – constitutes an experimental process. When treated as a field of experimentation, architecture and its sphere of influence cease to be only static objects and become spaces of events. In this perspective, experimentation reveals tensions between what is designed and how space is actually used. It may also serve as a tool for examining the boundaries of the discipline, provoking new ways of experiencing and sharing space, and deepening the understanding of reality, which often exceeds material and pragmatic design assumptions.
In the face of climatic and social challenges, it becomes necessary to reinterpret the notion of experiment as a search for balance between technology, ecology, and ethics. In this context, the Vitruvian triad – durability, utility, and beauty – gains renewed relevance, structuring the design process: from testing structural limits (firmitas), through redefining function (utilitas), to the development of aesthetics and symbolism (venustas). Sustainable architecture, operating within a closed loop and based on circular principles, is an example of an ethical experiment that redefines the designer's responsibility. Prototypes, installations, and pavilions – as laboratories of ideas – reveal a distinct relationship between the *experiment in space* and *the space of experiment*. Each architectural experiment functions in dialogue with its context: the landscape, history, and community – not rejecting tradition, but creatively reinterpreting it. Ultimately, the challenge of architecture is to create experiments that serve the human being – spaces that are meaningful, functional, ethical, and rooted in the world.
We invite you to reflect on the idea of Experiment in Space, explored across four thematic blocks:

1. EXPERIMENT IN ARCHITECTURE – MICRO-WORLDS OF INNOVATION

Architecture has, from the very beginning, functioned as a field of experimentation, grounded in the continuous search for new ways of shaping space that respond to the evolving needs of people. Architectural experiment encompasses not only form, but above all the relationships between humans, the environment, and technology – relationships that determine the quality of space.
Pushing the boundaries of materials, structures, and technologies forms the foundation of architectural development. Experimentation with material – its texture, weight, transparency, or durability – opens new design possibilities. Experimentation with form enables the exploration of innovative spatial configurations and the redefinition of relations between volume, light, and landscape. Working with light becomes a tool for creating atmosphere, enhancing perception, and enriching the symbolism of place. Experimenting with function allows us to break traditional programmatic divisions, leading to flexible, hybrid, and multi-purpose spaces that respond to changing models of living and community. Yet the most important architectural experiment remains the act of creating space itself – understood as an attempt to shape behaviours, experiences, and ways of coexistence among users.

2. EXPERIMENT IN URBANISM – LABORATORY: CITY

Experimentation in relation to the city may involve both the process of writing its spatial structure and reading the diverse impacts it generates. These two key approaches must be considered at different scales: entire urban units, selected building complexes, or specific fragments of public urban space. Urban design activities that create model solutions – building layouts, exhibition districts, or prototype functional-spatial systems – act as testing grounds whose results may be replicated and serve as proven answers to similar challenges elsewhere. In a similar way, architectural-urban competitions can be viewed as experimental searches for optimal living environments and well-organized, well-designed shared spaces. Conversely, reading urban space – its impact through form, functioning, and the meaning of hidden cultural codes – requires a rich, multidisciplinary research toolkit. It enables us to confront traditional urban theories with knowledge acquired through contemporary methods in cognitive psychology, cognitive science, and neurobiology.

3. EXPERIMENT IN EDUCATION – THE SPACE OF LEARNING THROUGH ACTION

Design experimentation is a key tool in shaping critical thinking within architectural education. Conscious testing of solutions enables students not only to generate alternative concepts, but also to analyse, question, and redefine existing design paradigms. Experiment reaches its fullest dimension in an interdisciplinary environment where different methodologies, languages, and research tools meet. It is at the intersection of diverse fields of knowledge that the most innovative didactic strategies emerge. Emerging technologies – digital tools, simulations, algorithmic form generation, additive prototyping, and process automation – are transforming the traditional understanding of the design process. Students of architecture can simultaneously explore ideas and test their material feasibility. Experimentation thus becomes not only a teaching method, but a comprehensive educational strategy: a tool for breaking established patterns, deepening the understanding of complex spatial and structural relationships, and fostering a critical and reflective approach to design in a dynamically changing architectural landscape. At the same time, every experimental action should be rooted in ethics and responsible practice – in reflection on the impact of design on the environment, local communities, and cultural heritage.

4. EXPERIMENT IN THE SERVICE OF SOCIETY

The contemporary human living environment – the city – functions as a complex system in which architecture and urban planning integrate inhabitants' needs with ecological, social, and infrastructural challenges while maintaining spatial coherence. Temporary interventions, prototypes, and grassroots initiatives make it possible to test new forms of use and coexistence, providing knowledge that supports design and planning processes. Architecture is not merely a physical object – it is also a platform for dialogue, co-creation of urban practices, and the shaping of social narratives. In this context, experiment becomes a tool for learning, communication, and building shared responsibility. This enables the city to adapt to the changing needs of its inhabitants while preserving its legible structure and the quality of public spaces. The social dimension of experimentation opens a space for reflection, research, and the presentation of new design approaches and new models for co-creating urban environments.

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Balancing Efficiency and Cognitive Flexibility: A Neuroarchitectural Perspective on Experimental Spaces

In the 21st century, architecture has evolved into a fundamentally experimental discipline that explores the interplay between physical environments, human action, and cognition. Beyond merely shaping material forms according to aesthetics, design paradigms dictate how individuals interact with their surroundings and navigate the world. Analyzing the built landscape as an active experimental process often reveals inherent tensions between pragmatic assumptions and actual human usage. This shift necessitates the emergence of neuroarchitecture to evaluate how spatial configurations directly influence cognitive health.

The research problem addresses the dichotomy between navigational efficiency and cognitive flexibility in spatial design. This paper aims to demonstrate how population diversity in spatial memory and resource retrieval strategies, often inherited by humans evolutionarily, should directly inform urban planning. A fundamental paradox emerges: environments designed for maximum efficiency inevitably diminish cognitive flexibility, and vice versa. Therefore, optimal architecture must deliberately balance these extremes to foster both operational efficiency and long-term cognitive health.

Recent neurobiological studies in animal models provide useful insights into how spatial memory networks adapt to environmental demands. Individuals spontaneously adopt different navigation strategies, either allocentric (relying on distal cues) or egocentric (schematic or habitual routes), which differentially engage the hippocampus and cortical areas (Czajkowski et al., 2020). Crucially, spatial information undergoes parallel processing; for example the retrosplenial cortex can establish its own memory trace independent of the hippocampus, specializing in landmark-based, schematic maps (Balcerek et al., 2021). Balcerek et al. (2024) also demonstrated that rigid regimens shift navigational burdens towards cortex-dependent schemata, while unpredictable, flexible environments strongly activate the dorsal hippocampus. This activation is supported by adult neurogenesis in the dentate gyrus, a process that Kim et al. (2022) highlight as a fundamental mechanism for resolving spatial interference and which is severely impaired in Alzheimer's disease. Furthermore, Hamed et al. (2024) established that spatial memory is linked to reward processing and emotional states, demonstrating how spaces become connected with emotional meaning.

Translating these findings to urban planning reveals the cognitive costs of hyper-efficient design. Modern cities prioritize egocentric, schematic navigation through clear, predictable layouts that minimize travel time. While efficient, such designs reduce the environment's cognitive demands, leading to "autopilot" navigation that relies on habit rather than active spatial mapping, bypassing the need for complex parallel processing (Balcerek et al., 2021). Over time, this lack of environmental experimentation diminishes cognitive flexibility.

Conversely, complex or ambiguous architectural configurations force users to rely on allocentric mapping, actively engaging the hippocampus to update spatial associations. Because adult hippocampal neurogenesis supports the separation of similar spatial patterns and its decline accelerates dementia symptoms (Kim et al., 2022), navigating challenging surroundings utilizes and sustains this neural plasticity. While this builds cognitive resilience, essential for safeguarding against the atrophy observed in dementia, it reduces functional efficiency. Additionally, the emotional valence of a space dynamically integrates with memory consolidation and updating (Hamed et al., 2024). Purely efficient spaces may fail to trigger reward-related memory traces necessary for psychological well-being.

Architecture must operate as a dynamic space of experiment rather than static functionality. To support cognitive health, designers must embrace the paradox of efficiency versus flexibility. By integrating predictable functional axes with complex, discovery-oriented zones, urban planning can stimulate the complementary neural circuits responsible for spatial navigation, cultivating environments that are practically efficient and cognitively enriching.

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Bauhaus in the Context of Contemporary Art Education

Every human being reacts to sounds and colors, is secure in tactile exploration of the surrounding space, etc. This means that initially every human being can participate in all the joys of sensory and sensuous experience. Laszlo Moholy-Nagy

The subject of our research is the relevance of Bauhaus pedagogy in contemporary art education. Our point of departure is the conviction that the fundamental principles of the Bauhaus Preliminary Course grounded in sensory experience, experimentation, and direct engagement with materials remain significant today, although they require reinterpretation within changing technological and cultural contexts. Our research examines selected original exercises developed at the Bauhaus and reflects on their relevance within contemporary art education. By revisiting these foundational pedagogical methods, we consider their potential for reinterpretation and adaptation in present-day teaching and artistic practice. The Bauhaus Preliminary Course, designed to foster experimentation with materials, form, and perception, continues to serve as an important reference point for many art and design programs, while simultaneously calling for critical reassessment.

The current state of research indicates that Bauhaus pedagogy, particularly in the work of László Moholy-Nagy, Johannes Itten, and Paul Klee, was oriented toward the development of perception through multisensory experience. Their approaches emphasized direct engagement with materials, sensory awareness, and the integration of intuition and analysis. Contemporary discourse in art education increasingly returns to these ideas, highlighting learning by doing, experimentation, and the processual nature of artistic practice, while also expanding them toward digital, auditory, and haptic dimensions.

Our research is grounded in the activities of the Intermedia Communication Studio. At the undergraduate level, the studio operates through the conceptual triad of the "3M": medium, message, and mediation, encouraging students to explore relationships between forms of expression and processes of meaning production. At the graduate level, the focus shifts toward the "3T": transgression, transformation, and transmedia, inviting students to question boundaries and engage in processes of change across media.

Within this framework, both the re-enactment and reinterpretation of selected Bauhaus exercises are undertaken, alongside the development of new experimental tasks rooted in sensory exploration. In the studio, a significant number of exercises focus on tactile experimentation exploring materials, textures, and space through direct physical contact, supporting the development of embodied awareness.
Equally important is the engagement with sound. Students work with listening practices focused on soundscapes and the audiosphere, cultivating attentive perception of acoustic environments and transforming auditory experience into artistic action. This reflects a broader shift toward multisensory strategies in contemporary art education, extending beyond the traditionally dominant visual paradigm.
The program emphasizes not only the realization of concepts but also the possibility of abandoning them in favor of intuitive action and improvisation. Students are encouraged to develop individual tools and modes of expression, enabling cross-sensory translation between media. An important component of our research is the experience gained during participation in the Open Studios program in Dessau, allowing engagement with Bauhaus pedagical models in situ and in dialogue with contemporary perspectives.

Ultimately, our research asks whether Bauhaus exercises can still function as effective tools in contemporary art education. Rather than offering definitive conclusions, it proposes a reflective framework grounded in practice and critical engagement, treating Bauhaus pedagogy as an open and evolving structure.

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Przemo Łukasik (born 1970 in Chorzów) is a co-founder of Medusa Group, an interdisciplinary design studio established in 1997 together with Łukasz Zagała. The practice works across architecture, urban planning, industrial design and visual communication. A graduate of the Silesian University of Technology in Gliwice, he also studied at the École d'Architecture Paris-Villemin in Paris and gained professional experience in Berlin and in the Parisian offices of Jean Nouvel Architecture and Odile Decq. He has served as a visiting professor at the École Spéciale d'Architecture in Paris.
Medusa Group has received numerous awards and distinctions, including the SARP Honorary Award 2024, three Grand Prix titles in the Architecture of the Year of the Silesian Voivodeship competition, the Architectural Award of the President of Warsaw, as well as nominations for the Mies van der Rohe Award and the Leonardo Award. Przemo Łukasik maintains a deep connection with the Silesian region. Outside architecture, he is an avid photographer and sportsman.

The Architecture of a Second Life: Szyb Krystyna, Szombierki Power Plant, and the Provocative Architect in the Moratorium on New Construction

In the face of today's multiple crises – climatic, housing, economic, and cultural – the idea of a "Moratorium on New Construction," proposed by Charlotte Malterre-Barthes, takes on particular significance. The French researcher and architect calls for a radical halt to the construction of new buildings in favor ting urban and industrial fabric. In this context, Silesian pragmatism, rooted in the region's industrial tradition, evolves into a radical, provocative design methodology.

Upper Silesia, with its rich yet often neglected post-industrial legacy, is becoming an ideal laboratory for such experiments. An analysis of two key installations – "Bytom Lighthouse" at the Szombierki Power Plant and "Krystyna's Beating Heart" at the Krystyna Shaft – clearly shows that when standard planning and investment procedures fail, the architect must go beyond the traditional role of designer and assume the function of negotiator, mediator, and social provocateur. Both site-specific interventions are realized in monumental, historic post-industrial structures. The Szombierki Power Plant in Bytom, a massive brick cathedral of the former energy sector, is transformed through an artistic and architectural intervention into the "Bytom Lighthouse" – a dynamic event space that attracts residents, tourists, and decision-makers. Meanwhile, the Krystyna Shaft, an iconic mining tower, comes to life as the "Beating Heart of Krystyna," becoming a vibrant hub of cultural and social activity. In both cases, dormant industrial structures – seemingly doomed to decay – are transformed into vibrant "event spaces" that generate new meanings and functions.
The key tool for this transformation is the "architectural pitch deck" strategy. Instead of a traditional construction project, the architect prepares a series of provocative, highly communicative temporary interventions to build attention capital around the site. Through spectacular, though often low-budget, actions, the site's potential is demonstrated, and various stakeholder groups are mobilized – from local communities and local governments to potential investors and grant providers. This approach radically redefines the architect's responsibility. The architect ceases to be a creator of new forms and volumes and instead becomes a "life-giver" – the initiator of a revitalizing infusion into the existing, often-forgotten urban and industrial fabric. Instead of consuming more land and materials, they focus on activating what already exists. This is not only economically and ecologically rational but, above all, an ethical approach to the profession. The entire process can be viewed as an ethical experiment. By building a capital of attention around post-industrial heritage, the architect makes a precise diagnosis of its cultural, social, and economic potential. He demonstrates that adaptive reuse is not an alternative, but the only rational and responsible path of development in an era of a climate crisis and limited resources.

Examples of "second-life architecture" in Silesia demonstrate that, during a moratorium on new construction, the creativity and determination of the architect-provocateur can become the driving force behind a genuine, sustainable transformation of space. This is not merely a survival strategy for post-industrial regions, but a model that can be successfully implemented on a global scale.

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Daan Roggeveen is a Dutch architect, urban researcher, and writer whose work explores the relationship between architecture, urbanization, and collective forms of city-making. He is the founder and creative director of MORE Architecture, a progressive international design agency operating across architecture, urban strategy, and spatial research. Through MORE, Roggeveen has developed an internationally recognized approach to urbanism combining participatory planning, contextual design, and interdisciplinary research to address density, inclusivity, and urban space.

Educated at TU Delft, Roggeveen has worked extensively between Europe, China, and Africa, focusing on the social and spatial consequences of rapid urban transformation. His projects range from urban masterplans and mixed-use developments to cultural and residential architecture, emphasizing adaptive, human-centered environments. Under his leadership, MORE Architecture has been included in the AD100 list of China's most influential design firms. Alongside his architectural practice, Roggeveen is active as a researcher. Together with Michiel Hulshof, he co-founded the Go West Project, investigating emerging urban conditions in China and Africa. Their research resulted in the acclaimed book How the City Moved to Mr Sun: China's New Megacities and exhibitions at the Louisiana Museum of Modern Art, the Shenzhen Biennale, and Storefront for Art and Architecture. He is the editor of Progress & Prosperity – the Chinese City as Global Urban Model and The Amsterdam Agenda – 12 Good Ideas for the Future of Cities. He writes columns on spatial development and hosts a podcast on high-rise and densification. A frequent teacher and lecturer at international universities, his expertise is regularly sought for debates and analyses on urban development, densification, and beyond.

Urbanism as a Laboratory: Experimental Approach to Inner-city Densification

Cities across Europe face housing shortages (Economist, 2025), and therefore the need for inner-city densification. This creates an opportunity to increase the diversity and dynamics of their urban frameworks (Roggeveen, 2018). The question is: how to plan for these dynamic conditions?

Traditional urban planning often relies on rigid masterplans that attempt to predict the future of a city in detail. However, contemporary urban conditions are increasingly complex and unpredictable. Complex ownership, historical heritage, red tape, conflicting stakeholder interests, rising construction costs and infrastructural limitations make it difficult to apply fixed solutions. Instead, cities should adopt a more adaptive and experimental approach to planning, one that allows urban plans to evolve over time rather than remain static.

This approach can be compared to jazz music: when the rhythm and the beat are clear, individual musicians can freely improvise and excel. In urbanism, this means establishing clear guiding principles while leaving room for flexibility and innovation. Instead of producing a final and rigid design for the city, we propose to create frameworks that can respond to future social, economic, and ecological changes. The next step is to fill these frameworks using a "laboratory" approach. A laboratory is a controlled environment where researchers can do experiments to test hypotheses or develop new products. It stems from the Latin word "labor", meaning work – thus, a workshop. In our practice, we use a similarly controlled workshop environment to work with groups of stakeholders on experiments in urban design.

Our method uses interactive design workshops as a medium for experiments and generating breakthroughs in urban planning. These workshops bring together architects, planners, municipalities, and local stakeholders to collectively shape urban proposals. Through dialogue and iterative design processes, participants can experiment and test ideas, respond to constraints, refine strategies in real time, and create high-quality output.

Two projects demonstrate the effectiveness of this approach. The first is the redevelopment of Buikslotermeerplein in Amsterdam, where a deteriorated shopping centre is being transformed into a vibrant and diverse urban centre (Stadszaken, 2024). Rather than presenting a single fixed vision for the area, the urban plan is developed as a concise set of "game rules" that guide development while still allowing adaptation over time. These game rules are developed in close collaboration with the municipality and the 80 (!) landowners in the area. This approach creates broad support for the plan, which is a more resilient and flexible urban structure capable of responding to future adaptation. In workshops, individual parts of the framework are developed in more detail, in close collaboration with stakeholders and their architects.
The second example is the "Tower in a Factory" project in Rotterdam's Rijnhaven district. In collaboration with municipal urban planners and local stakeholders, the building design – by MORE Architecture and Studio Nauta - evolved through a series of design workshops and discussions. The final design respects the existing industrial heritage while introducing a sustainable and flexible tower with around 400 new housing units into the city. The project illustrates how collaborative experimentation can lead to solutions that both preserve urban identity and address urgent housing demands.

Ultimately, the concept of "urbanism as a laboratory" redefines city planning as an ongoing and participatory process. By embracing experimentation, flexibility, and collaboration, cities can better respond to the complexities of contemporary urban growth.

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Since 2017, he has served as Head of the Chair of Architectural and Sculptural Design at the Academy of Fine Arts in Kraków. He was Vice-Rector for Cooperation and Internationalization from 2020 to 2024 and was reappointed for the 2024–2028 term. Between 2015 and 2020, he was a Visiting Professor at the Hubei Institute of Fine Arts. He also completed a teaching fellowship at the Zürcher Hochschule der Künste and participated in the Academic Leadership Program at the University of Cambridge. Since 2022, he has been affiliated with Florida Atlantic University as a research scholar.

Professor Struzik has lectured and participated in symposia at universities and institutions across Europe, Australia, Asia, and the United States. He is a member of the Steering Committee of the Transatlantic Dialogue Conference, the Society for Somaesthetics, and the European Cultural Parliament. He is also the initiator and Chair of the Somaesthetics and the Arts Center.

An award-winning artist, particularly recognized in Japan, Professor Struzik has received distinctions including the Kajima Sculpture–Architecture–Space Competition, the Kanazawa Machinaka Sculpture Competition, the Sakaide Art Grand Prix, the Kobe Biennale, and Wa-no Contemporary.

Somactive Art Practice: Embodied Attention as a Methodology for Artistic Research

The problem of overstimulation and the dominance of visual culture is not new. For decades, philosophers, psychologists, and neuroscientist have examined the fragmentation of attention and the gradual erosion of embodied forms of experience. Today, however, this condition appears to be reaching a critical and potentially dangerous stage. The acceleration of image-based communication, algorithmic distraction, and increasingly immersive technologically mediated environments—including the growing appeal of virtual reality—progressively isolate individuals not only from one another, but, more fundamentally, from their own sensory and bodily awareness. If contemporary culture fails to develop methods capable of restoring embodied attention, the consequences may extend beyond aesthetics into broader forms of social alienation, perceptual passivity, and the degradation of human experience itself.

My project, Somactive Art Practice, emerges as a direct response to this condition. It develops from both my long-term experience as a sculptor, selected Far Eastern traditions and the somaesthetics of Richard Shusterman. Working with materials such as clay, stone, wood, metal, and plaster—often in direct relation to changing environmental and open-air conditions—has fundamentally shaped my understanding of perception as an embodied, multisensory, and processual phenomenon. Sculptural practice requires continuous negotiation between the body and material resistance, weight, texture, gravity, movement, spatial orientation, and environmental conditions. Knowledge within sculpture is therefore not produced exclusively through vision, but through tactile engagement, kinesthetic awareness, rhythm, pressure, smell, sound, duration, and conceptual reflection.

This understanding emerged through years of material practice, but was also profoundly influenced by my early journeys to Japan and encounters with Zen philosophy, as well as later collaboration with Richard Shusterman. The synthesis of these experiences led me to recognize artistic practice as a form of embodied inquiry capable of generating both sensory and conceptual knowledge.

Based on this perspective, I propose Somactive Art Practice as a structured methodology for artistic research in which embodied attention functions as an operational condition of creative experimentation. Drawing from somaesthetics, embodied cognition, and selected traditions of mindful attention, the project seeks to transform embodiment from a primarily descriptive concept into a practical and research-oriented framework.

Somactive Art Practice is organized through six experimental modules focused on atmosphere, haptic perception, embodied scale, kinesthetic space, somatic memory, and temporal experience. Together, these modules investigate how bodily awareness shapes artistic production, design processes, and spatial understanding. Rather than functioning as prescriptive exercises, they establish open experimental situations in which perception and creativity can be examined through direct embodied experience. Importantly, this six-module structure is not conceived as a closed system, but as an adaptable framework capable of further development and refinement.

Through Somactive Art Practice, I aim to develop a transferable methodology capable of responding to the growing crisis of sensory overstimulation, perceptual disembodiment, and cognitive fragmentation. Ultimately, I argue that embodied attentiveness is not supplementary to artistic and design practice, but fundamental to how human beings perceive, think, create, exist, and relate to themselves, to others, and to the world around them.

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The Countryside as a Testing Ground for Architectural Experimentation

For centuries, the city has been the primary focus of architectural interest, the subject of experiments, providing answers to the most pressing questions. Against this backdrop, rural architecture is seen as conservative, repetitive and less interesting. An analysis of source materials and field observations in the context of Polish and European architecture suggests that rural areas are undergoing dynamic changes. This raises pressing questions that architects need to address.

The shift in the character from agricultural to residential is giving rise to a conflict that provides a significant context for design. Dynamic changes linked to economic development, environmental modernisation, social needs are leading to typological experimentation. Rural development is criticised for a lack of economic sustainability (Śleszyński, Kukołowicz, 2021). Detached housing, which dominates outside cities, is seen as a sub-optimal solution. Block flats evoke negative associations with the communist era; however, economic and environmental considerations argue in favour of a return to the multi-family housing concept. There are arguments suggesting that rural areas should not be viewed as lagging behind urban areas – for they may play a key role in the future sustainable development, food production, environmental protection, quality of life (Thorbeck, 2026). Design in rural areas offers potential for design exploration: architecture is characterised by a small scale, a natural setting. Buildings are subject to fewer regulations. The economic micro-scale allows for experimentation with materials, form, aesthetics and space, as well as the use of community participation and non-professional building techniques.

The new generation of projects reflects a drive towards personalised, innovative solutions. The “Farma toMy” project (Arche Group) demonstrates that rural architecture can serve as a testing ground for spatial and social experiments beyond the scope of detached houses. Robert Konieczny is conducting experiments, proposing a new typology of rural detached housing (Jodidio, Konieczny 2024). The “Dom Baby Jagi” project (Pole Architekci) highlights the potential for repetition and standardisation that has characterised rural architecture for centuries.

A set of questions is a base for experimentation: What does life in the countryside mean today – low costs, proximity to nature, the landscape, idyll and tradition, or distance from work and social isolation? How can we consciously harness local cultural and natural conditions, as well as conflicts of interest, to create a vision of the rural home of the future? What might a sustainable and accessible alternative to city life look like, one that is needed in the context of demographic challenges?

The objectives of the study are as follows: the conflict between the need to protect the landscape, agricultural land and local identity, and the pressures of development and investment, presents a serious challenge that may lead either to spatial chaos or to innovative typological solutions. It is of key importance to identify the dynamic mechanisms influencing rural architecture. It is proposed that a critical analysis be carried out of selected Polish projects, taking into account criteria such as: location in relation to urban centres, access to technical and transport infrastructure, natural and social conditions, local building traditions and their impact on the innovativeness of specific functional, spatial and organisational solutions (layout typology, architectural form, choice of materials, method of execution, details). The point of reference will be field research conducted in Switzerland and planned in Belgium as part of a research and educational project.

The experiment in urban residential architecture focuses on the search for “blue-green” ecological solutions that ensure privacy and a human scale within a dense, urbanised structure. In a rural context, however, the experiments focus on the development, investment and intensification of the built environment whilst preserving the qualities of the surroundings.

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Mycelium Morphologies Seminar as an Experiment in Teaching Future-Proof Design Skills

Large-scale Design-and-Build and Design-through-Research (RtD) courses have become a vital part of architectural education in recent years. They are additionally driven by the availability of digital fabrication infrastructure and by the proximity of material research conducted within architectural schools. In the German context, funded research projects in the development of mycelium materials for the built environment are usually driven by a rigid framework, decided years in advance during the grant application phase. Introducing research-related questions within the context of elective seminars with design students offers twofold benefits: it gives students insight into emerging areas of architectural materiality, and it gives researchers insight into the unexpected findings explored through RtD by the students.

The key challenge of conducting teaching units in connection with mycofabrication lies in the timeframe: design students need to understand the material and its constraints during all phases of cultivation to be able to tackle them creatively during the design process.

The Mycelium Morphologies seminar was conducted as a two-week block in March 2025 in the Future Material Laboratory of the University of Stuttgart. The seminar was attended by an interdisciplinary group of students of architecture, interior design and media technology from three involved schools (University of Stuttgart, HFT Stuttgart and TH Deggendorf). The goal of the seminar was to design and biofabricate the “shadow making” spatial installation for Salone Verde gallery, planning an appearance parallel to the Venice Architecture Biennale 2025, associated with the German national pavilion (Stresstest).

The know-how and the foundation for the seminar were drawn from previous works in mycofabrication and intentional textile hybridisation conducted in FuMaLab, both in framework of Fungilation (Biala, 2025) research project and the supervised master’s thesis (Amudhan, 2025). It restrained the design space to the solidification of elastic textiles by the growth of mycelium material.

The seminar positioned itself as an experiment in collective, integrative work by groups of students, which had to cross-inform one another on the findings from their iterative work on different aspects of the exhibition. The groups were: development of mycofabrication protocols (1), development of geometries of discrete modules by hands-on prototypes making (2), digital design og global geometry (3), development of hanging systems (4) and lightning scenarios (5).

The main challenge for the tutors was to moderate the information flow between the groups during the daily exchange sessions, where the installation’s design space was constantly redefined by each group’s findings. As the seminar also operated as the Design and Build endeavour, the tangible materiality of the available cultivation and drying chambers, pre-grown mycomaterials, textiles, and transportation requirements and costs, as well as the short timeframe, had to be taken into account.

The overall “experiment” of the seminar can be considered a great success. Within the allocated time, the fabrication protocol for doubly curved modules and their geometry was developed (Biala, 2025), and half of the targeted modules were fabricated. The subsequent steps needed to finalise the whole installation were taken over by another groups of students and institute’s employees and resulted with opening of the installation on time in Venice (Keep cool Salone Verde). Students reported high satisfaction from this unitary experience of working with emerging material, collective spirit and extreme exercise in interdisciplinary communication.

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Experiment as the Acceptance of Error: An Iterative Model of Design Education within the Framework of the "Space of Experience"

The Paradigm of Infallibility vs. Design Reality

Contemporary higher education in design and architecture faces a significant mental barrier at the threshold of professional training. The secondary education system, rooted in rewarding correct answers and discrediting mistakes, fosters an inherent fear of failure. However, the maxim of Seneca the Elder, "errare humanum est" (to err is human), constitutes the foundation of not only a philosophical truth but also a fundamental design methodology. Error is the most vital element of the pedagogical process, and its acceptance is a prerequisite for any authentic experiment.

Process as Iteration

Designing is not a linear pursuit of a goal but a complex iteration. Every proposed solution is subjected to verification, and the conclusions drawn - often derived from failure - become the basis for subsequent modifications.

In this light, error ceases to be an undesirable occurrence and becomes a research method. It allows for the precise mapping of areas to be avoided and identifies where the potential for true innovation lies. For a student of industrial design or architecture, an error occurring during the manipulation of material, light, or structure is the first signal of a genuine dialogue with the design problem.

Error as a Tool for Critical Thinking

Cultivating a sense of acceptance for error is a key task for the educator. It allows the young creator to fully "experience the process" rather than focusing solely on the aestheticization of the final result. Working with error teaches critical thinking - the ability to question one's own assumptions and ask the questions essential to creative processes.

Experimentation reveals tensions between what is designed and how space is actually used. It is precisely the moment when a solution fails that becomes most cognitively valuable. It allows for a deeper understanding of the essence of the medium and the specific needs of the user, which often exceed pragmatic assumptions.

Interdisciplinarity and the Ethics of Experiment

The educational experiment reaches its full dimension in an interdisciplinary environment. An error made within a team of diverse competencies allows for the confrontation of different perspectives and "design languages." In an era of climatic and social challenges, this approach takes on an ethical dimension. Sustainable design, based on circular principles, requires the courage to test risky solutions. Prototyping, as a laboratory phase of the "space of experiment," is de facto a proving ground for controlled errors.

Conclusion

Design education is a continuous experiment - a constant confrontation with the resistance of material and one's own cognitive limitations. For a future designer to create spaces and objects that are meaningful, functional, and ethical, they must first be granted the right to fail. It is through the lens of the mistake that the relationship between "experiment in space" and the "space of experiment" is most fully revealed. Ultimately, it is not infallibility, but the capacity for the creative interpretation of error, that defines the quality of architecture and design in the service of humanity.

Keywords:
design education, error as method, iterative process, design experiment, design, design process.

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Designing an Inclusive Café as a Didactic Experiment: Ordering Systems Supporting Employees

Addressing non-standard user needs and fostering collaboration between educational units is an important component of contemporary architectural education, as it develops sensitivity to users and practical design skills. In the summer semester of the 2025/26 academic year, students of the English-language Interior Design course at the Faculty of Architecture, Warsaw University of Technology, worked on the topic Exceptional Zone – a multifunctional café with an experimental program for graduates of a special education school. The classes were led by Dr Anita Orchowska and Dr Agnieszka Chudzińska.

The facility, planned for Chyliczkowska Street in Piaseczno, will employ graduates of the Special School Complex in Pęchery as service staff. The aim is to support their transition to independence and financial self-sufficiency. The project also has the potential to function as a space for social integration, hosting meetings and small events.

The educational objective was to incorporate inclusive design and address the needs of people with intellectual disabilities. The design experiment shifted focus from customers to café employees. Emphasis was placed on staff comfort and on developing a service system adapted to their capabilities. This approach follows ergonomic principles, including clear affordances, legible signifiers, and intuitive mapping, supporting orientation and decision-making.

The study was based on a didactic experiment involving 24 architecture students. The design process was iterative: initial concepts were developed, tested through analysis and consultations, and then refined in subsequent cycles. Each iteration increased in detail—from general spatial layouts to precise operational solutions. This process led to a set of strategies supporting employees, including simplified spatial organization, reduced sensory stimuli, clear zoning, and the integration of analogue service-support systems.

The results show that the ordering system, closely integrated with the spatial design, is a key component supporting employees with intellectual disabilities. Proposed solutions included pictogram-based menus, colour-coded systems, and limited product sets. Physical communication tools—such as order cards, table markers, and movable elements indicating readiness to order—were equally important, along with clearly defined service procedures. These measures reduced reliance on verbal communication, structured workflows, and supported repetitive task performance.

The findings suggest that architecture can function as a research tool exploring relationships between space, user, and work organization. Beyond accessibility, clarity and predictability of the environment are critical. Integrating spatial layout with service systems reduces cognitive load and simplifies interactions. The most effective solutions relied on limiting decision-making and maintaining clear spatial structures, while overly complex layouts led to disorientation.

The conclusions indicate that inclusive design in service environments requires an integrated approach combining spatial design, communication, and organization of work. Simple, analogue systems embedded within a clear spatial framework can significantly improve functionality and working conditions.

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Architectural Narrative as a Strategy for Addressing Challenges in Contemporary Urban Space

Zygmunt Bauman's analysis remains relevant, as he identified three types of postmodern "pathological" urban spaces: anthropoemic space ("expelling" others), anthropophagic space (based on forced assimilation), and "non-places" (devoid of distinct identity) (Bauman, 2006). Considered alongside Rem Koolhaas's concept of junk space and Manuel Castells' reflections, these frameworks illuminate the problems of contemporary urban environments. The research presented here aims to demonstrate that one method of eliminating their pathological effects lies in the creation of certain types of architectural narrative.

Based on observations and desk studies of selected case studies of contemporary buildings, model solutions in this field were identified, as well as individual architects' design methods. These "strategies" were deduced from the authors' project descriptions. The group of analyzed projects includes, among others, buildings by Bjarke Ingels Group (BIG), MVRDV, OMA, Diller Scofidio + Renfro, SelgasCano, Snøhetta, and UNStudio.

The study identified the following methods used by architects: (1) integrating the urban fabric into the building structure; creating buildings as (2) functional "hybrids" (Gyurkovich, 2013) or (3) "landscape elements"; and (4) using local tradition as a driving force of the concept. In the first two strategies, buildings that deeply incorporate urban spaces into internal structures, combined with additional functions (1) and new types of activities (2), offer opportunities for use and perception at different times of day and through diverse spatial practices, thereby activating a place. Examples include the Museum of Image and Sound in Rio de Janeiro (Diller Scofidio + Renfro, 2026), and CopenHill in Copenhagen (BIG, 2017). These approaches can be intepreted through Lefebvre's triad of the production of space, which frames urban space as a dynamic process in which everyday use is linked to cultural experience, namely, the reception of narrative (Lefevbre, 1992).

In turn, (3) transforming architectural structures into landscape elements (including elements of the social landscape) demonstrates a form of symbiotic integration between built form, context and history, fostering public awareness and engagement with the arts. Examples include the Delft University of Technology Library (Mecanoo, 1997) and the Oslo Opera House (Snøhetta, 2008). Meanwhile, solution (4) situating buildings as elements of the cultural landscape, shaped not only by physical surroundings but also by local tradition enables to reflect local character in architectural forms. Buildings become embedded in the sphere of imagination, symbolism, and the lived experience of residents. This occurs both through architects' adaptation to the site's rhythmanalysis and participatory design processes involving local communities, as seen in the Vistas de Cerro Grande Community Center in Chihuahua (Arquitectura en Proceso, 2011) and the Ofunato Civic Center and Library (Chiaki Arai Urban and Architecture Design, 2008).

The research demonstrates that these strategies lead to the introduction of narrative into architecture. However, narrative is not understood here as transmission of fixed symbolic meanings, but creating buildings in which users are guided through sequences of spaces, like readers of a story. The "events" created by architects influence the users' mood, much like the experience of reading a literary novel. Such an approach presents spatial narrative as a cognitive process of audience engagement, confirming Austin's theory (Austin, 2015). The study concludes that neither a specific architectural style nor the aesthetic values of space determine its "successful" functioning, but rather the act of "saturating" space with events, creating diverse encounters and experiences.

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User-Driven Reconfiguration of Public Space Through Movable Urban Furniture

Movable and reconfigurable urban furniture is increasingly explored as an approach flexibility into public space, enabling users to actively shape spatial arrangements and transforming public space from a static composition into a dynamic system through everyday use (Chew et al., 2021).

Research on movable furniture demonstrates its impact on spatial behavior. Studies in urban parks show that mobile seating enables the formation of "dynamic edges," where users cluster and engage in group activities, adapting configurations to group size. Similar effects have been observed in street interventions, where movable elements contribute to traffic calming, comfort and social interaction (Loo & Fan, 2023). A widely recognized example is the seating system Enzi seating MQ Vienna in Vienna's MuseumsQuartier (Enzo, n.d.). The ability to freely rearrange these elements leads to temporary spatial configurations, illustrating the role of users as co-creators of public space.

Complementary insights come from studies conducted in shared indoor environments, where numerical methods have been applied to analyze the use of movable furniture (Masaki et al., 2024). These studies demonstrate a strong relationship between the ability to rearrange space and the sense of autonomy. Spatial layouts are reorganized in response to group size, and territorial negotiation. Although focused on interior spaces, these findings reveal mechanisms that may be considered analogous in public space.

From a design perspective, flexibility and modularity are key principles in shaping contemporary urban furniture. Modular systems enable multiple spatial scenarios and support diverse patterns of use (Gulzar et al., 2025), while experimental and data-driven approaches, such as interactive installations and behavioral observation, expand the methods used to study user interaction in space (Chew et al., 2021).

These considerations regarding spatial variability and potential use scenarios were explored within the ROBOstudio course at the Faculty of Architecture, Warsaw University of Technology (ASK WUT). The project, developed by a team of six architecture students (Omotara Adebayo, Parisa Ebrahimi, Somia Hassan, Liudmila Rudz, Ewa Sokołowska, Philip Sule), focused on kinetic and modular urban furniture for a campus environment.

The design process followed a research by design approach (Roggema, 2017; Fraser, 2013), using the project itself as a tool for exploring potential spatial scenarios. An initial stage involved embodied scenario testing, in which students arranged chairs according to predefined use cases, such as individual work, group interaction, and informal gathering. This exercise revealed preferred spatial relationships, including distances and orientations.

Based on a site analysis of a campus square, the team developed a kinetic bench system composed of two overlapping layers capable of rotational movement around a fixed axis. This configuration allows for multiple spatial arrangements, ranging from linear seating to separated units and integrated table surfaces. A modular core enables further functional extensions.
The process included digital modeling and physical prototyping, allowing the verification of movement ranges and the identification of geometric conflicts. These tests revealed constraints related to spatial requirements and mechanical interactions, which significantly influenced the final design.

Although not implemented in a real environment, the experiment demonstrates how spatial variability can be explored through design as a means of anticipating potential user behavior. It suggests that the ability to reconfigure space alone does not guarantee user engagement; transformation must be intuitive and legible. In this sense, flexibility is not merely a formal property but a process negotiated between design and user.

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Experimentation in Education: MINIFORMA Student Workshops as Learning by Design (Editions 2023–2024)

Learning by design is an effective pedagogical framework for developing design competencies in architectural education. This article examines two editions of the MINIFORMA student workshops (2023–2024), co organized by SARP O. Wrocław and the Faculty of Architecture at Wrocław University of Science and Technology, addressed to first year (1st cycle) students. The aim is to analyse the impact of design build workshops on practical skills, critical thinking, creative agency, and environmental responsibility. The study frames learning by design as an integrated pedagogy combining problem based learning (PBL) and design thinking (DT), operationalized through targeted teaching aids, iterative prototyping, user centred inquiry, collaborative problem solving, and structured tools and templates within studio practice.

Theoretical foundations draw on experiential learning that posits knowledge formation through grasping and transforming experience (Kolb, 1984). Contemporary debates on experimentation in architecture and design pedagogy emphasise problem definition, prototyping, interdisciplinarity, emergent technologies, and testing as catalysts for pedagogical innovation (Kolhe, 2017; Nyka et al., 2020). MINIFORMA implemented these principles through modules comprising conceptual design, construction based prototyping, analogue and digital presentation, and an intensive work accompanied by documentation.

The research method is a qualitative case study of two consecutive workshop editions. First year students from the Faculty of Architecture at WUST responded to concise briefs for urban furniture in Wrocław. The 2023 brief concerned a seat incorporating bookcrossing elements; the 2024 brief concerned a recreational form. Stage I comprised intensive design sessions with presentations and consultations involving practitioner tutors and external critics representing potential user groups. Stage II required construction of full scale 1:1 prototypes from ephemeral materials chosen by participants (paper, cardboard, plywood, plastic), executed by student teams and culminating in final presentations and a public vernissage followed by an exhibition. Stage III envisaged selective realization in durable materials, with candidates chosen through self assessment surveys, discussion, and voting by participants and tutors.

In 2023 twenty seven students produced five prototypes and in 2024 fifteen students developed three prototypes. Students formed groups and independently assigned names to their projects, reinforcing ownership and project identity.

Analysis identified three principal effects. First, 1:1/1:10 prototyping reduced the distance between concept and execution, heightening awareness of structural and material constraints and enabling rapid testing of alternatives. Second, tutor involvement improved decision making and facilitated realistic selection of solutions. Third, iterative development and public presentation cultivated critical attitudes: students integrated accessibility, local context, material durability, and community impact into evaluative criteria. Workshops also functioned as sites of social experimentation, prompting students to survey urban sites, identify intervention potential, and test activation strategies.

Sequential stages and selective realization enhanced spatial understanding, scale perception, and contextual adaptation. By combining PBL and DT within a learning by design framework, students produced inventive, contextually sensitive solutions and demonstrated notable design maturity and engagement. MINIFORMA confirms that design build experimentation effectively integrates technical skill acquisition with ethical and social reflection. The study recommends systematic inclusion of 1:1/1:10 prototyping in early architectural curricula, supported by interdisciplinary training, technical pedagogical resources, and rigorous documentation to enable longitudinal evaluation and curricular refinement.

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Exploratory Acoustic Simulations of Underground Passage Geometries for Blind Echolocators

Blind travelers use multiple sensory strategies to orient in public space, among them echolocation (Thaler et al., 2020). Research has shown that click-based echolocation can support obstacle detection, spatial judgment, adaptive walking, and the formation of mental maps, especially in trained users (Norman et al., 2021). At the same time, the architecture-focused literature remains limited. While simulation and virtual navigation studies suggest that acoustic structure matters (Andrade et al., 2022), validated design rules for architectural spaces that support blind echolocators remain sparse (Bujacz et al., 2023). In particular, underground passages pose a relevant but underexplored case: they are common elements of transport infrastructure, often acoustically reflective, geometrically complex, and potentially disorienting even for sighted users. The aim of this research is therefore to explore the geometric forms of underground passageways and to simulate their influence on acoustic characteristics, identifying configurations that may generate acoustically legible changes relevant to blind echolocators.

The study uses Rhino software with the Pachyderm Acoustics plug-in to simulate a simplified "echolocator" moving through ten underground passage typologies. The model includes a sound source and two binaural receivers separated by 20 cm, with analysis focused on selected room-acoustic indicators, including EDT, T-30, clarity, definition, center time, and strength. The tested geometries include changes in corridor width and height, stairs, wall and ceiling deformations, simple intersections, forks, access paths, and galleries. The simulations assume highly reflective surfaces and a predefined exploration path representing the movement of a blind traveler through the modeled spaces. The method is intended as a comparative and exploratory tool.

The results suggest that some spatial transitions may be more acoustically legible than others. Width changes generated clearer and earlier shifts across several indicators than moderate height changes, suggesting that lateral expansion or contraction may offer stronger cues than vertical modulation alone. Stairs and abrupt volume transitions also produced distinct acoustic signatures, especially when the geometry introduced strong early reflections or sudden lateral openings. By contrast, some concave ceiling forms and complex intersection conditions yielded fewer stable patterns, suggesting reduced cue clarity and a higher probability of ambiguous echo information. The gallery-edge scenario appears especially important because standard geometry produced weak warning cues and would likely require additional architectural treatment. Overall, the simulations indicate that certain spatial transitions can be differentiated acoustically.

The study translates echolocation literature into architectural variables that can be simulated, compared, and later validated experimentally. It identifies several candidate strategies for further testing: using width modulation to announce major transitions, avoiding acoustically ambiguous fork geometries, and introducing dedicated reflective or diffusing elements near hazards such as stairs or gallery edges. At the same time, the study has clear limitations: the absence of background noise, head movement, and user-based validation. Future work should therefore combine binaural auralization, more realistic acoustic conditions, and trials with blind echolocators in real or virtual environments to determine which simulated cues genuinely improve mobility and safety. In this sense, the paper's main contribution is a framework for testing how architectural geometry may support echolocation in accessible transport spaces.

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Affordance Theory and Environmental Storytelling as an Alternative Framework to Access Unquantifiable Nature of Experimental Architecture

Experimental architecture exists on the crossroads between what's measurable using conventional metrics and the unseen relations between matter and its user. As such it remains open to a broad field of interpretation in transcribing and analyzing its results. Unified ways of measuring outcome of an architectural procedure such as POE, simply fail to apply when faced with an object that forfeits universally accepted norms of functionality and ergonomics.

Architecture seen as a geometrical sequence, unfolding information to a passing by user, can serve as a baseline to discuss the potential of such synopsis. Advantages of viewing experimental structures through the potency of its spatial narrative make it possible to shift the weight from the object to its connotations. Such effects can transpire through what's known as environmental storytelling - an arrangement of purposeful details, meant to tell a story in a less direct way. It enables observation of behaviors based on given clues rather than more straight forward principles. What emerges is something far different than standard user pathing, more resembling a chaotic network of human impulses and interactions.

If affordance theory relocates the locus from object to encounter, the evaluative task becomes one of mapping encounters rather than performances. What the experimental structure affords is readable only in the sequence of decisions. As Gibson established, meaning is never a fixed property of an object but an emergent relation between its material constitution and the capabilities of the perceiving subject. The absence of conventional affordances does not therefore represent a failure of architectural communication but its deliberate suspension. It is precisely this withholding that environmental storytelling makes productive: the arrangement of spatial clues that function as a musical score, which every user performs differently. The variation in performance is not chaos to be eliminated; it is the core of the experience.

A parallel to this can be found in Château La Coste in French Provance, where an ongoing architecture program transformed a vineyard into a network of experimental exhibits. Given a freedom for intervention, structures by Tadao Ando, Frank Gehry, Jean Nouvel, and others have emerged, distributed across a vast terrain without one unifying philosophy. Each architectural incision operates as an autonomous spatial proposition within the surrounding landscape, with no master narrative offered. Instead, what the cite offers is clues, giving a visitor multiple paths to choose and explore. Some end without a definitive resolution, others push the visitor further into the motion. Over time the behavioral traces accumulate organically, paths begin to emerge between pavilions, where people tend to linger, new routes are continuously discovered and rediscovered – around, under, over the pavilions themselves. Creating user experience that no occupancy survey could potentially predict. De Certeau observed that walkers route through the city, universally rewrite planner's map. Similarly, Château La Coste, in absence of one prescribed sequence, proves that emergent patterns of use are not incidental to designed experience but are created simultaneously to it.

Taken together, environmental storytelling and affordance theory work as a framework that does not resolve the tension between the measurable and the unseen inherent to experimental architecture, but add to it productively. The weight of assessment shifts from what the object performs to what the encounter produces. That pattern of use, however chaotic, is where the experimental argument finally becomes visible in nature.

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5. Legislation:
[1] Act of 7 July 2023 amending the Act on Planning and Spatial Development and certain other acts.
[2] The obligation to carry out a land absorption capacity assessment was introduced by the Act of 9 October 2015 on Revitalisation, amending Art. 10 of the Planning Act, without a precise methodology or effective enforcement mechanism.
[3] The original deadline (31 December 2025) was extended to 30 June 2026 by the Act of 4 April 2025; a further extension to 31 August 2026 has been proposed.

The Transitional Window and the Planning Rush: Local Spatial Plans in Poland in the Context of the General Plan Reform

1. Reform of the spatial planning system in Poland

The Act of 7 July 2023[1] introduced the municipal general plan (plan ogólny gminy) as a new spatial planning instrument replacing the spatial study (SUiKZP). Unlike the study, the general plan holds the status of an act of local law and designates planning zones (obszary uzupełnienia zabudowy) and infill development - the only areas within which zoning decisions (decyzje o warunkach zabudowy) may be issued once it enters into force. The primary aim of the reform is to curb urban sprawl and strengthen spatial order. Under the previous system, the spatial study did not constitute an act of local law, which limited the enforceability of its provisions, encouraged the dominance of individual zoning decisions and contributed to dispersed urbanisation (Chmielewski, 2001; Gzell, 2024). The oversupply of developable land was compounded for years by the absence of any constraint on designating residential areas in spatial studies. The 2015 amendment[2] first introduced a land absorption capacity assessment (bilans chłonności) linked to demographic projections, yet without an effective enforcement mechanism (Śleszyński, 2018).

2. Research aim and problem

This paper analyses the impact of the reform's transitional period on municipal planning practice and identifies phenomena that may lead to unintended spatial consequences. The focus is on local spatial plan (mpzp) preparation; zoning decisions constitute a separate research problem addressed elsewhere. The thesis advanced is that the transitional period has become a risky legislative experiment: it generated strong incentives for the mass 'production' of local spatial plans, representing an attempt to entrench existing - often excessive - land use designations before the more restrictive balancing rules of the general plan take effect. This phenomenon can be interpreted as a planning rush.

3. Context and method

The paper draws on an analysis of legislative changes in the Polish spatial planning system and observation of municipal planning practice during reform implementation - with findings being of a preliminary nature. The method is qualitative and interpretive, examining relations between law and institutional behaviour (Markowski, 2019).

4. The transitional window: mechanism and consequences

Under the transitional provisions, local spatial plans whose procedure had reached a specified stage before the new regulations entered into force may be continued under the previous rules - on the basis of the spatial study, which retains its force until the general plan enters into force, but no later than 30 June 2026.[3] After that date, initiating a new local spatial plan procedure without an operative general plan will be prohibited (Art. 67(4) of the 2023 Act) resulting in a surge of draft plans designating residential areas on the basis of frequently oversized spatial studies. Municipalities thus seek to entrench land use designations that, under the new system, could be restricted by the land absorption capacity assessment required in general plans - a mechanism binding land supply to demographic projections. Rather than mitigating accumulated planning backlogs, this "production" of local spatial plans may entrench them for decades. The spatial consequences are multifaceted. First, some newly designated residential areas may remain undeveloped without genuine demand. Second, the plans being adopted may perpetuate the dispersed development model the reform was intended to reverse. Third - and most significantly - plans adopted during the transitional period may permanently limit the capacity of general plans to shape a coherent spatial structure, since changing their provisions would entail the risk of compensation claims from landowners.

5. Conclusions

The reform generates a distinctive institutional experiment in which adaptive mechanisms of system participants become visible as new rules approach. The transitional window may paradoxically reinforce the very pathology the reform sought to remedy: oversupply of developable land and dispersal of development. Understanding this dynamic is essential for designing future regulations with built-in safeguards against undesirable transitional effects.

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Experiment – Transformation – Centrality in Post-Industrial Łódź

Post-industrial cities are places in which spatial change is visible. Former industrial areas are given new programmes, central districts are redefined, and smaller public spaces are adjusted to new patterns of use. Łódź is one example. Its urban structure is marked by transformations of industrial heritage and by interventions in streets, passages, courtyards, and inner-block spaces. The aim of the paper is to show how post-industrial transformation in Łódź produces new forms of urban centrality and how this process can be interpreted through the proposed triad: experiment–transformation–centrality. The main question is: how does this triad manifest itself throughout the city? It is assumed that major transformations are important and create strong urban attractors, but centrality can also be created in other urban spaces.

The proposed framework combines concepts of urban experiment and centrality. An experiment is understood as the intention to test and as uncertainty about the outcome. It may also be understood as a scientific experience undertaken to investigate a specific architectural phenomenon. This makes it possible to test new solutions and the effects of spatial transformation in areas adjusted to user needs. This understanding is close to the current literature in which experiments are treated as situated and change-oriented interventions (Karvonen & van Heur, 2014). Transformation refers to planned spatial intervention, especially adaptive reuse and the reprogramming of former industrial structures. Centrality is understood as a condition of space produced by concentration, access, movement, and attraction (Parr, 2017), and may be complemented by de Certeau's perspective, according to which planned urban space is reinterpreted through social actions, movement, and appropriation (de Certeau, 1984). Thus, the triad describes three moments of one process: experiment initiates, transformation gives spatial form, and centrality is the urban and social effect.

An interpretative case-study approach is adopted across three scales. At the scale of major interventions, Manufaktura, EC1, and Fuzja are clear examples of transformation. They redefine former industrial sites by introducing new programmes, meanings, and visibility. Manufaktura is one example of such an experiment. Once part of Izrael Poznański's textile factory, it became a commercial and cultural complex based on testing industrial architecture in a new form and meaning, together with the accompanying public space. Preserving the post-industrial aesthetics of the site was important for building its new image while maintaining its historical identity, as was testing a mixed-use model ensuring access to multiple services. The hybrid character of the space raises the question of whether it can perform as a centre functioning beyond the shopping gallery itself.

At the medium scale, one can observe interventions such as woonerfs, transformed public spaces, and the post-industrial interior of OFF Piotrkowska. These spaces form neighbourhood centres by improving pedestrian continuity, opening inner blocks, and creating conditions for staying rather than only passing through. At the micro scale, the same process becomes visible in smaller modifications such as remodelled courtyards or artistic insertions. These projects show that even subtle change can act as urban acupuncture and generate a wider ripple effect, including a new point of centrality (Marzi & Ancona, 2004).

The case of Łódź demonstrates that experiment, transformation, and centrality are interwoven processes. They create both central places and a sense of centrality, not only in large-scale projects but across the urban structure. In this sense, the proposed triad helps to describe post-industrial regeneration as a process that is both spatial and social.

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It may seem self-evident that every measure improving fire safety in historic buildings should be considered progress. The matter is less obvious when present-day regulations are applied to monuments and urban layouts formed in very different material, spatial, and social conditions. In such cases, heritage becomes a space for a conservation–technical experiment. This does not mean unrestricted testing on historic fabric, but rather a careful search for solutions that can protect people without unnecessarily weakening cultural value. Fire protection may thus become a paradoxical risk: it is introduced to prevent loss, yet, when applied mechanically, it may itself contribute to the erosion of authenticity, integrity, and material evidence of the past (Furmanek, 2025).

This problem is particularly clear in historic centres. Their layouts were shaped long before fire engines, evacuation standards, and present-day technical regulations. In many cases, these layouts and their specific features are part of the historical character of the place. Medieval urban structures of Polish historic centres cannot be reduced to obstacles awaiting technical correction; they are also elements of cultural identity and urban distinctiveness (Furmanek, 2025).

For this reason, fire safety in historic cities should not be understood only as a problem of emergency access. Wider roads, fewer barriers, or new traffic routes may indeed improve operational conditions for rescue services. At the same time, they may disturb the inherited spatial logic of the city. A technically efficient intervention may therefore weaken the very heritage value that it was meant to protect. A similar difficulty appears at the scale of individual buildings. Timber roof structures, old joinery, staircases, partitions, and interior finishes are constructional or functional elements, but they are also records of former technologies, uses, and aesthetic choices.

Recent research on standards for assessing and classifying fire hazards in historic buildings confirms that this field cannot be governed by regulation alone. Fire safety in a cultural heritage context requires a hierarchy of values and the ability to distinguish a necessary intervention from modernization carried out mainly because it is administratively convenient (Jurecki et al., 2025).

The issue can also be approached through crisis management. A historic building is then understood not simply as an object to be adapted to regulations, but as a vulnerable cultural asset requiring prevention, preparedness, response planning, and long-term supervision (Jaďuďová et al., 2025).

The conservation–technical experiment should therefore be seen as a responsible examination of intervention limits. Alternative solutions should be supported by both expert knowledge and conservation assessment. In many cases, such measures can provide an adequate level of safety without imposing standard solutions that are poorly suited to historic fabric. This is especially important in fire detection, where recent studies discuss point, linear, and aspirating smoke detection systems, together with minimally invasive and visually discreet solutions for heritage interiors (Mohai et al., 2026). Digital tools, including three-dimensional documentation, spatial analysis, evacuation modelling, and fire simulations, may also help to identify real risks and justify less invasive decisions.

Fire protection should be embedded in heritage management rather than added to it as a separate technical duty. in their work, architects and fire safety experts could devote even greater attention to the protection of the cultural values of monuments. In this way, historic buildings and cities can remain both safe and culturally meaningful.

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Making the Hidden Available: Problems with the Restoration of the Cloister at Tyniec Abbey

The centuries-old tradition and monastic values of Benedictine doctrine are enduring and resistant to change, remaining far removed from experimentation (Dąbek, Turowicz 2006). Yet contemporary conditions present Tyniec Abbey with challenges that require a redefinition of traditional cloistered places. The abbey is no longer an entirely closed institution: it now seeks to share its values, history, and knowledge with a wider public. It communicates through social media, operates a website and publishing house, promotes cultural heritage through the "To Protect Good" Foundation, and gradually reopens interiors previously inaccessible to outsiders. This raises the question of how the privacy and uniqueness of monastic interiors may be reconciled with broader public accessibility. Alongside Ora et Labora, one of the key Benedictine principles is hospitality, and welcoming guests is regarded as a form of encounter with Christ (Benedyktyni).

The issue of access to cloistered space was examined during the preparation of a restoration project for the cloister garth at Tyniec Abbey. Research began with archival studies concerning the cloister itself (Tomkowicz 1901), aiming to reconstruct the history of its transformations. Following the dissolution of the order at the turn of the XVIII and XIX centuries and a fire in 1831, the monastery fell into ruin, its possessions were dispersed, and the library was moved to Lviv, where it was destroyed during the Second World War (Gronowski 2009). The monks officially returned to Tyniec shortly before the outbreak of the war. Photographs from the 1950s and 1960s depict the ruined cloister site, with the western section missing. Traces of the original layout had largely disappeared and only limited written and iconographic sources survived (Gronowski 2010; Gawryszewska, Skibińska 2011). Consequently, iconographic studies of the cloister at Cluny Abbey — regarded as a prototype of European monastic architecture— were undertaken, alongside comparative studies of other medieval cloisters.

The earlier renovation project, prepared by a team led by Janusz Bogdanowski in the late 1980s and early 1990s (Myczkowski 1993), introduced a columnar arborvitae surrounded by geometric planting beds arranged along diagonal axes. Over four decades, the tree grew disproportionately large in relation to the small interior. At present, the deteriorated limestone surfaces and poor condition of the retaining walls require urgent restoration, while the cloister remains inaccessible to visitors. Following an assessment of the site's condition, it was decided to prepare a new restoration design incorporating public accessibility while preserving the site's historic value. Research also addressed the intensity of use, as well as the programme and timeframe for opening the cloister to visitors.

The design phase verified the project assumptions considering the research findings. Four alternative restoration concepts were prepared to test solutions in relation to several objectives: restoring the significance of the cloister at Tyniec Abbey; ensuring accessibility for visitors with limited mobility; improving the durability of the structure under intensive use; introducing an educational programme; and preserving the cloister's intimate character. The final design adopted a restrained planting composition centred on a low-growing shrub, such as a shaped yew, accompanied by groundcover planting and a flexible arrangement of herbs and potted plants. Simple wooden benches with backrests and armrests completed the composition. The project proposes replacing the uneven crushed limestone surface with limestone paving slabs to improve accessibility. The retaining wall will undergo extensive conservation and stone repair works, while the entrance will require the removal of a low parapet and the enlargement of the existing opening to form an entrance doorway.

The restoration of the cloister garth at Tyniec Abbey constitutes both a reconstruction and an interpretative intervention, owing to the lack of direct evidence concerning the site's historical transformations and the need to ensure accessibility for visitors outside the monastic community. The reopening of the cloister, styled to evoke the interwar restoration period, together with the introduction of a new educational function, represents an experiment that may transform traditional patterns of monastic use whilst simultaneously redefining the monastery's role within the local community. The long-term success of the project, however, can only be assessed through its future use.

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Implementing Building Energy Modelling in a Master's Level Architectural Design Course: Preliminary Insights from a Pilot Study

Civilizational challenges related to environmental degradation, fluctuating energy prices, and the resulting social problems require new approaches to the design of buildings. In EU countries, a redefinition is becoming necessary, driven not only by evolving economic conditions but also by legal ones, which will shape the design standards in the coming years (Ormuž, 2025). Since the greatest environmental impact of standard buildings occurs during their operational phase (Sartori & Hestnes, 2007), the design optimization methods, including energy optimization, are becoming increasingly important [Pan et al., 2023]. However, this requires specific competencies that have not yet been widely integrated into architectural education in Poland.

The educational goal was to create an architectural design course within the specialization at the Faculty of Architecture, Warsaw University of Technology, introducing building energy modelling as a core element of the design process. The related scientific objective was to evaluate the implemented methods in terms of their effectiveness (achieving target energy parameters of buildings) as well as their impact on the design process and student satisfaction. It was anticipated that these methods would not only improve the projects in terms of efficiency but also foster further creativity and innovation.

The project's brief involved designing an off-grid mountain lodge in a highly protected area with minimal environmental impact. This required the use of passive architectural and construction solutions, as well as the implementation of efficient HVAC systems, particularly those based on renewable energy sources. To achieve the intended goals, appropriate teaching methods were introduced: group work (allowing for specialization within groups), data-driven design (based on simulation outcomes), learning through teaching, and brainstorming. The building's energy and environmental optimization was planned at every stage of the design development process. The results of individual analyses were intended to form the basis for selecting the more promising concepts out of the numerous initial ideas presented by all the students (based on their intuition), as well as for determining the direction of their development. Final detailed analyses were meant to facilitate the refinement of specific design elements. Direct Sun Hours and Incident Radiation simulations were introduced at the initial stages of the course. In subsequent stages analyses of interior lighting (Daylight Factor, Daylight Autonomy, Useful Daylight Illuminance), energy performance of the building (Usable Energy Demand, Final Energy Demand), indoor comfort, and finally, occurrence of thermal bridges in structural nodes were performed. The main tool chosen for simulations was the EnergyPlus engine supported by the Ladybug plugin for Grasshopper.

The course and its methodology were evaluated based on pilot classes through an analysis of work results, discussions among process participants, and anonymous student surveys. The completed architectural projects were characterized by well-thought-out, generally coherent pro-environmental solutions, as well as diversified, and in some cases unconventional approaches to form and materiality. In the opinion of both teachers and students, the new simulation approach used not only helped achieve more environmentally optimized designs (although not always meeting the initially assumed parameters) but also broke established patterns of the design process, allowing for a deeper understanding of physical phenomena in buildings and the environment, even changing previously held beliefs about their nature and the potential effects of specific architectural interventions.

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Research by Design as Architectural Experiment: Designing the Warsaw Skyscraper between Form, Function and Urban Context

Contemporary architecture operates less and less under conditions of straightforward problem-solving. More often, it works within a field of shifting relationships, conflicting requirements, and incomplete data, which has become a typical condition of architectural practice. The skyscraper intensifies this condition with particular clarity. Its scale, technological intensity, and impact on the structure of the city make it more than a building type; it becomes a laboratory of design decisions. Within such a design environment, the status of the architectural concept becomes crucial, along with the question of whether it is a ready-made solution adopted at the outset or rather a construct developed iteratively through the design process. In the design of a Warsaw skyscraper in particular, research by design transforms the concept from a predefined form into an operational workflow system in which spatial, functional, and urban decisions emerge through feedback between design intent and contextual constraints (Goncikowski, 2024; Schön, 1983).

Two important shifts are now clearly visible. First, design is no longer understood as a linear sequence moving from analysis to solution; it begins to function instead as a reflective practice in which knowledge is produced through action (Schön, 1983). Second, the building itself is no longer understood as a stable and closed object; it becomes a relational assemblage negotiated between matter, technology, use, and context (Latour & Yaneva, 2008). In relation to skyscrapers, this means not only thinking of the tower as an autonomous figure, but also shifting attention toward its urban agency. In Warsaw, this has particular significance because the skyscraper operates here between the individuality of form and the discontinuity of urban context (Goncikowski, 2024).

Research by design should therefore be understood not as an illustration of the author's thesis, but as a means of producing it. The project becomes a research instrument: it tests variants, compares consequences, and reveals dependencies that cannot be captured through description alone. The analysis is organised around four interdependent parameters: mass, function, core, and urban interface. Mass defines at once the building's silhouette, its relationship to the skyline, and the way it descends to the scale of the street. Function defines not only programme, but also its degree of hybridity and capacity for adaptation. The core orders the structural, circulation, and economic logic of the building. The urban interface, in turn, determines whether the tower remains an isolated object or becomes part of urban continuity (Goncikowski, 2024). Understood in this way, design takes the form of a digitally assisted experiment in which the generation of variants serves not formal arbitrariness, but controlled parametric change (Kolarevic, 2005).

The most important outcome of this approach lies in a changed understanding of the skyscraper. Form is not the end point of composition, but the record of a negotiation between structure, environment, and the image of the city. Function no longer signifies stable assignment, but a system capable of intensification and transformation. Context, meanwhile, does not remain a background condition; it becomes an active field of influence that constrains, enables, and redefines design decisions. In this sense, the skyscraper enters the logic of the city as a durable yet dynamically reinterpreted structure of relationships (Rossi, 1982). Research by design therefore reveals not only solutions, but above all a field of tensions: between efficiency and flexibility, autonomy and integration, icon and infrastructure.

The conclusion is twofold. First, the architectural concept of a skyscraper should not be understood as a closed figure, but as a structure developed iteratively. Second, high-rise design in Warsaw gains research value only when form, function, and urban interface are developed simultaneously. Architecture does not precede knowledge here; architecture must produce that knowledge.

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(Re)writing the City: The Impact of Selected Contemporary Large-Scale Multi-Family Residential Developments on the City's Urbanscape

Multi-family housing remains one of the most rapidly developing architectural typologies in Poland, particularly in the largest cities. Consequently, multi-family housing, and in particular complexes of multi-family residential buildings, have a significant and lasting impact on the urban landscape. The dynamic growth of this construction sector and the number of new buildings being constructed do not always go hand in hand with an attempt to create high-quality urban space. The lack of government and local authority programmes for the development of multi-family housing is most often linked to the construction of individual multi-family residential buildings or small complexes of such buildings. In such cases, the urban structure of these developments is largely dictated by local conditions such as: the shape of the plot, existing and potential neighbouring developments and the constraints arising from them, as well as regulations and restrictions under local law. An additional factor shaping the final urban design is the drive to utilise the plot area to the fullest extent possible in order to maximise private investors' profits. Urban connections with the neighbourhood, or thinking in terms of the overall urban form and context of both the planned and existing development, often takes a back seat. Amidst the mass of small-scale multi-family housing developments, large-scale complexes or districts of multi-family housing are also being built. The author's hypothesis is that large-scale multi-family residential complexes have greater potential for the conscious and high-quality shaping of the urban form, as well as for reshaping or rewriting the city's urbanscape. How, then, are contemporary large-scale multi-family residential complexes being shaped in Poland? What is, if so, the impact of contemporary large-scale multi-family on the city's urbanscape?

This article attempts to analyse and assess the impact of large-scale urban planning schemes for residential developments on the urban landscape, based on selected projects in two of Poland's largest cities: Kraków and Warsaw. To this end, the urban planning schemes for multi-family residential developments by two design studios were selected: JEMS Architekci in Warsaw and B2 STUDIO in Kraków. These firms are also among the largest in both cities. The research objective of this academic work is: G1 Selection and analysis of large-scale multi-family residential development schemes in Kraków and Warsaw. G2 To examine the characteristic features and principles shaping the urban form of selected multi-family residential development plans. G3. To define discernible trends and patterns in their formation and to identify their impact on the city's urbanscape. G4. To formulate conclusions and validate the proposed thesis. A further aim of this article is to examine the impact of the largest architectural firms on the overall character of the city's landscape.

In order to obtain results that are as objective as possible, the research methodology was based on the use of a variety of research methods and tools, including: 1. review and critical analysis of the state-of-the-art, 2. statistical analysis (quantitative and statistical research methods), 3. qualitative research, 4. analysis and visualisation of results. The conclusions were based on logical argumentation and the research results were based on the method of logical argumentation.

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A Temporary Research Model of a Sensory Garden as a Tool for Spatial Experimentation

Contemporary urban design is increasingly recognizing sensory environments as instruments supporting health, accessibility, and spatial inclusion (Krzeptowska et al., 2023). However, translating environmental psychology into concrete design parameters remains difficult, particularly in projects requiring flexibility and non-permanent solutions.
This article presents a temporary sensory garden as a research-oriented prototype intended to operationalize theoretical assumptions into a spatial system prepared for future empirical studies. The study examines how spatial organization, sensory zoning, and vegetation strategies can enable subsequent evaluation of user comfort, movement, and perception, proposing such installations as adaptable, low-risk bridges between theory and evidence-based design.

The study adopts a Research-through-Design approach, contextualized within contemporary architectural research methodologies (Frayling, 1993). Following epistemological framework, architectural practice is recognized here as a primary mode of knowledge production (Fraser, 2013). Consequently, the temporary physical installation is not treated as a finalized, purely aesthetic artifact, but as a full-scale spatial prototype and an active medium of investigation. Within this paradigm, the modular garden acts as a physicalized, structured design hypothesis. This framework posits that theoretical assumptions regarding sensory integration, universal accessibility, and spatial legibility can be encoded directly into tectonic and geometric parameters (such as structural path layouts, architectural thresholds, and modular volumetric boundaries) and subsequently tested. The temporary urban prototype thus functions as a dynamic research apparatus designed to yield empirical, evidence-based data regarding user trajectories and human-spatial interactions.

The physical layout of prototype garden is structured around three gently meandering parallel paths and three functional zones. The paths form a comparative observational matrix enabling analysis of user density, duration of stay, and behavioral patterns in future research. Quantitative data will be complemented by surveys, interviews, and simulation workshops addressing diverse functional limitations.
The prototype is organized around calming affordances that support stress reduction. It includes three pathways – Sight, Hearing, and Touch – connected to the Sensory, Rest, and Silence zones - ergonomic benches, hammocks, and a “silence pavilion” define points of pause, orientation, and optional vestibular engagement. A rain garden introduces water as both an ecological and sensory element, supporting retention and seasonal variation.
Vegetation functions simultaneously as a compositional, perceptual, and climatic medium, with larger plant masses structuring space and shaping sensory experience. Selected shrubs and grasses provide structural rhythm and seasonal change, while fragrant and dynamic species enhance olfactory and kinetic perception. Dense greenery and bamboo function as acoustic and visual buffers, particularly within the Silence Zone.
Sensory experience is further shaped through spatial sequencing. Visual effects are achieved through light, shadow, and layered planting, while tactile engagement is introduced via material variation at body scale. A small lateral walking strip with differentiated textures offers an optional, low-impact haptic experience without disrupting circulation.
The infrastructure will be extended with site-specific artistic installations developed by artists from the Academy of Fine Arts in Wrocław, each addressing a selected sensory modality and introducing controlled spatial stimuli. The temporary character is ensured through modular Air-Pod container planting, allowing reversibility and reconfiguration. The garden is conceived as a modular, testable research apparatus.

Conclusions
The project should be understood as a research instrument rather than a concluded therapeutic landscape. Its value lies in transforming theoretical knowledge into a spatially testable framework for future observation and comparison. By combining modularity, sensory differentiation, and accessibility, the prototype establishes a foundation for subsequent empirical assessment of inclusive urban environments.

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Urban Climbing as an Architectural Experiment: Alternative Vertical Mobility in the City

Urban climbing, also known as buildering, refers to the practice of climbing elements of existing urban architecture such as buildings, bridges, walls, and infrastructural structures without the use of installed climbing holds. While often perceived as an extreme or recreational activity, it can also be understood as a form of interaction with the built environment and a manifestation of experimental spatial practice. Compared to other urban sports, such as parkour or skateboarding, urban climbing remains relatively underexplored in architectural and urban studies.

This paper aims to investigate the relationship between urban morphology and the emergence of climbable structures, framing urban climbing as a form of architectural experiment. The main research question concerns how architectural form, material, and function influence the possibility and character of climbing activity in the city. The central thesis proposes that urban climbing reveals latent spatial potentials within the built environment and can be understood as an informal method of testing architecture through use.

The study draws on existing discussions on urban practices and the "right to the city" (Lefebvre, 1991), as well as research on user-generated spatial practices and informal urban activities. It also relates to the concept of architecture as a field of experimentation, where the relationship between designed intention and actual use becomes a subject of inquiry.

The research is based on the development of a catalogue of urban climbing routes in selected European cities, currently including Bern and London. The catalogue records individual routes together with spatial and architectural parameters such as the type of structure, materiality, height, and morphological characteristics of climbed surfaces. Data were collected from climbing guides, online maps, and visual documentation, and supplemented with cartographic tools such as satellite imagery and street view.

The analysis reveals that urban climbing is closely linked to the material and morphological characteristics of the city. In Bern, historical stone structures and retaining walls dominate, resulting in climbing routes characterized by slab movement and the use of edges. In contrast, London's more diverse material palette, including concrete, steel, and brick, enables a wider range of climbing typologies, including overhangs and varied grip techniques. These findings indicate that differences in urban structure directly influence the forms of climbing that emerge.

Beyond its physical dimension, urban climbing can be interpreted as a form of alternative vertical mobility. Unlike conventional mobility systems, it does not serve efficient transportation, but rather functions as an experimental practice. Climbers test the limits of their bodies, the structural capacities of materials, and the social boundaries of acceptable behavior in public space. In this sense, urban climbing becomes a way of exploring the relationships between the individual, architecture, and the city.

The paper concludes that mapping urban climbing routes can serve as a methodological tool for analysing urban morphology and uncovering hidden spatial potentials. By focusing on non-intentional uses of architecture, the study contributes to a broader understanding of how urban space is experienced, reinterpreted, and activated by its users. It also highlights the importance of incorporating user-driven practices into architectural discourse and future design strategies.

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Hybrids at the Zoo – Experiments Not Just in Architecture

The term hybrid refers to a crossbreed resulting from the cross-pollination of two parent forms. It's adoption by the theories of architecture and urbanism, alongside terms such as biomorphism and folding, used increasingly widely in reference to both individual buildings and fragments of urban space is visible in the 21st century. Hybrids, according to Koolhaas, originate from American skyscrapers of the late 19th century. They can be considered both at the level of functional, structural and technological systems, and at the formal level, where (as in linguistics) an object or complex combines elements drawn from different styles – different architectural languages. Finally, in hybrid spaces, in a manner hitherto unseen, the external public space overlaps with the internal spaces of the buildings. Postmodern society is constantly craving new experiences and sensations, seeking them out in both the virtual and the real worlds. It has been proved that such could be provided by experimental architectural hybrids.

Modern zoos occupy an important place within the functional structures of cities, metropolises and regions, fulfilling complex roles as research centres, educational institutions, and nature conservation facilities for endangered species. But they also continue to fulfil, drawing on Enlightenment concepts of menageries, the functions of places for recreation, relaxation and contact with nature – third places. These functions are embodied both in open spaces – parkland areas featuring animal enclosures and aviaries – and in buildings located within the zoo grounds.

Experimental buildings that bring together, under one roof, displays of numerous animal species from different taxonomic groups, as well as from different habitats and climate zones, are extremely complex technological and functional structures. Undoubtedly, visiting them provides a more intense and richer experience, especially as, unlike the outdoor areas of a zoo, they are accessible regardless of the weather and the time of year. In turn, they require designers to create ideal conditions for the presentation of various biotopes, in which not only visitors but, above all, their animal inhabitants will feel comfortable and safe.

These evolved from aquariums and oceanariums, where, as early as the last decades of the 20th century, exhibitions of other (non-aquatic) species of animals associated with the marine environment began to appear, such as at the Monterey Bay Aquarium (1984), the Florida Aquarium (1992), and the New England Aquarium in Boston (mid-1990s), Barcelona (1992–2001) and Lisbon (1998). L'Oceanogràfic in Valencia designed by S. Calatrava and F. Candela (2003), also holds a special place, presenting a variety of freshwater and saltwater habitats. All of these were built outside the zoos.

Experimental hybrid facilities in zoos also include an oceanarium component. Furthermore, they showcase other species and systematic groups of animals not associated with aquatic environments. Two such facilities have been built in Poland to date: the Africarium at Wrocław Zoo (2014) and the Orientarium at Łódź Zoo (2022), designed by teams led by Dorota Szlachcic. The architect herself describes these facilities as "biotope architecture", as they are dedicated to presenting the fauna and flora of Africa or Asia and the surrounding seas, and house over a dozen biotopes with varied climatic conditions. Adjacent to the buildings are also outdoor enclosures used seasonally. These experimental structures, created as a result of competitions, have been successful both in terms of visitor numbers and have gained recognition as a new way of protecting and presenting biodiversity.

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Experimental Framework for Rapid Response Housing with 3D Concrete Printing: Comparative Analysis and Design Directions

The rising frequency of crises, from wars and violent conflicts to natural and climate-related disasters, has driven displacement and intensified housing problems worldwide. UNHCR reports that 123.2 million people were forcibly displaced by the end of 2024 (UNHCR, 2024), increasing the demand for rapid-deployment housing systems that meet social requirements. Conventional emergency models, including tents and container units, are introduced as temporary solutions in response to crises but often remain in place long after the initial response period has ended. Their limited adaptability, poor thermal performance, low repairability, and restricted cultural suitability reduce the overall livability of these shelters in prolonged displacement contexts. As emergency conditions increasingly transition into long-term settlement, the gap between temporary shelter and permanent housing becomes a critical challenge, requiring systems that can respond both immediately and incrementally over time as conditions change.

Recent studies have explored the applicability of 3D Concrete Printing (3DCP) for emergency and post-disaster housing, positioning it as a potentially scalable alternative construction method in crisis conditions (Koç et al., 2022) (Almomani et al., 2023) (Mehdizade et al., 2025) (Subramanya & Kermanshachi, 2022). However, the existing body of knowledge on 3DCP for rapid response, focuses on isolated advantages rather than evaluating 3DCP against the full range of rapid-response housing requirements. This limits a broader understanding of the extent to which 3DCP can, in practice, respond to these demands.

This paper addresses that gap through a comparative framework that maps rapid-response housing requirements against current 3DCP capabilities across three parameter groups: production and deployment, resources and materials, and spatial and social performance.
The framework draws evidence from built demonstrators, shelter proposals, and peer-reviewed studies, which are comparatively examined and synthesised into a SWOT analysis. This analytical structure enables a systematic assessment of the relationship between housing requirements and the technological capabilities of 3DCP.

The analysis suggests that 3DCP's strengths lie predominantly in the production domain. Automated fabrication enables rapid deployment, geometric precision, and structural durability while reducing labour dependency and material waste through controlled extrusion processes. However, material sustainability and supply-chain dependency remain unresolved challenges, although ongoing research into low-carbon material mixes and locally sourced materials continues to address these constraints. Its limitations, by contrast, are primarily architectural. Most existing 3DCP shelter prototypes remain spatially fixed, optimised for printability rather than habitation, with limited capacity to accommodate cultural practices, household growth, or the gradual transition from emergency shelter to permanent housing. These limitations highlight the gap between technological production capabilities and broader habitation requirements.

These findings suggest that although technological feasibility has advanced significantly, the key challenge lies in translating 3DCP's manufacturing potential into architecturally viable housing systems. Based on the SWOT analysis results, this study proposes three architectural design guidelines for rapid-response housing: modular configuration to enable phased expansion, adaptable spatial layouts responsive to changing household structures, and context-sensitive material strategies prioritising locally available resources and climate-responsive construction. Together, these design directions reposition 3DCP shelters from fixed emergency units into evolving architectural systems capable of supporting long-term displacement scenarios.

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Features of Housing with Defensive Functions in Border Areas

Russia today is attempting to occupy Ukraine, threatens to break through the Suwałki Corridor to Kaliningrad through the territory of Poland, has ambitions regarding Estonia, Latvia, and Lithuania, and aspires to dominate Europe. Such a potential military threat requires a rethinking of approaches to the design of the residential environment.Five years of war in Ukraine have shown that Russian forces advance along roads leading to settlements that are convenient for defense. Therefore, the planning of settlements bordering the Russian Federation, as well as the buildings themselves, must be adapted for defensive purposes.

Among contemporary examples, the experience of Israel is well known: the integration of protected spaces such as mamad and mamak into residential buildings, designed to protect civilian lives from shells, bullets, fragments, blast waves, and toxic gases. Unlike Israel, in Ukrainian settlements attacked by Russian forces, civilian residents are evacuated. In their place are military personnel who use residential buildings for defense needs: as observation posts, communication hubs, remote drone control points, command centers and headquarters, storage facilities, stabilization points for the wounded, rest areas for soldiers, and strongpoints.
Therefore, housing must be designed to be comfortable for residents in peacetime while also being adaptable for defensive use in the event of war.

At the urban planning level, coordination between architects and security specialists is essential when designing border settlements. Spatial organization should take into account potential directions of threat, ensure mutual coverage between buildings, control of approaches, and the minimization of «blind spot».

Russia's war against Ukraine, characterized by the widespread use of strike drones, robotic systems, and artificial intelligence, has generated fundamentally new methods of warfare. The continuous front line has been replaced by dispersed defensive nodes with only a few soldiers. On both sides of the line of contact, to a depth of up to 50 km, there is a "grey zone" that is continuously monitored from the air – day and night, in all weather conditions – by strike drones. Movement within this zone is extremely dangerous for both vehicles and individuals. Rotation of personnel is nearly impossible; therefore, soldiers may remain at defensive positions for up to six months or longer.
Accordingly, residential buildings must incorporate autonomous systems for water supply, sanitation, and basic energy provision.

An essential element of a residential building used as a strongpoint is a basement divided by solid structural walls into several compartments, with reinforced floor slabs above. The basement should be equipped with a backup ventilation system in case of chemical attacks, as well as emergency exits beyond the potential debris zone in the event of building collapse – preferably an underground exit leading to an auxiliary outbuilding.

It is also desirable for buildings to have flat roofs and the possibility of earth berming on the side exposed to potential shelling. Measures are required to make enemy intrusion more difficult: armored doors, firing slits facing the entrance, and protective screens or mesh on windows to prevent drones, grenades, and chemical agents from entering the building.

For multi-apartment buildings, additional features should include vertical clusters of mamad units integrated with stairwells, as well as designated platforms for receiving logistics drones. Under conditions of total drone dominance in the air, underground connections between individual buildings and rear support structures are highly relevant. Thus, the development of housing with integrated defensive functions is an interdisciplinary task that combines architecture, urban planning, and security strategies. Its implementation requires the development of new approaches and typological solutions adapted to contemporary threats.

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The Library – A Space for Experimentation?

When considering experimentation in architecture, we rarely think of libraries. Still, we often view these institutions – which have been present in cities and lives of respective communities for centuries – as repositories that gather knowledge and experience from all eras, yet themselves remain outside of time and its destructive effects (Palej, Homiński, 2007).

However, experimentation is an integral part of the library. The bold use of four slender cast-iron columns resulted in the reading room of the National Library in Paris (1862–1868, Henri Labrouste), which to this day impresses with its spaciousness. In turn, the Jagiellonian Library owes its architectural character to an innovative welded steel columns that are hollow inside (1929–1940, Stefan Bryła), and integrate structural, mechanical, functional, and spatial considerations. The first adaptation of the large open-stack library model may seem like an experiment in hindsight, too, but it was radical for Poland at the time: the University Library in Warsaw (1994–1999, Marek Budzyński and Zbigniew Badowski).

Libraries emerging in the 21st century – amid uncertainty about their future role and speculation regarding possible scenarios – impress with the diversity of solutions. The call for diversity is not new; it has been present in the discourse on library spaces since the 1980s (Faulkner-Brown 1979, McDonald 2006). However, today it addresses not only the diversity of users and their various needs; it also represents a departure from established typologies and forms in favor of hybridization, the expansion of services offered, the blurring of boundaries between the library and other programs, the city, and the landscape, as well as the search for synergies within the social and institutional environment.

The hybrid Fondo building in Barcelona (2012–2014, Pich Architects) brings together under one roof such diverse functions as a market hall, a supermarket, a preschool, and a public library, with an essential culinary laboratory that complements the library's collection of culinary literature. A communal cooking kitchen can also be found in the Oodi Library in Helsinki (2013–2018, ALA Architects) and the Gabriel García Márquez Library in Barcelona (2022, Suma Arquitectura) – both of which have been honored with the title of IFLA Public Library of the Year.

Adaptability, mobile furniture, and the blurring of boundaries between seemingly contradictory zones are hallmarks of many contemporary libraries. To this end, designers make generous use of open spaces, treating them as a new library resource on par with the materials stored there. Some buildings go further, seamlessly blending exterior and interior spaces, with interiors modeled after the landscape, as seen in the Rolex Learning Center in Lausanne (2007–2010, SANAA) and the Tama Art University Library in Tokyo (2004–2007, Toyo Ito).

An inherent consequence of experimentation must be a willingness to make mistakes. The drive to boldly incorporate new functionalities into library programs and blur the boundaries between them sometimes results in spatial conflicts that, in the past, were successfully neutralized through design. Evidence of this can often be found in critical reviews regarding acoustics, spatial conflicts, or the atmosphere of many contemporary libraries, left by dissatisfied users on popular websites. Nevertheless, as one of the few public institutions accessible to everyone for only a nominal fee, libraries of today serve as a testing ground for the use of public spaces and as prototypes for future institutions of knowledge.

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Revitalization of Mining Waste Hills as an Experiment in Space (Using the Example of the "Vizeyska" Mine in Lviv Region, Ukraine)

Given today's environmental, social, and climatic challenges, revitalizing degraded mining areas is essential for achieving a balance between ecology, aesthetics, and technology. Reclaiming post-mining landscapes is a globally significant strategy. The Zollverein project in Germany exemplifies successful transformation and provides cultural, recreational, educational, and commercial services, demonstrating the potential of such reclamation efforts (Patsyuk, Kazakov, 2024).

Lviv region is the most powerful mining region in western Ukraine. The concentration of coal-based man-made objects in a relatively small area has caused environmental degradation (Bryk et al., 2019). Mine waste heaps (stoves) are usually located near or a short distance from mining sites, and they occupy different areas and vary in height. They are factors of negative consequences of urbanization - in addition to environmental hazards, waste heaps disrupt the attractiveness and aesthetics of nearby cities (Voloshchin, 2023, p. 136). In particular, the waste dump of the Vizeyska mine in the Lviv region has been in operation since 1960 and was decommissioned in 2009. The mine territory periodically attracts the attention of stalkers and industrial tourism enthusiasts. Since this territory is already used by separate unorganized groups of industrial tourists, despite the existing danger of landslides and the lack of appropriate infrastructure, it is urgent to develop a project for the landscape and spatial organization of the territory as an open public space.

As part of the student workshop "Space of Ideas" (2025), students of Urban Planning Department of Lviv Polytechnic National University, under the guidance of teachers, developed a revitalization project for the Vizeyska mine dump near the village of Silets, Sokalsky district, Lviv region. The developed project consists of adapting the dump area to modern challenges and giving it a new function. In our opinion, the transformation of the Vizeyska mine dump into a functional space for recreation and social interaction should be carried out with the stabilization of the terrain and prevention of slope erosion. First of all, this is to ensure the preservation of the vegetation that occupied this area in the process of developing the pioneer stage of overgrowing the dump and carrying out biological reclamation measures. During the spring of 2025, teachers of Urban Planning Department of Lviv Polytechnic National University conducted field surveys of the Vizeyska mine dump area. During which, on the lower parts of the slopes of the mine dump of the northern and eastern exposure, natural pioneer groups of Betula pendula Roth - Pinus sylvestris L. - Calamagrostis epigejos (L.) Roth were identified, and groups of Populus tremula L. - Calamagrostis epigejos (L.) Roth - Achillea millefolium L. - Chamaenerion angustifolium (L.) Scop are sporadically found. On some slopes, self-growth of Salix caprea L., Robinia pseudoacacia L., and Alnus glutinosa L. is observed, and Quercus robur L. is occasionally found. The results of our study confirm previous studies by scientists in 2007 (Ivanov, 2007, p. 334).

In our project, the area of the defunct "Vizeyska" mine is being reclaimed by allowing natural overgrowth. Also, we are adding new plant species to the existing landscape. This method supports biological reclamation. Over time, the site will become suitable for creating a cultural landscape complex. The renewed space will attract people from the region, helping transform a depressed area into one attractive for the economy and tourism.

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National Identity in Architecture and Ways of its Creation

The issue of preserving national identity is relevant for many countries in the context of globalization. The purpose of the study is to identify the defining features of national identity in the architecture of Ukraine, because there is no single catalog of Ukrainian identity in different historical periods and in different fields of knowledge.

A new methodology for researching and implementing national identity in architecture is proposed, based on classifiers of national identity features in different historical periods. According to the method of system-structural analysis of style classifiers in different historical periods, a database of national features in architecture is created using artificial intelligence, which is open to addition with new features of national identity. Such a database significantly reduces the time for scientists and designers who deal with national identity in scientific works and implement it in modern architecture on a national basis. Similar databases can be created separately for architecture, fine arts, decorative and applied arts, and intangible culture. One of the tasks of such a methodology is to identify the process of genesis of national identity in various spheres of social life and the place of architecture in this process. It is determined how the Ukrainian national identity integrated into itself and transformed more ancient identities (Trypillian, Kyivan Rus) and national identities of other peoples (Polish, Slovak, Romanian, etc.).

Argued by the period of concentrated expression in architecture, as well as in other spheres of Ukrainian life, the period of the 17th-18th centuries is called. This is the maximum surge of phenomena united by the common term “national identity”. Since the greatest influence on the lands of Ukraine was exerted by the Polish Baroque (through which, in fact, European Baroque traditions penetrated), it is necessary to characterize the features of the Polish Baroque in more detail. Polish Catholic Baroque went through traditional stages of development from simplicity and limited decoration in the early period to emphasized decoration in the mature period, when a large amount of stucco decoration was used on facades and in interiors, and expensive finishing materials such as marble, gilding, and monumental paintings were used in the interiors. In the second period, earlier churches were decorated. Thus, national identity in architecture is presented as a complex multicultural phenomenon at the intersection of many spheres of culture and art.

The issue of loss of identity always causes a surge of national consciousness and national forms in architecture. The proposed methodology, on the one hand, analyzes the established characteristic forms of Ukrainian national identity in architecture, and on the other, uses the latest scientific and technical achievements (artificial intelligence, new methods of accumulating and processing materials, new materials and structures that allow modernizing and contemporizing traditional Ukrainian forms and methods of decoration).

The scientific novelty of the research is as follows:

- a new scientific direction has been initiated, and a methodological apparatus has been developed for the affiliation of architectural objects and cultural phenomena to the Ukrainian national identity, foreseeing their ability to develop and be supplemented with new features of the Ukrainian national identity for the first time;

– the hypothesis that the most significant expression of identity as such is architecture, which integrates manifestations of identities in various spheres of human activity and represents a constant process of transformation with the integration of different identities, is confirmed.

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Between Project and Process - Incorporating Tensions of Urban Fringes in the Design: The Case of Park Aktywności Rodzinnej in Wawer, Warsaw

Cities have no fixed image or final state. They are living systems, continuously evolving and interacting with their surroundings. Their edges are the most dynamic zones - sites of constant change, shaped through informal negotiations over space, function, and use. Rigid design imposed on such fluid environments creates tension between established balances and externally introduced interventions.

Park Aktywności Rodzinnej (PAR) in Wawer, Warsaw - a publicly commissioned family activity park at the fringe of an urban and natural environment (JAZ+Architekci, 2026) - serves as the case study for this paper. The research problem concerns how to integrate diverse values and continuous change, characteristic of urban fringes, into the design process. The aim was to introduce the qualities expected of public space into an ecotonal territory without destroying its functional openness or ecological potential.

Holland, Risser, and Naiman (1991) describe the significance of the boundary between ecological systems - the ecotone - as a dynamic and complex ecosystem that affects biodiversity on a scale disproportionate to its geographical range. In such spaces, Reed and Lister (2014) propose treating design as an adaptive, open-ended process that anticipates transformation and works with dynamic ecological systems rather than imposing fixed spatial outcomes. This approach resonates with the location of PAR: a zone in-between, with vast potential regarding flexibility in terms of usage, adaptation, and ecological impact.
The experiment therefore adopted a process-oriented design methodology, treating the ecological environment as a stakeholder alongside human users. Pre-design neighbourhood consultations revealed expectations of high anthropopression regarding the future functions of the park (Urząd Dzielnicy Wawer m.st. Warszawy, 2022). Current space usage was identified through desk research, site visits, spatial analysis, and landscape value assessment, forming the framework for the design process and its anticipated adaptation.
Flexible programmatic elements - functionally open yet precisely located - were introduced to channel institution-driven activities into designated zones while leaving other areas unprogrammed. The spatial structure comprises dispersed activity islands and a biocenotic zone with controlled circulation, including a walkway that enables shifting perspectives and direct engagement with nature. Zoning permits limited modification while maintaining a clear distinction between high-activity areas and zones of ecological succession.

The project is evidenced by spontaneous community use, incremental spatial changes, and ongoing programme adaptation to site conditions. It operates as an interface between human activity and ecological processes, sustaining a balance between use and succession. The resilience of the space lies in its capacity to accommodate not only planned, institutionally anticipated scenarios but also spontaneous activities - both those emerging from the project's own dynamic and those rooted in the site's pre-existing history. Anticipating this capacity is a defining quality of socially resilient design.
The flexibility proposed in the project was intended to foster balance, understood as a continuous, ongoing process rather than a static state. New initiatives continue to emerge, demonstrating that the process is functioning in practice.

Urban fringes - transitional zones between areas of distinct character - embody the potential of both and exhibit the highest degree of dynamism. A process-oriented approach to public space design enables designers, institutions, and users to co-create space according to collectively negotiated rules. The careful formulation of these rules fosters users' identification with the place, which in turn reinforces the process and ensures the long-term resilience of the designed space.

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Urban Planning as an Experimental Process: Adaptation, Uncertainty, and Knowledge Production in the City

The city can be understood as a complex and dynamic system characterized by non-linear relationships between social, economic, environmental, and spatial factors. Under such conditions, the predictability of planning decisions is limited; therefore, urban planning increasingly takes the form of hypothesis testing in real-world settings – through pilot projects, temporary uses of space, or experiments with new mobility solutions. In this sense, the city becomes a "laboratory" in which the effects of implemented interventions are observed and evaluated. From the perspective of planning theory, this implies a shift away from the classical linear model (diagnosis – plan – implementation) towards adaptive or reflexive planning, based on the cycle: design – implementation – evaluation – adjustment. Planning thus resembles an experimental process in which decisions are continuously tested and revised.

Contemporary research emphasizes that urban experiments must account for their spatial context and the challenges of transferring solutions between places. Recontextualization is therefore crucial – adapting experiments to local conditions, from simple replication to complex reconfigurations (Novalia et al., 2026). The more complex the solution and the more distinct the context, the greater the need for adaptation.

A key concept is also "urban experimentation" and the "living labs" approach, involving the deliberate implementation of innovations in controlled urban conditions to observe their social and environmental effects. These initiatives are often co-produced by multiple stakeholders, enhancing their experimental character. Even small, temporary urban projects can have transformative potential, as they build networks of trust and engage users, while their impact is strengthened through communication and knowledge flows (Jóźwik, 2026).

Experimental planning is closely linked to a "trial-and-error" approach, in which decisions are made under incomplete knowledge and adjusted based on observed outcomes. Many urban parameters – such as building height, density, or functional composition – result from a compromise between design assumptions and empirical testing of their consequences.

From an epistemological perspective, urban knowledge is contextual and situated rather than universal. Each city represents a distinct case; therefore, experimentation in urbanism serves both as a research method and a means of knowledge production through practice.

Experimentalism also has a social and political dimension, as it involves interventions in residents' lived environment, raising questions of participation, spatial justice, and responsibility. Contemporary approaches therefore combine experimentation with participatory processes and impact evaluation.

In the Westend district of Frankfurt am Main, protests in the 1960s and 1970s opposed the transformation of a residential area into an office district driven by modernist planning, land speculation, and demolition of historic buildings. Residents and activists organized demonstrations, formed the Aktionsgemeinschaft Westend, and carried out building occupations, including the 1970 occupation of a building at Eppsteiner Straße 47 to prevent evictions and demolition. These actions became part of the broader "Frankfurter Häuserkampf," influencing later approaches to heritage protection and participatory planning (Lefèvre, 2018). The protests highlight the experimental nature of urban planning, as high-density development was tested on the living city, producing unforeseen effects, while citizen resistance acted as a corrective force, exposing the limits of top-down approaches and supporting more adaptive and participatory practices.

In conclusion, urban planning can be understood as an experimental process grounded in iteration, testing, and adaptation under uncertainty, while also serving as a means of understanding and co-producing the city as a dynamic system.

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Repair and Memory as Design Tools: Architecture of Care as Experimental Practice

Contemporary architectural discourse continues to privilege novelty, the replacement of the existing by the new and tabula rasa approach to design. This paper challenges that assumption by proposing that repair and memory can function as operative design categories, constituting a rigorous experimental approach rather than a peripheral practice. The research examines whether architecture grounded in care, continuity, and attentive transformation can operate as a form of experimentation within the discipline.

The central research problem asks how repair and memory can be employed as active design tools. The thesis argues that care-based design, understood through the Architecture of Care as an experimental paradigm, constitutes a form of architectural experimentation that redefines authorship, temporality, and the relationship between intervention and existing conditions.

The theoretical framework synthesises four positions. Joan C. Tronto's ethics of care defines care as maintaining and repairing the world. Stewart Brand conceptualises buildings as evolving processes. Yuriko Saito's everyday aesthetics legitimises the ordinary as a site of aesthetic value. Aldo Rossi situates architecture within the domain of collective memory and permanence. These perspectives form a basis for understanding repair as a primary design operation.

The "House of Care" a long-term art-research study of an old single-family dwelling used as a laboratory, is framed as an experimental environment in which a method combining art-research, autoethnography, and geometric analysis is tested as a primary design operation. Art-research conceives design as a mode of knowledge production in which making, observing, and interpreting are interdependent, while autoethnography situates the researcher within the spatial and temporal processes under study. Geometric analysis, developed in earlier urban-scale research and adapted here to a single dwelling, examines proportional systems, spatial sequences, and relationships between built form and landscape. This enables the identification of latent spatial logics that inform non-invasive and continuity-based design strategies. The study builds on findings published in 2022, extending the framework into a new spatial and methodological context.

Three spatial elements of the „House of Care" are analysed as operative evidence. The staircase, assembled from repurposed industrial crane components, inscribes material memory into domestic space, where prior structural life persists as spatial form rather than representation. The pathway, whose geometric order has been slowly colonised by moss, exposes temporality and non-human agency as co-constituents of design completion. The annual cycle of closing and opening the house, understood as a choreography of maintenance, defines architecture as a relational and temporal practice rather than a corrective intervention. Geometric analysis conducted in parallel reveals consistent proportional relationships and spatial hierarchies that precede individual interventions. This suggests that design can operate through recognition and extension of existing spatial orders rather than their replacement. In this framework, architecture is no longer the design of objects, but the disciplined cultivation of long-term spatial relations.

The findings demonstrate that repair and memory are conditions of architectural innovation. The Architecture of Care repositions experimentation as a disciplined engagement with existing structures, materials, and temporal processes. This approach aligns with contemporary concerns of sustainability and circularity by prioritising reuse, longevity, and reduced intervention. In this sense, the experiment addresses contemporary environmental and social challenges by redefining responsibility in architectural practice. As such, it proposes a model of architectural practice that is ethically grounded, methodologically rigorous, and scalable across different spatial contexts.

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Feedback and the Production of Space: Toward a Noise-Based Architecture

Architecture is often conceived as the organization of stable forms in space. However, contemporary urban environments increasingly behave as dynamic, unstable systems shaped by recursive interactions between material, infrastructural, and social processes. This creates a mismatch between architectural design methods and the phenomena they attempt to organize. In no-input mixing and feedback-based systems, there is no external signal. Sound emerges from the circulation of energy within a closed circuit. The practitioner does not compose in a traditional sense but tunes parameters – gain, filtering, thresholds – through which structures emerge and destabilize. These systems are highly sensitive: minor changes can trigger amplification or collapse into silence.

This approach is operationalized in my work, where feedback systems are spatialized rather than confined to audio circuits. In no[*]input (Program/Abakus, Warsaw 2025), a quadraphonic setup distributes feedback loops across the room, allowing architectural features – walls, distances, bodies – to function as components of the circuit. Using piezoelectric microphones, material space becomes an unstable system shaped by both audience presence and self-reference, producing localized standing waves, runaway gain, and sudden acoustic dead zones. Sound is co-produced by space, both in its static materiality and the internal movement which continuously reconfigures the system. The work treats space as a feedback instrument, where instability and resonance become directly perceptible.

This way of thinking about spatial design resonates with Graham Harman's concept of withdrawn objects. Entities never fully coincide with their relations; something always exceeds interaction. Feedback systems make this audible: even in closed loops, components do not resolve into transparency. Their nonlinearities produce emergent behaviors that cannot be fully predicted. Architecture, viewed this way, is not a transparent system of control but a field of partially inaccessible elements whose interactions generate spatial phenomena. At the level of perception, Juhani Pallasmaa's emphasis on embodied, multisensory experience becomes crucial. Feedback foregrounds acoustic space as immersive and physical – vibration, pressure, resonance – rather than representational. A feedback-oriented architecture would not simply contain space but participate in its production, shaping how bodies are situated within evolving sensory environments.

Cybernetics provides a bridge between these domains. Norbert Wiener defined feedback as the basis of control and communication in machines and living systems, while Stafford Beer extended this to complex organizations, emphasizing adaptation over stability. Cities and buildings can thus be understood as large-scale feedback systems in which information, matter, and energy circulate through recursive loops, where deviations may be amplified as much as corrected.

Noise theory sharpens this insight. As argued by Ray Brassier, noise is not merely disturbance but a fundamental dimension of reality that resists integration into meaning. In feedback systems, noise emerges internally when circuits exceed their operational thresholds. Urban phenomena such as congestion or interference can therefore be understood as intrinsic effects of dense, recursive systems rather than external disruptions.

Design, in this context, becomes the shaping of loops rather than objects. Architectural intervention can be understood as the insertion of delays, filters, and points of saturation within existing systems. Rather than aiming for equilibrium, such an approach embraces controlled instability, allowing space to evolve through use and interaction.

Reframing architecture as the shaping of feedback systems shifts attention from form to process. Design becomes a practice of tuning dynamic, unstable interactions through which space is continuously produced.

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Temporary Urbanism as Experiment: The Case of Jazdów Settlement in Warsaw

Introduction
The notion of experiment is linked to the intentional decisions of the creation of environments enabling the verification of planned dynamics. In the space of the city, being a highly complex system, this intentionality is reduced (especially in the neoliberal economies and public policy governing neoliberal contexts). Nevertheless, some procedures can be observed along the decades, such as generational changes, appropriation processes, behavioral changes, urban decay, local identity social phenomena etc. In urban planning the rule of law plays a capital role in shaping the physical order of streets, buildings, public spaces etc., but the inhabiting and use practices impact the surroundings.

Background context and research aim
The concept of temporary urbanism is linked to the changing dynamics of urban planning processes. The aim of the presented research was the exploration, explanation and illustration of notions of temporality in urban planning and architectural interventions of larger scale.

The urban experimentation
Synthetic literature review allows to question terms from the semantic areas of: temporary territorial management, urban planning, urban revitalization, gentrification, urban experimentation, social changes, social capital, tactical urbanism, economic growth, urban economics, neighborhood identity, function evolution, adaptive reuse and many others. All the mentioned terms shape a cloud, trying to grasp and describe the ongoing evolution of spaces changing far beyond the image given by local communal urban governance prescriptions. The play of human actors, from within and beyond of the local project shapes practices, trespassing political agreements. Ars longa, vita brevis.

Research method
Qualitative approach to gathered data was focusing on the query of explanation of observed phenomena, search of reasons and factors impacting both intended and non-intended development directions. The research action was aimed also to collect both internal and external drivers of change. The hybrid interdisciplinary method included literature search (scientific articles, social media content, texts of the culture), qualitative (semi-open) interviews, field and theme expert consultations, site visits, documentation query etc.

Analytical procedure, interpretation and discussion
The example of Jazdów Finnish Houses Settlement, although reduced by infrastructural investments (Trasa Łazienkowska) can be treted as un-finished project of renovation of natural and architectural tissue, composed in intertwined within the administrative headquarters of the Capital city center of Warsaw. Its functioning is, in the last two decades (2013-2026), an experimental battlefield of melting activities and policies. Since its beginnings, in 1945, it is intended to function as a temporary urban solution settlement. The fate continues.

Synthetical Conclusions
The rapidly changing international situation, in a world impacted by global economy in the context of climate changes require adaptability of built structures and spaces to adapt to changing needs. This way the opening to mixed use, adaptive reuse and mobile solutions might enhance the sustainability of urban areas. The experimental approach, encompassing different paths of local development is an approach, opening the adaptability to yet unknown scenarios.

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Forming a New Approach to the Creation of Public Space and Sacred Architecture on the Example of the Pilgrimage Center of Simeon Lukach in Starunya Village

The contemporary architectural environment is shaped by technological innovation and changing social expectations. Public spaces are no longer understood solely as functional components of the urban environment; instead, they increasingly operate as platforms for communication, community formation, and collective experience. Within this context, sacred architecture is also being reinterpreted. Contemporary sacred spaces can move beyond the traditional model of isolated religious objects and become open public environments that respond not only to spiritual needs but also to contemporary social and psychological demands.

The research question of the study concerns how contemporary sacred architecture can function simultaneously as a pilgrimage center and an open public environment while preserving the spiritual depth of sacred space. Particular attention is also devoted to identifying new typological approaches to pilgrimage architecture introduced through the Pilgrimage Center of Blessed Symeon Lukach. To investigate these issues, the research applies methods of comparative architectural analysis, phenomenological interpretation of place, and spatial-functional analysis. Emphasis is placed on the relationship between architectural form, landscape integration, public accessibility, and the sequential emotional experience of space.

Contemporary international practice demonstrates a variety of approaches to the formation of sacred pilgrimage environments. One significant example is the Sanctuary of Lourdes (France), which represents a large-scale integration of religious, landscape, and public functions within a continuously evolving pilgrimage infrastructure. Another important case is the pilgrimage complex of St. Pio in San Giovanni Rotondo (Italy), designed by Renzo Piano, where sacred architecture is interpreted as an open and emotionally oriented public environment that combines monumentality with a human-centered spatial experience.

In present-day Ukraine, marked by social crisis and wartime challenges, the role of such spaces has become especially significant. Places associated with prayer and pilgrimage increasingly function not only as religious destinations but also as environments of emotional recovery, stability, silence, and mutual support. Under such conditions, new design approaches to sacred complexes are required, allowing them to operate as integrated elements of the public environment capable of connecting spiritual meaning with everyday social interaction.

The Pilgrimage Center of Blessed Symeon Lukach proposes a new typological interpretation of sacred architecture in which the pilgrimage complex is conceived not merely as a religious object but as an open public environment oriented toward social interaction, emotional recovery, contemplation, and everyday human presence. Rather than dominating the surrounding landscape, the architectural composition is integrated into the natural environment of the Carpathian foothills through principles of spatial continuity and environmental complementarity.

A phenomenological understanding of place forms the basis of the architectural concept, where spatial perception emerges through the interaction of silence, natural landscape, material restraint, light, movement, and emotional sequences of experience. Spatial organization within the complex is structured as a gradual transition from open collective spaces toward more intimate and contemplative zones, thereby forming an emotionally differentiated architectural environment.

The presented research demonstrates the possibility of transforming contemporary pilgrimage architecture into a hybrid spatial model that combines sacred function, public accessibility, phenomenological depth, and landscape integration. Such an approach expands the social and typological potential of sacred architecture within contemporary architectural discourse. In this way, the project contributes to the development of a contemporary typology of pilgrimage centers based on openness, environmental integration, and phenomenological experience rather than on exclusively liturgical function.

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The Church of the Holy Spirit in Tychy by Stanisław Niemczyk as the Realization of Professor Włodzimierz Gruszczyński's "Architecture of Emotion"

In a series of drawings entitled "Architecture of Emotion" (Architektura Wzruszeniowa), Gruszczyński searches for new, original forms for the "cities of the future," drawing inspiration from the architecture of the Podhale region. The expressive sketches depict monumental forms, resembling pyramids in size and massively scaled-up highlander huts in shape. As Tomasz Węcłowicz writes, Gruszczyński "wanted to articulate a synthesis of traditional form in a new size, even on a geoplanistic scale, and in a new material – reinforced concrete." Gruszczyński was a charismatic professor, and in his classes, students experimented with tradition, regionalism, and form inscribed into the landscape. One of them was Stanisław Niemczyk.

Niemczyk (1943–2019), born in Czechowice-Dziedzice, began working at Miastoprojekt – Nowe Tychy after his studies in Krakow. At the time, the city was a massive construction site. In the late 1970s, he was commissioned to design a church in Tychy-Żwaków. Completed in 1982, it has since become one of his most recognizable works. The Church of the Holy Spirit is deeply imbued with Christian symbolism, visible both in its overall form and in its architectural details.

The article, based on the case study method, addresses the issue of the proximity between the forms created by Gruszczyński in his sketches and the form of the Church of the Holy Spirit.

The church appears as an abstract form of a mountain in the landscape and, despite its scale, remains in harmony with its surroundings. It thus approaches the ideal sought by Gruszczyński – "architecture with the structure and scale of the landscape."

The modesty and purity of the church's sharp angles evoke the archetypal shape of highlander houses. Borrowing the form of a highlander cabin is a characteristic motif in the designs of Gruszczyński's students and his sketches. The arcades surrounding the church recall the soboty typical of Polish wooden churches, though in Tychy they appear in a radically transformed state. Scaled up to accommodate crowds of worshippers, they have materialized in reinforced concrete – a structural choice that also dictates the interior. There, massive concrete beams organize the space, acting as a contemporary iteration of the raw timber framework that Gruszczyński considered to be quintessentially Polish.

Finally, Prist Franciszek Resiak, the builder of the church, writes that Niemczyk believed he had "created a work in a biblical-Polish style." This is another convergence with the professor's explorations, who heard in the regional architecture the "still living melody of Polish architecture," which was meant to inspire him and his students.

Can the Tychy church be considered the realization of "Architecture of Emotion"? Niemczyk openly acknowledges the influence of Gruszczyński, whose designs largely remained on paper. The Church of the Holy Spirit in Tychy was the first to be realized by Niemczyk. Gruszczyński passed on to Niemczyk a way of thinking about architecture which ultimately allowed him to build one of the most expressive architectural oeuvres in Poland.

When we consider that Gruszczyński was, in turn, a student of Adolf Szyszko-Bohusz – co-founder of the Faculty of Architecture in Cracow, and a seeker of a truly "Polish architecture" – we can deduce the profound role of the master. Gruszczyński served as the guardian of an idea, and the university as its laboratory: a place where this vision awaited the opportune moment and the right person to finally materialize.

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Water in Public Space as a Medium for Artistic and Social Experimentation

Introduction

The presence of water in public spaces co-creating and accompanying architectural objects, urban furniture, and works of art – significantly enhances the social and aesthetic value of this space (Langie et al. 2022) and also has significant potential for creating a place and strengthening its identity. Using this medium for experimentation by designers working in public spaces seems not only a natural artistic endeavor but also a social activity aimed at engaging audiences and allowing communication between them and the creators.

Subject and method of work

The subject of the analysis are buildings and artistic installations located in public spaces, accompanied by bodies of water – both artificial and natural. The empirically collected material served as a starting point for the authors' research, based on literature studies, on the artistic techniques and methods of influencing audiences employed by the creators of the selected works.

Discussion

An example illustrating the use of water in a symbolic context in architecture is the Embassy of the Republic of India in Berlin (Léon Wohlhage Wernik Architekten, 2001). The designers deliberately employed the water motif as an experimental transposition of the significance of the Ganges River for the nation and its people. In all spaces the presence of water is strongly emphasized, guiding the user from the front of the site, through the entrance atrium, to the courtyard in front of the residential wing (Elmenhorst 2010).

In the City of Arts and Sciences complex in Valencia (Santiago Calatrava, 1998-2009), artificial water bodies surround all the buildings, each with its organic forms. As conceived by the designer, the mirror image of the Planetarium allows viewers to see the full shape of an eye (Tola and Vokshi 2013). Furthermore, the water's context takes on particular significance at night, when the interplay of light and reflections creates a true spectacle of architecture and its multiplications before the viewer's eyes.

The 75-meter-long, open-air stone staircase "Sea Organ" (Nikola Bašić, 2006), descending into the Adriatic Sea, serves as a stepping wall, a seat, and a musical instrument. Thanks to an internal, integrated system of 35 pipes of varying lengths and diameters, they produce different sounds depending on the direction and wavelength of the waves, offering audiences on the seafront a surprising acoustic experience. The stairs are divided into seven segments, each tuned to a G major or C major chord with an additional sixth (Krvavica et al. 2018).

"The London Mastaba" (Christo and Jeanne Claude, 2018) is a 20-meter-high temporary art installation composed of 7,506 oil barrels painted in four colors, located on a floating platform on the Serpentine Lake in London's Hyde Park. The scale of the structure can be fully appreciated by approaching it by boat allowing for a more profound experience of the object's impact on one's sense of balance. The artist himself described the geometry of the mastaba this way: "It's a movement, an explosion of force. When you stand sideways and look at the sloping walls, you feel the entire structure is about to explode" (The Mastaba).

Summary

The analysis demonstrates the extraordinary diversity of ways in which water is used as a creative medium by designers whose works are publicly available. At a time when experience – both physical and intellectual – is central to design, it can provide a rich source of inspiration for further experimentation.

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The Concept of Adaptive Governance as an Urban Experiment

The implementation of urban planning reform in Ukraine is being pursued through the draft Urban Planning Code of Ukraine. Over the past decades, Ukrainian urban planning has struggled with at least two key challenges. How to translate the provisions of a Master Plan into subsequent detailed planning of urban (or rural) fragments at scales of 1:500–1:1000. Frequently, the provisions of the Master Plan came into conflict with the finer scale of territories, as they could not account for the full complexity of local conditions. As a result, they often became declarative in nature, with lower levels of planning attempting to bypass them. In the development of detailed planning documentation, the principles and spatial solutions of Master Plans were often substantially ignored, and instead, self-serving and corrupt decisions were adopted, initiated by local developers.

This situation was rooted in a deeper and insufficiently recognized fundamental problem: the disjunction between the static nature of urban planning documentation (paper and digital plans, zoning) and the dynamic evolution of urban life. The war has significantly intensified these challenges: the emergence of refugees and internally displaced persons, the destruction and occupation of cities and villages, land contamination with mines, ongoing hostilities, and missile attacks across the entire territory of the state.

A response to this condition is the idea of creating a spatio-temporal hinge – a transitional space between the static nature of plans and the implementation of flexible, tactical solutions through the creation of a Digital Twin: a digital representation of the city based on accessible data infrastructures, managed with the support of artificial intelligence.

First, in the formulation of Urban Planning Conditions and Restrictions (UPCR) (evidence-based UPCR), municipal architecture departments should utilize the Digital Twin to substantiate planning constraints and requirements related to engineering preparation, public realm design, and urban greening, while fixing specific parameters through the application of Nature-based Solutions (NbS).

Second, in order to obtain UPCR, a developer submits pre-design proposals (urban planning calculations) within the framework of a regulatory sandbox (pre-application screening). The city establishes a voluntary fast-track procedure: if a developer tests preliminary design solutions through the Digital Twin and integrates NbS, they receive guaranteed expedited issuance of UPCR along with administrative support. Such a model is resilient to political change and can function independently of specific configurations of urban planning legislation.

Third, the legal consolidation of UPCR is ensured through the Design Brief (Terms of Reference): all parameters embedded in UPCR on the basis of Digital Twin data become mandatory for designers and are subsequently verified through state expert review.

In summary, the Digital Twin formalizes the rules, the regulatory sandbox makes them economically attractive, and the city grants them legal force.
The Digital Twin effectively acts as an urban planning regulator by introducing a dynamic dimension. It enables the formation of a practice of evidence-based tactical urbanism even within imperfect regulatory environments, substantively enhancing the existing powers of local self-government bodies in implementing both Comprehensive Spatial Development Plans of Territorial Communities and national legislation.

Digital Twin and the Regulatory Sandbox as an adaptive tactical urban hinge allows for the direct integration of dynamic, data-driven parameters into Urban Planning Conditions and Restrictions (UPCR) and the Design Brief. This, in turn, ensures the implementation of Nature-based Solutions (NbS) in urban development and renders them legally binding already now – without the need to wait years for the completion of national spatial planning reform.

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Selected Spaces in Post-Industrial Sites as an Experimental Approach Towards Playful Users' Activities – Axonometric Models

In a reality of automatization and statistic data, this research claims that equally important are observations of human life without looking for a regularity. Public spaces are used according to repetitive and predictable behaving. Users posses habits and rituals dictated by daily routines. However, occasionally people act freely and use unobvious potential of space. Alice Covatta (2018) writes that urban environment filled with "setting for playful activities such as fun, relaxation, exercise or escape becoming a qualitative manifestation for a healthy human experience". Having this in mind, following thesis was defined: due to their compound set of shapes and forms, post-industrial sites constitute a peculiar type of a testing ground. What can be observed there, can serve as guidelines in two perspectives: prototyping in analogous sites or adapting into entirely different environment.

Revitalization of former industrial sites is a broadly developed topic. Literature covers analysis of revitalization approaches (Gasidło, 2008), good practices descriptions (Pancewicz, 2012), problems and concerns statements (Wyrzykowska, 2025), historical and social significancy recognition (ICOMOS, 2011). There is a gap in terms of synthetic catalogue of architectural solutions linked with specific impact on users' behaviour. The research problem is to capture things that may be rare but happen. Investigation aimed to deliver simplified axonometric models of spaces with potential of playful users' activities to be used in conceptual thinking about revitalization possibilities of former industrial sites in Poland.

First step of this research was post-industrial sites selection based on the criteria: location: Ruhr/ Saar region, revitalization process: ended/ ongoing, original function: mine/ ironworks, operation time: 19th/20th century, structure: complex of several buildings. Second step was in situ visits: Zollverein Coal Mine Industrial Complex in Essen (2023), Landscape Park Duisburg-Nord (2023), Erlebnisort Reden in Schiffweiler (2025), Nordsternpark in Gelsenkirchen (2025), Phoenix West in Dortmund (2025). Observations included: behaviour mapping, selecting places with playful behaviour, collecting data about: paths, greenery and water, post-industrial structures, new structures. Third step was measuring in Google Earth program on 2D aerial view. Fourth step was modelling with AutoCAD program. Finally, created axonometric models of spaces were sorted according to similar activities and compared.

As a result, 16 axonometric models of spaces were created. 8 types of activities were observed: group exercises (dancing, fitness), climbing (a post, a chimney, short walls, 3D sing), group amateur photo shooting, recreation with dogs (group training, playing), playing group games (football, frisbee, tag), riding scooters, sitting and talking/ listening to music from a speaker (on a spoil heap, on steps, while having a picnic) and catching plankton. Most of them were noted on a square (limited by tangible or intangible boundaries) with a rectangular shape. There was a great variety of dimensions with max. 0.2ha for static activities and 0.3ha – 1ha for dynamic ones. It was marked that observed spaces offered multiple-directions walking possibilities, 'domesticated' and accessible post-industrial structures and well-arranged greenery of mixed height and density at close proximity.

This research contributes to the complex process of integrating vast spaces of former industrial zones with citizens' daily life. Further steps include collecting more models of spaces and creating a catalogue of examples. With simplified graphic representations, it is possible to test potential solutions at the very early stage of designing. Using such catalogue would allow to experiment not only in post-industrial sites but also in other public spaces.

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Development and Certification of a Retractable Anti-Terrorist Gate: A Technical Study on Impact Resilience

In recent years, there has been a growing demand for effective security measures for critical infrastructure against threats posed by vehicle-borne terrorist attacks. Solutions that combine high impact resistance with aesthetics, ergonomics, and the ability to integrate with existing architecture and fencing systems are becoming particularly significant. Fancy Fence, a company specializing in retractable gates, has undertaken the development of a new type of anti-terrorist gate that meets the requirements of ASTM F2656 and IWA 14-1 standards, while maintaining its key product feature – the ability to fully retract the structure into the ground in the open position.

The aim of this work is to present the design process, structural analysis, prototype construction, and certification of the new Fancy Fence anti-terrorist gate, designed to stop an N2B class vehicle weighing at least 7.2 t traveling at speeds up to 48 km/h. The research problem concerns the evaluation of whether it is possible to achieve a P1 effectiveness class (penetration ≤ 1 m) while maintaining a compact design that allows the gate to be hidden underground and cooperate with existing lifting mechanisms. The thesis of the paper is to determine whether a properly designed steel structure, supported by Finite Element Method (FEM) analysis and correctly selected dynamic model assumptions, allows for the required anti-terrorist resistance without sacrificing the functionality typical of Fancy Fence products.

The design methodology used in the study was supported by numerical calculations using the Finite Element Method. Preliminary static analyses were performed in the SolidWorks Simulation environment, while crash calculations were carried out in the ANSYS LS-DYNA program using the explicit method. The numerical model included 20 steel profiles with a length of 2385 mm and a 200×100×10 mm cross-section, anchored at a depth of 585 mm. The impact of a 7.5 ton vehicle was simulated at a speed of 50 km/h.

Analysis of the virtual crash test demonstrated that the integrity of the bollards was maintained without shearing, confirming that the key condition for stopping the vehicle was met. Ultimately, main profiles with internal reinforcements were used, resulting in a gate mass of approximately 3500 kg. This necessitated the use of a counterweight of similar mass, which allowed the lifting mechanism's functionality to be preserved using a standard drive.

The actual crash test was conducted at the CTS crashtest-service.com laboratory in Germany. The impact conditions involved a 7.2 t vehicle traveling at 48 km/h. A P1 class according to ASTM and a V/7200[N2B]/48/90: -1.0 m result according to IWA were achieved. The recorded impact energy was 0.67 MJ. Following the collision, the vehicle was deemed non-roadworthy.

The applied design methodology and FEM analyses allowed for the development of a gate that meets anti-terrorist standards while maintaining retractable functionality. This work confirms that even simplified numerical models, given correct material and boundary assumptions, can effectively predict structural behavior under extreme conditions, reducing the need for multiple, costly physical tests.

Furthermore, the results of this study have significant practical implications for the design of modern perimeter security systems, particularly for critical infrastructure facilities such as airports, embassies, or military sites. The retractable gate concept can be further developed toward increasing resistance to heavier vehicles and higher impact speeds, as well as integration with monitoring and building automation systems. Future research also plans to account for variable soil conditions, which may influence the actual behavior of the structure during impact.

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Performative-Installational Model of Visual Communication: Integrated Design and Archetypal Stylization in the "Vertep 2025" Project

Within the contemporary architectural paradigm, the "space of experiment" functions as a dynamic tool for recontextualizing cultural codes. The Vertep 2025 project, developed at the Department of Visual Design and Art of Lviv Polytechnic National University, establishes a foundation for transforming the traditional Christmas narrative into an integrated design system. Responding to socio-cultural challenges, architecture and design are shifting beyond static objects to evolve into ecosystems of communication, enabling new forms of identity transmission through interactive and performative practices.

Theoretical Framework and Novelty

The research is grounded in Nicolas Bourriaud's concept of Relational Aesthetics (1998), treating art as a space for social action, and Yuri Lotman's semiotic theory of culture (1990). The novelty lies in developing a model where the architectural environment functions as an "event field," integrating static objects, live performance, and digital communication. A central methodological element is the strategy of semantic reduction, inspired by Kazimir Malevich's Suprematist principles. This choice represents a performative homage to the restoration of the Kazimir Malevich Prize in Ukraine and Poland in 2025. Suprematism operates as a universal avant-garde language, enabling a departure from descriptiveness by directing attention toward profound spiritual and philosophical meanings.

Archetypal Stylization

The project's core innovation is the radical shift from anthropomorphic iconography to avian archetypes. This strategy of "archetypal stylization" transforms each character of the Nativity play into a geometric architectonic object: Angels appear as doves (peace), the Holy Family as storks (protection), and the Three Kings as peacocks (dignity and honor)... Such symbolic translation allows the narrative to be reimagined through universal archetypes, opening new interpretive horizons.

Multivector Implementation

The integrity of the project is ensured through three interdependent vectors, unified by a single visual design code:

- Architectonic-Semiotic Level (Space): conceptual modules – the White Cube (space of idea, following O'Doherty, 1999) and the Black Cube (space of reflection) – form the structural framework. Within them, avian archetypes appear as large Suprematist compositions, symbolizing a return to fundamentals and serving as intellectual resistance to global upheavals.
- Performative-Anthropological Level (Body): the project shifts from static observation to mobile architecture. Bird codes are embodied in kinetic costume-modules acting as micro-spaces for the actor. Through bodily plasticity, these modules transform into "live sculptures," activating architectural form and engaging the viewer as a co-participant.
- Communication-Media Level (Interface): typography, media design, and branded merchandise extend the project's reach. Merchandise functions as a carrier of memory, embedding experimental ideas into everyday life and ensuring continuity beyond the exhibition space.

Conclusions

The Vertep 2025 project validates the performative-installational model as a catalyst for social interaction. Integrating the White/Black cube dichotomy with kinetic costume-modules, it builds a semiotic bridge between national memory and contemporary audiences, shifting passive observation toward active engagement. Beyond theory, the project shows practical value in exhibition design and cultural programming, ensuring relevance for academia and wider publics. This multivector synthesis—combining architectonics, corporeality, and digital media—creates a holistic ecosystem able to adapt complex narratives to global contexts. Semantic reduction transcodes traditional imagery into universal symbols, actualizing heritage within contemporary design. Thus, integrated design and performative practices enable identity transmission, redefining architecture as a medium for interactive exploration.

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Experimenting with modular spatial models as a tool in the education of architecture students.

In higher education, experiments – such as student projects conducted in a competitive format, in real time, and under time pressure – are an invaluable teaching tool for helping students gain experience that can benefit their future professional careers. The use of experiments in the education of architects is particularly important. This can be achieved through new forms of work and original teaching programs (Strand, Nielsen 2025; Pires, 2024).

This article describes the results of conducting an experiment over a 10-year course cycle. The experiment involved a single spatial modeling workshop repeated annually. The starting point for developing the original program was to create a tool for manipulating the six principal elements of Vitruvius – site, area, division, wall, roof, and opening – within the models built by students. The material used in the experiment was a set of solids and planes based on a basic module the size of a cube, with a side length of one inch and its multiples. The sets described above consisted of solids in 16 sizes ranging from 1 inch to 10 inches (rectangular prisms and cubes) and planes in 8 sizes – compatible with the solids.

In this article, one workshop from the program of spatial impressionistic compositions, used in teaching at the University of Warsaw in urban planning courses between 2012 and 2022, is analyzed. Ultimately, 274 built models were analyzed, in which teams of three students from four, and in some cohorts five, student groups participated (a total of 836 students, depending on the number of students in a given group). The workshop centered on a text-based task that participants were required to complete in the form of a three-dimensional model on a 1:20 scale. The task was to construct an impressionistic composition illustrating the emotions accompanying the contemplation of one of three specified (selectable) masterpieces of European painting: (1) The Oath of the Horatii (Jacques-Louis David), (2) The Bar at the Folies-Bergère (Édouard Manet), and (3) Family of Saltimbanques (Pablo Picasso). The design brief specified that the space should be no larger than 200 m2 and serve as part of a larger complex (such as an art gallery). It was to be a special section of the gallery dedicated solely to this one painting, where it would be temporarily displayed. The goal was to create – through architectural means – a unique and epic setting for its topic. It was necessary to determine how it would be displayed to gallery visitors. No accompanying elements (such as seats, benches, or historical artifacts from the period) were permitted beyond the shape and proportions of the walls and floor themselves.

The research hypothesis posits that a well-defined workshop task within an experimental system serves as a synergistic element in the education of future designers and architects and significantly accelerates the process of learning and developing a spatial imagination.

As a result of the analysis, the research hypothesis was confirmed, and a number of very interesting findings were observed in the results. The distribution of topics was as follows: 1 – first topic 48%, 2 – second topic 38%, 3 – third topic 14% of students, which indicates that the most frequently chosen theme was that of a strong and unambiguous emotional message (the drama of choices in the face of death), while the least frequently chosen was the theme of feelings that are difficult to articulate and partially abstract (the theme of social and professional exclusion in topic 3). This experimental workshop was deliberately selected for this article because it also received very high scores in the evaluations of individual teams. The authors of all 274 mock-ups passed the course; there was not a single failing grade, and a total of 11.6%, or 32 teams, received excellent grades.

The fact that all 274 student teams participating in the experiment competed without forfeit (no failing grades - they didn't give up without a fight) clearly demonstrates the appeal of this experiment to students. The inclusion of a second format of visual arts – in this case, outstanding works of European painting – proved to be an added value, inspiring students to explore new spatial forms in their own work (Duzenli et al, 2019).

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Kinetic Photovoltaic Roof Systems with Biocomposite Components: From Simulation to Field Experiments

Contemporary building envelopes are increasingly conceived as active environmental systems rather than static protective layers (Gonçalves et al., 2024; Nagy et al., 2016). Within this shift, kinetic photovoltaic (PV) systems offer a promising approach to improving solar energy harvesting by adapting panel orientation to changing solar conditions. Their architectural relevance is particularly strong in adaptive roofs and façades, where energy generation may be combined with environmental control and expressive building performance. Yet despite the growing body of simulation-based research on kinetic systems, the translation of such systems into full-scale architectural prototypes remains limited (Olszewski et al., 2017; Zhang et al., 2022), This gap is evident in case of solar tracking systems located in Central European conditions, where the balance between direct and diffuse radiation, seasonal variation, and architectural constraints complicates the assessment of their efficiency.

This paper presents an interdisciplinary study on kinetic roof-integrated photovoltaic systems developed through collaboration between the Faculty of Architecture at Warsaw University of Technology, the Faculty of Physics at Warsaw University of Technology, and the department of Biobased Materials and Materials Cycles (BioMat) at the Institute of Building Structures and Structural Design (ITKE) at University of Stuttgart. The project combined architectural design, parametric simulation, prototyping, material development, and field measurements. To reduce the system's embodied carbon trace, selected structural components were fabricated from biocomposites.

The study origins from previous simulation-based research (Dąbrowska-Żółtak et al., 2024), which evaluated static, single-axis, and dual-axis tracking strategies for Warsaw. That study demonstrated that appropriately configured single-axis systems can approach the performance of dual-axis tracking while remaining mechanically simpler and economically attractive. The aim of the present research is to examine how the performance of kinetic PV systems changes when moving from simplified numerical models to full-scale roof prototypes. The research problem concerns the discrepancy between theoretical gains predicted in simulation and the actual performance of a spatially integrated system operating under real conditions (geometry and motion constraints, shading, construction durability, and environmental exposure).

The research combined simulation-based design with field experimentation. Detailed simulations were developed reflecting the actual geometry of the prototypes, including panel spacing, axis inclination, and restricted angular ranges introduced to reduce mutual shading. Three roof-based configurations were then fabricated: a static reference system, a horizontal single-axis tracking system, and a polar-aligned single-axis tracking system. The prototypes were installed and tested at the test site in Wilga, in central Poland, where measurements were collected for selected sunny days in August 2024.

The comparison of simulation and field results confirmed the general performance trend identified in the earlier study: tracking systems improved energy yield most clearly in the morning and late afternoon, when low-angle solar radiation could be captured more effectively. At the same time, the research revealed that spatial and mechanical constraints substantially affect performance. The need to maintain sufficient spacing between moving PV elements increased the required system footprint, while mutual shading, calibration differences, and prototype manufacturing tolerances influenced the measured gains. These findings suggest that full-scale experimentation is essential not only for validation but also for redefining design priorities in kinetic PV architecture.

The study shows that kinetic roof-integrated photovoltaic systems should be understood as interdisciplinary architectural-energy systems.

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Acoustic Intervention as Experimental Design Practice: A Case Study from an Architectural Design Studio

The acoustic quality of educational environments is both a measurable physical condition and a first-person spatial experience. In architectural schools, students encounter acoustic problems before they can name their physical causes: a room may feel tiring, unclear, resonant, or unsuitable for focused work. Studios and classrooms are usually assessed through functional, visual, and spatial criteria, while acoustic performance is treated as a compliance issue. Recent acoustics-education research argues for a broader model in which students connect listening, measurement, simulation, and design decisions, so that acoustic space becomes both an object of technical assessment and a design material (Milo, 2020; Xiao et al., 2022).

This paper examines an acoustic intervention in teaching spaces at the Faculty of Architecture, Warsaw University of Technology, and asks how a studio-based process can function as an experimental framework for assessing acoustic improvements.

Existing research offers partial frameworks for acoustic intervention in architecture. Soundscape-oriented studies prioritize listening, perception, and situated experience (Milo, 2020; Xiao et al., 2022). Architectural acoustics and retrofit research emphasize measurable parameters such as reverberation, intelligibility, background noise, and acoustic privacy (Yang et al., 2025). Acoustic performance-based design adds simulation feedback for early design decisions (Badino et al., 2020; Peters, 2015). Less explored is their combination in architectural education within one coherent cycle. This paper examines a framework in which students' first-person acoustic diagnosis is tested in situ, developed through simulation-informed design, materialized as a full-scale prototype, and verified by pre- and post-intervention measurements. It asks whether such a process can produce educational value and empirically verifiable improvement of acoustic spatial quality.

The study was conducted with 30 students in the course Case Study 2025, focusing on excessive reverberation in a selected teaching room.

The process followed an evidence-based design logic in six stages: first-person acoustic experience, baseline measurement, simulation-informed design iteration, full-scale prototyping, post-intervention measurement, and comparative evaluation. The design phase was supported by digital modeling in Rhino and Grasshopper, with acoustic simulations performed using Pachyderm Acoustics, allowing students to compare intervention strategies before prototyping.

The baseline assessment identified excessive reverberation, rather than background noise, as the dominant acoustic problem. Mean RT60 was 1.94 s across 63–8000 Hz and 1.51 s in the speech-relevant 500–2000 Hz band. Ten concepts were simulated, and four full-scale prototypes with different geometries and placements were built and retested in situ. Simulations predicted speech-band RT60 reductions of 30–49%, but post-intervention measurements revealed uneven performance: one prototype matched its prediction at about 31% reduction (mean RT60 about 1.05 s), one underperformed at 28% against a predicted 49%, and two produced no measurable speech-band reduction despite favorable simulations. The combined installation reached about 1.12 s, corresponding to a 26% reduction. This gap between predicted and measured performance showed that simulation outputs require empirical validation, turning studio design into an empirically tested research process.

The case study shows that architectural education can serve as a productive environment for experimental spatial research. The combination of diagnosis, simulation, prototyping, and measurement enabled students to move beyond intuitive problem-solving toward a performance-informed design process. The study remains limited to a single teaching context and should be further developed through comparative applications in other settings.

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The Future of Post-War Urban Reconstruction: Experimenting with AI Tools to Restore or Preserve Local Identity

Artificial Intelligence is rapidly making inroads into various fields that require critical thinking, creative action and specialist knowledge. In accordance with Copernican law, inferior currency will likely drive out superior currency; in other words, simple, cheap and efficient applications will quickly gain financial backing and replace well-educated people in the workplace. Therefore, a more ambitious goal for science should be to seek out and support applications that do not replace thinking, but open up new possibilities for it.

There is also space for experimenting with AI in urban planning. Analysing complex phenomena described by countless variables is a task impossible to carry out using analogue methods. It is therefore difficult to grasp the subject and create an algorithm that will harness the computational power of digital machines to achieve the awaited objective. Tools that can recognise the complexity of the problem themselves and seek the right solution can help here. In such a situation, an expert can retain control over initiating the process and evaluating its results.

A city is an extremely complex structure, comprising numerous subsystems operating in the realms of the inanimate, the living, and the immaterial. Simply analysing the interdependencies between these subsystems is a challenge in itself. Meanwhile, the destruction of such a city, brought about by natural disasters or armed attacks, introduces an additional layer of complexity to these phenomena. The city undergoes degradation in terms of its built environment, infrastructure networks, community functioning and natural environment. Along with the destruction of the space, the values, meanings and bonds that were embedded within it disappear. Material losses are merely a visible sign of the death of urban life.

Syrian cities became the most interesting research field to examine this problem. After more than a decade of armed conflict, these face unprecedented challenges in urban reconstruction. Urban areas, such as Al Qaboun in Damascus, face a range of challenges arising from extensive physical destruction, fractured networks of infrastructure, severed social unity, and diminished city identity. Conventional reconstruction methodologies frequently neglect to comprehensively document and analyse the entire spectrum of root causes of issues inherent in conflict-affected situations, instead proceeding directly to design or planning solutions. This phenomenon of avoidance has been demonstrated to have a deleterious effect on the effectiveness and cultural applicability of reconstruction processes.

The recovery process demands methodologies that go beyond traditional approaches to accommodate the intertwined complexities of physical, social, and economic destruction. The research proposes an innovative theoretical framework called “Reconstruction Priority Intelligence” (RPI), which integrates explainable artificial intelligence techniques with community participation to guide urban recovery phases in an adaptive and sustainable manner. The framework relies on gradient boosting algorithms (XGBoost/GBDT) and interpretation techniques (SHAP) to analyse the multiple factors influencing reconstruction priorities, while incorporating a “Human-in-the-loop” mechanism to ensure the alignment of technical decisions with local needs and cultural contexts.

This theoretical idea has been applied to the Al-Qaboun district as a conceptual case study. It demonstrates how the proposed framework can balance economic, social, and environmental priorities in a post-war context. The research contributes to bridging the knowledge gap between artificial intelligence applications in urban planning and the specific needs of reconstruction in conflict-affected contexts. It presents a conceptual framework applicable to other Syrian cities and post-war regions globally.

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Make Space for Experiment to Come

Experimentation – the deliberate act of testing whether something works within a specific spatial context – lies at the core of architecture. For thousands of years, architectural innovation has unfolded in an evolutionary, incremental manner. As Bjarke Ingels observes: “architecture, as anything else in life, is evolutionary. Ideas evolve; they don’t come from outer space and crash into the drawing board” (TEDBlog, 2009). At the same time, twentieth-century experiments in architecture and urbanism demonstrated that experimentation in the built environment comes never without risk. Architecture does not disappear once tested; it remains inhabited, capable of improving or diminishing lives, and at times of being abandoned. This reality often encourages reliance on familiar forms and established aesthetics in order to limit the risks of intervening in complex socio-spatial systems.

The aim of this research is to explore the following questions: How can public space be used as a field of experimentation? How such experimentation can be carried out in a socially responsible and economically sustainable manner? What its benefits are? And how cities can create conditions that support it?

To address these questions, a comparative analysis of tactical urbanism initiatives in Milan, Barcelona, San Francisco, and Warsaw was conducted.

Despite inherent risks, making space for experimentation in contemporary cities appears increasingly necessary. Urban environments marked by diversity and rapid change require practices capable of engaging uncertainty rather than avoiding it. Experimental practices in public space involve a wide range of actors: residents, municipalities, experts and businesses, and thus become a form of collective practice that reshapes both space and social relations.

This dimension is particularly evident in tactical urbanism, where interventions are typically fast, low-cost, and reversible, allowing spatial hypotheses to be tested in real conditions without the burden of permanence. Such approaches correspond to the idea of the “prototype”, inspired by Bruno Latour theories, understood as an open proposition that provokes further inquiry rather than closure.

Examples from Milan (Piazze Aperte), Barcelona (Superblocks), San Francisco (Parklet & Shared Spaces Programs), and Warsaw, including temporary interventions such as “Dotleniacz” at Grzybowski Square, show how public space can function as an urban laboratory. These initiatives enable observation of user behaviour, assessment of impacts, and collection of evidence under real-life conditions.

The analysed cases indicate that experimentation can be socially responsible and economically sustainable when based on reversible, low-cost, and incremental interventions. Temporary installations and pilot projects allow outcomes to be evaluated before large-scale investment. Equally important is the involvement of residents and stakeholders through consultation and co-design, making experimentation a process of shared learning rather than risk transfer onto users.
The benefits are multiple: reduced risk, support for innovation in mobility and public space and improved understanding of how urban environments are used and experienced. Tactical interventions often enhance spatial quality and social activity. Finally, successful experimentation depends on supportive institutional and cultural conditions, including flexible procedures, political willingness to test solutions, acceptance of iterative learning, and support for community-led initiatives. Temporary interventions are not only tools for testing ideas, but also mechanisms for generating knowledge that supports more effective and resilient urban development. Such an approach redefines the role of architecture itself: rather than producing definitive forms, it engages in processes that remain open, adaptive, and responsive. Architecture, in this sense, becomes less an object and more an ongoing inquiry – a practice of learning from the world it helps to shape. Architectural experimentation can no longer be understood as a singular, isolated act, but rather as a process unfolding within a dynamic field of relations. It is precisely this relational complexity that calls for new modes of experimentation – ones that are iterative, reversible, and collectively enacted.

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Public Space as a Laboratory of Accessibility: Experimental Interventions and Inclusive Urban Scenarios

Contemporary public space is no longer expected only to organise movement, leisure, or representation. It is increasingly understood as a dynamic field of social relations, everyday practices, temporary events, and changing user needs. Architecture does not end with the creation of physical form, but continues in the way a place is used, adapted, or rejected by its users. This is particularly important in the context of demographic ageing, social diversity, accessibility requirements, and the need for urban spaces that support participation and autonomy.

Many contemporary urban interventions focus on visual attractiveness, formal novelty, or event-based activation. However, a space may be lively and recognisable, while remaining difficult to use for older people, persons with disabilities, users with sensory limitations, children, caregivers, or people with reduced mobility. The central question is therefore not only whether a public space attracts users, but whether it is accessible, legible, comfortable, flexible, and open to different forms of participation. In this context, the architectural experiment should be understood not merely as a search for new form, material, or technology, but as a method of testing the relationship between spatial framework, action, and diverse users.

The paper aims to examine how temporary interventions, urban activators, participatory actions, and low-budget spatial prototypes may function not only as tools for revitalising problematic urban areas but also as instruments for recognising and verifying the real needs of different user groups. A temporary installation, pavilion, urban garden, or adapted fragment of an underused site may be treated as a spatial test: a provisional framework that makes it possible to observe how people enter, occupy, avoid, modify, or appropriate a place.

The study is based on a qualitative, exploratory, and interpretative methodology. The analysis follows three steps: a review of literature concerning temporary urbanism, tactical interventions, social participation, universal design, and inclusive public space; the intentional selection of case studies representing different models of experimental spatial action; and the development of a preliminary analytical framework for assessing experimental public spaces in relation to accessibility and inclusion.

This framework includes six interrelated categories: physical accessibility, spatial legibility and orientation, sensory comfort and safety, flexibility of use scenarios, participation, and the possibility of correction after observing actual use. These categories make it possible to analyse experimental interventions not only as aesthetic or formal gestures, but as spatial situations that may either enable or restrict participation in urban life.

The paper argues that the value of experimental public space should not be assessed solely through its visual distinctiveness, popularity, or capacity to generate events. Its significance also lies in its ability to reveal hidden barriers, test alternative forms of participation, and support the gradual transformation of urban environments into more accessible, readable, adaptable, and socially responsible spaces.

The analysis leads to three conclusions. First, the activation of public space does not automatically mean its inclusiveness. Second, temporary actions may reduce the risk of long-term spatial exclusion by allowing design assumptions to be tested and corrected in real conditions. Third, participation should be understood not only as consultation, but also as observing, negotiating, and modifying spatial solutions in response to actual use. In this sense, public space becomes a laboratory of accessibility: a place where inclusive urban scenarios can be tested, corrected, and developed in response to changing social needs.

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Accessible and Adaptable Housing as a Condition for the Independent Living of Persons with Disabilities

The article addresses the issue of autonomy among persons with disabilities (PwD) in the context of housing and examines the effectiveness of minimum standards for accessible and adaptable dwellings. The aim of the study is to identify solutions that support independent living, as well as best practices in housing development, with particular emphasis on multigenerational and adaptable housing models, as well as planning principles applied in European cities.

The analysis is situated within the context of key social and political challenges, including the housing crisis, demographic changes, and the need to ensure the right of persons with disabilities to independent living and full participation in society. Particular attention is given to the growing role of supported housing, integrating personal assistance services and assistive technologies – from simple solutions facilitating everyday functioning to advanced smart home systems. This direction reflects a broader European shift from a care-based model to one that supports autonomy.

The article identifies significant research gaps, including the lack of a systemic evaluation of the effectiveness of assistive technologies in addressing the actual needs of users, insufficient recognition of design standards for adaptable housing, and limited analysis of the integration of personal assistance services with housing infrastructure – both in organizational and legal terms. It also highlights the insufficient inclusion of end-user perspectives in design processes, resulting in the dominance of top-down approaches over participatory models.

An important element of the discussion is the economic analysis of implementing accessibility standards. Based on data from Habinteg, it is shown that the absence of the accessible and adaptable housing standard (M4(2)) may generate significant social and individual costs – residents of lower-standard dwellings (M4(1)) may incur expenses higher by up to approximately £27,000 due to the need for subsequent adaptations.
A recent survey shows strong public support for accessible housing in the UK: 72% of adults believe all new homes should be suitable for all ages and abilities, while nearly half think society does not do enough to support independent and safe living as people age. Despite this, awareness of “lifetime” or adaptable homes remains low, and more effort is needed from government and industry to promote their benefits.

Importantly, building to the Category 2 (accessible and adaptable) standard involves only a small additional cost – estimated at around £1,387 per home – which is minimal compared to overall housing prices. This modest upfront investment can help avoid much higher long-term social and economic costs associated with inaccessible housing, such as healthcare expenses from injuries and increased demand on public services.

The article also raises the question of the scope and methods of implementing adaptability principles in European countries. It indicates that adaptable housing, designed to allow flexible adjustment to changing user needs, can provide an effective solution both in the context of an ageing society and diverse household structures, including multigenerational families.

From a public policy perspective, the study contributes to the discussion on the role of national and local authorities in shaping housing environments that support independence, safety, and dignity of life – in line with Article 19 of the UN Convention on the Rights of Persons with Disabilities. The identification of barriers and good practices provides a basis for developing systemic and legislative recommendations at both national and European levels.

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Reading the Palimpsest. The Medieval Location Layout of a Small Town as a Laboratory of Spatial Forms. The Case of Subcarpathia

A city can be read like a palimpsest, a text that has been repeatedly overwritten, in which each era leaves its own layer of meanings, structures, and tensions. This is particularly evident in small towns with medieval origins, where the original layout has not been erased by subsequent transformations. In this sense, they become not only a testament to the past but also a unique laboratory of spatial forms, a place where one can study both the process of "writing" the city and its "reading." The urban layouts of small towns in Subcarpathia, shaped in the 14th century as part of a planned settlement campaign, were an experiment, though not in the modern sense of the word. They were rather an attempt to establish an order: geometric, social, and economic. The market square as the center, the street network as a communication structure, and plots as modules of everyday life all created an extremely coherent spatial model that emerged from a prototype replicated in various locations. However, what was intended to be repeatable always underwent transformations in practice under the influence of local conditions: topography, social relations, history. It is in this tension between the model and its implementation that the experimental nature of the city is revealed. It is not a closed project, but a process constantly negotiated between the past and the present. Interpreting this process requires a perspective that extends beyond technical analysis. It demands sensitivity to traces, not only material ones, but also symbolic ones. The layout of the market square, the proportions of the building blocks, and the relationships between public and private spaces all speak to the ways of life, hierarchies, and values of past communities.

The present poses new challenges to heritage. Small towns are losing their functions, residents, and often their sense of identity. Therefore, a return to their medieval "legacy" is not a gesture of nostalgia but an attempt to understand the spatial and functional mechanisms that fostered their durability and cohesion. They can be treated as archives of solutions, not for direct replication but for reinterpretation. An urban experiment does not necessarily create new radical forms. It can consist of carefully reading what already exists. The small towns of Subcarpathia offer a unique opportunity in this regard: they are sufficiently "legible" to discern their original layout, yet sufficiently transformed to reveal the dynamics of change within them. In this sense, they become a laboratory not only for designing the future but also for reflecting on the very nature of the city. They show that space is not merely a physical arrangement but a carrier of memory and experience. An experiment, rather than breaking with the past, can be a dialogue with it. The concept of an ethical city, developed by Zuziak, integrates earlier approaches to urban planning, combining aesthetics with ethics and highlighting the need for harmonious spatial design. In this context, urban development should respect cultural, social, and environmental values, which, in the case of small historic towns, is particularly significant for the preservation of their identity. Reading the urban palimpsest is thus an act of interpretation, but also of responsibility. The way we read these layers determines which new layer is added next.

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Creating Common Space and Good Neighbourliness

A Case Study of the OSBB Kniazhe Residential Community in Lviv, Ukraine.

The quality of residential environments depends not only on architectural solutions but also on the social structures that shape everyday life between buildings. In Ukrainian cities, condominiums known as OSBB (Associations of Co-owners of Multi-apartment Buildings) represent a unique form of collective governance that directly influences the creation and maintenance of shared spaces. This paper examines OSBB Kniazhe, a residential complex at 5K Kniahyni Olhy Street in Lviv, built in the early 2000s on a former military compound site by the Karpatbud corporation. The study explores how shifts in management models and community-driven spatial interventions foster good neighbourliness and improve the quality of common space.

Despite several locational advantages – proximity to schools, medical facilities, Horikhovyi Hai Park, and well-developed public transport and cycling infrastructure – the residential complex suffered from significant spatial deficiencies. The courtyard environment was dominated by individual garages with separate entrances, creating large impermeable asphalt surfaces, visual monotony, and so-called "dead zones" devoid of social activity. The children's playground was neglected, and there were no spaces for teenagers or adults. The six-storey buildings lacked elevators, severely limiting accessibility for elderly residents, families with small children, and people with disabilities. Additionally, the absence of commercial premises on the ground floors and persistent technical problems – roof leaks, heating and sewage malfunctions – contributed to residents' dissatisfaction and social isolation.

The management structure initially followed a vertical model, where a single person made all decisions with little transparency or community involvement. This hierarchical approach led to poor communication, mutual distrust, indifference, and even aggressive interactions among neighbours. The turning point came with the transition to a horizontal, democratic management model based on teamwork, open discussion, and shared problem-solving. Residents began to unite around common concerns, forming initiative groups that addressed both governance and spatial issues directly.

Several community-driven spatial interventions illustrate this transformation. The guest parking area, which had become an informal car storage zone blocking garage exits, was reorganized through a simple yet effective solution: painted parking lines resolved ninety percent of access conflicts. A more ambitious project involved the creation of a small public square with a safe play environment for children, achieved by closing a vehicular road between two courtyard areas. This intervention expanded the safe pedestrian territory, allowing children to play without constant parental supervision and reducing traffic-related hazards. Tree pruning and new pathway design improved visibility, comfort, and the overall aesthetic quality of the courtyard landscape.

The introduction of a digital governance platform called "Dakh" – a web-based and mobile application for all OSBB participants including board members, accountants, and co-owners – further strengthened transparency and communication. This tool enabled residents to participate in decision-making processes regardless of their physical mobility or schedule constraints.

The outcomes of these combined spatial and organizational interventions are significant: the formation of an active residential community, co-financing of improvement projects, creation of recreational and leisure spaces, enhanced safety, reduction of toxic communication patterns, and the emergence of a core group of engaged residents. The case of OSBB Kniazhe demonstrates that good neighbourliness is not merely a social phenomenon but a designable condition, achievable through the integration of participatory.

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Between Block and House: Atrial Dwellings as a Social Experiment in the Polish People's Republic, 1960s–1980s

In PRL Poland, housing was divided into two categories: the prefabricated estate and the individual house. Industrialised block construction – driven by speed, cost and scale – had priority; detached housing was secondary. In the early 1960s, the atrium house entered Polish discourse through Danish work (Utzon). Architects led by Adamczewska-Wejchert (1976) adopted it as a tool for densification and ordering residential districts. The phenomenon remains poorly recognised in historiography.

Thesis: Polish atrial dwellings of the 1960s–1980s formed multidimensional social experiment – typological (testing compact low-rise housing as a tool for urban densification and order), organisational (bottom-up coalitions delivering housing outside mass prefabrication), and negotiated (mediating dwelling form between social groups and central planning). All three dimensions return today as answers to contemporary challenges: densification, housing co-operatives, and planning consensus.

Literature is sparse: the only Polish monograph (Adamczewska-Wejchert, 1976) is promotional-projective. Recent accounts of PRL detached housing sit within broader syntheses or are regional (Napieralska, 2020). Edwards (2006) provides the comparative frame. The method combines historical-architectural analysis with tools of the new humanities: Actor-Network Theory (Latour & Yaneva, 2008), treating the architectural object as a network of heterogeneous actors, and oral history, reading architecture through designers' and inhabitants' testimony.

The Za Cytadelą estate in Poznań illustrates the thesis. Built 1978–1990 on the northern edge of Cytadela Park, its central, topographically difficult parcel holds single-storey terraced atrial houses combining a residential wing with a studio wing. The estate was initiated by Zbigniew Bednarowicz, president of the Poznań chapter of the Union of Polish Visual Artists (ZPAP): plots were granted at symbolic prices to artists and faculty of the State Higher School of Fine Arts (PWSSP) – Krzymańska, the Kopczyńskis, Teisseyre – hence the informal name "Artists' Estate" (Osiedle Twórców - after Błaszczyk, 2011). This was a co-operative model: a milieu-based coalition replacing the large housing co-operative. Future residents collectively secured land, design, and materials; houses were then built individually. The architect Jerzy Szmidt, who designed the atrial section, was simultaneously designing atrial houses for the "Osiedle Młodych" co-operative – evidence of the typology circulating across actors.

Through an ANT lens, all three dimensions intersect here: typological (the atrial layout as mediating object, joining domestic intimacy with studio productivity), organisational (the ZPAP/PWSSP coalition), and negotiatory (architect-as-mediator, state policy favouring creative workshops, traditional construction against the concrete panel). The estate was never labelled an "experiment" in official policy – yet this peripheral status allowed it to test forms of shared dwelling unavailable to prefabricated estates. An analogous mechanism appears at Warsaw's Orężna (1969-1974), co-designed by Adamczewska-Wejchert for Warsaw University of Technology faculty, and in Wrocław atrial houses of the 1960s: professional groups negotiating dwelling form recurs geographically.

The analysis confirms the thesis: the Polish atrial house was a social experiment in a threefold sense – typological, co-operative, negotiatory. Its significance lies less in form than in the configuration of actors that briefly opened a third path between block and house. These dimensions carry present-day relevance: the compact low-rise model answers calls for densification and for taming suburban sprawl; the co-operative mode returns to Polish housing discourse; negotiatory design is becoming a path to planning consensus. The atrial legacy is thus not only heritage but a working resource for today's housing policy.

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The Public Space of the River's Surface in the City as a Space for Experimentation.

Human coexistence with the river has always been an experiment. The existence of urban structures on the edge of the water element has been linked to pushing and shifting the boundaries of technological capabilities. At the same time, each era has introduced its own distinctive way of utilizing public space along the water's edge. Thus, the aquatic space of experimentation combines technological and social experience (Matusik, 2025). It thus reflects the functioning of the contemporary social model. The postindustrial era has blurred the physical boundary between the aquatic environment and the urban landscape. This has opened up new opportunities for social experimentation in direct contact with the aquatic environment. This fact also poses risks associated with excessive development of certain river sections in cities, which may lead to a loss of landscape and ecological value and, consequently, social value (Hysa et al., 2025). At the same time, there is a strong trend toward creating a natural river landscape within the city – a return to the river's natural state. This trend, linked to a shift in perceptions of environmental issues such as the renaturalization of river channels, has both economic and social implications (Fernandes et al., 2025). At the same time, research indicates a steady increase in the public's need for contact with water spaces in their various forms (Wilczyńska et al., 2026).

This article focuses on the social aspects of new spatial solutions along the river's urban surface. The aim of the study is to identify the relationship between the water's surface and social preferences regarding the choice of recreational areas. The primary objective is to identify social models that can help optimize design decisions. The research focuses on examples implemented in European cities such as Kraków, Berlin, Copenhagen, and Graz.

The characteristics of the river were of particular importance for this study. The study took into account the river's flow dynamics, the specific nature of the natural landscape, and the role of urbanized sections, which influence the stimulation of social perceptions. The research method, based on a comparative analysis of examples, enabled the creation of a typology of objects that accounts for how they function in the social sphere. The following indicators were adopted to construct the typology: social presence (dynamic / static), accessibility (physical / visual), contact with the water surface (direct / indirect / distance), connection of the element to the waterfront edge (fixed / dynamic), function (floating volumetric structure – building; floating spatial structure – platforms, green islands, artificial swimming area/beach; pedestrian footbridge/pedestrian-bicycle bridge).

The research identified three groups of social models: the dynamic model, the static model, and the hybrid model. These models illustrate shifts in social preferences regarding access to riverfront spaces in the city. At the same time, applying them in the design of structures along the riverfront enables the optimization of the planning of its public spaces.

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Architectural Experimentation under Limited Professional Autonomy

Within design pedagogy, the architectural project is described as an exploratory act aimed at solving spatial but also social challenges (Leatherbarrow, Mostafavi, 2002), whereas in practice it is frequently constrained by economic pressures, search for efficiency and quick realisation. This article examines how such conditions affect early-career architects and contribute to the gap between educational ideals of experimentation and professional realities.

This study adopts a qualitative, literature-based methodology. The argument is developed through a critical review of academic and professional literature on architectural labour and education. The article also draws on secondary data from published surveys concerning graduates' expectations and their professional experiences.

Experimentation in the context of architecture requires that designed spaces and buildings come into existence in order to observe the results and draw conclusions. Without this materialisation, it is impossible to fully assess the impact of a project and its consequences, as it remains merely theoretical. In the current neoliberal structure of the market, which emphasises efficiency and standardisation, the experiment remains the domain of the theoretical sphere as in practice it encounters numerous constraints. Experimenting requires time, research, and general acceptance of failure and reiteration; thus, it is often considered to be inefficient or commercially risky. Client expectations, tight schedules, and regulatory requirements frequently prioritise predictable outcomes over exploration.

During university education, students are being taught values and principles both in the context of design and morality and are encouraged to see themselves as agents of change. While this pedagogical model aims to produce responsible and creative individuals, it often promotes an unrestrained design approach and supports maintaining the design studio class as a bubble separated from the real-world constraints and the limited, instrumentalized reality (Deamer, 2020). Although architects are encouraged to view their ideas as socially transformative, their actual workplaces rarely allow the development of such ambitions. Architecture students are told that their ideas can solve pressing problems in the world and yet, in practice they are limited in their capacity to develop those ideas by the same pressures of neoliberalism that shape all other industries - the coercion to monetise all activities and maximise efficiency (Cortright, 2021). This discrepancy is clearly noticeable as students move on to start their professional career. In a survey conducted by CAA, 35% of respondents reported a "Perceived Disconnect between Education and Practice" and, in addition 50% of respondents indicated "Limited Professional Influence and Agency" in their current place of employment (Commonwealth Association of Architects, 2026). These experiences point to a structural misalignment between how architecture is taught and how it is practiced.

Experimentation undoubtedly remains a crucial component of architecture, as it enables progress and supports the development of innovative ideas. However, its implementation is clearly limited in professional environments. Architectural education must acknowledge that architecture is a form of labour rather than a higher calling. Defining it as such does not diminish its significance but allows to shift the perspective of young architects regarding how they perceive their work and their role in contemporary world. Such shift can help reduce the gap between their expectations and professional reality.

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The modern construction industry is facing increasing environmental challenges, related to, among other things, excessive energy consumption and high carbon dioxide emissions. It is estimated that this sector is responsible for approximately 35% of global CO_2 emissions and generates between 45% and 65% of waste deposited in landfills (Narloch, 2025). Mycelium-based composites are among the materials showing great potential. These materials are formed through the colonization of lignocellulosic plant residues by mycelium, which binds the substrate into a lightweight, porous, and biologically cohesive structure with low thermal conductivity (Miśkiewicz et al, 2023). Although the use of mycelium in architecture is becoming increasingly popular, and such materials are gaining growing market interest, their application is still associated with numerous challenges. They are primarily due to the unconventional appearance and specificity of the material, as well as the negative perceptions associated with fungi [Lewandowska, 2024]. Mycelium-based products can be fabricated, among other things, from agricultural waste (Appels et al., 2019), thereby reducing waste and supporting the principles of the circular economy. In addition, mycelial composites are lightweight, biodegradable and have good thermal and acoustic insulation properties (Appels et al., 2019). Thanks to these features, mycelium-based materials can be used both as structural elements and insulation. The properties of the materials produced depend largely on the species of fungus used to produce the composite, but also on the type of substrate on which the mycelium develops. Research by Elsacker et al. (2019) showed that the type of fiber (hemp, flax, or wood waste) has less influence on the compressive stiffness of the material than the form of the fiber, such as loose, cut, in the form of packs or pre-compressed (Elsacker et al., 2019). This means that proper substrate preparation and control of the mycelium growth process play a key role in shaping the properties of the final material.

The research presented in this paper involved the production of a composite material using the mycelium of Ganoderma sp. grown on agri-food waste. During the production process, the fungus was cultivated on a properly prepared substrate and then placed in molds to achieve the desired shape. The resulting composites were additionally coated with a natural biopolymer, the use of which was to increase their mechanical strength and improve the durability of the material. The produced composites were characterized by good insulation properties and satisfactory mechanical strength. A thermal conductivity coefficient of around 0.07 W/(m·K) (test procedure in accordance with standard PN-EN 12667:2002) and a compressive strength of an average of 280 kPa (test procedure in accordance with standard PN-EN ISO 29469:2023-05) were achieved, indicating the high potential of this type of biomaterial in construction applications. Additionally, the produced material was found to be safe for humans and animals, free from toxic elements such as mercury, lead, arsenic, cadmium, and chromium, and contained valuable macro- and microelements, indicating its potential for further applications (including Ca, K, Mg, and P). The results suggest that mycelium-based composites could be a promising direction for further research into ecological and sustainable architecture.

Further research aimed at enhancing durability, water resistance, and mechanical properties has the potential to expand the application of fungal biomaterials in sustainable architecture, green building practices, agriculture, horticulture, and other sectors. The increasing interest in mycelium-based composites in low-carbon construction indicates the potential for these materials to become a viable alternative to traditional building materials in the future.

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Mosaics and Ceramic Panels as a Way of Shaping the Urban Space of Lviv (Second Half of the 20th Century – Early 21st Century): Issues and Prospects

The urban space of Lviv has historically evolved through a synthesis of architecture and monumental art. Mosaics and ceramic panels occupy a special place within this ensemble – they are a living embodiment of the city's aesthetics, preserving the memory of the past whilst responding sensitively to the challenges of the present.

The aim of this article is to examine mosaics and ceramic panels not merely as decorative elements, but as complete works of art that shape the urban space, influence visual culture and construct our collective memory.

In Lviv during the post-war period of the second half of the 20th century mosaics and panels adopted the role of aesthetic focal points and even dominant features of the urban landscape, counterbalancing the uniformity of standardised Soviet architecture. Under the totalitarian Soviet regime, monumental and decorative art was subject to the ideological influence of Socialist Realism, although in some cases it evolved into a distinct form of creative expression: despite ideological pressure, monumentalists from Lviv managed to develop a unique artistic language. The use of abstraction and generalised imagery enabled artists to preserve their national identity (Cherkes, 2008), whilst integrating these elements into the canons of the official tasks of the time.

Since the declaration of Ukrainian independence in 1991, particularly following the start of the full-scale Russian invasion in 2022, the issue of protecting the monuments that define Lviv's authenticity has become extremely relevant. Some works, such as the mosaics by Volodymyr Patyk ("Fishes"), Bohdan Soika ("Information Space") and Roman Patyk ("Impulse and Control"), have been successfully restored following their destruction. However, others, notably Borys Gorbaliuk's work on the building of the Lviv Ceramic and Sculpture Factory, have been lost forever.

The issue of preserving mosaics and ceramic panels in Lviv is complex and multifaceted, as these works of monumental art lie at the intersection of historical memory, ideological heritage and contemporary urban changes (Posatskyi, 2007). A significant proportion of the mosaics and panels created in the second half of the 20th century are now at risk of destruction or defacement due to building renovations, façade insulation, changes in the functions of urban spaces and the lack of adequate conservation status (Piddubna, 2017). An additional challenge is the ambiguous perception of these structures as relics of the Soviet era; they are often not regarded as genuine cultural value. At the same time, the mosaics and panels serve as important testimonies to their era, embodying artistic practices and local identity, and forming part of the city's visual landscape.

Today, the preservation of these sites requires a comprehensive, experimental approach – ranging from documentation and research to legal protection and promotion. To this end, in 2024, the Lviv City Council established a working group on the preservation of mosaics, comprising art historians, architects and councilors. An important step was the marking of the monuments with the international "Blue Shield" symbol, as well as the creation of an interactive map showing the locations, names, dates of creation and photographs of the most significant works.

The preservation of mosaics and ceramic panels is not merely a matter of restoring the physical structure, but above all of safeguarding the city's multi-layered memory. Only through professional reinterpretation and active promotion can these works be fully integrated into the urban space of the 21st century, whilst remaining an integral part of Lviv's authentic character.

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An Experiment in the Public Interest: Intuition ahead of Theory in Healthcare Architecture

The ageing of the 21st century society presents a challenge for contemporary architecture. In future, an increasing proportion of the population will rely on healthcare and care facilities, and the quality of their design will determine their quality of life. Today, this type of architecture comprises a category of highly complex healthcare facilities that lack features designed to provide comfort. This has been the subject of a long-standing debate amongst doctors and architects alike regarding the humanisation of healthcare architecture (Bates, 2018). The concept of "architecture of care" posits that a multisensory space rich in familiar patterns supports the recovery process. In describing this phenomenon, Charles Jencks even went so far as to use the term "architectural placebo" (Jencks & Heathcote, 2010). Nowadays, however, the psychological effect provocatively named by Jencks, along with other phenomenological propositions, is finding systematic support in neuroscientific research, which provides rigorous evidence for what was previously assumed intuitively.

Indeed, the history of healthcare architecture provides fascinating examples of how effective design solutions are often implemented long before a coherent scientific theory has been formulated, transcending the boundaries of the discipline. Since ancient times, water has been used for therapeutic purposes, even though the field of balneology did not provide evidence of this until hundreds of years later. In this case, intuition preceded theory by centuries. The practice of spa therapy developed before scientists could describe its mechanisms. Intuition drew, of course, on the immense symbolic and cultural potential of water, which, being a prerequisite for human existence, served as a repository for people's dreams, hopes, fears and reverence. In particular, immersion in water symbolised complete renewal (Eliade, 1958/1997) and possessed a symbolic power that was internalised by spa architecture and practice. Physical contact with water became the equivalent of a ritual of rebirth.

An example of the intuitive experimentation applied to contemporary architecture can be seen in the Maggie's Centres, small psychological support centres located within oncology wards across the UK. These are not strictly medical facilities; they do not offer medical treatment but instead provide a safe and welcoming space for patients. Their architecture is imbued with familiar patterns, such as a central kitchen with a large table reminiscent of a home, or a garden offering contact with nature. Jencks emphasises that whilst architecture cannot cure people, well-designed spaces can instil hope, offering comfort, sparking curiosity and inspiring a desire to keep going and not give up. For a long time, this effect eluded quantification, yet it was consistently implemented in design practice.

There are many measurable benefits of spending time in natural surroundings or a carefully designed relaxation area, such as increased dopamine levels, reduced cortisol levels and lower blood pressure (Eberhard, 2009). Jencks points out, however, that some of the positive effects architecture has on the viewer cannot be measured – and that this is precisely why the designer's sensitivity must sometimes take precedence over the researcher's methodology. "Buildings will certainly evolve towards hybrid Maggie's Centres; they will become more complex, ambiguous, spiritual, humorous, friendly, daring and full of life. Architecture will return as a normal part of healthy living" (Jencks, 2013). Imagination and following one's intuition – provided it is rooted in concern for people – can be an immensely effective form of social experiment.

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Record. Message. Experiment.

Methodological observational research, employing an interpretative approach, demonstrates the relationship between the constructing of architectural space and philosophical interpretations of concepts, as well as established theories of perception, including Marshall McLuhan's theory. By doing so it indicates that the essence of Architecture, when interpreted as a medium, lies in the transmission of information (message). In this case, an architectural object can be interpreted in two ways: in terms of content or material form, i.e. its specific formal "packaging". Both spheres, regardless of the assessment of their mutual semantic adequacy, constitute a whole, forming a coherent or incoherent structure. Record – concerns content or matter. The same applies to the message. Both cases are the inspiration for a supra-architectural Idea, influencing the activities of the building's users or the urban development concept. Record – creates an archive of information. Message – its presentation. Record and message are subject to interpretation depending on the social, historical and spatial context.

History is, in a sense, created by chroniclers and historians, who describe and interpret events and imbue them with a subjective narrative. Architecture – recorded in a unique archive of history, created by interpretation of the whole world of concepts and taking form in material manifestations – is not solely created by architects. In this sense, anyone who records it becomes a subjectively interpretative archivist – a historian. The record – inherently ephemeral in its very nature – concerns the content or substance of Architecture. The recording of content affects the structure of the matter. Apart from the record, what is essential to interpreting the meaning and understanding the concept of the experiment is its message. In the case of an architectural object, a complex of buildings or urban development, the record takes place in two stages. The first is the Concept design. The second is its material manifestation – the resulting building or complex of buildings. In both cases, the record can be presented as the conveyance of the Idea's message. It is, however, addressed to diverse audiences. In the first case, these are professional and decision-making bodies. In the second, the recipient is, in a sense,"everyone" – for, as Rem Koolhaas put it, an architectural object constitutes a universally accessible artefact in the material archive of social education, becoming the currency of social enlightenment.

"If you change the way you see the world, you change the world you see." These words, used in an advertisement for a programme that blends the real world with holograms and virtual reality – and which also promotes Microsoft's HoloLens and virtual assistant Cortana – herald the dawn of an AI-driven world. Presented over ten years ago, in 2015, at a conference at the Moscone Center in San Francisco, they can still serve as inspiration for creating experiments in many fields. As the foreshadowing of potential creation of architectural spaces in areas previously inaccessible, they constitute a unique conceptual experiment – an antithesis to the meaning of the Vitruvian Triad… When it comes to conceptualising architecture, we are faced with two possibilities. On the one hand, it is a kind of replication of the interpretations of existing meanings. On the other hand, it is a negation of these meanings, arising in a specific semantic opposition to existing models and their corresponding symbolic meanings.

The experiment may concern the record itself or the message of that record. Consciously created, it arises in the realm of supramaterial creation of the Idea of the concept of Architecture. It is a paradigm shift in the interpretation of commonly known signs, symbols, relationships, and their connotations. It suggests an understanding of these that differs from the existing one, or creates entirely new, previously unknown interpretations from new signs and symbols. Therefore it constitutes a specific palimpsest – a semantic overwriting within the sphere of adopted Ideas. It expresses the New inscribed upon previously existing content. It suggests interpretations of this "New" that differ in meaning – and, consequently, their adequate expression in material manifestation, that is, an implemented object. In architecture, interpretation of the archetype is crucial. It is created in the record and interpreted in the message. The matter of interpretation plays a key role. Its construction replicates established meanings, creates new ones, or redefines existing ones.

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Spatial Experiments for Children's Creativity

In the context of preschool education, spatial design may play an important role in supporting children's creative development and independent exploration. As Henry Sanoff notes, one of the primary goals of a learning environment is to encourage children to generate novel responses by providing materials that enable manipulation, improvisation, and sustained involvement (Sanoff, Henry, 2016). Research on preschool environments further indicates that adaptable and changeable spaces may positively influence creativity. According to Şahin and Dostoglu, a flexible environment can encourage children to think differently and propose new ideas (Şahin, Neslihan, 2021). One way of achieving this is through the introduction of spatial elements that children can independently modify, changing the appearance, function, or organization of their surroundings.

The topic of enhancing creativity through kindergarten design emerged during PhD research focused on Polish public kindergartens. Preliminary observations revealed a significant degree of uniformity between the analyzed classrooms, particularly in terms of furniture and interior equipment. Most spaces contain relatively few elements that allow children to actively shape or transform their environment. Interactive and changeable architectural features remain uncommon.

This paper aims to identify practical design solutions that may stimulate creativity through adaptable spatial elements in kindergarten architecture. The study is based on selected case studies and serves as an introduction to further research within the context of existing Polish preschool buildings.

One of the most notable examples is the Montessori School in Delft, designed in 1966 by Dutch architect Herman Hertzberger. The kindergarten hall contains a square recessed area filled with loose wooden blocks that children can rearrange freely. The blocks function as stools, construction elements, or toys, enabling children to create seating arrangements, towers, trains, or secluded play areas. Hertzberger emphasized that the recessed space evokes a sense of retreat and enclosure, encouraging imaginative use and spatial exploration (Hertzberger, 2016). This example demonstrates how a simple architectural intervention can support both creativity and social interaction.

A more contemporary example is the kindergarten in Tromsø designed by 70ºN Arkitektur in 2006. The project introduces flexible "play walls" functioning as interactive spatial dividers. Each side of the wall offers different activities and incorporates foldable furniture, closable openings, chalkboards, and pin-up surfaces. Their movable and multifunctional character allows children to continuously adapt the space to changing needs and activities. Such solutions also introduce children to the concept of adaptable architecture and resource-conscious design.

Another example is the Kekec Kindergarten extension in Ljubljana designed by Jure Kotnik in 2010. The façade consists of revolving vertical slats that can be manipulated by children. One side retains the natural wood finish, while the other is painted in bright colors, allowing children to visually transform the appearance of the building. Beyond its technical role as shading and protection, the façade becomes an interactive play element and a tool for user participation.

The presented examples demonstrate that creativity-supporting spatial solutions can be implemented both in new buildings and renovated preschool facilities. Although such approaches remain relatively rare, further research and systematization of design methods may increase awareness among architects, educators, and decision-makers regarding the role of adaptable space in preschool education.

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A Reinterpretation of the Museum's Historic Courtyard for an Experimental Creation

Introduction

In contemporary museums located in historic buildings, exhibition space is insufficient. It often fails to meet contemporary standards and needs. Permanent exhibition spaces are designed as the institution's main exhibition spaces, often with the intention of presenting them unchanged for decades. Creating temporary exhibitions, however, is a multi-stage process combining curatorial concepts with logistics and spatial arrangement (Lorenc J. Skolnick L. Berger C., 2008). Therefore, temporary exhibitions are mobile and adapted to the location where they are presented. In this area, space for experimentation is essential, as exhibition design is a creative process. The author's analysis indicates museums' efforts to consciously migrate selected exhibits to provide exhibition spaces that transcend closed walls. Such efforts are intended to open up to contemporary audiences.

The aim of this article is to analyze the adaptation and reinterpretation of existing Polish museum courtyards into new exhibition and cultural centers in historical spaces, based on selected objects.

Case Studies and Adaptation Directions

For the purposes of the analysis, six museum facilities located in historic buildings were reviewed. The basis for the review was their internal courtyard, entrance plaza, and green area, which extend the exhibition and cultural functions of museums (Pudelska K., Mirosław A. 2012). The role of heritage in the social and cultural significance of contemporary spaces cannot be overlooked (Sowińska-Heim, 2018). The democratization of the exhibition space at the "Playground" at the WRO Art Center, as an example of demusealization, is a thread of the social experiment (Wicher C. 2022). The evaluation criteria for the examined facilities were: protection of the historical fabric, multifunctionality, and the space as a local center.

- State Archaeological Museum in Warsaw (PMA)
- Jacek Malczewski Museum in Radom (MJK)
- Courtyard of Culture in Warsaw (NDK)
- National Museum in Krakow (MNK)
- Museum of Architecture in Wrocław (MA)
- Wawel Royal Castle in Krakow (ZK).

Results

Contemporary museum courtyards serve as multidimensional spaces integrating educational, social, and cultural activities. Adapting them to new needs requires an interdisciplinary approach combining legal, conservation, and architectural. MMA organizes outdoor exhibitions, craft workshops, and the 30th Science Festival, as well as music festivals and classical music concerts performed by the Sinfonia Varsovia orchestra. MJK showcases a variety of green spaces that have been transformed into venues for cultural events. NDK demonstrates how courtyards can be used for concerts and events. The MNK Museum Forum aims to create a meeting space for artists, the public, and city residents. Planned renovations and expansions of the MA Museum by 2026 will enrich the exhibition space and promote social dialogue in Wrocław. ZK focuses on adapting historic arcaded courtyards for tourism, concerts, and exhibitions, while preserving their historical character. Overall, the article highlights the integration of cultural activities into the urban space to foster community engagement and artistic expression.

Conclusions

Reinterpreting the historic museum courtyard requires a paradigm shift: from a passive backdrop for exhibits to an active narrative space. Key directions in this process include the incorporation of ephemeral forms (sound, light, performance), the synthesis of historical rigor with modern methods of participation and treating the space as an educational "third educator." Breaking the mold of stationary exhibitions in favor of "site-specific" activities transforms the viewer into an active participant, strengthening the emotional connection with cultural heritage and giving it a contemporary dimension.

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Between Iconic Status and Accessibility: The Implications of Spatial Experiments for Universal Design

For many designers, architecture is a unique field in which each project is treated as an individual, one-of-a-kind work, often a formal or technological experiment. Projects of this kind, often designed by “star architects”, aim to influence the cultural landscape and, at times, become its icons. In such cases, the artistic value and recognizability of a building may be placed on par with its functionality, and sometimes even above it. The remainder of design practice, which some do not classify as architecture but only as construction (Piątek, 2024), encompasses structures that are technically and functionally sound but do not aspire to be considered iconic works of architecture. This dichotomy marginalizes accessibility: pursuing formal originality often neglects universal design principles, limiting space access for users with special needs. This article analyzes the tension between architectural spectacle and universal design, using selected projects from the last 25 years.

Case Studies and Universal Design Principles

Analysis applied seven universal design principles (equitable use, flexibility in use, simple and intuitive use, perceptible information, tolerance for error, low physical effort, size and space for approach and use) as evaluation criteria (Preiser, Ostroff, 2001). Selected objects were analyzed in their context and in context of detailed guidelines for universal design (Warsaw City Hall, 2017).

- Daniel Libeskind, Jewish Museum, Berlin, 2001, JM
- Jean Nouvel, Torre Agbar, Barcelona, 2005, TG
- Foster & Partners, Apple Fifth Avenue, New York, 2006, AP
- Daniel Libeskind, Royal Ontario Museum, Ontario, 2007, ROM
- Santiago Calatrava, Ponte della Costituzione, Venice, 2008, PdC
- Herzog & de Meuron, Elbe Philharmonic Hall, Hamburg, 2018, EP
- Steven Holl Architects, Hunters Point Library, New York, 2019, HPL

Results

Major accessibility shortcomings in the analyzed objects (codes from the list above are in parentheses):

- Inaccessible key spaces due to stairs without alter natives (e.g., ROM, HPL, EP).
- Objects are not adapted to the needs of people with special needs, main entrances unfit for special needs users; separate routes required (e.g., Ap, original version of PdC).
- Sloped walls, low ceilings, unprotected sharp edges (e.g., ROM, TG).
- Long/narrow corridors lacking rest areas (e.g., JM).
- Collision-risk glass surfaces (e.g., Ap).
- Spatial solutions disorienting users, lack of solutions facilitating orientation in space, e.g., guide paths (e.g., JM, EP, ROM).
- Poor surface contrasts hiding edges (e.g., EP, ROM).
- Slippery or glare-causing surfaces (e.g., PdC).

Conclusions and Recommendations

The tension between pursuing iconicity and implementing universal design principles is not inevitable; it stems from specific priorities in the design and decision-making processes. Many identified issues could be resolved without harming aesthetics. Key steps for better accessibility:

- A paradigm shift: universal design should not be seen as a constraint on creativity, but rather as a way to expand it, a challenge, or even a field for innovation.
- Education/awareness: architects, investors, and users require better understanding of the long-term social and economic benefits of inclusive solutions.
- User participation: involving diverse-needs individuals from project outset.
- Regulatory support: enforce standards beyond minimums for true universal design.
- New value criteria: redefining what makes architecture “outstanding” by including inclusivity as a core quality criterion.

21st-century architecture must dazzle with democratic access and empathy for varied needs. Though embedded in handbooks, education, and discourse (Kuryłowicz, 2024), universal design demands constant reinforcement.

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Flood as a “Natural Experiment”

Flood can be interpreted as a natural experiment, enabling an empirical assessment of the functioning of urban systems under extreme conditions. Unlike numerical simulations, such events provide direct data on water levels, flow dynamics, and the relationship between historical hydraulic infrastructure and contemporary spatial development.

In September 2024, the Genoese Lowlands led, among other things, to flooding in the Czech-Polish border area, with Nysa being one of the most affected cities. Field studies conducted from the onset of flooding until the outflow phase provided significant empirical data. The analyses focused, among other things, on: The possibility of ultimately raising the water level in the city’s canals to the level it has maintained since the Middle Ages, i.e., by approximately 2.0 meters.

The proposal to raise the water level is justified in several areas. First, it concerns the safety of historic buildings, which, due to high groundwater levels, have been founded on wooden piles since the Middle Ages. Second, the reactivation of the degraded hydrological system could contribute to increased stormwater retention. Third, the presence of water in urban spaces could positively impact the microclimate and promote increased biodiversity.

Historically, the durability of buildings and structures in Nysa was ensured by stable groundwater levels, which remained constant until 1945. In the 1950s, during intensive urban transformations, the existing hydraulic system, including the drainage system, was compromised. A significant lowering of the water level was adopted as a countermeasure. This resulted in the exposure of the upper sections of the wooden foundation piles and their exposure to oxygen, which initiated destructive biological processes. The observed cracks on many historic buildings may indicate that the material, which previously functioned in anaerobic conditions, may have lost its natural durability. Furthermore, the lowering of the groundwater level could have led to changes in the geotechnical properties of the subsoil, including the consolidation of organic soils. An example is the observed cracking of the walls of the Church of the Assumption of the Blessed Virgin Mary in Nysa, which may be due to differences in foundation settlement.

The issue of lowering water levels is one of the key challenges in protecting the cultural heritage of a growing number of historic cities. In Venice and Amsterdam, to maintain the stability of structures built on piles, water levels are artificially regulated, foundations are reinforced, and continuous geotechnical monitoring is conducted.

The 2024 flood, which severely affected Nysa, revealed the important role of historic canals and moats as elements of the water retention and distribution system. In this context, this event can be interpreted as a natural stress test of the urban system, simultaneously serving a diagnostic function. The field studies conducted allowed for an assessment of the impact of restoring historical water conditions on post-war development, as well as verification of previous modeling assumptions. It can be argued that both excessive and excessively low water levels pose a significant threat not only to structures founded on wooden piles. In the context of safety, cultural heritage protection, and spatial planning, it is essential to examine the durability of wooden piles and conduct continuous water level monitoring.

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Consistency and Experiment: The Work of Daniel Libeskind

Between the poetry of the hand and the mind's eye.

– Niall Munro on the work of Daniel Libeskind

Daniel Libeskind constantly experiments in creating space, and in shaping architectural forms, he employs his own individual visual language. He has developed a uniquely individual language of expression. H. Read, recalling Goethe, observed that style "is one of the fundamental characteristics of personality. It is a visual record of the encounter between spirit and matter that takes place in the psyche, and testifies to what extent, on this plane, the spirit is capable of shaping matter to satisfy its own need for externalization, that is, for expression" (Read, 1981, p. 279).

In this context, analyzing Daniel Libeskind's work, we readily perceive that the architect leaves a trace of his individuality in every creative endeavor. And although his work is most often categorized as deconstructivist, he himself defends himself against such opinions. He believes that it is not even a style, but rather a kind of intellectual movement, particularly relevant to the last two decades of the 20th century. What others attribute to deconstructivism in his work, he himself believes that the individually shaped building forms and details are rooted in tradition, yet simultaneously challenge many assumptions. Libeskind's work is based on a constant search for meaning, not significance, in architecture. He rejects the idea of architectural neutrality, confronting history and the disordered reality by transforming them in a uniquely individual way into spatial forms.

The renowned architect Moshe Safdie observed that there are very few fixed rules in contemporary architecture (Safdie, 1993, p. 44). This allows for freedom of decision, creative freedom, and at the same time, encourages the creation of one's own, recognizable architectural world. This idea can undoubtedly be applied to the work of Daniel Libeskind, who, with each successive project, demonstrates that he is an artist of extraordinary artistic sensitivity, possessing extraordinary intuition, and shaping exceptionally individual architectural forms. When thinking about his work, we can also refer to the words of Stanisław Fijałkowski, who assigns a unique role to intuition: "Intuition, like reasoning, is the discovery of something unknown, but in a different mode – it can surprise us differently than reasoning, which never surprises us..." (Taranienko, 2012, p. 145). It triggers the discovery of new, unknown, surprising structures in the creative process, which are uniquely individual.

It is no accident that I refer to the artist. If we look at contemporary art, we see its chaos, which is often apparent, and within it we can discern a certain order. Many artists work in a planned manner, building a specific structure for the work, often seemingly hidden. Observing the development process of outstanding artists, we can also discern the deeply intellectual foundations of their work. And this is also the case with Daniel Libeskind. And yet, his work extends far beyond the framework of architecture. The forms that create his buildings and structures, which are not compact, create the apparent impression that his works belong to the deconstructivist movement. However, Daniel Libeskind disagrees with this perception of his work. Deconstructivism presupposes the redistribution of structure, while an architect is primarily concerned with constructing architecture that speaks expressively. Each subsequent form is also an experiment with space. For him, architecture is also the art of communication. Architecture tells stories that can reach deeply hidden dreams and desires. His architecture tells stories is a mirror of society, reflecting the history of a place. He believes that the power of architecture lies in its complexity. Architecture should evoke surprise. It cannot be taken for granted. Therefore, each of Libeskind's projects is an experiment that creates an architecture full of tensions, pushing its own boundaries, and the architect constantly emphasizes that "providing meaningful architecture does not mean parodying history, but articulating it" (Tenenbaum, 2015).

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The Antifragile City: Constraint as an Architectural Resource

Architecture is never only about what we are able to imagine. It is about what reality allows us to build – and whether what is built can withstand its encounter with that reality. In the twenty-first century this tension has sharpened to an extreme. On one side, capital and generative tools allow us to design cities at continental scale, outpacing any prior experience. On the other, the planet, the climate, history, human communities and material resources increasingly refuse to accommodate these visions. The gap between what is possible in a project and what is possible in the world has become a defining field of contemporary architecture: it determines what proves viable.

NEOM's The Line in the Saudi desert, announced in 2021 as a 170-kilometre linear city for nine million residents, was suspended by the Public Investment Fund in September 2025: 2.4 kilometres of foundation completed, the projected population cut from 1.5 million to under 300,000. Masdar City in Abu Dhabi, launched in 2006 as the world's first zero-carbon settlement for 50,000 residents, has gathered roughly four thousand permanent inhabitants over nearly two decades – despite an effectively unlimited budget and the absence of any external threat. Both projects show that the capacity to imagine and even finance a city does not amount to the capacity to populate it.

Kharkiv was home to roughly 1.5 million people before February 2022. Within the first months of the full-scale invasion that figure fell to approximately 300,000. By early 2023, however, it had returned to around one million – and the city has held this level under daily shelling. More than 350 heritage structures have been damaged and require restoration without an approved master plan, without stable funding and with no guarantee that what is restored will survive the immediate future.

In such conditions, architecture becomes a practice of working with uncertainty. The city requires not only large strategies but, above all, adaptive decisions and small interventions – ones that can respond quickly to shifting conditions, address residents' basic needs and at the same time generate prototypes for further scaling. It is these actions, rather than finished images, that begin to define urban reality.

This shift has theoretical grounding. Taleb's concept of antifragility (Taleb, 2012) describes systems that do not merely withstand stress but become stronger because of it. Meerow, Newell and Stults (Meerow, Newell & Stults, 2016) translated this logic into urbanism through the notion of urban resilience; Dovey and Kamalipour (Dovey & Kamalipour, 2018) showed how it operates in neighbourhoods that grow without a master plan. The shared idea is simple: architecture tested by constraint answers genuine needs – and that is the criterion that separates a megaproject from a living city.

If the twenty-first century has a central architectural problem, it does not lie in scale or ambition. It lies in the relation between what is spent and what is grown – in the capacity of an environment to give rise to life that was not fully anticipated by its design. In that sense, megaprojects have very little to say. Cities formed under the pressure of constraint have almost everything.

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An Urban Aquaponic Farm in Wrocław as a Spatial and Infrastructural Prototype

Urban food production is increasingly discussed not only as an environmental or economic issue, but also as a spatial one. Within this context, small-scale, site-specific interventions can provide valuable insight into how productive systems interact with existing urban structures. This paper presents a concise case study of an urban aquaponic farm implemented in Wrocław, Poland, understood as a spatial and infrastructural prototype embedded in a real urban setting.

The farm was designed and constructed between 2021 and 2024 as part of the research and implementation project Urban Stormwater Aquaponic Garden Environment (USAGE). It is located approximately 4 km east of Wrocław's Market Square, within the revitalised "Na Grobli" complex, an area characterised by a mix of post-industrial buildings, technical infrastructure, educational facilities, and housing. The site lies between the Oder and Oława rivers and forms part of a broader landscape shaped historically by water management and hydro-technical functions. This context influenced both the functional assumptions and the architectural expression of the project.

In spatial terms, the farm occupies a semi-public plot owned by the municipal waterworks. The installation consists of three adapted shipping containers arranged as a freestanding structure. Each container accommodates a distinct function: aquaculture, plant cultivation, and auxiliary and educational spaces. The total footprint of the installation is modest and intentionally compact, allowing it to fit within an existing paved area without major interference in the surrounding greenery or circulation patterns. The choice of container construction enabled precise control over dimensions, ease of assembly, and the possibility of future relocation or expansion.
From an urban design perspective, the project operates at two interrelated scales. At the local scale, it introduces a new programme into an underused fragment of the site, transforming a technical service area into a place that supports limited public access, educational activities, and informal meetings. The positioning of the structure was determined by solar exposure, visibility from pedestrian routes, and its relationship with nearby buildings, including a historic water tower. Architectural decisions – such as the colour differentiation of containers and the introduction of a lightweight pergola and green roof – were intended to improve legibility and soften the technical character of the installation.

At the broader urban scale, the farm functions as a test case for integrating food production into existing infrastructure networks. While the system relies on electricity and water supplied from the city, its internal organisation is based on closed-loop circulation of water and nutrients. This arrangement reflects current discussions on circular economy principles and urban resilience, but the significance of the project lies primarily in its spatial implementation rather than technological novelty.
The planning process combined standard urban and architectural analyses with qualitative methods, including site observation and interviews with selected local stakeholders. This approach made it possible to align technical requirements with expectations regarding accessibility, aesthetics, and use. As a result, the farm was conceived not as a purely utilitarian facility, but as an element of the urban landscape that remains readable and approachable for non-specialist users.

In conclusion, the Wrocław aquaponic farm demonstrates how small, precisely located interventions can contribute to the ongoing "reading" and "writing" of urban space. Its value lies less in its production capacity and more in the concrete knowledge it provides about siting, scale, form, and governance of productive infrastructure within the contemporary city.

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Walking Through Memory: Movement and Spatial Experience in Contemporary Urban Memorials

Memorials play a crucial role in shaping collective memory and mediating relationships between historical events and contemporary urban space. Traditionally, monuments were conceived as monumental and figurative objects intended to communicate clear and stable narratives about the past. However, contemporary memorialization practices increasingly depart from this model, shifting toward experiential forms that emphasize individual perception, bodily movement, and contextual integration.

The aim

The aim of this paper is to identify and systematize spatial and architectural strategies through which contemporary II World War memorials mediate individual experiences of memory, with particular emphasis on movement, scale, and site-specific context.

The method

The research employs a qualitative, multi-method approach combining literature review, critical analysis, site visits, and elements of autoethnographic method. The study is based on a comparative analysis of selected European World War II memorials, examined through interpretative spatial analysis. The inclusion of site visits and autoethnographic reflection introduces an embodied and experiential perspective, emphasizing how individual perception and bodily movement shape the interpretation of memorial spaces.

Results and discussion

The theoretical framework draws from architectural theory and memory studies. Kevin Lynch highlights the role of spatial structure in shaping urban experience, while Juhani Pallasmaa emphasizes the multisensory and embodied nature of architectural perception. James E. Young's concept of the counter-monument demonstrates how contemporary memorials often reject traditional monumentality in favor of forms that provoke reflection and critical engagement. Together, these perspectives frame memorial architecture as a spatial situation that invites active participation rather than a purely representational object.

Contemporary forms of memorialization extend beyond the traditional monument associated with a freestanding form, pedestal, and monumental materials. Instead, they adopt diverse shapes, scales, and modes of interaction with the viewer. In this paper, several parameters are identified that characterize contemporary memorial forms in public space: Scale: Macro scale / Micro scale; Atmosphere: Sublime / Everyday; Space: Urban / Green; Movement: Around / Among; Position: On the way / Away; Role in space: Primary / Secondary. These parameters constitute key tools for interpreting the conceptual framework of memorials. The case studies include the Memorial to the Murdered Jews of Europe in Berlin, the decentralized Stolpersteine project present in numerous European cities, and other memorial interventions integrated into urban or landscape settings.

As an extension of this approach, a case study of counter-monuments is included, focusing on experimental forms of interaction with the viewer. These examples serve as a reference point for identifying emerging trends in memorial design.

The analysis demonstrates that movement plays a central role in structuring the experience of contemporary memorials. Large-scale memorial environments encourage movement, creating immersive conditions in which perception evolves through exploration. In contrast, micro-scale interventions embedded in everyday urban routes introduce moments of reflection within routine movement. In both cases, walking becomes an interpretative process through which individuals construct personal understandings of the past. Rather than imposing a singular narrative, these memorials enable plural and individualized modes of remembrance.

Conlusions

Contemporary II World War European memorial architecture increasingly operates through spatial experience, embodied movement, and sensitivity to the genius loci of a site. These approaches highlight the growing importance of experiential and interpretative strategies in shaping meaningful memorial landscapes in contemporary cities.

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Scenographic Micro-Worlds: Architecture as a Space of Experimentation

Within museums, galleries, pavilions, and installations, architecture frequently appears not as a permanent structure but as a temporary micro-world. These environments enable testing relationships between body, atmosphere and movement. Designed to be perceived and experienced, such spaces are assembled for a limited time and often dismantled once the event concludes. Film production operates under similar conditions: sets are constructed for specific sequences, briefly inhabited, and subsequently removed. Despite their temporality, such sets create fully readable architectural worlds within the duration of the film image.

Scenography enables architectural experimentation that is not possible within the limitations of permanent construction. Scenographic micro-worlds released from the requirements of durability and conventional programming can test new relationships among users, environments, and technological systems. They may experiment with light as atmosphere, movement as a spatial condition and temporality as a design tool. In this way, they address contemporary architectural concerns related to flexibility, hybrid use and sensory experience.

This perspective has a substantial theoretical lineage. Henri Lefebvre argued that space is socially produced through practice, movement and encounter. Bernard Tschumi extended this idea within architectural theory. He emphasised that architecture is inseparable from the events occurring within it - from human activity rather than static form. Giuliana Bruno further applied this concept to cinema. She described the interaction between bodies and spatial images as both emotional and kinetic. Scenography intensifies these conditions by making experimentation visible. Space becomes an environment in which perception and behaviour can be shaped, observed, and reconfigured. Historically, architecture has long relied on scenographic strategies. Renaissance perspective stages trained designers to guide viewers' movement through illusionistic space. Nineteenth-century exhibition structures, such as the Crystal Palace, treated architecture as a full-scale proposition built for a season and then demolished. These examples demonstrate that some of architecture's most innovative developments have developed through temporary formats rather than permanent monuments.

This approach continues in 20th and 21c spatial practice and film. Fritz Lang's Metropolis (1927) transformed visionary urban drawings into a city defined by machinery and circulation. It tested new relationships between scale, infrastructure, and movement patterns. Four decades later, Jacques Tati's Playtime (1967) constructed an entire district of glass and modular repetition. It revealed how modern environments shape behaviour through transparency and standardisation. In Dogville (2003), Lars von Trier reduced architecture to chalk lines and sparse objects on a horizontal plane, demonstrating that minimal spatial cues can still create social relations and emotional intensity.

Comparable strategies are evident in contemporary architecture. Cedric Price's unrealised Fun Palace (1964) proposed architecture as a flexible grid for changing programs, movement and participation. The project replaced permanence with continuous transformation. Diller Scofidio + Renfro's Blur Building (2002) replaced conventional walls with mist. It transformed architecture into an exploration of atmosphere and climatic perception. Elements of Architecture, curated by Rem Koolhaas at the 2014 Venice Biennale, presented architecture as a scenographic sequence of rooms, each dedicated to a single building element. In these examples, temporary structures function as experimental laboratories where designers test ideas that permanent buildings cannot easily accommodate. Here, scenography operates not as decoration but as a design method. Some of the most significant contemporary architectural experiments arise within such temporary environments.

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Design Workshops in Sztynort as an Example of Experimentation in Architectural Education – The Application of the Learning-by-Doing Method

Contemporary architectural education must redefine its methods to address the increasing complexity of social, economic, and environmental conditions. Traditional models, relying on simulated tasks, often fail to prepare students for real-world constraints (Salama, 2015). In this context, academic experimentation becomes a vital tool for integrating theory, practice, and reflection.

The aim of this paper is to analyze the learning-by-doing method applied during the 2025/26 winter semester workshops at the Faculty of Architecture, Warsaw University of Technology. The research investigates how real investment and functional constraints influence the learning process and the quality of design solutions.

The paper advances the thesis that an experimental approach, based on professional-like conditions, transforms the educational environment into a space of experiential learning. While architectural studies are inherently vocational, the traditional educational model often relies on a static, predefined programme. The experimental nature of the Sztynort workshops lay in the introduction of a "living", external decision-maker, which forced students to move away from safe simulation toward a dynamic and fluid process typical of real professional practice, yet rarely encountered in controlled academic settings.

Contemporary theories, such as Donald Schön's "reflective practitioner" (Schön, 1983) and David Kolb's experiential learning (Kolb, 1984), emphasize active participation and iterative testing. In Sztynort, students designed a 150-room hotel and recreational complex within a historic site. The functional programme was modified in real-time through close cooperation with an investor. This introduced a genuine decision-making context where proposals were shaped by negotiations between architectural vision and economic, social, and conservationist constraints.

The qualitative analysis of this experiment showed that students identified design limitations more effectively and demonstrated greater responsibility. Although balancing creativity with pragmatism proved challenging, the non-linear process fostered critical thinking (Pallasmaa, 2012). The realism of the task, the opportunity to present work to a potential investor, and the confrontation with recognized architects during a post-workshop competition significantly increased student motivation.

Furthermore, the experiment shifted the tutor's role from a traditional evaluator to a moderator, supporting students in formulating individual strategies. In conclusion, the Sztynort workshops confirm that integrating external stakeholders and real-world unpredictability enhances not only design competencies but also the communication and social skills essential for the architectural profession.

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Prescriptive and Performance Based Daylighting Regulation as an Architectural Experiment: Parametric Simulation of Polish Residential Standards

Daylight in buildings can be regulated in two fundamentally different ways: through prescriptive geometric rules that define minimum glazing dimensions, or through performance-based metrics that measure the luminous result directly. In Poland, the regulations (Warunki Techniczne) require window area to be at least 1/8 of the floor area (Ministerstwo Rozwoju, Pracy i Technologii, 2021) – an efficient rule that captures aperture size but not room depth, glazing height, or the overcast-sky conditions dominant in the Polish climate.

The Daylight Factor (DF), defined as the ratio of indoor to outdoor illuminance under a standard overcast sky, offers a simple performance-based alternative. A 2% average DF is widely accepted as the threshold at which a space is perceived as adequately daylit without continuous reliance on electric lighting (Chartered Institution of Building Services Engineers, 2015). Prior research has shown that Polish prescriptive standards inadequately represent actual luminous conditions across residential typologies (Lose, 2015), with similar gaps identified in non-residential contexts (Gorzelewska & Kwieciński, 2025).

This paper treats that gap as the subject of an architectural experiment. Rather than evaluating compliance in isolated cases, it constructs a controlled micro-world in which the assumptions embedded in regulation are made visible, variable, and testable.

Methods

The study employs an automated parametric workflow (Grasshopper, Honeybee, Radiance) to construct a controlled experimental field. Methodologically, it adopts a Design of Experiments (DoE) logic, treating room width, depth, and window height as controllable factors and average DF as the measured response.

The parametric model systematically varies these parameters across ranges representative of contemporary residential rooms, generating a continuous design space rather than discrete cases. This allows regulatory assumptions to be tested under controlled variation, analogous to experimental procedures in laboratory settings.

Rather than producing isolated performance evaluations, the model maps relationships between geometry and daylight, identifying which parameters govern performance and where prescriptive rules diverge from performance-based criteria. Results are analyzed using spatial heatmapping, distributional statistics, and regression modelling.

Findings

The experiment's central finding is structural rather than numerical: prescriptive compliance does not constitute a reliable proxy for daylight performance. The two regulatory models operate on different logics and produce systematically divergent outcomes.

The results reveal that daylight performance is governed by structured relationships between geometric parameters. Certain variables act as limiting conditions-most notably room depth – suppressing performance regardless of other factors. Others function as enabling conditions, expanding the range of achievable outcomes. Window height introduces a nonlinear effect: for a fixed glazing area, vertical proportions influence how deeply light penetrates a room, making aperture shape as significant as size. These relationships are not visible within prescriptive rules, which reduce a multidimensional spatial problem to a single parameter.

Conclusions

The contribution of this study lies in treating parametric simulation as a mode of architectural inquiry. By constructing a controlled experimental environment, it becomes possible to interrogate regulatory logic across an entire design space rather than through individual cases. The 1/8 rule functions as a coarse safeguard against under-glazing, but fails as a predictor of daylight performance. A revised framework should incorporate geometry-sensitive relationships, particularly between room depth and window height. More broadly, the study demonstrates how simulation can operate as an experimental method in architecture-linking design, analysis, and regulation and aligning regulatory intent with measurable spatial performance.

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Feedback Loops and Open Forms: Architecture, Algorithmic Control, and Human Agency in the Digital City

The contemporary city no longer operates solely as a physical structure composed of streets, buildings, and infrastructure. Increasingly, it functions as a responsive system of data extraction, prediction, and behavioral modulation. Sensors, algorithms, recommendation systems, surveillance networks, and platform economies continuously collect information about human activity and transform it into feedback mechanisms shaping perception, movement, and social interaction. In this context, architecture expands beyond material construction and becomes entangled with invisible digital systems organizing contemporary life. This study investigates the relationship between architecture, algorithmic environments, and human agency through the concept of feedback loops derived from cybernetics and media theory. The research problem addresses how digital infrastructures reproduce the modernist ambition to regulate social behavior through spatial organization. The central thesis argues that contemporary algorithmic systems function as a new form of "closed architecture," where predictive technologies optimize and condition human experience while simultaneously reducing unpredictability, spontaneity, and the openness traditionally associated with democratic urban space.

The discussion is situated within the intellectual legacy of modernism and postwar architectural theory. Le Corbusier's vision of the city as a "machine for living" established one of the dominant paradigms of spatial rationalization in twentieth-century urbanism. His projects proposed environments organized according to efficiency, circulation, and functional order. In contrast, Oskar Hansen's theory of Open Form challenged the totalizing tendencies of modernist architecture by emphasizing participation, indeterminacy, and the active role of inhabitants in shaping spatial experience. Hansen rejected architecture as a completed ideological statement and instead proposed space as a dynamic framework for human interaction.
The text suggests that contemporary digital systems increasingly resemble the logic of modernist spatial control rather than Hansen's participatory openness. Algorithmic platforms predict behavior before actions occur, navigation systems eliminate uncertainty, and interfaces optimize attention through continuous behavioral feedback. The contemporary subject inhabits environments that are not only observed but actively recalibrated in real time according to patterns of data extraction and prediction. The smart city emerges not as a neutral technological development, but as an extension of cybernetic governance into urban life.
Methodologically, the study combines critical architectural theory, media studies, and visual culture analysis. It employs a comparative interpretative approach connecting modernist urban theory with contemporary examples from architecture and digital art. The analytical framework draws upon concepts developed by Michel Foucault, Gilles Deleuze, and Shoshana Zuboff concerning surveillance, control societies, and data capitalism. Simultaneously, the study examines artistic and architectural practices exposing the hidden infrastructures of algorithmic power. Particular attention is given to practitioners such as Forensic Architecture, Hito Steyerl, Trevor Paglen, and Liam Young. Their works expose the invisible operations of surveillance systems, machine vision, satellite infrastructures, and data-driven urbanism. In these practices, the artist and architect no longer function solely as creators of objects or buildings, but as interpreters of systemic conditions shaping contemporary existence. Art becomes a diagnostic instrument capable of visualizing forms of control otherwise imperceptible within everyday life.

The analysis concludes that the convergence of architecture and algorithmic systems produces a new spatial condition defined by continuous feedback and behavioral prediction. As digital infrastructures increasingly mediate urban experience, the question of openness becomes critical once again. Hansen's Open Form gains renewed relevance as a conceptual counterpoint to the closed operational logic of algorithmic governance. Future architectural thinking must therefore move beyond technological optimization and reconsider how space might preserve ambiguity, autonomy, and the possibility of unprogrammed human experience within increasingly data-driven environments.

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Open-ended Play: The Afterlife of Late Modernist Playgrounds in Warsaw and Tychy

In postwar Poland, playgrounds formed an integral part of the landscape of housing estates. Rather than consisting of standardised sets of equipment with predefined functions, they were often conceived as carefully designed open spatial compositions. Combining art, utility, and free activity, these environments encouraged imaginative engagement and allowed young users to explore space in diverse and unpredictable ways. Today, their experimental and non-programmatic character reveals a notable capacity to adapt to changing modes of using urban space.

In recent years, late modernist playgrounds have attracted growing scholarly attention. Well-known examples include the playgrounds designed by Aldo van Eyck in Amsterdam, the sculptural landscapes of Isamu Noguchi, the expressive solutions found in British estates, and the space-age forms developed in the Soviet bloc. Against this background, Polish examples remain largely unrecognised, even though they aligned with (and in some cases anticipated) international trends.

The paper examines selected surviving examples of sculptural playgrounds in Warsaw and Tychy - cities with a significant number of socmodernist housing estates, which provided the primary context for these designs. The study focuses on playgrounds with a strong artistic character, predominantly made of concrete and integrated into the spatial composition of residential environments. The survey included an analysis of original design concepts and spatial intentions, as well as an assessment of the current condition of their form and material substance. The present state was documented through photography and 3D scanning. An important component of the research was the observation and recording of contemporary patterns of use.

The selected case studies reveal a wide range of design approaches, from comprehensive spatial compositions to small-scale interventions. They were created not only by architects and landscape designers but also by sculptors invited to collaborate on housing estate projects. This diversity is reflected in the forms themselves, which range from unique, individually designed structures (e.g. “Bielany Shell” in Warsaw, Maria and Kazimierz Piechotka, 1950s), to forms adapted for serial production (e.g. “Bison Climber” in Tychy, Andrzej Musialik, 1957). Some examples remain well preserved and protected, including the “Labyrinth” and associated playgrounds in the Szwoleżerów estate in Warsaw (Alina Scholz, Halina Skibniewska, Zbigniew Hummel, 1971). In other cases, the original spatial composition has been partially lost or obscured by later additions, as seen in the play sculptures in Tychy’s Estate D (Wojciech Firek, Andrzej Getter, 1979).

What unites these examples is not only their sculptural quality but, more importantly, their openness to user interpretation. Unlike contemporary playgrounds, which are often highly regulated, standardised, and enclosed, these earlier forms did not impose a single mode of use. Field observations indicate that despite physical degradation, many of these spaces continue to be actively used. Their current functioning has adapted to contemporary patterns of life: alongside traditional forms of play, new ways of occupying space have emerged, including children spending time together online via smartphones.

At the same time, demographic changes mean that these spaces are increasingly used not only by children but also by older residents. Their indeterminate form allows for such functional shifts without requiring physical transformation. These playgrounds can be understood as enduring urban experiments, whose potential has not been exhausted by changing social conditions. Their survival may also be linked to their aesthetic value, which positions them not only as functional environments but also as works of public art.

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In Search of a New Park Form: The Central Park of Culture and Recreation in Warsaw as an Urban Experiment of the 1950s

The Central Park of Culture and Recreation (Polish: Centralny Park Kultury i Wypoczynku, CPKiW) in Warsaw was one of the most significant investments of the city's post-war reconstruction period and a flagship realization of the Socialist Realist era. The implementation of such a vast undertaking in the city center was possible only due to specific historical conditions: the extensive destruction of Warsaw during World War II and the new political reality of the Polish People's Republic. Of crucial importance was the communalization of urban land, which enabled radical spatial transformations and large-scale public investments. In this context, the CPKiW became a large-scale urban experiment, stemming from the authorities' utopian approach to treating urban space as a tool for social engineering.

Conceptual designs for the park began in 1950 at the "Zieleń" Studio of the Warsaw Urban Planning Office under the direction of Alina Scholtz. The preliminary design and documentation were later developed by the "Stolica" Central Office of Municipal Construction Projects, with Zygmunt Stępiński as the general designer. The park, planned on an area of 308 hectares – covering area between 3 Maja Avenue, Ujazdowskie Avenue, the Royal Łazienki Park, and the Vistula River – was intended to become a "space of events", creating a new quality of life for residents upon the ruins of the former Powiśle district. Although the project was only partially realized (today's Rydz-Śmigły Park), the process serves as a unique study of an experiment that utilized post-war conditions to implement radical planning visions.

The "park of culture and recreation" concept originated in the USSR during the 1920s and 1930s, where a new type of public green space was developed, combining mass recreation with cultural, educational, and sports programming. In early 1950s Poland, this concept was a entirely new phenomenon, which lent the design and implementation works the character of a "laboratory of ideas." The first Polish publications on the subject sought to understand and adapt this form, revealing tensions between Soviet models and the local context.

The primary objective of this research is to trace the evolution of the CPKiW projects as a model urban experiment. The study aims to answer how architects adapted an alien park typology to the specific situation of a post-war city while navigating the expectations of the authorities and the limitations of the reconstruction period. A key aspect is tracing the conceptual evolution – from the monumental visions of Socialist Realism to the pragmatic solutions developed during the political "Thaw." This allows the CPKiW to be seen not merely as a landscape design, but primarily as a testing ground where a new relationship between designed form and the mass user was explored.
The methodology follows an interdisciplinary approach, combining critical archival analysis with compositional, spatial, and urban studies. The research is based on an extensive inquiry into successive design variants, supplemented by a discourse analysis of the contemporary press and field research of preserved fragments. This allows for the confrontation of theoretical assumptions with their actual spatial impact. The study demonstrates that the CPKiW was a process of "learning by doing" – a dynamic response to changing political and social guidelines.

This analysis frames the park as an experimental research field in which the boundaries of landscape architecture were tested in service of the utopian idea of a "new world".

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Participation as a Tool of Urban Experiment: Prototyping Dialogue, Space, and Decision Making

Contemporary cities operate as complex socio-spatial systems in which effective urban design requires integrating legal frameworks, professional expertise, and residents' lived experiences. Within this dynamic environment, social participation functions as a form of experimentation in the service of society, enabling the testing, verification, and refinement of planning and design solutions under real urban conditions. Its growing significance stems not only from regulatory obligations – particularly those outlined in the Spatial Planning and Development Act (Ustawa, 2003) – but also from the need to enhance transparency, strengthen public trust, and build social legitimacy for planning decisions.

Participation encompasses a wide spectrum of collaborative tools, including workshops, research walks, expert panels, geo surveys, and charrette methods (Pyziak, 2024). These instruments act as social participation prototypes, allowing stakeholders to explore alternative scenarios, identify conflicts and opportunities, and co-create spatial strategies. Their shared purpose is to merge expert knowledge with user experience, producing more accurate diagnoses of local needs and more resilient design outcomes.

However, effective participation requires more than procedural compliance. It demands soft skills such as communication, mediation, and teamwork, as well as the ability to conduct design processes in an open, iterative, and adaptive manner. Despite the increasing availability of training in this field, practice often diverges from legal assumptions, as illustrated by the accelerated preparation of general plans in Poland. These discrepancies highlight the need for participatory methods that are not merely formal obligations but experiential processes capable of generating new knowledge and fostering shared responsibility for urban development.

The concepts for Skawina developed by the team of T. Jeleński, K. Racoń-Leja, and M. Leja, demonstrate how intensive, workshop-based collaboration enables analyses, consultations, and design work to occur simultaneously. This approach not only increases the relevance and precision of proposed solutions but also strengthens social acceptance by embedding community perspectives directly into the design process. Such experimental, participatory practices highlight architecture's role as a platform for dialogue, negotiation, and co-creation – an arena where temporary interventions and collaborative prototypes help shape long-term spatial strategies.

In the educational sphere, social participation constitutes a fundamental component of training future architects and urban planners. Integrating participatory methods into academic curricula – through urban labs (Mensig-de et al, 2020), workshops, stakeholder engagement, activity mapping, and context analysis – helps students understand the complexity of socio-spatial environments. It also supports the identification of local "urban DNA," encompassing place identity, cultural heritage, and everyday spatial practices. Participatory education fosters ethical awareness, social responsibility, and the recognition that urban design is inherently a co-creative process rather than a purely expert-driven endeavour. In this sense, education itself becomes a laboratory of experimentation, preparing students to navigate and shape the evolving challenges of contemporary cities.

In conclusion, social participation should be regarded as a foundational experimental tool for shaping sustainable, inclusive, and socially grounded urban environments. By linking legal requirements, design practice, and education, it establishes a framework for meaningful social dialogue and contributes to the quality, resilience (Widzisz-Pronobis, 2023), and public reception of planning documents and implemented projects. Participation thus emerges not as a formal obligation but as a strategic instrument that enables cities to evolve in response to the needs of their inhabitants.

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Experimental Architectural Education: Learning by Doing through Bridge Load-bearing Tests in Secondary Schools

Architectural education in the compulsory education system in Poland often involves students passively listening to lectures on historic buildings or preparing presentations. These methods fail to convey the complex structural forces and spatial relationships, creating a gap between theory and practical application in the built environment. Consequently, this leads to low student engagement and rapid knowledge loss.

This qualitative case study evaluates an experimental architectural curriculum for secondary students. The aim is to evaluate the effectiveness of load-bearing bridge tests as a superior alternative to traditional lectures. The research investigates how transitioning from passive lecturing to learning-by-doing influences the students' understanding of structural stability. This study posits that experimental prototyping driven by gamification provides a more durable cognitive foundation for architectural literacy than passive learning. Data were collected during secondary school workshops (2022-2025) through participant observation and qualitative feedback analysis, comparing student engagement during manual tasks with conventional teaching methods.

Academic research emphasizes active methodologies over passive instruction in architectural education. Panchariya (2018) as well as Güzelçoban Mayuk and Cosgun (2020) highlight learning-by-doing as essential for bridging theory and practice, while Pak and De Smet (2023) validate experiential learning through live projects. Gamarra et al. (2022) address similar issues, recognising competition as a key factor in developing problem-solving skills and teamwork. In the book The Eyes of the Skin, Pallasmaa (2012) argues that a purely intellectual approach is insufficient, necessitating multisensory engagement for a deep, embodied understanding of space.

The research centered on a structural challenge: designing and constructing bridge prototypes for empirical load-bearing analysis. The session followed a structured sequence: theoretical briefing, physical modelling, experimental testing, and collective discussion. The objective was to optimize structures for maximum capacity. While groups developed distinct typologies, all models maintained a uniform span to ensure comparability. Each bridge underwent a load test until reaching maximum available load or structural failure. Finally, students measured deflection, recorded ultimate loads, and conducted a comparative analysis.

Observational data revealed significantly higher cognitive focus during manual prototyping than during theoretical instruction. Peak engagement occurred during inter-group competition, where gamification acted as a catalyst, evidenced by intense participation in the load-bearing trials. These findings corroborate Gamarra et al.'s (2022) assertion that gamification enhances problem-solving efficacy. Crucially, structural failure was perceived not as a setback, but as a pivotal learning moment. This immediate feedback prompted reflection, aligning with Pak and De Smet's (2023) research on live projects. Furthermore, model analysis showed that students correctly applied composition with reliance on notes. This practical mastery supports Panchariya (2018) and Güzelçoban Mayuk and Cosgun (2020) regarding validating theory through practice. Qualitative feedback confirmed the experiential approach was perceived by participants as highly engaging and educationally valuable.

The study confirms that integrating empirical experimentation into architectural workshops significantly enhances student engagement and knowledge retention. Physical prototyping of bridges facilitates a cognitive transition from abstract concepts to embodied spatial understanding. Ultimately, this experimental methodology proves to be a superior educational pathway, fostering an intuitive grasp of the forces acting within building structures – a level of comprehension that traditional, passive instructional methods are unable to replicate.

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Contemporary Architecture as a Space of Perceptual Experiment: Between Compositional Intent and User Experience

Contemporary architecture increasingly operates as a field of perceptual experimentation in which space is no longer understood solely as the outcome of formal or functional decisions but as a medium of experience constructed through human perception. In this context, architectural experimentation should not be reduced to novelty, expressive form, or technological innovation. Instead, it can be understood as a cognitive and experiential process that investigates the relationship between compositional intent and the lived experience of architectural space.

The aim of this paper is to analyze contemporary architecture as a "space of perceptual experiment," where meaning emerges through the interaction between architectural composition and the user's sensory, bodily, and interpretative engagement. Architecture is approached here as a temporal and performative phenomenon, realized not only at the moment of design completion but continuously through perception, movement, and use. Such an understanding shifts the focus from architecture as an object toward architecture as an event, shaped by sequences of spatial impressions and embodied encounters.

The paper examines selected principles of architectural composition – such as proportion, materiality, light, spatial sequence, and enclosure – through the lens of perceptual experience. These principles are treated as instruments that deliberately structure attention, orientation, and emotional response. Rather than functioning as autonomous formal devices, they operate as experimental variables whose effectiveness becomes evident only through user interaction. The gap that often exists between compositional intention and actual spatial experience is therefore treated as a productive field of inquiry rather than a design failure.

Methodologically, the study draws on analytical reflection supported by experimental representational tools, including physical models, spatial simulations, and immersive environments. Such tools allow architectural space to be tested, interpreted, and re-experienced beyond traditional drawings, revealing perceptual qualities that are not immediately accessible through conventional means of analysis. In this sense, experimental methods become extensions of architectural cognition, enabling a deeper exploration of spatial atmosphere and phenomenological effects.

The discussion emphasizes that viewing architecture as a perceptual experiment introduces an ethical and disciplinary shift. By foregrounding the user's experience, architectural composition becomes a responsibility as much as an aesthetic or technical endeavor. Experimental architecture, understood in this way, does not seek provocation for its own sake but aims to heighten perceptual awareness and deepen the understanding of how space is experienced, interpreted, and remembered.

The conclusions suggest that treating contemporary architecture as a space of perceptual experiment offers a robust framework for rethinking compositional strategies and evaluative criteria. Such an approach broadens architectural research beyond formal analysis, integrating perception, experience, and interpretation as central components of architectural knowledge. Ultimately, this perspective positions the user at the core of the architectural process and reinforces architecture's role as a mediator between intention, experience, and meaning.

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From Dialogue to Site Design: The Experiment as a Tool for Urban Co-creation - Case of Zieleniec Wielkopolski and Skwer Sue Ryder, Ochota, Warsaw

Cities function as complex, living systems in which architecture and urban planning must integrate human needs with ecological and infrastructural challenges. Accordingly, the design process is shifting away from top-down delivery of ready-made solutions towards co-creation with end users (Healey, 1997). The growing recognition that residents are experts in their own lived environments has led to a redefinition of participation - from a procedural obligation to a key contributor to design quality.

This paper argues that the effective transformation of public spaces with strong community attachment requires moving beyond reactive consultation towards active co-design. Such an approach enables the direct translation of users' needs into spatial solutions. The study, conceived as a design experiment, proposes a participatory model for the modernisation of a space of high communal significance. It addresses the following research question: how do co-creation tools influence the alignment of spatial development concepts with real user needs?

Literature review and team experience identifies several risks associated with participatory design processes. These include information asymmetry between designers and users, inconsistent vocabulary arising from differing perspectives, lack of skills in moderating dialogue, a focus on predefined solutions rather than underlying needs, difficulties in translating outcomes into design decisions, and the misinterpretation of user needs (Miessen, 2010; Pękała & Kaźmierczak, 2023, Sanoff, 2000). The proposed methodology was designed to minimise these risks.

The process was structured in three sequential stages. The first stage involved defining boundary conditions and selecting participatory diagnostic tools based on the client's brief and on-site observations. The second stage consisted of a diagnostic walk with users, organised around four thematic pillars derived from the Lens Method: safety, nature, play, and social interaction (Waśniewska et al., 2025). Participants worked with structured diagnostic worksheets, sharing knowledge about the everyday functioning of the space. The walk, facilitated by a professional moderator, enabled dialogue between users and designers. Each group, comprising up to fifteen participants, was supported by two designers. The third stage involved workshops in which diagnostic findings – formulated in the language of needs – were translated into spatial solutions through dialogue with an interdisciplinary team of designers and social researchers. This ensured continuity between diagnosis and design and reduced the risk of losing user input in later stages of the process.

Several factors contributed to the effectiveness of the process. Direct, in-situ engagement of participants proved particularly valuable, as did structured moderation and the involvement of a large interdisciplinary design team. The applied method had a measurable impact on design outcomes: decisions were grounded in real user needs, embedded in both social and spatial contexts, and responsive to place identity. Additionally, participants developed a better understanding of the design process, which increased acceptance of proposed solutions.

The conducted research demonstrates that a properly structured participatory diagnosis is fundamental to successful co-design. By integrating community-driven insights directly into the workshop phase, we ensured that the final design was not merely a professional vision, but a reflection of local needs. The experiment we present highlights that the process itself proves to be a critical element which serves to mitigate inherent tensions within the community and allow creative work with users. However, as the potential for design flaws remains high, this model requires further testing to ensure its scalability and resilience in diverse urban contexts.

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Connectivity Index as a Criterion for the Quality of the Pedestrian Network

This study is devoted to establishing quantitative indicators of urban pedestrian networks and examining the correlation between these indicators and network quality. The aim of the research is to experimentally model the optimization of a pedestrian network, identifying scenarios in which minimal interventions produce a significant effect.

The research methodology is based on the analysis of spatial data (GIS), which underpins the derivation of quantitative indicators of the studied objects, as well as on Graph Theory [1;2], enabling the determination of topological connectivity within network segments, including pedestrian infrastructure.

The initial dataset consists of indicators of pedestrian space obtained through cartographic analysis for three Ukrainian cities - Lviv, Kharkiv, and Kryvyi Rih. It has been determined that Lviv contains 11,122,222 m^2 of pedestrian space, which, relative to its population of 729,038 inhabitants, corresponds to 15.25 m^2 per capita. Kharkiv has 18,700,000 m^2 of pedestrian space, equating to 12.88 m^2 per capita for a population of 1,451,132 [3]. Kryvyi Rih has 301,500,000 m^2 of sidewalks, corresponding to 46.23 m^2 per capita for a population of 652,137. However, the amount of pedestrian space per capita is not a decisive indicator of pedestrian network quality when considering urban parameters such as the significantly different city diameters - 17, 23, and 64 km, respectively. The surplus of pedestrian area in Kryvyi Rih, in combination with its low density and large urban diameter (64 km), emphasizes the priority of connectivity indicators over purely physical area measures.

The study models the connectivity of pedestrian network segments by introducing a minimal number of new links (a “minimal intervention scenario”) and evaluates the resulting effect. The experiment is not tested through implementation but through measuring changes in system parameters.

Using the example of Lviv, the connectivity index (Beta Index), which reflects network complexity, is calculated. Treating the problem of pedestrian network connectivity as a graph-analytical task, it is established that approximately 28% of pedestrian surfaces in Lviv form a continuous network at the urban scale, while the remainder exist as isolated segments that do not ensure overall connectivity.

Within the set of pedestrian network segments, 52 connected groups forming closed-loop structures are identified. The number of stable connections between them is six. Accordingly, the connectivity index of the pedestrian network, defined as the ratio of links to network segments, is 0.11. As an experimental scenario, if the intervention is limited to only eight strategic additions -specifically, introducing eight new links between identified closed-loop groups - the connectivity pattern changes substantially. Network fragmentation decreases by 13–15% relative to the initial state, while the connectivity index increases to 0.27. This demonstrates that targeted strategic interventions produce a nonlinear effect: an increase in the number of links by only 15% results in a 2.5-fold improvement in overall system efficiency.

The outcome of the study is the introduction of a new metric for assessing the quality of pedestrian networks - the connectivity index (links/nodes).

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Emergent Architectural Detail - The Importance of Design-Build Experiments within the Postdigital Curriculum

In the postdigital paradigm, design, material, and fabrication cannot be understood as separate fields of inquiry. Instead, they operate as interrelated processes forming a continuum between the architectural idea and its realisation. Despite this shift, architectural pedagogy continues to focus primarily on CAD methods and advanced digital models. Materiality has acquired agency and, as Achim Menges argues, has ceased to function merely as a passive carrier of form (Menges, 2016). Fabrication methods, particularly within the context of numerical control, have expanded the boundaries of both imagination and production. Nevertheless, fabrication is still perceived mainly through the lens of CAD/CAM integration and push-button fabrication (Kolarevic, 2003), understood merely as a seamless realisation of a predetermined form in a selected material. Yet both contemporary digital practice and the broader history of architecture demonstrate that tools themselves can generate emergent detail.
An example can be found at Tverrfjellhytta, where the top surface retains machining traces from the CNC milling process (roughening), likely as a cost-saving measure. In the Snøhetta project, the ribbed texture produced was not intentionally designed. In contrast, the Armadillo Vault, presented by the Block Research Group at the 15th International Architecture Exhibition – La Biennale di Venezia in 2016, transformed fabrication constraints into a conscious, though still emergent, architectural detail through an informed use of CNC circular-saw technology (Block et al., 2017). These examples reveal the limitations of relying exclusively on increasingly advanced CAD models, which often create the illusion of a complete digital representation.

This paper briefly presents experiences from a Problem-Based Learning (PBL) design studio conducted in 2025 within the Information Architecture specialisation at the Faculty of Architecture, Warsaw University of Technology. Students were asked to explore, through physical experimentation, methods for designing and producing additively manufactured ultra-thin concrete formworks known as Eggshell Formwork (Burger et al. 2020).

The first phase of the studio consisted of a series of interdependent models testing the constraints of FFF printing and concrete casting. The experiments addressed questions such as: What are the geometric and technical parameters of FFF printing with a custom 1mm nozzle? What is the minimum achievable detail within the formwork, and what level of detail can structural concrete C25 reproduce? These questions were explored through physical experiments involving variations in cylindrical concrete compression samples. The collected data established the boundary conditions for six parametric column designs, each approximately 30 cm in diameter and 240 cm in height.

The emergent detail revealed itself only during the full-scale test casting, when it became clear that, despite prior simulations, the formwork was too flexible and cracked under concrete pressure. To solve this issue, an algorithm was developed that added vertically winding ribs across the formwork surface, visually reminiscent of buttresses. Their aesthetic qualities became apparent only after the formwork was removed, when the ribs left a delicate surface pattern resembling an additional displacement map applied to the columns.

The key argument presented is that without physical experimentation and the continuous resolution of emerging problems, such qualities would never have been discovered. Following the notion of a continuum between design methods, material, and tools, architectural curricula should be structured to facilitate the generation of both explicit and tacit knowledge across all three domains. Within the postdigital condition, this may be more important than before.

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Camouflage Reborn: Community Painting at Fort Mistrzejowice as a Participatory Approach to Heritage Activation

The ring of Austro-Hungarian fortifications encircling Kraków represents one of Central Europe's most extensive heritage systems. Despite their physical presence, many structures remain culturally invisible to local communities. Fort Mistrzejowice, situated within a post-socialist housing estate, represents this latent heritage. Recent restoration activities by the Municipality (ZBK Kraków) revealed an additional aspect of this invisibility: the restored surfaces were vandalised shortly after completion.

In 1914, the Austro-Hungarian military issued instructions for camouflage painting, as documented in the Kriegsarchiv in Vienna. Laboratory analysis of surviving paint traces on the fort's surfaces helped identify the original mineral pigment palette, meaning the specific combination of naturally occurring compounds used to make the paints. This analysis also ruled out post-historical interventions. The material evidence, verified and approved by the heritage conservation authority responsible for protecting historical sites, became the chromatic and compositional foundation for the event.

This study considers a contemporary participatory event: the communal repainting of Fort Mistrzejowice in its documented camouflage colours, engaging residents from adjacent housing estates. Organised by ZBK Kraków, the event provided protective equipment, purpose-mixed paints, and on-site safety training. The camouflage composition was outlined in chalk prior to painting; two teams worked concurrently, one applying dark green and the other ochre, resulting in a collectively authored outcome grounded in archive documentation. The research question is whether the collective recreation of a historical material intervention can serve as an effective tool for both historical heritage activation and informal social guardianship.
Participatory camouflage painting functions as an outdoor artistic and educational event that fosters place-identity through embodied experience, and as a conservation strategy that transforms potential vandals into self-appointed site custodians.

This study utilises laboratory paint analysis, conservation documentation, direct observation of the event, qualitative analysis of participant responses, and post-event monitoring data from a mobile camera station at the site.

The event is analysed using the six-step participatory framework proposed by Djabarouti (2024). The process started with holistic project mediation, coordinating among ZBK Kraków, the heritage conservation authority, and the local community. Place-based users with embodied experience were identified among residents of adjacent housing estates. The fort served as the site for participatory knowledge transfer, including safety training, pigment mixing according to 1916 formulas, and chalk-sketched camouflage composition, all conducted on-site. Laboratory paint analysis and archival documentation informed the project brief, grounding aesthetic decisions in material evidence. The historical camouflage narrative was thus enacted through the physical act of conservation. The event also functioned as a celebration of the conserved structure as a symbol of partnership: the painted parapet became a collectively authored object, visually distinct and recognised by its creators. Post-event camera monitoring documented a significant behavioural shift: site activity was limited to walking and recreational use, with no new graffiti or tags recorded after the painting event. The camouflage logic, based on the transition between figure and ground, served as a metaphor for the fort's urban condition: simultaneously present and overlooked, made visible by collective effort.

The Mistrzejowice experiment shows that participatory material practice can activate heritage identity and extend social protection. Camera monitoring supports this outcome. The camouflage, once meant to hide the fort, now helped it reappear in collective consciousness and under public stewardship.

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The Projective Mirror Method: From Persuasion to Relationship. A New Paradigm of Architect–User Communication in Contemporary Interior Design

Contemporary interior design practice increasingly departs from the linear model in which the architect, relying on personal aesthetic patterns and professional expertise, imposes a predefined spatial vision on the user. The persuasion-driven, top-down approach that dominated for decades is proving insufficient in the face of complex human needs and growing societal awareness of the role space plays in wellbeing. The aim of this article is to present a model in which the designer – user relationship is not a project stage but its central axis. The thesis argues that the architect may adopt the role of a "projective mirror" – a medium enabling the user to reveal hidden or unarticulated needs through dialogue, observation, and experiment.

The key research question is: to what extent can the architect–user relationship function as a cognitive tool rather than a mere exchange of initial requirements? The article assumes that by acting as a mirror, the designer does not merely interpret user statements but helps refine them – through visual stimuli, narrative tools, material tests, and prototyping.

Christopher Alexander's model is grounded in the premise that every person is, in essence, a designer who operates through personal patterns. Each individual makes design decisions, regardless of how well-formed or coherent these internal patterns are. The task of the "professional" designer is to refine existing patterns that may lead to ineffective outcomes and to coordinate them into a coherent whole. Each subsystem of criteria corresponds to a different set of patterns. The architect incorporates them and weaves them into a unified structure – one that ultimately becomes a "technical project."

Research on architectural communication and the designer–user relationship appears across architectural theory, environmental psychology, and design studies. Norberg-Schulz (1979) emphasizes the existential experience of place, highlighting the need to "read" the user through cultural grounding. Christopher Alexander (1979) develops the most comprehensive participatory model, assuming that everyone possesses patterns that should be acknowledged and improved rather than ignored. Lawson (2010) frames design as a cognitive activity in which the architect learns the user iteratively. A shared assumption across these perspectives is that design constitutes dialogue; however, this article proposes shifting from "dialogue as a phase" to "dialogue as method."

The "Mirror Method" was developed as a practical tool for working with clients in interior design. It comprises three layers of interaction:

1. The declarative mirror – analysing user statements, metaphors, emotions, and recurring keywords;
2. The visual mirror – testing user reactions to images, materials, moodboards, colours, and spatial scenarios;
3. The experiential mirror – prototyping micro- fragments of the interior and working with light, texture, and scale to extract authentic preferences.

The method assumes that users rarely articulate their needs directly; only well-constructed "mirrors" enable these needs to surface.
Case analyses demonstrate that user declarations often function as superficial communication strategies ("I want minimalism", "I don't like colour"), which reveal their true meaning only when confronted with visual or spatial stimuli – such as the need for order, control, calm, or conversely, expression. The Mirror Method modulates the design process so that the architect does not impose patterns but supports users in understanding themselves through design. This shifts the focus from aesthetics toward the psychology of inhabitation and positions the designer not as a transmitter but as a catalyst.

The article argues for moving design practice toward a relational approach in which the architect acts as a mirror – reflecting the user's potential rather than replacing it with personal preferences. Such a model enables the creation of interiors that are more adequate, therapeutic, and enduring. The development of dialogic and experimental tools may become a key pathway toward a new quality of interior design practice.

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Strangeness: Narrative Experiments with Form in Architecture

Architectural narrative, here defined as the conceptual layer of an architectural work that enables the shaping of space according to the ideas an architect wishes to convey (Psarra, 2013), can be understood as a process of "writing" through form and remains fundamentally experimental. Indeed, this tool of the design proces (Wierzbicka, 2013) does not involve mindless repeating familiar spatial forms or copying existing building types; rather, it requires developing an individual language of expression, sometimes called the poetics of architecture. Notable examples of architectural narrative include Daniel Libeskind's Jewish Museum in Berlin (2001) and Steven Holl's Knut Hamsun Centre in Hamarøy (2009).

Completed works of architectural narrative present ideas in different ways. Some use classical or rational forms to discuss ideas in a serious manner. In other cases, architects choose extraordinary forms and create space as an experience. They seek innovation and push the boundaries of established aesthetics and symbolism, as seen in the theses of this conference. From these works, one can build a catalogue of narrative strangeness in architecture.

The catalogue is grounded in the author's interpretation of Mieczysław Wallis's theory of aesthetic experience (2004), applied to the narrative of built space. Wallis divided positive aesthetic experience into two categories: soft and harmonious, providing immediate pleasure; and sharp, partially disharmonious, in which repulsion, disorientation, fear, or depression is initially felt, followed subsequently by consolation. Harmonious experiences are provided by beautiful objects; disharmonious ones by expressive, comic, sublime, or tragic objects, depending on whether the recipient initially feels repulsion, alienation, amazement, or fear.
Wallis categorised expressively ugly objects as irregular structures marked by asymmetry and stylistic heterogeneity. The Musée des Confluences in Lyon (Coop Himmelb(l)au, 2015) is a clear example. It lacks orthogonality, rhythm, and visible constructional logic. The irregular mass sits against a neglected riverbank and spatial chaos, yet forms a distinctive image. This aligns with the architect's vision of a distorted experiment that reflects the modern world and stimulates curiosity.

Ruins, as another species within the category of expressive objects, "tell" the history of a place. At the Chilean Architecture Biennial (2025), Anagramma Arquitectes and Óscar Aceves Álvarez inserted a temporary pavilion into the burned ruin of the Church of San Francisco de Borja in Santiago. This project shows architecture as the reactivation of existing places, where layers of time and material create new narratives.

By contrast, grotesque objects, in Wallis's terms, depart from common types and norms, striking the imagination of viewers. Furthermore, comic objects are purposeless and absurd. Both these qualities come together at the Inntel Hotel in Zaandam (Molenaar & Van Winden Architecten, 2010). This twelve-storey tower is composed of stacked replicas of traditional Zaan houses. Its comic quality stems from the surprise of seeing normally separate houses transformed into something resembling a clumsy child's drawing.

Regardless of aesthetic beauty, even when perceived as strange or ridiculous, narrative buildings have value as storytelling spaces. In this approach, urban and architectural space becomes both the medium and the result of social practices. Furthermore, it shifts from a static entity into a space of events, tied to time and sometimes to surprise and novelty: to experiment.

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New Sacred Spaces in the Context of the Secularization of European Society

The process of secularisation in Europe is beginning to shape new typologies and forms of experimentation in religious architecture. In the new districts on the outskirts of major cities, 21st-century churches are being built. The process in Europe has been underway for decades. At the same time, one can observe greater involvement on the part of the remaining believers in the church. A preliminary contribution of a new typology for 21st-century churches: in my view, begins with the scale factor of the buildings and the implications this factor has for their sacred character.

The scale of the church. Churches have been structures that dominated the landscape, towering in scale above the secular buildings. The sacred character of these buildings was created by: porticoes, rose windows, the rhythm of high windows, domes and slender towers. Scaled-down buildings – referencing spatial archetypes but on a much smaller scale. Examples of such churches, referencing historical forms, include: the Notre-Dame du Lac church in the Ginko district of Bordeaux, France; the New Church in Boulogne-Billancourt, Paris, France; and the wooden rotunda in Nesvacilka, the Czech Republic. Within this group, there are projects derived from modernism like the church by Álvaro Siza in Rennes, France.

Sculptural abstraction is the second type of building that draws on Christian symbolism in its form, creating a new sacred quality, often through contrast with its surroundings, for example the Church of St Monica in Madrid and the Church of St Joseph in La Bienveillante, Versailles, France.

The building as a complex of functions. The church, as the main space for celebrating the liturgy, is integrated within the building's volume with ancillary functions, giving the building a diverse architectural expression under one roof. Examples of this type of building include: the Church of St John Paul II in Páty, Hungary, near Budapest, and the new church in San Sebastián de los Reyes, Madrid, Spain.

Churches on the ground floors of buildings with other functions. This type of space is created to bring the Eucharist closer to the faithful at a low cost, in the context of purchasing or renting such premises. Examples of such projects include the church in the Łacina district of Poznań Poland, and in Paris, Church of Our Lady of Lourdes in the Menilmontant.

Churches of Christian unity. This type is particularly associated with interior architecture. It stems from a growing interest in the West in Icons and the art of the Eastern Church, which has preserved a sense of mystical continuity. Interiors are being created, particularly in Central Europe, which combine the post-synodal character of the interior with icons and art from the Eastern tradition. For example, the church in Międzylesie near Warsaw or the new seminary chapel in Warsaw.

A preliminary conclusion from the emerging typology is that many initiatives arise in connection with dialogue and the actual needs. It may be assumed that this is a reference to the Church in dialogue and on a journey. The question that arises in conclusion is whether we have the beginning of "Synodal architecture".

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Architecture and Digital Technologies as Complementary Tools for Enhancing Museum Accessibility: A Case Study of the Silesian Museum

Contemporary museums are transforming the way they operate by encouraging visitors to interact and actively participate, rather than passively viewing exhibition content. This, in turn, enhances inclusivity and accessibility in this field. Digital technologies such as apps, multisensory devices, and location systems enable accessibility to extend beyond the physical space, encompassing cognitive and sensory aspects. Thanks to new technologies, the participation of visitors with varying physical and cognitive abilities can be enhanced, ensuring inclusivity for diverse groups of visitors. Research by Garcia Carrizosa, H. et al. (2020) demonstrated that their effectiveness stems from the integration of these technologies with the space and the needs of the audience. The Silesian Museum in Katowice is an interesting example in the context of implementing digital technological solutions in the built environment.

The purpose of this article is to summarize the role of modern digital technologies in shaping the accessibility of museum spaces. The research question formulated is: How does the integration of modern digital technologies with architecture affect actual accessibility for users with diverse needs and abilities? and the thesis: Digital technologies increase user engagement in the reception of exhibition content.
A review of the literature on the subject points to the concept of the smart museum, which emphasizes the importance of technology for accessibility. The study by Kasemsarn and Nickpour (Kasemsarn, Nickpour, 2025) presents a model for integrating technologies (DST) with the principles of universal design learning (UDL), aimed at engaging seniors in active participation in content reception. Indoor positioning systems play a significant role, as they help users - including those with disabilities - navigate spaces more easily and obtain information about their surroundings (Bracco et al. 2020). Research also points to the need for design that incorporates participatory approaches and simultaneous, diverse communication channels to ensure universal accessibility (Marty, 2008).

This study employs a case study approach focusing on the Polish example Silesian Museum, with particular attention to the digital technologies used and their relationship to the design and organization of space.

The Silesian Museum, located on the site of the former “Katowice” mine, is a cultural facility with a complex above - and underground structure. The dispersed layout, where spatial orientation can pose cognitive challenges, is supported by digital technologies. The museum offers, among other things, a mobile app with an audio guide function, a geolocation system, a platform for virtual tours and video walls.
Multisensory forms of communication, audio description, content in sign language, and translations into foreign languages were implemented, aligning with the concept of a smart museum and enhancing the inclusivity of the museum space. The facility is equipped with a building management system that includes e.g control of lighting, temperature, and access to restricted areas.

This analysis provides valuable insights into the integration of digital technologies with architecture, but it also highlights existing limitations. Some solutions require active engagement, which may be a barrier for certain visitors.

The assumption that digital technologies play a significant role in improving museum accessibility, including architectural accessibility. Taking the Silesian Museum as an example, it should be noted that technological and architectural innovations do not replace one another but rather complement each other, and the complementary nature of these components of the museum environment should be preserved.

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Problems of Territorial Planning and Architectural Design (Form Generation)

The purpose of this study is to consider defense territorial planning as an experiment in space and time. Ukraine inherited a significant territorial defense infrastructure that had originally been intended for offensive operations in the western direction, whereas today protection from the east has become the primary concern. Beyond purely military aspects, this represents a large-scale task of developing a new troop deployment system, requiring architectural solutions in territorial planning and volumetric-spatial design aimed at creating an effective defense system against not only military attacks but also terrorist actions.
There is an urgent need to investigate the problems of formation, regulation, and standardization, as well as the development of territorial planning and urban planning solutions for defense clusters during the transition to a new defense development concept in Ukraine aligned with international standards.

The concept involves creating a fundamentally new and modern system of territorial troop deployment within specialized clusters that will function as regional mobilization and training centers, as well as facilities for assembling marching battalions during special periods. A regional military mobilization and training center should provide a complete range of modern residential, social, and specialized facilities for accommodation, organization, training, physical and moral preparation, recreation, and leisure activities for territorial military personnel.

In addition to the above, the research process includes fundamentally new functions, definitions, and terms inherent in Ukrainian culture and traditions.
These clusters may be referred to using historically traditional names such as “Sich’s” (for example, Dnipro Sich, or Chernihiv Sich) and may include centers for patriotic youth education. For this purpose, it is advisable to locate such facilities near historical fortifications – for example, the Transcarpathian Sich near Palanok Castle in Mukachevo, based on the existing military settlement.

The expected results of the study are achieved through obtaining additional findings regarding the identification of potential opportunities resulting from the application of advanced construction and architectural technologies, calculations of structural strength and spatial stability of buildings, optimization of structural materials, and engineering communication systems of building complexes.

The cluster approach also performs an important social function by promoting employment opportunities for residents of rural and adjacent territories and contributing to local budget revenues through the most efficient use of the available resource potential and competitive advantages of the entire region.
The study also involves the creation of a Building Information Model (BIM model) of the viability and functioning of the object throughout all stages of its lifecycle, from conceptual design to disposal.

The developed findings, particularly the elaboration of principles for the design and construction of territorial defense complexes, are of significant importance for strengthening Ukraine’s defense capability due to the possibility of their application in military training and preparation, as well as in supporting the effective organization of troop deployment areas and defensive positions.

The next stages of the research will focus on volumetric-spatial planning solutions and design, functional schemes, integration with the urban environment, adaptability of the military cluster, and the influence of such facilities on urban planning, architecture, and the economic and cultural conditions of the region.

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Streets as a Design Experiment on the Example of Comprehensively Planned Urban Development, Seestadt Aspern in Vienna and Miasteczko Wilanów in Warsaw

The layout of urban public space – streets and squares – constitutes the most legible and recognisable component of a settlement's morphological structure. It can be defined as a network of interconnected places, taking the form of urban interiors.
In the literature, the street is treated as a specific environment for the everyday activities of urban communities. It serves not only as a movement space but also as a venue for most social activities. "Streets and their sidewalks, the main public places of a city, are its most vital organs" (Jacobs, 1961). Consequently, Jan Gehl observes that urban design should begin with life and user activity: "first life, then spaces, then buildings – the other way around never works" (Gehl, 2010).

The perceived image of urban space, as experienced by users, is undoubtedly influenced by objective spatial parameters. Thus, physiognomy related to urban form and the functioning of urban life's corridors are two key features that determine the qualitative perception of public space.
Recognising these two aspects, Rapoport (1987) defines the street as a linear space lined with buildings on both sides and describes it as a place – based on use, serving as a setting for specific activities.
Research on the interrelationship between street characteristics – quantitative (related to form) and qualitative (related to the level of social activity) – is conducted at two scales: a large scale, where the street is understood as a directed urban interior, and a small scale, where the street is a fragment of a network of urban spatial-temporal flows.

To describe features related to the physiognomy of urban public space, the following parameters were used: (1) street width within boundary lines, (2) the ratio of building height to street width (H/W), (3) the share of continuous frontages, (4) the proportion of pedestrian and vehicular routes, (5) green strips, (6) clusters of street furniture, and (7) the share of publicly accessible services in ground-floor programs.
The identification of qualitative characteristics of urban social activity is associated – in addition to physical parameters and individually perceived levels of comfort, safety, and attractiveness – with factors such as: (a) imageability, (b) the degree of spatial enclosure, (c) the preservation of human scale, (d) transparency between public and private space, and (e) complexity (the visual richness of a place), which most researchers consider key qualities in the assessment of streets (Tandon, Sehgal, 2020).
Contemporary, comprehensively designed development layouts in new districts – created "in cruda radice" are places where a fully formed design concept confronts real functional challenges of the city. The street becomes a testing ground for ideas concerning the effectiveness of systems (natural and technological connections), social relations, and the aesthetic quality of public space.

The aim of this short paper is to provide a preliminary identification of the spatial and functional characteristics of streets in new European housing developments, understood as experimental spaces, based on two comparable yet distinct case studies: Seestadt Aspern in Vienna and Miasteczko Wilanów in Warsaw.
A key distinction between these two examples lies in Aspern's clearly separated traffic flows and a greater hierarchy of space. According to strategic documents, the district was designed as a compact, mixed-use urban district where public space and mobility concepts are tested as part of a long-term urban experiment (City of Vienna, 2014).
By contrast, Miasteczko Wilanów represents a different model of urban extension, based on a classical street grid and perimeter-block development.

Preliminary research results indicate that streets in new developments constitute a key field of urban experimentation, where cultural, planning, and political differences become evident.

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The Multifunctionality of Green Spaces in Architecture: A New Design Approach in Poland Considering Nature-Based Solutions

In Polish cities, we are witnessing an experiment - introduction of a new approach to designing greenery in architecture, expected to fulfill functions from mitigating the effects of climate change, through improving user comfort, to influencing human psychology. This trend reflects a broader paradigm shift in the designing of built environments, involving a move away from focusing exclusively on the needs of human users toward consideration of the environment surrounding the designed space – bound by a network of dependencies and fundamentally influencing human users.

This change affects design approaches consolidated over recent decades, in which greenery was treated mainly as plantings confined to botanical considerations and often delegated to specialists outside the core architectural team. Nowadays, multifunctional greenery systems are being designed – frequently supported by complex technologies, part of wider spatial network of dependencies, and belonging to the concept of Nature-based Solutions (NbS). In Poland during the era of the centrally planned economy, extensive Urban Green Spaces (UGS) were rarely assigned functions other than recreational ones; in the free-market reality, this function has mostly been replaced by rainwater retention on small areas designated by planning guidelines.

Over the past two decades, the multifunctional effects of UGS have been extensively studied and effective solutions developed, now being implemented in Polish cities. However, due to its novelty, this process still appears experimental, with unpredictable outcomes.

Investment pressure in city centers limits space for greenery, but solutions enable effective UGS even in very small spaces. Integrating the function of greenery into architectural design requires identifying the needs and capabilities that urban green spaces (UGS) can fulfill and access to applicable solutions. The multifunctional effects of UGS require interdisciplinary analyses that define, for each location and application, necessary functions and a feasible set of design solutions.

OBJECTIVE. A new, multi-criteria approach requires a revision of the process and tools used in greenery design within architecture.

THESIS. Designing greenery in architecture to fulfill multifunctional functions requires interdisciplinary analyses that identify needs and available technological solutions

LITERATURE REVIEW. Extensive taxonomic research has been conducted on the effects of UGS on improving living standards in cities, defining the main challenges – economic, ecological, and social. A specific issue is adaptation to climate change; for Polish cities the main challenges are the urban heat island effect and rainwater management. An important aspect is biophilia and aesthetic impact.

METHOD. Comparative analyses and case studies were conducted in selected Polish cities, including analyses of planning documents, realised projects and their planned functions, and compliance with planning guidelines and scientific assumptions reported in the literature.

ANALYTICAL AND INTERPRETIVE SECTION. An overview of the design process was conducted, considering scientific assumptions and planning documents and identifying interdisciplinary data required for the design. The results of a comparative analysis of planning documents in the studied locations were presented.

CONCLUSIONS. Municipal planning documents are expected to define the purpose of greenery in the architecture; to establish requirements, they must cover a significantly wider area than the area addressed by the architectural design.

Functional targets for greenery in the architecture can be defined only through interdisciplinary analyses at the architectural design stage, due to time frames and often cost constraints, particularly in smaller-scale projects. The need to integrate various planning documents and supplement them with design guidelines for greenery in architecture has been identified.

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Experimentation in Sacred Architecture – Between Innovation and the Continuity of Tradition

Generally speaking, sacred architecture is attributed with characteristics such as stability, durability, continuity of tradition, and the use of relatively repetitive formal and symbolic solutions. Despite this, buildings serving this function have been, and continue to be, the subject of experimentation. This article analyzes experimental solutions, with particular emphasis on post-Vatican II projects. An attempt is made to highlight both the benefits and the risks associated with their implementation.

Architecture – including sacred architecture – reflects the spirit of the era. It undergoes transformations alongside social, ideological, and religious changes. For a long time, the Roman Catholic Church served as a significant patron of art and architecture. The most distinguished architects were given the opportunity to experiment with, among other things, new materials, construction methods, forms, and the organization of interior spaces. As a result, pioneering structures and solutions emerged, which influenced subsequent sacred works and the style of the era.

The spreading avant-garde trends in art and architecture of the early 20th century, along with the ideas promoted by advocates of liturgical renewal, brought about new experimental solutions. The documents of the Second Vatican Council and subsequent guidelines, as well as the reflections of eminent theologians (including R. Guardini and J. Ratzinger), expressed the Church's openness to contemporary trends in art and architecture, provided – as is often emphasized – that they lead to faith. This is particularly emphasized in selected passages from documents such as Sacrosanctum concilium, the General Introduction to the Roman Missal, Paul VI's Gaudium et spes, and John Paul II's Letter to Artists. In post-conciliar churches, experimental solutions have been and continue to be implemented, aimed at realizing in architectural form ideas such as, for example, the full and active participation of the entire congregation (so-called participatio activa), liturgical renewal (so-called aggiornamento), or the Church's openness to the modern world. For example, the latter was clearly emphasized in Alvar Aalto's design for the Church of the Assumption of the Blessed Virgin Mary in Riola. The Finnish designer employed sliding main entrance doors that allow the nave to open directly onto the square in front of the church. More radical and experimental solution – also from an engineering perspective – appeared at the Church of the Sacred Heart of Jesus in Munich (designed by Allmann Sattler Wappner Architekten). Its front glass façade serves as a massive gate, considered one of the largest ever constructed in the history of architecture.

Experimental initiatives include the construction of temporary pavilions during global events. During the World Expo 2000, the Christ Pavilion designed by M. von Gerkan was erected, which was later relocated to the grounds of the Volkenroda Monastery in Thuringia. One of the most striking examples of so-called "experimental spaces" in recent years is the Vatican Pavilion – Vatican Chapels – presented as part of the 16th International Architecture Exhibition at the Venice Architecture Biennale (2018). The Holy See's participation in this event was unprecedented. Ten chapels were erected as temporary installations/pavilions/sculptures, loosely inspired by E. Gunnar Asplund's chapel at the Stockholm Forest Cemetery. Those chapels showcased the diversity of contemporary design trends.

The results of the analysis indicate that the benefits of implementing innovative ideas are often indisputable, but the risks of their application can raise many concerns. The use of experimental solutions inherently carries the risk that the final result may be misunderstood and rejected by the public and users.

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Experimentation through Making: The Importance of Design-Build within Academia

The concept of "experimentation through making" has existed within architectural education for centuries, especially through three-dimensional exploration of scale models. This pedagogy was synonymous with one of the most influential design institutions of the twentieth century. Black Mountain college (1933-1957), located in the eastern United States and led by Josef and Anni Albers and Walter Gropius, taught future artists and architects – such as Buckminster Fuller and Cy Twombly – through a 'learn by doing' mentality that emphasized the necessity of construction at full scale. This pedagogy still exists today in architectural education, particularly through design-build studios where students work as a team to construct projects at full scale for local clients. This multi-year research chronicles the design-build studios offered at thirty schools of architecture within the United States.

The chronicled design-build projects rely on experimentation to tackle context-specific challenges of technical solutions, ecological sensitivity, and societal complexities through ethical design. Design-build projects are usually built for local communities, which means the constructed spaces must be meaningful, functional, and ethical. Depending on the project, students meet with the client – oftentimes non-profit organizations who lack the funds to hire professional architects – and community members to effectively design for their needs. Projects must also explore material and construction experimentation. Scorcia, a material derived from volcanic stone, was tested and used by students from the University of Arizona in their Sentinel House project, while a complete gabion wall façade was implemented by students from the University of Colorado at Denver for bathrooms along a hiking trail at 4,000 meters elevation in the Rocky Mountains. Another project, a floating camping dock that can only be reached by kayak, had to consider material construction hierarchy to ensure that local alligators would not terrorize campers. Constant combinations of community outreach, ecological sensitivity, and localized material use yields ethically conscious projects. Since experimentation reaches its fullest dimension at the 1:1 scale, students learn valuable teamwork skills as they create space for real users. This serves as a monumental moment not only in their education, but in their lifelong critical thinking practice.

Chronicled projects range in size, from outdoor classrooms for elementary schools and a kayak storage facility for a state park, to workforce housing and net-zero private homes. Regardless of scale, each project required students to critically experiment through both design and full-scale construction. The research was conducted through interviews of design-build professors and was supported with survey responses from over 230 former design-build students between the years of 2014-2024. Through the survey results, it is evident that this experience shaped students' behavior towards the profession, technical proficiencies, ecological and financial design sensitives, and greater empathy through client interaction. The survey also proved that such design experimentation was key to shaping critical thinking capacities and questioning existing design paradigms. Above all, the survey yielded near-unanimous support for future architecture students to enroll in design-build studio options. This research has been presented in a similar format at the 85th annual Texas Society of Architects Conference in 2024 and at New York Build Expo in 2025. A manuscript for book publishing is currently underway.

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The Need for Contact as a Lens for Reading Urban Space: A Case Study of EKOPark, Ursus, Warsaw

Design decisions in public space are typically grounded in existing-condition analyses and professional intuition, yet they often fail to capture how places actually meet users' needs. This study argues for a research-based approach using a matrix of "user needs lenses" supported by systematic data collection. Treating EKOPark as an urban laboratory where methodology is tested and findings can inform interventions elsewhere, it shows how evidence-based diagnosis can precede redevelopment decisions.

The case study is the diagnosis of EKOPark in Warsaw (Jaworski et al., 2025), located at the interface of an established housing estate and a rapidly urbanizing district. The area combines an award-winning designed park with spontaneously developed fragments and a marketplace slated for relocation amid major road reconstruction. The research problem regarded ways the spatial configuration satisfies the need for social interaction across user groups; the aim was to identify potentials and deficits and inform redevelopment decisions.
Gehl (1971) describes eight psychological needs in living environments, based on her observations of urban life. Among them, the need for Contact: "to look at, to listen to, to speak with others, to do things together with others" (p.167). Building on Gehl's definitions, we adjusted them for public space specificity, added three further needs, and operationalized them into measurable indicators in the IAS tool (Jaworski et al., 2025).

IAS methodology was applied to the EKOPark diagnosis for its capacity to read space through the lens of human need. Designed as a replicable toolkit, IAS draws on an interdisciplinary framework combining urban design, sociology, and behavioural research, enabling the same space to be examined from multiple disciplinary perspectives (Waśniewska et al., 2025). The study combined inventory, activity and trajectory mapping, systematic observations, interviews, a research walk, and in-depth interviews (IDIs). Data triangulation allowed cross-referencing of conclusions, confronting spatial evidence with lived experience in ways single-discipline approaches cannot achieve.

The need for contact is defined as participation in social life through observation or interaction with other users, expressed through shared activities in pairs or groups and shaped by the organization of space. EKOPark primarily supports interaction in small groups; infrastructure enabling larger gatherings is absent. User groups coexist in distinct zones with minimal conflict. The interdisciplinary methodology provided insights about two vulnerable groups, teenage girls and seniors, whose needs remain invisible to conventional spatial analysis.

The park functions as a local space for everyday encounters: walks, children's play, outdoor exercise, and marketplace visits. Although teenagers are a visible presence, closer analysis reveals this applies overwhelmingly to boys. The current spatial arrangement lacks elements supporting interaction among teenage girls. Although the site includes facilities for seniors such as an outdoor gym, planned changes will significantly affect their ability to socialize. The marketplace sustains daily social relations among older adults, not only as a place for shopping but as a setting for regular contact; its planned relocation poses a direct risk to these ties.

Capturing the complex needs of diverse users requires analysis grounded in research on human needs combined with systematic, interdisciplinary observation. EKOPark as an urban laboratory shows this approach does not merely describe existing conditions: it confronts traditional spatial analysis with insights from cognitive and social sciences, producing evidence that is locally specific yet methodologically transferable, forming the basis for decisions that genuinely serve the fundamental function of public space, satisfying the needs of its users.

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The Model as a Space for Experimentation: Interuniversity Workshops on Sensory Garden Design at the Intersection of Architecture and Art

1. Introduction and Research Problem

Sensory gardens play an increasingly significant role in the therapeutic shaping of urban space, supporting the emotional self-regulation of their users. However, translating neurobiological and environmental theories into practical design guidelines remains a considerable challenge. The design methods employed to date have seldom accounted for the subjective and diverse needs of particular user groups, which justifies the adoption of participatory tools.

3. Theoretical Background

The theoretical foundation of this study rests on J. Ayres' theory of sensory integration, which describes the perception-processing-response feedback loop (Ayres, 2023), and J. J. Gibson's theory of affordances (Gibson, 2014), which distinguishes between activating and calming environmental stimuli. Research in environmental psychology indicates that the simultaneous stimulation of two to three senses alleviates cognitive fatigue, whereas fully multisensory environments may lead to sensory overload. Pallasmaa (2012) draws attention to the architectural dimension of sensory perception, criticizing the dominance of vision in design and advocating for a more comprehensive engagement of touch, hearing, and proprioception in the experience of space. The methodological framework for the workshop process is provided by Brown's design thinking concept (Brown, 2008), which assumes an iterative progression from empathy with the user, through rapid prototyping, to the testing of solutions.

4. Research Method

A 9-hour interdisciplinary co-design workshop was conducted at the Eugeniusz Geppert Academy of Art and Design in Wrocław. The workshop involved 31 students from the Faculty of Architecture at the Wrocław University of Science and Technology and the Faculty of Interior Architecture at the Academy of Art and Design. Participants were divided into six interdisciplinary teams working in two separate rooms, so as to combine the perspectives of both disciplines. Each pair of teams was tasked with designing a zone with a different sensory emphasis: visual, tactile, or auditory. Rapid physical prototyping was employed using traditional materials (paper, foam board) at a 1:20 scale. Data were collected through non-participant expert observation and transcriptions of the final project presentations.

5. Design Workshop Results

Qualitative analysis of the six models confirmed the successful translation of theory into architectural solutions. Given full design autonomy, students consistently rejected strong stimulation in favor of calming affordances conducive to stress deescalation:

1. Acoustic zones - sound-dampening structures, including "silence pavilions" and sound-reflecting screens;
2. Tactile zones - a shift in focus from the architectural macro-scale to tactile stimuli accessible at body scale, such as textures, moss, sand, and varied surface granularity, producing a grounding effect;
3. Visual zones - the use of shadow, translucency, and spatial sequencing while avoiding intense stimuli.

While the final garden prototype was developed independently by domain specialists to establish a rigorous framework for tracking empirical variables, via Facial Expression Recognition (FER) and Visual Analogue Scales (VAS), the student workshops functioned as a separate, complementary stream of evidence. The spatial correlation between the students' intuitive designs and the experts' structural concepts confirms that theoretical neuroscience and empirical user-experience design converge on similar environmental outcomes.

6. Discussion and Conclusions

Interdisciplinary co-design workshops proved to be an effective tool for the operationalization of user experience (UX) in architectural design. The combination of macro-spatial architectural thinking with the micro-topographic sensitivity of interior architecture made it possible to bridge academic theory and the design of sensory therapeutic environments that support well-being.

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AI and the City as Laboratory: Experiments with Urban Space

The beginning of the 21st century has been marked by the significant impact of ICT on various aspects of life. High levels of digitization contribute to the transformation of many knowledge and economic sectors. These changes are also evident in urban planning and architecture. Digital design tools and 3D modeling enabled architects to experiment with form and materials. The adoption of advanced technologies in architecture has given rise to new design trends, characteristic of Asymptote, HDA-X, Morphosis, and Zaha Hadid Architects, setting a new standard in urban space.

Today, with AI, experimentation goes further, introducing data, algorithms, and simulations into the design process to optimize and explore solutions. In this context, AI redefines the creative process and the role of the architect, enabling experimentation in the laboratory that cities have become. The application of AI to identifying patterns of local architecture, generative design, optimization, and efficiency also extends to the search for spatial forms and urban layouts. The work of Studio Tim Fu, who describes himself as an "Architectural Futurist," breaks boundaries between human and machine collaboration, revealing a new approach, as exemplified by Bled Lake Estate in Slovenia (2025).

The development of digital twins as design tools supporting smart city and living lab solutions introduces AI into urban development. This approach enables planning and design at all scales—from buildings to cities, and potentially regions. AI-supported digital twins create a dynamic environment in which algorithms enable large-scale modeling, scenario testing, and simulation of climate change, disasters, and planning decisions and their impact on residents' lives.

This raises questions about the architect's traditional role and the design process, as both are undergoing fundamental changes. Experiments with AI are shifting the boundaries of authorship. In the design process, AI analyzes and presents data, eliminates suboptimal solutions for the city and building performance, proposes forms and solutions appropriate to the given parameters, and showcases its own visual appeal. The architect continues to make key decisions, though, by delegating a range of processes to the system, increasingly becomes a curator of the process, an interpreter of data, a mediator, and an intermediary between space, society, and the logic of the algorithm. AI's support for creative and decision-making work leads to integrated authorship that is difficult to separate. This situation raises the issue of responsibility for space and the quality of people's lives.

Data-driven design decisions are transforming the city into a laboratory with real-time experimentation. Furthermore, exploration involves people in the process. The Cognitive City trial planned in Abu Dhabi up to 2027 seems to push the boundaries of urban planning. The central Aion Sentia system is consolidating all data from public and private institutions as well as individual users to offer access to knowledge, information, and services. Beyond this centralized system, the analysis of user behavior patterns, habits, and preferences will create a city of the future where integrated AI algorithms would make decisions and evaluate their consequences on the user's behalf.

As a result, cities and buildings are not only physical but also become dynamic systems, accessible in real time. The experiment shifts from the creator's computer to the scale of the city and the lives of its inhabitants. This raises a natural question about the limits of technological intervention in creating human living spaces. Will the city of the future remain human?

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Innovative Use of Mycelium in Construction – Mycotecture Research for Sustainable Development

The aim of the interdisciplinary research project presented in this article was to identify innovative applications of mycelium in construction, particularly in residential buildings. In the era of growing ecological awareness and the need to reduce CO2 emissions, biological materials are becoming an important sustainable alternative to traditional construction materials (Lewandowska et al., 2025; Alaneme et al., 2023). Mycelium stands out due to its unique properties: high mechanical strength, excellent thermal and acoustic insulation, low weight, and full biodegradability (Ghazvinian & Gürsoy, 2022; Lewandowska et al., 2025).

The project was implemented in 2026 through close cooperation between researchers from Warsaw University of Technology and experts from the Łukasiewicz Research Network – Lodz Institute of Technology. The experimental studies focused not only on verifying the material's technical parameters but also on assessing its potential for use in real construction conditions and public spaces. A key assumption was that mycelium should be treated not only as an experimental raw material, but also as a material with aesthetic and design value (Wierzbicka & Ławińska, 2024). The research methodology was comprehensive and multi-stage. It began with a thorough analysis of international literature on mycotecture (Ghazvinian & Gürsoy, 2022; Alaneme et al., 2023). Next, research goals were defined, substrate formulations were developed, and laboratory tests were conducted on compressive, bending, and moisture-resistance properties. After the laboratory phase, the team moved on to the key public space experiment. One of the most visible outcomes of the project is a prototype wall made entirely of mycelium bricks and blocks. The wall was installed in Łowicz on a public outbuilding. This field experiment aimed to test the material's durability under real conditions – exposure to rain, frost, UV radiation, temperature fluctuations, and urban pollution. The wall's design draws inspiration from hunting tradition – the entire structure was painted with characteristic "hunting stripes" that refer to the local cultural heritage of the Łowicz region. Thanks to this, the wall serves both research and artistic-educational functions, attracting attention from residents and tourists.

The results obtained so far are promising. Mycelium-based material demonstrates good load-bearing capacity at very low density. It also shows natural self-regenerating and antibacterial properties (Wierzbicka et al., 2025; Lewandowska et al., 2025). During the public experiment, the wall's durability has been monitored in central Poland's climate. After several months of exposure, no significant structural damage has been observed, confirming the material's high potential for application. Despite its advantages, mycotecture still faces challenges, including long-term resistance to biological and atmospheric factors, as well as the standardization of production processes (Alaneme et al., 2023). Nevertheless, the authors emphasize that further development of this technology may enable "growing" entire structural elements, or even simple buildings, directly on construction sites in the future.

In conclusion, mycotecture opens new perspectives for sustainable construction. By combining the innovative biological properties of mycelium with design and cultural aspects, it can become an important element of the green transformation of Polish and European architecture.

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Narrative and Participatory Teaching Strategies – The Role of Experimentation in Architectural Education in the Age of AI

Experimentation in architectural education remains a key tool for fostering critical thinking. Conscious testing of solutions allows students to generate alternative concepts, analyze, question, and redefine existing design paradigms (Wierzbicka, 2013). In the era of rapid AI development, narrative and participatory teaching strategies take on particular significance, especially at the Faculty of Architecture of the Warsaw University of Technology, within the Department of Architectural and Urban Design. In this Department, the narrative method treats architectural and urban design works as multi-threaded stories embedded in cultural, social, and spatial contexts. While working on semester assignments (including those with themes such as "between"), students construct spatial scenarios in which the design tells the story of a place, its inhabitants, and future transformations (Wierzbicka, 2013; Szpakowska-Loranc, 2023). A key element of this strategy is participation – the narrative ceases to be the designer's monologue and becomes a shared, co-created story. Students conduct narrative workshops, field interviews, mapping of residents' histories, and participatory sessions, and the collected local narratives directly shape the design concept (Nazidizaji et al., 2015).

The development of AI opens up new possibilities for educational experimentation. Artificial intelligence tools support the generation of form variants, usage simulations, and rapid prototyping, while allowing a focus on the project's narrative and social layers. At the Department of Architectural and Urban Design, students combine algorithmic generation with participatory storytelling – AI helps test hundreds of variants. Still, the final selection and interpretation belong to human narrative and dialogue with the community (Havik, 2012). As a result, the educational experiment becomes hybrid: technological and deeply humanistic. Such narrative and participatory teaching strategies transform traditional instruction into an active, reflective process. Students not only learn to design space, but also to build shared stories, develop empathy, and take responsibility for architecture's impact on the environment and local communities. In the age of AI, the role of experimentation lies in maintaining a balance between technological tools and humanistic values – narrative and participation become an antidote to unreflective design automation (Wierzbicka, 2013).

The introduction of narrative and participatory strategies, combined with AI-assisted experimentation, reinforces the modern character of architectural education at universities. At the Faculty of Architecture of the Warsaw University of Technology, particularly in the Department of Architectural and Urban Design, students prepare not only for the role of designers but, above all, for the role of mediators and narrators of the future of space. Architecture ceases to be a purely technical artifact – it becomes an ethically grounded, collectively created, and technologically supported narrative about the city and its inhabitants.

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Democracy in Space: Temporary Architecture and Learning by Design in the Liberec School of Design and Practice – Creating the City through Experiment

The study examines temporary architecture as a methodological tool for spatial experiments and participatory urban development within the framework of research-by-design. In the Liberec case study, the design-and-build pedagogical approach serves as both an educational method and a catalyst for urban innovation. Contemporary architecture is evolving from static structures to experimental, process-based methodologies. Temporary architecture is central to this shift, offering reversibility, flexibility, and cost-effectiveness for rapid prototyping and real-time spatial evaluation in urban settings. It functions as an applied research method, using spatial interventions as experimental probes to observe real-life interactions between users, environment, and built form. Thus, urban landscapes become practical laboratories, establishing architecture as a methodological research approach.

The research presents an overview of architectural design and research approaches, based on projects in Liberec over the past 25 years, using unstructured interviews and university archival records. Design-based methodologies merge conceptual thought with physical construction, making building central to knowledge creation. This challenges traditional architectural education by blurring the division between idea and material.

In Liberec, this experimental approach is shaped by architects like Martin Rajniš and Jiří Suchomel and the Technical University's Faculty of Arts and Architecture. Their work emphasises material exploration, construction innovation, and direct involvement. Students participate in large projects, gaining real-life experience in collaboration, negotiation, and decision-making. This develops key competencies, transforming education into practice-based research and learning. Knowledge is created through iterative design testing and refinement.

Temporary Architecture and Democratic Practice
Temporary architecture fosters democratic practices through participation, flexibility, and co-authorship, enabling open use and reinterpretation. Unlike persistent structures, this allows for open use and reinterpretation, as in the case of "Vlna" pavilion. This architectural approach serves as a platform for interaction. Similarly, temporary interventions, such as Mjölk architects' sauna at Harcov Dam, activate public spaces and introduce new social practices.

Liberec as a spatial laboratory.
Liberec blends experimental architecture with education and urban life. Public spaces, particularly around Liberec City Hall, host temporary art installations that continuously test spatial concepts. The city's identity is also defined by permanent experimental projects like Karel Hubáček's Ještěd Tower. Its unique hyperboloid form exemplifies innovative construction, bridging temporary spatial activities with enduring architectural achievements. Hubáček's SIAL team also played a key role, shaping architects such as Martin Rajniš, John Eisler, and Jiří Suchomel, who based their design practice on experimentation and a relationship with construction.

Conclusion.
Temporary architecture, combined with design-based learning methods, is an effective tool for exploring spatial configurations and fostering democratic participation. This In Liberec approach is transforming architectural education by actively engaging with urban contexts, making architecture a process of inquiry, collaboration, and responsibility. Liberec serves as a testing ground for successive generations of architects trained in the spirit of putting theoretical principles into practice, experimentation and experience within the laboratory of public space, which has led to the development of an original teaching method that subjects even the most insidious ideas to constant critique and confronts them with reality. This has led to a synergy of content within the educational process. Therefore, "democracy in space" refers not only to physical construction but also to the processes of creation.

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Spatial Experiment and the Formation of Social Space in Modular Housing Architecture

Contemporary housing architecture increasingly faces the challenge of responding to rapidly changing social structures, environmental conditions, and patterns of everyday life. Standardized residential models frequently fail to address the growing need for adaptability, spatial flexibility, and socially responsive living environments (Gehl, 2011). Within this context, modular housing architecture and experimental spatial strategies emerge as important tools for redefining collective living structures and strengthening social interaction through adaptive architectural systems.

The paper examines the role of spatial experiment in the formation of social space within modular housing architecture. The research investigates how experimental architectural strategies based on variable environmental and spatial parameters may influence social interaction, behavioral dynamics, and the adaptability of residential environments. The research problem focuses on the extent to which modular and transformable housing structures can function as socially responsive architectural systems. The main thesis assumes that spatial experiments in modular architecture can function as analytical and design tools that actively influence social interaction, behavioral patterns, and the formation of adaptive social environments.

The theoretical framework of this paper combines research related to proxemics, human-centered behavioral architecture, placemaking theory (PPS), and experimental sustainable architecture. Within modular housing systems, architecture increasingly operates not as a fixed structure but as an adaptive spatial framework capable of transformation according to environmental, social, and functional conditions. The concept of social space is therefore understood as a dynamic process generated through spatial relations, flexibility, and user participation. Furthermore according to Habraken (2007), flexible housing should enable users to actively participate in the organization and transformation of their living environment rather than remain constrained by fixed and standardized residential systems. This perspective redefines housing architecture as an adaptive spatial framework capable of responding to changing behavioral patterns, social interaction, and individual spatial preferences.

The methodological framework combines analytical and interpretative methods with spatial prototyping and scenario-based design research. The study applies selected principles of Habraken's support and infill methodology, emphasizing user participation and adaptive spatial organization within modular housing systems. The experimental process was conducted during the Experimental Architecture Design Studio (2025/2026) at Bydgoszcz University of Science and Technology and focused on three spatial variables: sunlight exposure, foldable partition systems, and recombinable modular structures. Students tested alternative spatial configurations through iterative design scenarios examining the formation of socially responsive and adaptable residential environments, adaptability, privacy gradients, behavioral interaction, and the transformation of collective and individual living space.

The analysis demonstrates that transformable modular structures may significantly contribute to the formation of adaptive and socially responsive residential environments. Flexible housing configurations allow users to modify spatial relationships according to changing patterns of privacy, community interaction, and collective use. Variable sunlight conditions influence behavioral perception and spatial activation, while foldable partition systems support temporary transitions between private and shared space. Furthermore, modular combinations and adaptive spatial arrangements increase the long-term usability and resilience of residential architecture by enabling continuous spatial transformation without complete structural replacement.

The study indicates that spatial experimentation within modular housing architecture may support integration, adaptability, social interaction and participatory spatial organization. Experimental architectural strategies therefore become not only formal design explorations but also interdisciplinary research tools supporting the shape of socially sustainable housing structures and responding to contemporary urban and behavioral challenges.

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Prefabrication as an Architectural Experiment - Scale of Integration and Circular Potential

Contemporary architecture is increasingly shaped by the principles of the circular economy, which emphasize resource efficiency, material reuse, and design for disassembly (DfD). At the same time, prefabrication and modular construction are widely promoted as inherently sustainable solutions. In design practice, however, prefabrication is often treated as a homogeneous strategy, without sufficient differentiation between its forms and their actual contribution to circularity (Crowther, 2005).

The aim of this paper is to examine the relationship between the scale of prefabrication and the circular potential of buildings, understood as their capacity for disassembly and reuse. The research question is: how do different levels of prefabrication—from volumetric modules, through panel systems, to low-integration structural elements—influence the reversibility of building systems? The paper advances the thesis that the most technologically advanced prefabrication systems are not necessarily the most circular; rather, circular potential is primarily determined by the reversibility of connections between components.

The study adopts a comparative analytical approach based on a simplified typology of prefabrication. Three main categories are identified. First, volumetric modules (3D), delivered as fully equipped spatial units, offer high efficiency and reduced on-site work, but rely on integrated systems and permanent connections, which hinder disassembly. Second, panel-based systems (2D) represent an intermediate level of integration, balancing factory-controlled production with assembly flexibility. Third, prefabrication based on low-integration structural elements allows for high adaptability and disassembly potential, at the expense of increased on-site labor and reduced production control.

The analysis is framed through the principles of Design for Disassembly (DfD), including reversibility of connections, material separability, and accessibility of building layers (Durmisevic, 2006). Research emphasizes that designing buildings as material banks requires careful detailing of joints to enable the future recovery of components (Witeczek, 2023). This approach is further developed in the CIRCON framework, which introduces circularity indicators related to reuse potential, spatial reversibility, and lifecycle performance (Pikoń et al., 2023).

The proposed approach was applied to a case study of a small-scale development of single-family houses in Toszek, designed by the author of this paper and developed as a design experiment in collaboration with the investor. The project employs prefabricated steel frame panels with infill, assembled as flat elements. A key innovation is a proprietary interlocking mechanical joint system ("click" connection), enabling dry assembly without adhesives or permanent bonding. The system was tested in a production hall to evaluate assembly efficiency as well as acoustic, thermal, and waterproof performance.

During the design process, a critical modification was identified: the separation of the structural panel from the external façade layer, including the wind barrier membrane. This adjustment ensured adequate envelope performance while maintaining reversibility and enabling material separation.

The findings demonstrate that the most technologically advanced prefabrication systems are not necessarily the most circular. High levels of integration may limit disassembly and reuse due to system complexity. In contrast, panel-based systems combined with reversible mechanical joints provide a more balanced approach, enabling both efficient construction and long-term adaptability.

It can therefore be concluded that the circular potential of prefabricated architecture is determined less by the scale of prefabrication than by the logic of connections between components. Micro-scale innovations in joint design and layer separation can significantly enhance disassembly and reuse, redefining prefabrication as a flexible strategy within circular architecture.

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Railway Communication in Urban Structure: Modernisation of Line R7 in the Warsaw District of Falenica.

Issues related to the urbanisation of areas associated with railway infrastructure constitute an important subject of research and analysis in the academic literature. They highlight the city-forming role of railways and indicate their beneficial influence on the development of surrounding areas. The railway, as the backbone of urban and regional transport, offers the possibility of achieving functional and spatial cohesion. Properly planned railway communication helps to counteract the processes of urban sprawl, contributing to urban compactness and protecting agricultural, open, and natural areas. Transit-Oriented Development (TOD) can serve as an effective tool in transforming cities (Calthorpe, 1993), particularly those possessing a well-developed railway network. Research highlights that access to a railway station is a key factor enhancing the investment attractiveness of cities. It enables the realisation of convenient interchange nodes and transport hubs, providing users with access to various modes of transport and associated services (Bertolini, Spit, 1998). Sustainable mobility and the independence of rail-based transport from road traffic are of particular importance within agglomerations and major cities (Suchorzewski, 2010). In an era of road transport dominance, increasing railway competitiveness is a prerequisite for its development and remains an imperative.

On 29 August 1877, the Droga Żelazna Nadwiślańska (Kolej Nadwiślańska) was inaugurated, running from Mława through Warsaw to Kovel. The line underwent numerous modernisations and transformations over the subsequent decades (Ajdacki, 2025).
Today it constitutes Railway Line No. 7, connecting Warsaw with Lublin and the border crossing with Ukraine at Dorohusk. The ongoing modernisation of the Warsaw Wschodnia – Lublin – Dorohusk section aims to bring the line into compliance with European Union requirements in the field of rail transport, including the adaptation of infrastructure to basic quality standards, the improvement of rail transport attractiveness, and the enhancement of its competitiveness relative to other modes of transport. Agglomeration and long-distance train services will be separated, with infrastructure adapted to speeds of 160 km/h for passenger trains and 120 km/h for freight trains (SISKOM, 2026).

In the late nineteenth century, numerous halts and stations began to emerge along the railway line, and the areas situated in their vicinity underwent dynamic urbanisation, forming what became known as the „Pasmo Otwockie". The railway and its associated built environment expansively encroached upon forested areas, permanently altering the character of the terrain, its function, and its spatial image. In 1895, Karol Jakub Hanneman carried out the parcellation of lands belonging to him in the vicinity of the then-existing railway halt, thereby initiating the establishment of the Wille Falenickie settlement. The present-day district of Warszawa Falenica is surrounded by forests, well connected to the centre of Warsaw and to neighbouring districts and towns, and is equipped with numerous commercial, service, cultural, educational and healthcare facilities.

The ongoing modernisation of Line No. 7 in Falenica is intended, upon completion, to improve the flow of long-distance and agglomeration trains, to introduce grade-separated road, pedestrian and cycling connections between the eastern and western parts of Falenica (and other districts and towns), and to deliver a range of additional infrastructure improvements. However, the works are currently causing considerable hardship and disruption for local residents. The modernisation will irreversibly alter the district's spatial layout, visually separating the parts of the district on either side of the railway line by means of acoustic screens, and will introduce changes to the organisation of road, public transport, pedestrian and cycling traffic. It will also result in the demolition of existing structures, the elimination or relocation of current functions, and the mass felling of trees within the railway corridor and on adjacent land.

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The Static Scheme and the Aesthetics and Functionality of Architectural Form

An inseparable component of every architectural work is its structure. Over the centuries, building structures have evolved from simple statically determinate schemes to the more sophisticated implementations of recent decades. This article focuses on the influence of the static scheme on the aesthetics and functionality of a building. The mathematical logic of construction enables the optimal arrangement of structural materials and influences the size of the created space in the form of large-span roofs. The research was conducted during annual classes in the Architecture and Technology of Structures course. As part of the classes conducted at Warsaw University of Technology, students verify their acquired knowledge of structural mechanics during the design process and, most importantly, during the independent implementation of small-scale wooden structures designed in the Royal Łazienki Park in Warsaw. These classes were very helpful in confirming the thesis about the relationship between the clarity of a building's structure and its aesthetics and functionality. The authors' own research was supported by literature, particularly the works of Wacław Zalewski and Edward Allen.

Simple static diagrams of beam, vaulted, arched, or shell roofs have been used since antiquity to bridge large spans and thus create larger spaces; see the Colosseum and the Pantheon.

With the introduction of more advanced materials such as concrete, iron, and steel, it became possible to apply static diagrams that more fully utilized the achievements of mathematics and the theory of structural mechanics. The works of Robert Hook, Leonard Euler, and Maxwell Mohr enabled the precise determination of forces in both statically determinate and indeterminate systems, as well as the analysis of structural stability.

In the 19th century, the category of iron suspension bridges emerged, with the first European implementation at the Menai Bridge in Wales in 1826. The silhouettes of other suspension bridges, such as the Golden Gate and the Brooklyn Bridge, owe their expressiveness to accepted logical static diagrams utilizing tensile forces. Along with bridge construction, bold static models were also used in public buildings. A pioneer in the development of suspended structures was the Polish architect Maciej Nowicki. The Paraboleum hall in Releigh, USA, based on his concept, utilized a balance of forces in a scheme of two arches transmitting compressive forces and suspended suspension cables. Similar solutions were also employed by the renowned architect Erno Saarinen. The structures designed at that time were characterized by a dynamic silhouette and covered large-span spaces while sparingly using construction materials. Public buildings from the People's Republic of Poland in the 1960s and 1970s also provided outstanding examples of the use of clear static models. Engineer Wacław Zalewski's projects, such as the Katowice Spodek Arena and the Supersam in Warsaw, are excellent examples of the impact of static models on the visual and functional aspects of architectural work. Similarly, though on a different scale, students sought analogous relationships during construction experiments conducted at the Royal Łazienki Park.

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From Sports Infrastructure to Third Places: Reading Shared School Sports Fields in Warsaw

Contemporary cities face rising competition for urban land and limiting opportunities to create new public space and sports infrastructure. In this context, existing underused assets become strategically valuable, with school sports fields being one such resource (Infrastructure Victoria, 2024). Due to their even distribution across residential neighbourhoods, typically within walking distance of homes, and accessible without special equipment or membership, they have the potential to enhance access to spaces for physical activity. The municipal Open Sports Fields programme in Warsaw has significantly expanded community access to school sports infrastructure: approximately 68% of schools now keep their sports fields open outside school hours, although the actual accessibility and intensity of use vary considerably between locations (UM Warszawa, 2026).

This study treats the school sports field as an urban laboratory: a bounded, observable space where design, governance and everyday social practice interact and can be systematically read. The aim was to understand how management practices and spatial design influence the intensity and diversity of community use of school sports fields outside school hours.

The study draws on interdisciplinary, socio-spatial methods examining how spatial configuration shapes diverse use of public space (Waśniewska et al., 2025), and on research into gender dynamics in school sports fields and coined footballcentrism term (Architektoniczki, 2025). Findings are interpreted through Oldenburg's third place concept, informal gathering spaces outside home and work, whose decline carries documented wellbeing consequences in residential neighbourhoods (Finlay et al., 2019).

This qualitative case study examined three shared school sports fields in Warsaw, selected from a pool of eight observed during a verification stage. Each case was investigated through desk research, spatial inventory, activity mapping, field interviews with users, and interviews with managing staff. Findings were further developed through a focus group with designers and a workshop with diverse stakeholders involved in the governance and everyday use of school sports fields, reinforcing the study's urban laboratory framing.

The findings show that school sports fields function as genuine third places: informal, low-threshold grounds where users return regularly, treat the space as their own local territory, and blend physical activity with socialising, with friends as well as strangers.
This informality coexists with high demand, which produces continuous spatial negotiation between user groups and between spontaneous and organised use - revealing how overcrowding forces waiting, improvised surface-sharing, and in extreme cases withdrawal from activity altogether. Currently, this third place belongs most strongly to children and teenagers: boys visit independently four times more often than girls, and unaccompanied exercising adults are rare. Football pitches attract a narrow user group - predominantly boys engaging in a single activity - while multifunctional courts and running tracks draw a far wider range of users and unconventional activities. Integrating a playground shifts the profile toward families, while active edges - benches, residual greenery, fence margins - sustain watching, resting and socialising, pointing to their potential as deliberately designed social infrastructure.

Management, involving consistent maintenance and deliberate non-interference, signals that everyone is welcome to simply show up and play.
Despite football's dominance, school sports fields already function as third places - thus vital social infrastructure. Treating them as urban laboratories has surfaced transferable principles of inclusion that, given the systemic character of municipal school governance, can be scaled across an entire city, making them a replicable, low-cost lever for improving neighbourhood wellbeing.

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Experimenting in Space: Threats and Opportunities for Using Local Spatial Development Plans to Protect the Rural Landscape in Warsaw's Suburban Zone.

The rural landscape is a key element of the identity of many countries, not just Poland, though it is of significant importance there. Its established position stems from a cultural history deeply rooted in rural life. The history of rural landscape transformation, particularly that driven by political changes, demonstrates how readily rural landscapes change. This phenomenon is intensified around large cities, where these processes are more dynamic. Villages in such locations rapidly change their purpose and character, taking on residential functions characteristic of the suburban zone. They lose their identity – historically shaped settlement patterns are blurred, and above all, their agricultural function is extinguished. Local plans, one of whose goals should be to protect cultural and environmental values, can unintentionally exacerbate negative phenomena when poorly formulated; when well formulated and based on knowledge, they can help protect many values, including the rural landscape. Currently, there is a noticeable prioritization of urban space, both in legislation and in implementation. Too little attention is paid to protective regulations for rural areas. These negative planning trends will be irreversible in the coming years, and, in the face of environmental challenges, they appear to be a missed opportunity for pro-environmental optimization in spatial planning. Green infrastructure (GI) in rural landscapes is also of paramount importance for ecosystem services and should therefore be treated as an equal element of the national ecosystem.

This study focuses on the use of local spatial development plans (which function as local law in Poland) as a planning tool influencing the rural landscape in Warsaw's suburban area. The aim is to assess their effectiveness in controlling the rural landscape. The aim is to answer the question of how far local spatial development plans actually shape the rural landscape. It also aims to determine whether local plans promote its protection or lead to scattered development and the degradation of spatial order.

The problem under investigation is multidimensional and concerns both legal norms and their material spatial effects. A mixed-methods approach was employed, combining historical-legal analysis, spatial analysis (GIS and historical cartography), and social analysis. Legal and spatial data were compiled, supported by the identification of local conditions. A case study was conducted, comparing selected examples, leading to the presentation of synthetic conclusions.

The study demonstrated that this tool can produce various spatial effects that can permanently transform the rural landscape. Experience shows that many decisions are left in the hands of planners, who, in turn, often lack specialist knowledge, a solid legal foundation, and the support of local governments as representatives of residents' will, and forgo overly restrictive regulations, thus allowing for arbitrary transformation of the rural landscape. Planning trends in rural space management were demonstrated using examples from local spatial development plans from Serock and the surrounding area. It has been shown that the current protection system is insufficiently coherent and requires systemic changes.

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Urban Ecological Interactions and Their Impact on the Vitality of Urban Ecosystems

Within the context of sustainable development, the resilience of urban ecosystems is defined as the ability to adapt to internal and external factors, withstand threats, and maintain structural and functional integrity (Gabrel, Gabrel, 2022). This is particularly relevant for recreational settlements, whose development occurs under conditions of increased ecological sensitivity to natural and anthropogenic changes, driven by natural-geographical, economic, and sociocultural factors. In the Ivano-Frankivsk region, urbanization and the development of tourism infrastructure have led to the transformation of the natural environment (Zhyrak, 2022), necessitating a reconsideration of urban-ecological interactions as a key factor in ensuring balanced spatial development.

The study was conducted using the example of 10 recreational settlements in the Ivano-Frankivsk region: Bohorodchany, Verkhovyna, Vorokhta, Deliatyn, Dolyna, Kosmach, Mykulychyn, Nadvirna, Rohatyn, and Yaremche.
Based on the analysis of structural models of urban ecosystems by A. Kostrovitsky, V. Kucheryavyi (Kucheryavyi, 1999), M. Golubets (Golubets, 2001), and M. Gabrel (Gabrel, 2004), the author suggests his own model of resilience. It is based on the all-round assessment of the territory's ecological potential, encompassing natural resource characteristics, the ecological capacity of the environment, ecosystem stability, biotic productivity, socio-ecological aspects, and recreational value. The model anticipates the distinguishment between two interconnected subsystems – natural and anthropogenic. A comprehensive assessment of ecological potential is crucial for formulation of sustainable development strategies and provision of the long-term resilience of settlements.

Methodologically, the study uses the indicative approach based on indicators of environmental safety, social cohesion, infrastructure development, transport accessibility, and demographic stability. The assessment was conducted on the five-point scale using evaluated coefficients, which enabled the creation of the integrated resilience index and the comparative analysis of the territories.

Environmental factors determine the nature of the interaction between the natural environment and human activity, based on the influence of spatial organization and quality of life. The most significant factors are abiotic (climate, relief, hydrological network), biotic (forest cover, greening), and anthropogenic factors (development, recreational pressure, logging). The main risks include erosion, flooding, mudslides, and soil instability.

It has been established that the spatial organization of settlements is determined by the type of planning structure (linear, radial-ring, clustered, dispersed, combined, compact, and linear-node, density, and fragmentation of development.
Urban-ecological interaction is defined as the systemic exchange between the natural and anthropogenic components of the environment, encompassing ecological, architectural, social, infrastructural, and security aspects. They are classified by the definite outlines (direct, reverse, horizontal, vertical), functional characteristics (resource-based, protective, regulatory, destructive, recreational), and degree of ecological balance (stable, strained, critical).
There are four models of resilience proposed: spontaneous-adaptive, planned-adaptive, institutionally-managed, and innovation-self-sufficient.

The proposed models of urban-ecological interaction enable a systematic assessment of territorial potential and the formulation of effective spatial development strategies.
The resilience of urban ecosystems is formed as the result of complex interaction of environmental, social, infrastructural, and security factors.
Resilience ensurance requires integration of interdisciplinary approaches to urban planning, architecture, and design with the focus on regional context, natural conditions, and contemporary challenges, including military risks.

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Bucharest as an Urban Laboratory: Experimental Approaches to the Encounter of Old and New Architecture

Bucharest presents a complex architectural landscape shaped by discontinuity rather than gradual evolution. Political ruptures, war-related destruction, and large-scale transformations during the communist period have produced a fragmented urban fabric in which contrasting building typologies coexist without clear spatial mediation (Ghenciulescu, 2020). In this context, the city may be understood as an urban laboratory, where the relationship between historical structures and contemporary interventions is continuously tested.

This paper examines how architectural projects in Bucharest negotiate the relationship between pre-existing fabric and new construction. It argues that contemporary interventions operate as intentional spatial experiments that redefine the relationship between continuity and rupture through three primary strategies: adaptive reuse, façade reconstruction, and contrast-based insertion.

The study is based on a comparative analysis of selected case studies, interpreted through the lens of architectural conservation theory. While Ruskin (1849) emphasizes the preservation of authenticity and opposes restoration, Viollet-le-Duc (1854–1879) supports interpretative reconstruction. Contemporary approaches further stress contextual integration and continuity (Brolin, 1980; Semes, 2009), while post-socialist studies highlight the fragmented and layered condition of Bucharest (Ghenciulescu, 2020).

The Marmorosch Hotel exemplifies adaptive reuse, transforming a historic bank building into a contemporary hotel while preserving key spatial and decorative elements. Continuity is achieved through transformation, allowing the historical identity of the building to coexist with new functions.

In contrast, the Novotel Bucharest City Centre represents façade reconstruction. A reconstructed façade of the former National Theatre functions as a representational layer for a modern structure behind it. This approach raises questions regarding authenticity and the distinction between preservation and simulation, and remains debated within conservation discourse (Kyriazi, 2019).

The headquarters of the Union of Romanian Architects illustrates contrast-based insertion. The preserved historic façade serves as a structural shell, while a contemporary volume is inserted behind and above it. The juxtaposition of old and new emphasizes temporal layering and makes architectural transformation legible. These examples demonstrate that contemporary architecture in Bucharest operates through diverse and often conflicting strategies. Adaptive reuse prioritizes continuity, façade reconstruction produces a mediated image of the past, and contrast-based insertion foregrounds difference. Rather than resolving the city's fragmented condition, these approaches articulate it.

Bucharest may therefore be understood as a space of experiment, where architectural interventions actively negotiate the tensions between preservation and transformation. This perspective aligns with the New European Bauhaus framework, emphasizing context-sensitive and forward-looking approaches to the built environment.

EXPERIENCING SAFETY - A SYSTEM OF PLACES IN URBAN SPACE

Micro-Urban Interventions within the Warsaw University of Technology Campus Bis

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3D models developed in collaboration with and printed courtesy of the Faculty of Mechatronics:
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Academic year 2025/2026

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Cities are experienced not only through their physical structure, but also through the perceptions, emotions, and interactions they generate. Among these experiences, the sense of safety plays a fundamental role. It determines whether public spaces are actively used, shared, and appropriated by their users, or whether they remain merely circulation routes within the urban fabric.

This design studio investigates how architectural and spatial interventions can enhance the perception of safety in everyday urban environments. Rather than treating safety as a technical issue associated with control or surveillance, it is understood here as an outcome of spatial and social relationships—visibility, orientation, continuity of movement, and the presence of diverse users.

The project is situated within the Campus Bis of the Warsaw University of Technology, in the vicinity of Politechnika Square. This area is characterised by intense pedestrian flows and the coexistence of academic, residential, service, and transportation functions, creating a complex and dynamic urban context.

The studio proposes the design of a system of interconnected places organised along a pedestrian route linking key locations within the campus. These places are conceived not merely as functional nodes, but as points of spatial intensity that support presence, orientation, and interaction.

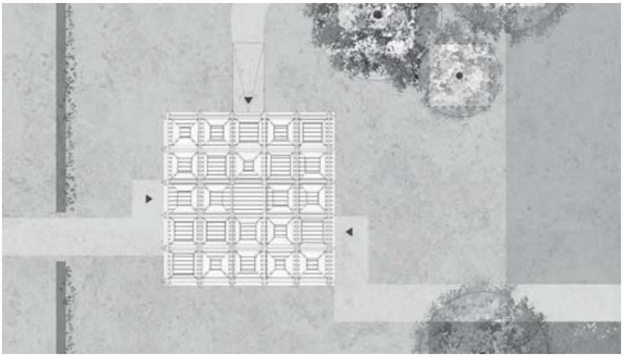
Within this system, students identify locations where the spatial configuration weakens perceptions of safety and other basic needs (Desmet, Fokkinga, 2020). In response, they design a micro-urban intervention—a small architectural element capable of transforming the spatial experience.

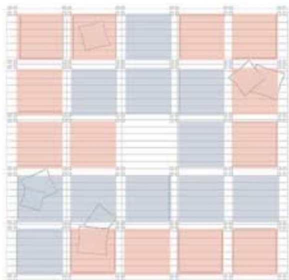
The intervention may take the form of a place for pause, an orientational element, urban furniture, or a small, shared space. The scale of the project is intentionally limited to highlight the transformative potential of small, precise actions. These actions are further differentiated into those supporting active and passive modes of use, both individually and in groups. The design process follows a research-through-design methodology. Students begin by observing, mapping, and analysing relationships among movement, visibility, accessibility, and social presence. This is followed by the development of a simplified physical model of the site, used as a tool for testing spatial scenarios.

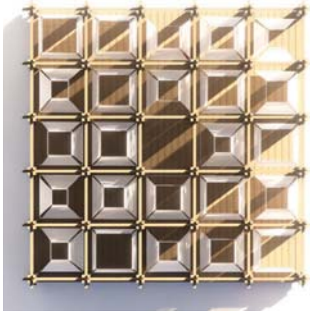
The outcome consists of both a spatial proposal and a critical reflection on the role of safety as an architectural quality of urban space. The project contributes to a broader discussion on the future of shared environments and extends the principles of the New European Bauhaus (NEB, 2021) by proposing safety as its fourth pillar. The project also commemorates the 200th anniversary of Warsaw University of Technology and aligns with UN Sustainable Development Goal 11 (UNSDC 11), which calls for cities that are inclusive, safe, resilient, and sustainable.

Through small architectural gestures and deliberate spatial design, the studio explores how design can support the everyday experience of safety and a sense of belonging in the contemporary city. The design process follows a research-through-design methodology. Students begin by observing, mapping, and analysing relationships among movement, visibility, accessibility, and social presence. This is followed by the development of a simplified physical model of the site, used as a tool for testing spatial scenarios.

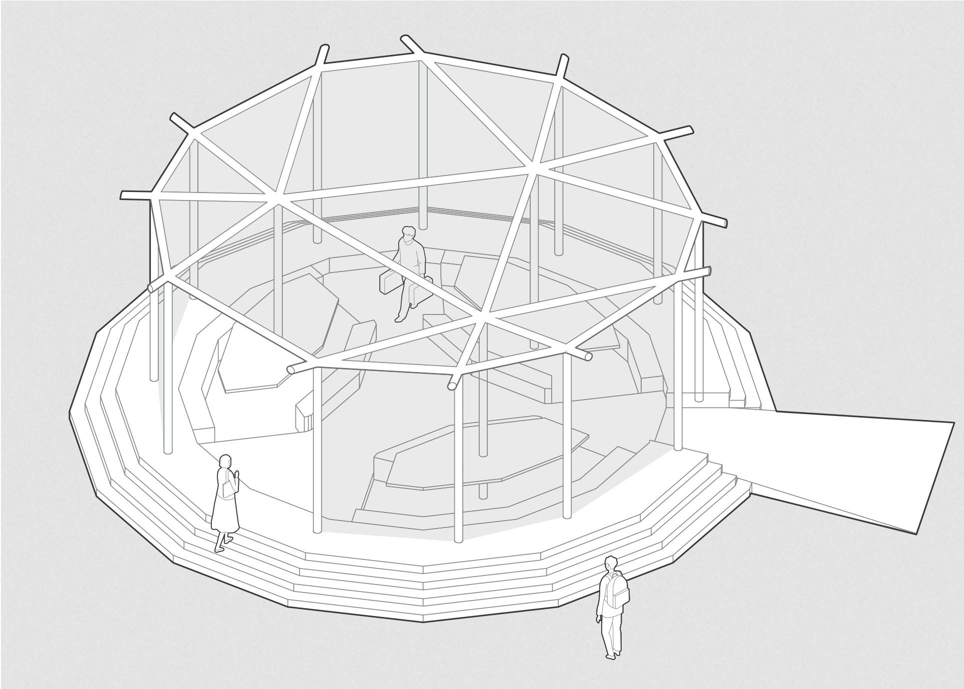
Place	Need I	Need II	Need III	Function
I	Comfort	Community	Purpose	Outdoor library
II	Comfort	Relatedness	Impact	Outdoor class
III	Comfort	Beauty	Autonomy	Healthy path
IV	Comfort	Beauty	Stimulation	Rest & Play
V	Comfort	Fitness	Purpose	Active bench
VI	Impact	Relatedness	Simulation	Relaxation place

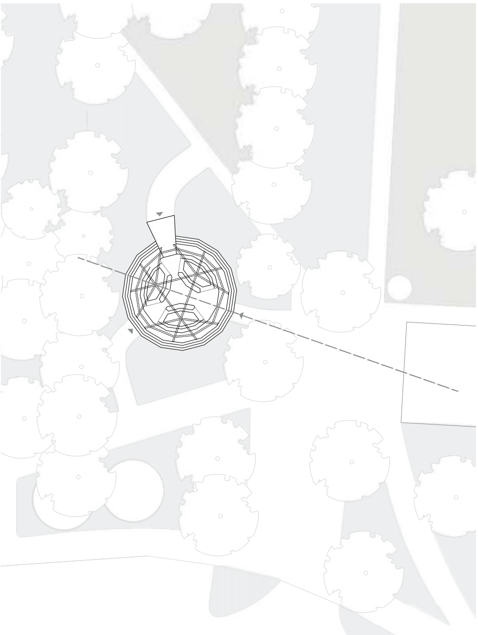


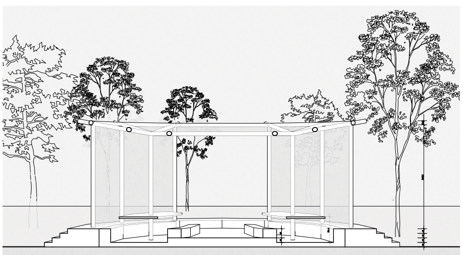




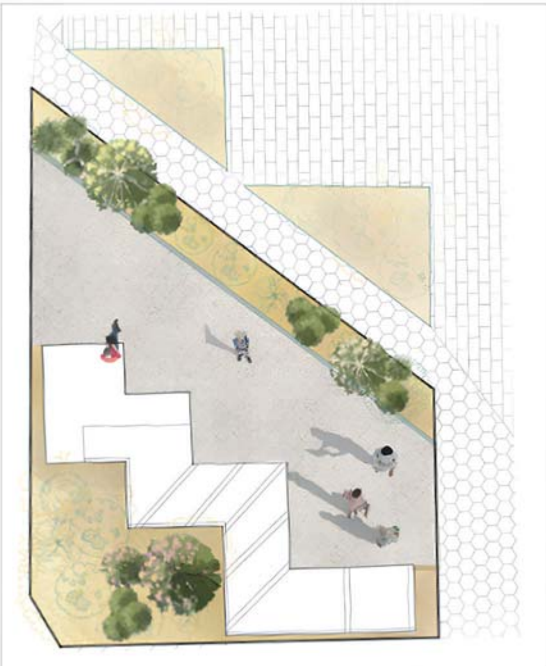
The Resting Grid - Alicja Igielska, Karolina Lechowicz

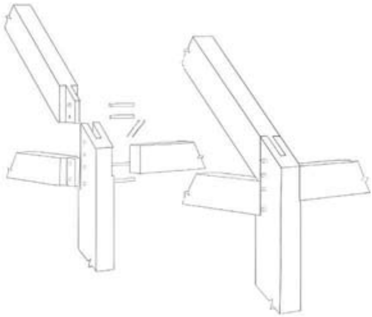


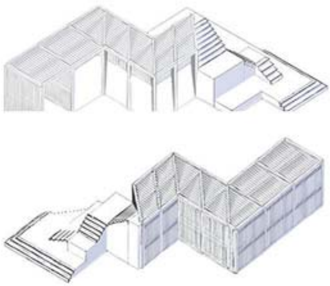




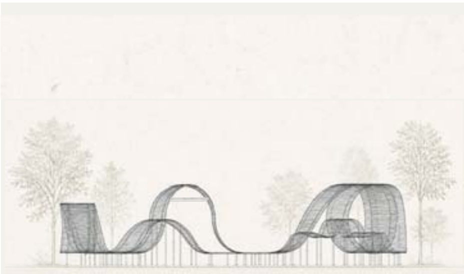
Outdoor class pavilion - Aleksandra Leszczyńska, Filip Zatorski

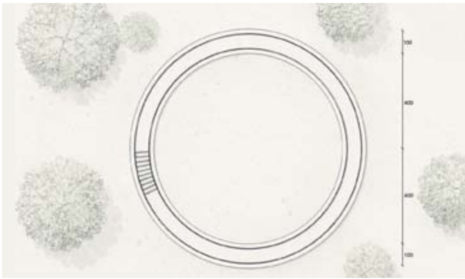


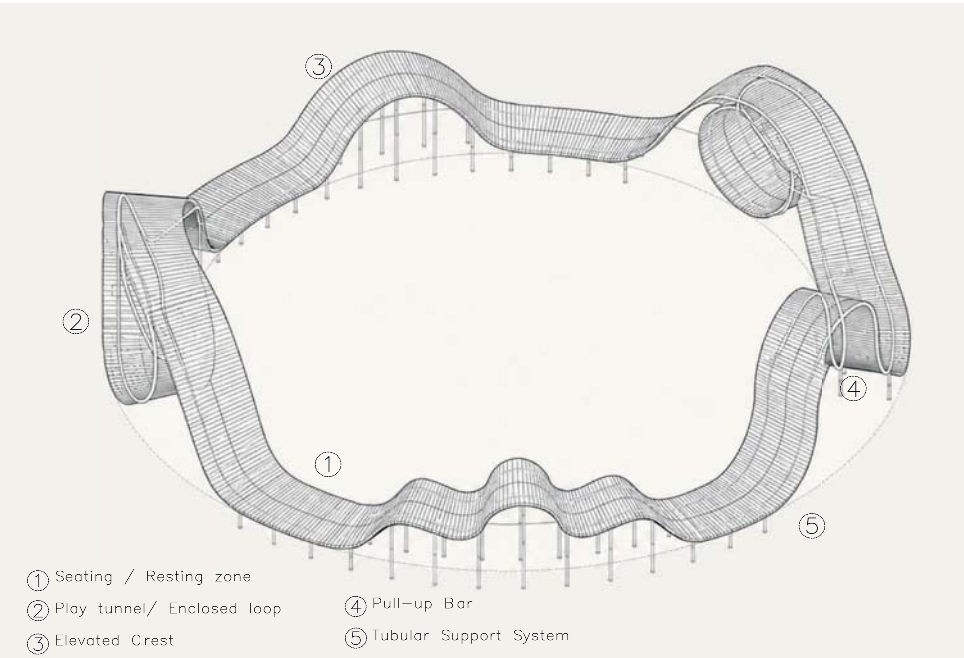




Outdoor Library - Oleksandra Holubenko, Jakub Radlicki







Active bench - Michalina Konefał, Sara Loperena Monzon

EXPERIMENT IN SPACE - THE SPACE OF EXPERIMENT

Music as inspiration

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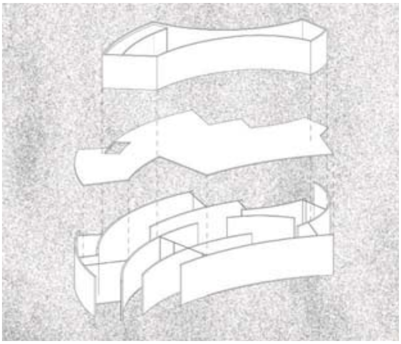
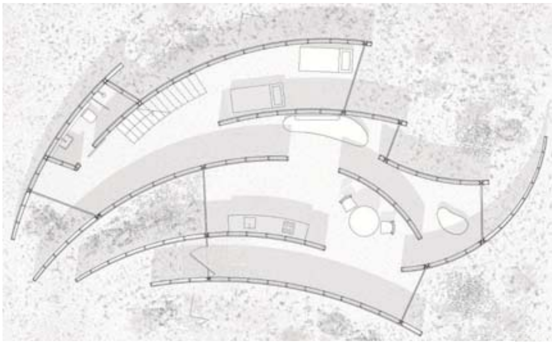
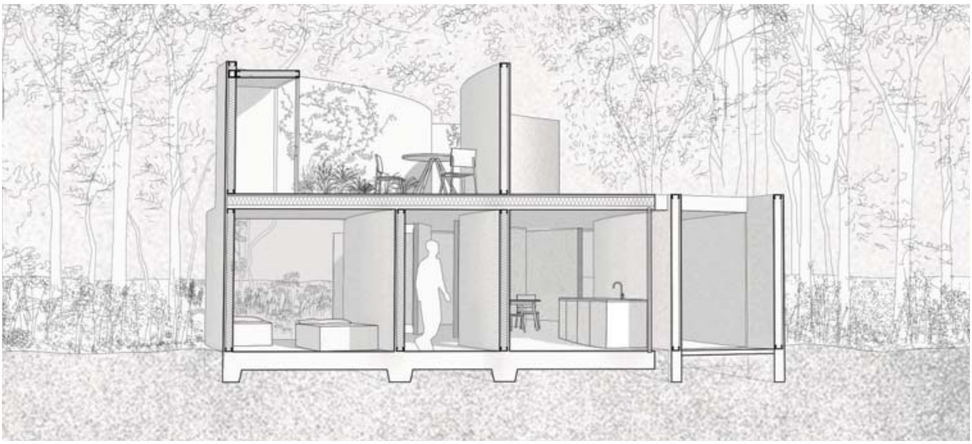
The Faculty of Architecture at WUT implements a curriculum based on the "learning by doing" method. Students begin by formulating a clear "idea"—a theme that unifies the entire design concept. Building a narrative and consciously shaping a form that interacts with the landscape are key. The theme of the project is a year-round retreat (home-studio) integrated into the natural surroundings. The house is intended to be a place of rest, contemplation, and deep engagement with the landscape. The starting point was an individually selected piece of music. Music, as a medium that strongly influences emotions, mood, and memories, served as the primary source of inspiration for the project. The goal was not to illustrate the piece literally, but to translate the accompanying emotions and structure into the language of architecture and physical space.

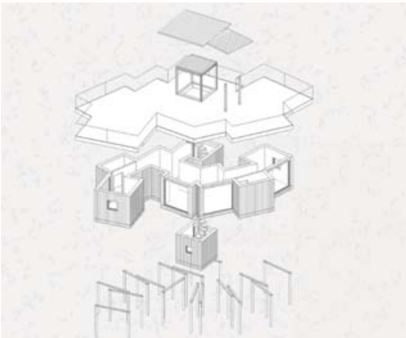
Narrative methods and spatial experimentation play a central role in the project. Narration allows for the conscious construction of a sequence of experiences—from the first impression, through tension and climax, to the resolution. Experimentation with space involves testing various relationships between form, material, light, and the user's movement. The space becomes a laboratory where the abstract qualities of music (rhythm, dynamics, silence, tension) are embodied in concrete architectural solutions. Every design decision is the result of an iterative experiment—drawings, models, mockups, and field tests.

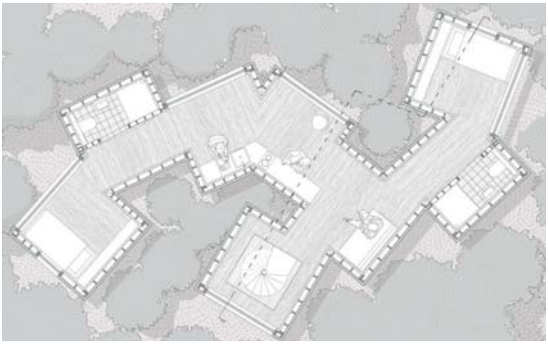
The work began with a detailed analysis of the site's context—the topography, rocks, vegetation, exposure, and acoustic qualities. Based on this, the background, transparency, density, and diversity of landscape elements were assessed. At the same time, the musical piece was interpreted in terms of rhythm, dynamics, tonality, tension, and climax. The form of the shelter results from a negotiation between the natural context and the individual interpretation of the music. The resulting structure opens the user to previously discovered ideas, while ensuring functionality for year-round use with distinct seasonal variations.

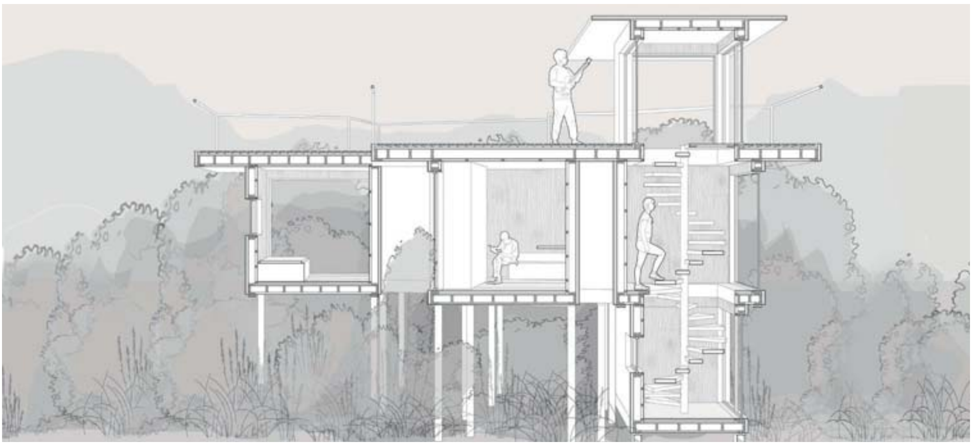












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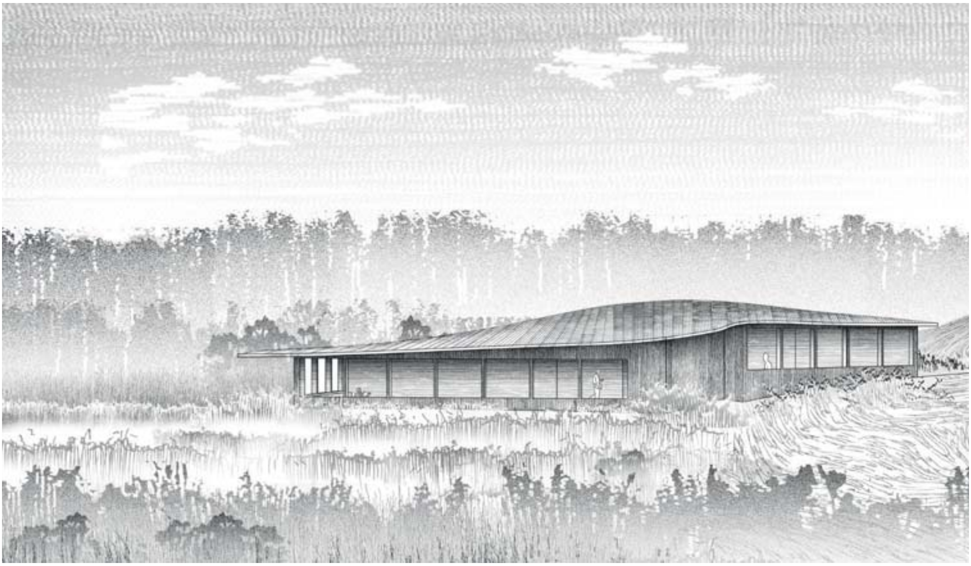

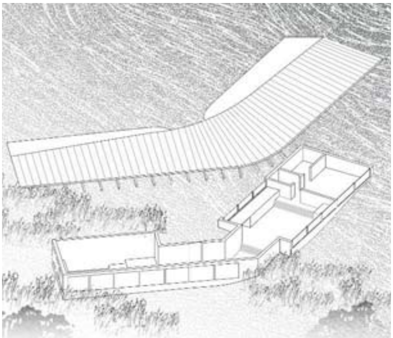

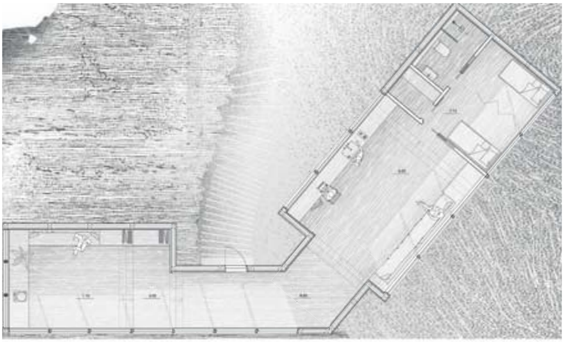

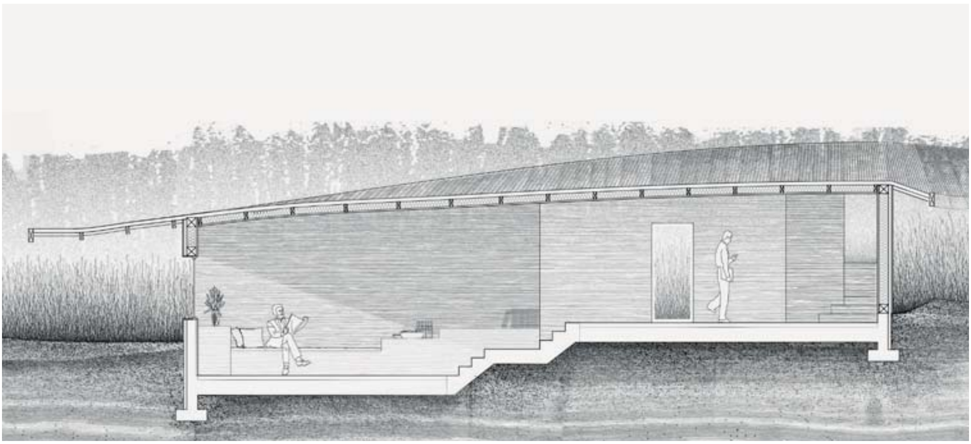

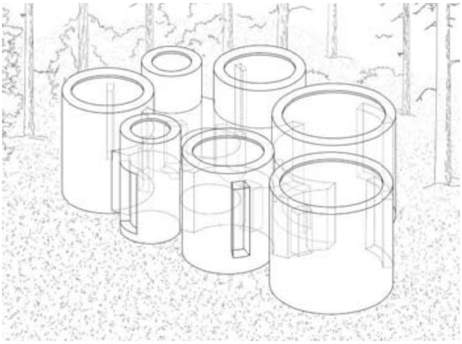

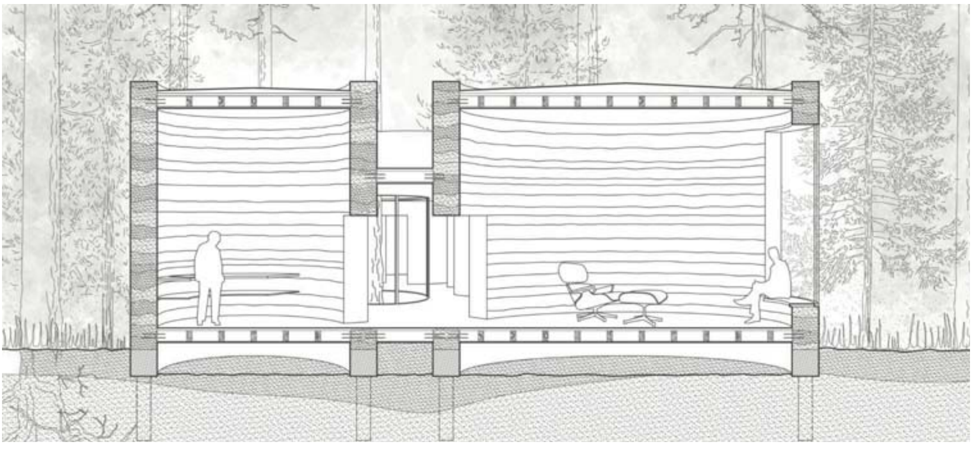

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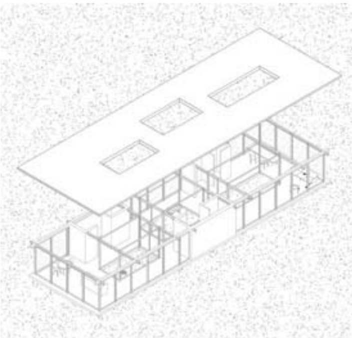

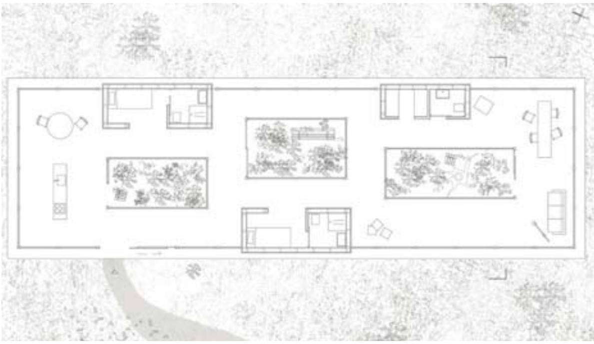

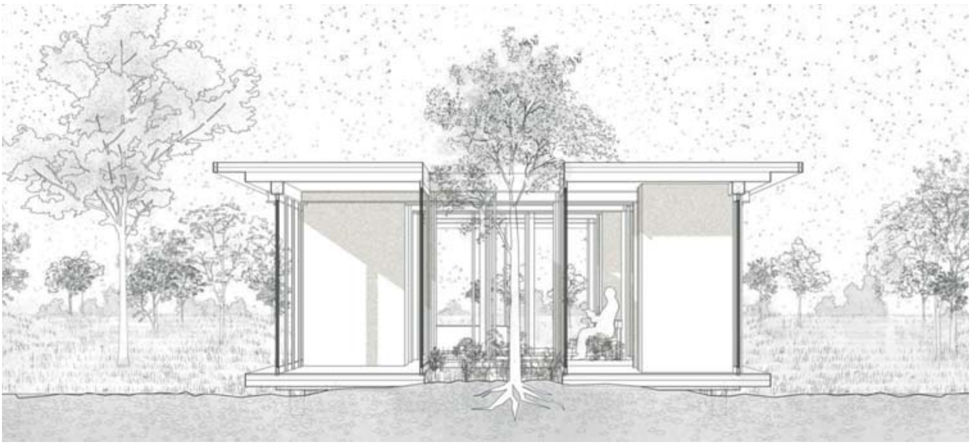

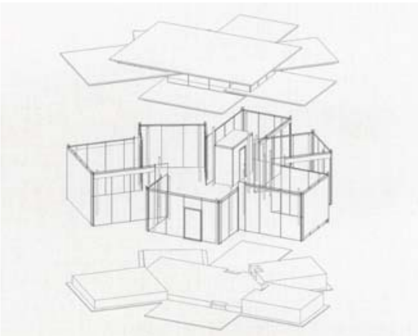

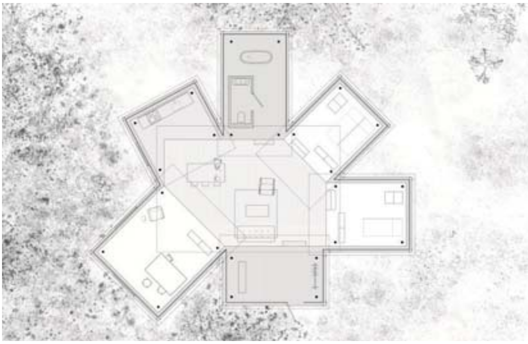

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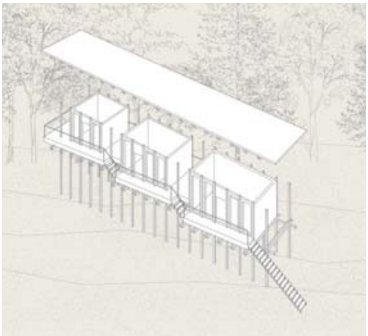

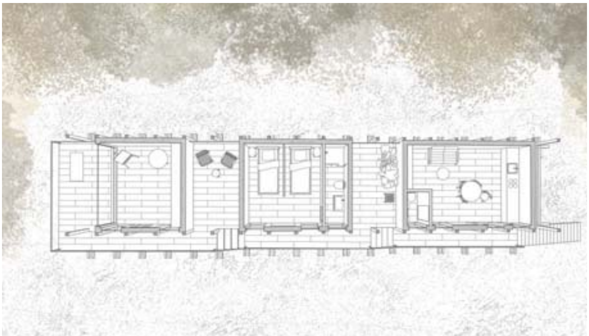

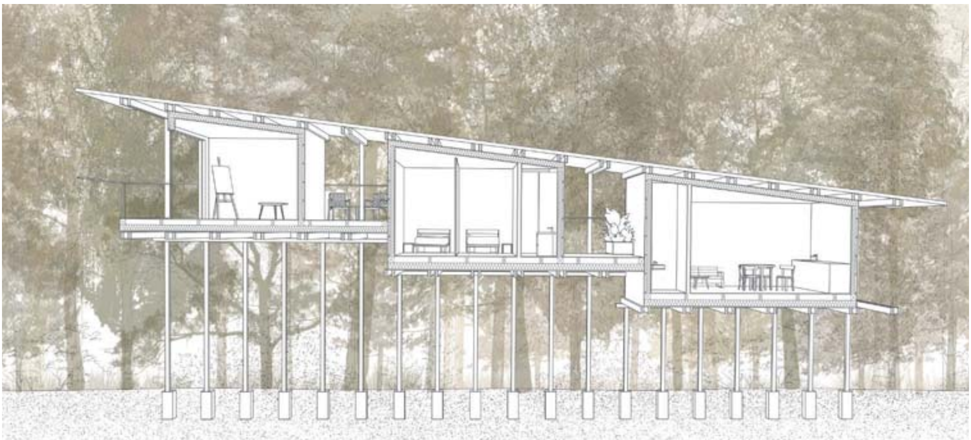

EDUCATIONAL LABORATORY – FALENTY HOUSING ESTATE

An Experiment in Urbanism within Architectural Education Based on Student Design Projects

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Experiment in urbanism, understood as the conscious testing of models of the city's spatial structure, has long constituted an important component of architectural education. In this perspective, the city is perceived as a complex system of spatial and social relationships in which urban structure co-creates the identity of a place (Domaradzki, 2016). Design is not limited to the creation of forms; rather, it involves the interpretation of the existing context, spatial hierarchies, and the meanings attributed to shared spaces by their users.

This paper presents educational experiences derived from a semester design studio conducted with fifth-semester students in the Studio of Applied Urban Design at the Faculty of Architecture, Warsaw University of Technology. The task focused on developing a housing-dominated development project located in Falenty - Raszyn Municipality near Warsaw. A key premise was designing within the existing spatial structure of the settlement, considering already completed development occupying approximately half of the project area. This requirement constituted a deliberate didactic and cognitive experiment, linking theoretical reflection with design practice (Chmielewski, 2001).

Students were required to analyse the existing urban layout, including the identification of built-up areas, the relationships of building scale, landscape connections and the natural system, the potential of public spaces, and the transportation network. The objective was to achieve continuity of the urban fabric shaping public spaces through the introduction of new development and a system of landscaped green areas while respecting the existing land-use structure (Domaradzki, 2016). The new urban structure was intended to continue and reinterpret the existing spatial order rather than negate it. Such an approach assumes that the legibility of the layout, spatial hierarchy, and the quality of public spaces constitute fundamental tools for shaping the living environment and its identity.

The projects were carried out in three- or four-person teams as complete design concepts for the entire 80-hectare area bounded by the streets Zygmunta Opackiego, Falencka, Willowa, and Hrabska Road. The design process encompassed both the configuration of building blocks and the system of public and semi-public spaces, including local streets, squares, pedestrian pathways, and green areas. The plans identified the main centres of smaller residential clusters with higher-density multi-family buildings, as well as areas with low-density single-family housing of varied forms. In addition, zones with key social facilities generating communal life were proposed, alongside the organization of a public transport system. The designed housing estates were treated as a coherent urban whole, in which the relationships between blocks, street layouts, and public spaces create a legible structure of the neighbourhood (Mazur & Trębacz, 2025). Particular emphasis was placed on the comprehensiveness of solutions within the designated area, concerning the relationship between built-up areas and public spaces, as well as on linking the designed structure with its surroundings, in particular the historic palace and park complex in Falenty and the Stawy Raszyńskie Nature Reserve.

Falenty Housing Estate functioned in this process as a design laboratory, allowing for the testing of various models of habitation and forms of social organization. Individual teams proposed different compositional strategies – from more compact and intensive structures, based on clearly defined blocks and squares, to more open layouts emphasizing landscape relations and continuity of shared spaces. The selected project area was treated as a city-like dynamic urban system, subject to continuous transformations and negotiation of meanings (Gzell, 2024).

An important extension of the educational experiment was the involvement of actual users of the space in the design process. The ideas developed in the housing estate concepts were confronted with the views and expectations of Falenty residents through an open public debate.

EDUCATIONAL LABORATORY – FALENTY HOUSING ESTATE

Urban design housing estate - Arcadian - Falenty Estate in Raszyn Municipality near Warsaw

the idyllic

the axis of the historic palace

the green backbone

the tripartite division of the layout

co-creation with existing buildings and context

THE OBJECTIVE IS: spatial structure of the city as a cognitive, didactic & design experiment

public spaces

transportation

greenery

AN URBAN-ARCHITECTURAL EXPERIMENT IN RADOM

An Interdisciplinary Approach to Downtown Space and Architectural Education
Transformation of a central district

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Wierzbicka, A. M., Trębacz, P., Duda, M., & Heidgerken, T. (2025). Public Space as the Fabric Connected to an Evolving City Structure: Case Study of Powiśle Północne, Warsaw, Poland. Housing Environment, Article 53. https://doi.org/10.2478/he-2025-0029

The transformation of transport nodes into intermodal hubs essential for the functioning of metropolitan areas constitutes one of the key challenges in the transformation of inner-city areas. The area surrounding the main railway station in Radom covers approximately 50 hectares and for decades constituted a functional boundary of the city centre. Creating both a physical and psychological barrier in the perception of residents, they separated the city centre from peripheral districts. As a consequence of the disrupted continuity of the urban fabric and limited transport connections, the development potential of attractive centrally located land remained largely underutilised.

In response to these challenges, an interdisciplinary educational experiment integrating both urban and architectural scales was organised within the architectural-urban design studio at the Warsaw University of Technology. The project enabled students to test their design proposals within a real urban context. The exhibition presenting the results of the project at the Mazovian Centre for Contemporary Art Elektrownia was made possible through cooperation between the Warsaw University of Technology, the City of Radom and the Municipal Urban Planning Office in Radom, which coordinated activities supporting the actual needs of local residents.

The aim of this paper is to present an urban–architectural educational experiment as a teaching tool enabling students to develop critical thinking, design complex urban structures and work within a large design team.

The research problem concerns the possibility of linking urban-architectural education with design experiments carried out in a real urban context, allowing alternative scenarios for urban development to be tested and design proposals to be verified through interaction with actual users of urban space.

The research thesis is formulated as the following question: can an interdisciplinary design experiment conducted within a large design team enhance the understanding of spatial relationships in the city and support the development of innovative, socially and spatially responsible urban solutions?

Experimentation in urban design includes both the process of creating urban structure and subsequently testing its functioning and its impact on users (Domaradzki, 2013). Observations of Polish cities containing railway areas show that railway infrastructure has historically reinforced spatial divisions and separated different urban zones, while at the same time creating opportunities for regeneration once appropriately transformed (Duda & Trębacz, 2021). The tunnelling of railway lines in city centres enables investment in previously problematic areas, increases the cohesion of the urban fabric and supports social activities, contributing to the creation of high-quality public spaces.

In architectural education, experimentation allows students to generate alternative concepts, analyse and redefine established design paradigms. The design process in Radom was carried out in several stages. Initially, four groups of three students developed analyses and preliminary concepts. The groups were then merged, and two final concepts were selected, each developed by six students.

The analyses addressed issues related to the identification of the broader spatial context of the site. Context plays a key role in evaluating design decisions (Trębacz, Jóźwik & Duda, 2025). Spatial structures of the city are particularly important.

Among them, the most significant for the creation of architectural form and urban composition is the system of public spaces (Wierzbicka, Trębacz, Duda & Heidgerken, 2025). Other important systems requiring analysis include natural and infrastructural networks.

AN URBAN-ARCHITECTURAL EXPERIMENT IN RADOM

An Interdisciplinary Approach to Downtown Space and Architectural Education
Transformation of a central district

Radom on Track

Central Hub

THE OBJECTIVE IS: interdisciplinary approach to central district and architectural education

The first concept – "Central Hub" – proposes a comprehensive integration of transport infrastructure with revitalised public spaces around the Radom Main Railway Station. The key idea is to connect the two sides of the railway tracks through a new railway station building and two pedestrian footbridges providing clear, safe and continuous pedestrian connections above the rail tracks.

On both sides of the railway line, two semicircular green corridors were designed, performing recreational, ecological and spatial-ordering functions while softening the barrier effect of railway infrastructure. External vehicular traffic is redirected through two tunnels running parallel to the railway tracks, which significantly reduces car traffic at ground level and allows the creation of pedestrian- and cyclist-friendly public spaces.

The new railway station is complemented by a group of new public and service buildings, including a bus station integrating regional and urban transport, a railway museum emphasising the identity of the area, a heritage conservation centre, a hotel and a church. The second concept – "Radom on Track" – addresses the physical and mental division of the city created by the railway line. The authors emphasise the drastic separation into two distinct zones: the city centre and the district known as "Zatorze". They observe that this infrastructural "scar" generates noise, hinders mobility and wastes the investment potential of highly attractive urban land.

The project "Radom on Track" responds to the need to restore the continuity of the urban fabric and to reintroduce a human scale to the area. The railway tracks are placed in a tunnel along a 1.5-kilometre section, allowing nearly 30 hectares of central urban land to be reclaimed. The roof of the tunnel becomes a new urban platform.

The compositional focal point of the design is a Multifunctional Green Bridge, combining pedestrian, recreational and cycling routes with small leisure "pockets" and viewing points. The surroundings of the bridge are complemented by buildings with commercial, cultural and transport functions, forming a coherent continuation of the existing urban fabric.
As a result, residents from both sides of the railway line gain a shared meeting space – a so-called "third place" – supporting the development of local identity and neighbourhood integration. The transformation of former industrial land into high-quality public space also brings measurable economic benefits and contributes to the improvement and further development of surrounding areas. "Radom on Track" represents a vision of a compact, ecological and pedestrian-friendly city.

Both concepts allowed students to test relationships between form, function and public space within a real urban context.

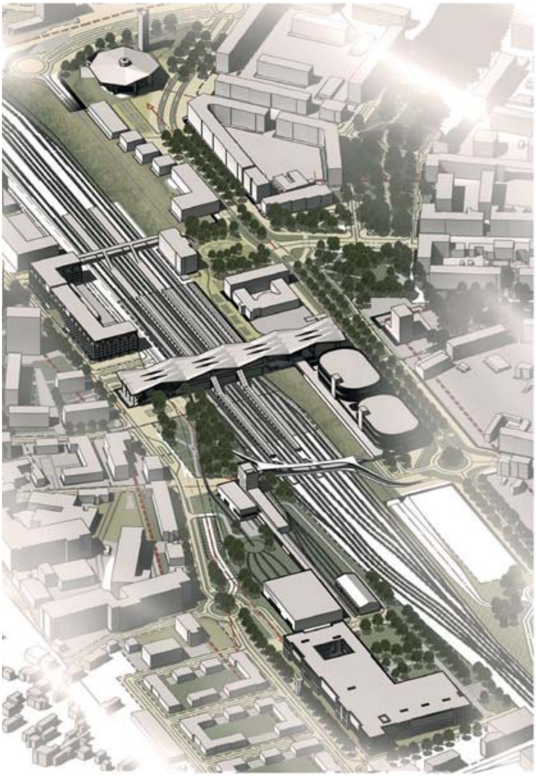
The experimental teaching process based on solving real spatial problems demonstrated the potential for:
• developing critical thinking and reflection on complex spatial relationships;
• testing alternative design concepts within a real urban context;
• understanding the importance of integrating architecture, urban design and public space;
• preparing innovative and socially responsible urban solutions.

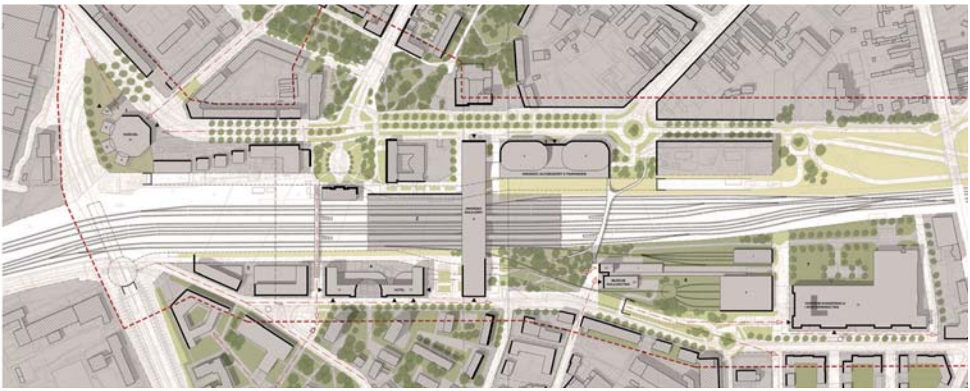
The project demonstrates that an experimental approach in architectural and urban education, supported by cooperation with the city, can become an effective teaching tool integrating analysis, design and the evaluation of real spatial consequences.

The experiment involved three key educational aspects. Firstly, cooperation with real users – city residents – including consultations with municipal authorities and the presentation of projects at a public exhibition, allowing students to confront their ideas with social expectations. Secondly, interdisciplinarity, achieved through the integration of architectural and urban design projects, enabling students to develop a coherent vision of downtown space while creating urban regulations for specific architectural interventions. Thirdly, work within a large design team – six students developing a shared concept while simultaneously designing individual architectural projects and respecting the urban regulations of the entire area. This required continuous negotiation between individual creativity and collective responsibility for the overall design.
Such a structure of work supported the development of collaboration skills, critical analysis and reflection on design consequences across multiple spatial scales.

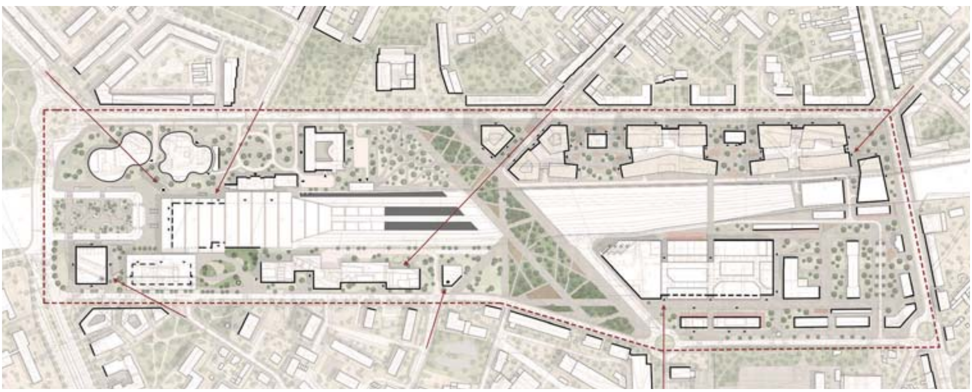
AN URBAN-ARCHITECTURAL EXPERIMENT IN RADOM

An Interdisciplinary Approach to Downtown Space and Architectural Education
Transformation of a central district | Concept CENTRAL HUB





An Interdisciplinary Approach to Downtown Space and Architectural Education
Transformation of a central district | Concept RADOM ON TRACK



THE NEW HEART OF RADOM

The surroundings of railway station as Transit Oriented Development (TOD). Educational Laboratory – an Experiment in Urbanism within Architectural Education Based on Student Design Projects

TUTORS
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Renata Jóźwik, PhD Eng. Arch.

Course: Architectural and Urban Design
Project – Transformation of a central district

Long – cycle Master's degree programme
semester 11/11 WUT
Academic year 2025/2026

STUDENTS | Team 1 GREEN RIBBON
Daniel Boguszewski
Anna Kaczkowska
Zuzanna Koc
Award-winning student project
of the SARP Radom competition

STUDENTS | Team 2 STAPLES
Natalia Dominiak
Emilia Oniszczuk
Maria Wolska

STUDENTS | Team 3 GREEN PROMENADE
Oliwia Błoniarczyk
Natalia Nawrot
Weronika Sowińska

STUDENTS | Team 4 FOUR SQUARES
Antoni Banach
Matylda Boryna
Katarzyna Durmaj

STUDENTS | Team 5 GREEN BELT
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Transit-Oriented Development (TOD) projects concentrate urban development around accessible and efficient public transportation hubs. Compact, mixed-use districts are created within walking distance, combining residential, commercial, office, and recreational functions. Key principles include reducing car dependency, prioritizing pedestrian and bicycle traffic, ensuring high-quality public spaces, and integrating various transport modes. TOD supports sustainable urban development, improves quality of life, and enables more efficient land use (Cervero & Kockelman, 1997; Ibraeva et al., 2020).

Dense, pedestrian-friendly development around transport hubs influences the entire city by creating interconnected urban clusters. Instead of isolated projects, TOD promotes polycentric development, where different districts form local centers of activity. This improves access to services and workplaces, shortens commuting times, and lowers transportation costs. TOD projects also stimulate the revitalization of degraded areas, attract investment, and increase the value of urban space. In the long term, they limit urban sprawl and encourage more rational use of infrastructure and spatial resources (Vale, 2015).

The area surrounding Radom's main train station is an important element of the city's structure, combining transportation, commercial, and public functions. However, the district remains neglected, difficult to access, and in need of modernization. Connecting the two sides of the city currently divided by railway tracks could become a catalyst for development and help retain young residents who increasingly migrate to Warsaw.

After the political transformation, the collapse of parts of the industrial sector negatively affected Radom's labor market. In 2020, unemployment reached approximately 12.6%, compared to only 1.7% in Warsaw. Recently, new jobs have emerged due to the growth of the service and logistics sectors and infrastructure investments, especially in transportation. Faster rail connections, new roads, and the airport opened in 2023 have strengthened links with the Warsaw metropolitan area. Nevertheless, economic conditions and geographic proximity to Warsaw continue to drive outmigration. Radom's population declined from 240,000 in 1990 to around 205,000 today, while population aging and low birth rates intensify this process (Główny Urząd Statystyczny, 2021).

The New Heart of Radom

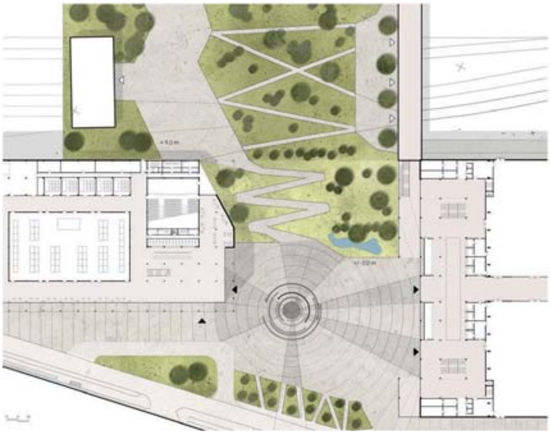
Following an analysis of local conditions and urban problems, five student teams developed design proposals for the area around Radom's main train station. Their aim was to improve urban quality and create conditions encouraging young people to remain in the city while attracting investment and new residents. Using TOD principles, the projects envisioned a new city center with improved accessibility and stronger integration of urban functions. The students proposed several distinct urban structures attractive to residents, users, and investors. Introducing diverse residential and community-oriented functions would also stimulate service-sector growth, one of the key drivers of Radom's economic transformation.

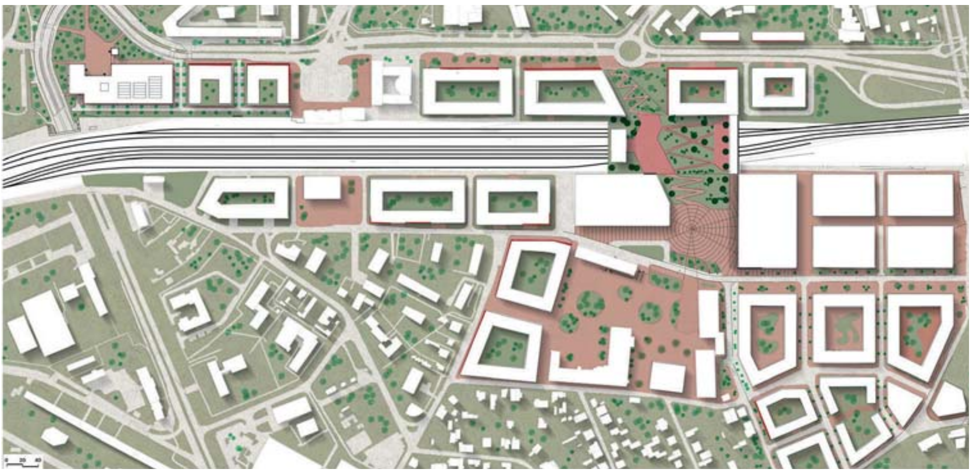
Each proposal improved degraded public spaces, introduced a clearer urban structure, and strengthened social interaction through attractive public areas. By creating a more inclusive and accessible environment, the projects aimed to counter depopulation and aging while increasing Radom's competitiveness within the Warsaw metropolitan area.

An important aspect of the exercise was its experimental character. Five teams developed different concepts for the same site under identical spatial and programmatic conditions, enabling comparison of various TOD-based design strategies. The projects explored alternative relationships between buildings and public spaces, as well as different levels of development intensity. This allowed students to analyze multiple urban transformation scenarios and their impact on spatial organization, social integration, and city structure. The exercise demonstrated that urban design in intensively revitalized city centers requires iterative processes and critical evaluation rather than singular, definitive solutions.

THE NEW HEART OF RADOM

The surroundings of railway station as Transit Oriented Development (TOD)
Team 1 GREEN RIBBON | Award-winning student project of the SARP Radom competition

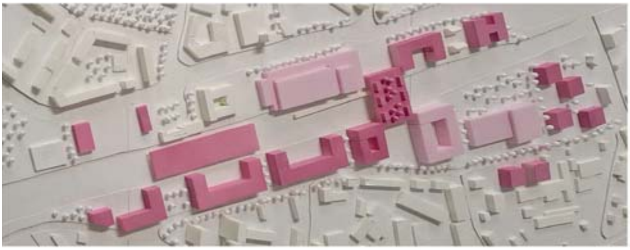


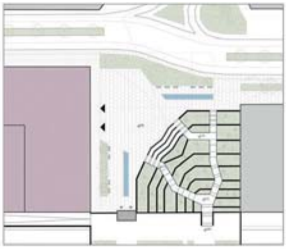


The surroundings of railway station as Transit Oriented Development (TOD)
Team 2 STAPLES | Team 3 GREEN PROMENADE

THE NEW HEART OF RADOM

The surroundings of railway station as Transit Oriented Development (TOD)
Team 4 FOUR SQUARES | Team 5 GREEN BELT





The surroundings of railway station as Transit Oriented Development (TOD)
Exhibition

SPECIAL ZONE PROJECT

A multi-purpose café with an experimental programme for special school leavers in Piaseczno

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English Bachelor's Degree Programme
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STUDENTS | Concept PAINTERS' GAMBIT
Barbara Sobiepanek
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Maja Tucharz

Warsaw University of Technology
Faculty of Architecture, Poland

The 'Special Zone' project – a multi-purpose café with an experimental programme for special school leavers. At the venue, which is soon to be built on Chyliczkowska Street in Piaseczno, pupils from the Special School Complex in Pęchery will find their first jobs as staff serving in the communal areas of the club and café. The aim of this initiative is to help young people leaving school achieve self-reliance and financial independence. The venue will become not only an attractive dining destination but also a place for social integration during the chess and draughts meetings and tournaments held there.

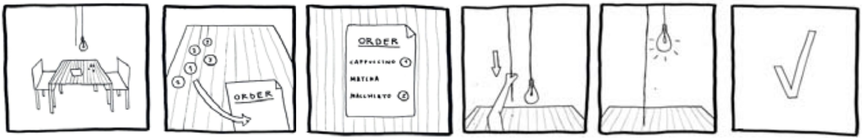
The Joinery

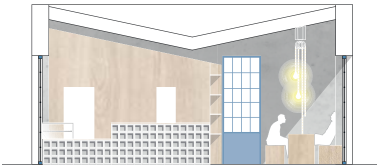
The project originates from the need of creating a calm yet interactive place for gathering. This flexible space, characterized by raw and industrial materials, reflects its double function, serving both as a café and as an arts and craft workshop, still permitting other uses. An interactive pulley-system way of ordering facilitates the waiters' work, while making the clients' experience memorable.The use of recycled plywood and concrete enhances the workshop's spirit and the blue window details make the space feel more clear and joyful. As a whole, the project wants to be a simple yet effective solution for bringing Piaseczno's citizens together.

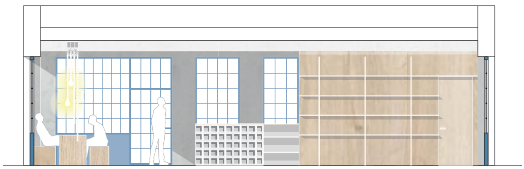
Painters' Gambit

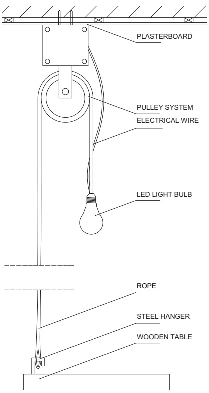
This project presents a chess-centered café designed as an inclusive and welcoming space, with a focus on accessibility for all users. The interior combines raw, minimalist concrete surfaces with the warmth of natural wooden furniture, creating a balanced contrast between industrial and cozy atmospheres. Lush greenery and subtle metal details soften the space and add a sense of calm and connection to nature. Chess is the heart of the café not only as a game but as a shared experience. Guests are invited to engage creatively by painting their own small chess boards, making each visit personal and interactive. The design encourages both quiet concentration and social interaction, fostering a community built around strategy, creativity, and inclusivity.

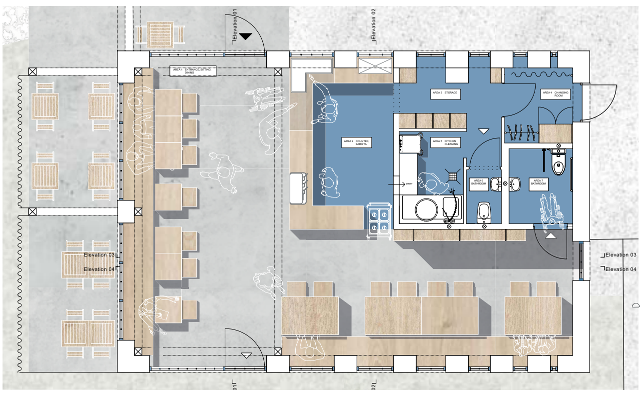
A multi-purpose café with an experimental programme for special school leavers in Piaseczno
Concept **THE JOINERY**





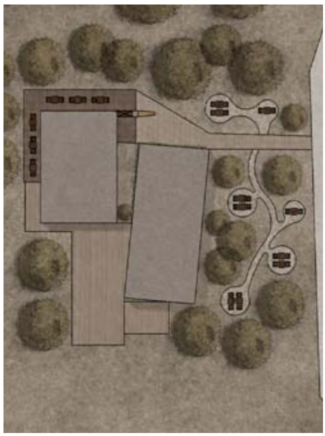






SPECIAL ZONE PROJECT

**A multi-purpose café with an experimental programme for special school leavers in Piaseczno
Concept PAINTERS' GAMBIT**



A multi-purpose café with an experimental programme for special school leavers in Piaseczno
Concept PAINTERS' GAMBIT

ARCHITECTURAL WORKSHOP IN SZTYNORT

An Architectural and Urban Planning Concept for a New Hotel and Recreation Complex in Sztynort

ARCHITECTURAL DESIGN TUTORS:
Anna Maria Wierzbicka, DSc PhD Eng. Arch., Assoc. Prof.
Jakub Pieńkowski, MSc Eng. Arch.
Filip Strzelecki, MSc Eng. Arch.

URBAN DESIGN TUTORS:
Renata Jóźwik, PhD Eng. Arch.
Pawel Trębacz, PhD Eng. Arch.
Magdalena Duda, MSc Eng. Arch.

CONSULTATIONS
Kinga Zinowiec-Cieplik, PhD Eng. in Landscape Architecture
Mariusz Wrona, MSc Eng. Arch. – Construction
Alicja Szmelter, PhD Eng. Arch. – History of Architecture

MASTER'S STUDENTS
Zofia Całka, Eng. Arch.
Aleksandra Sułecka, Eng. Arch.
Natalia Szewczyk, Eng. Arch.

Warsaw University of Technology
Faculty of Architecture, Poland
Specialisation – A3 Architecture of Technology and Structures – Context and Meaning
Master's degree (second-cycle studies)
sem. 3/4 WUT

THE ORGANIZERS OF THE WORKSHOP
King Cross Developments Sp. z o.o.
Nowy Sztynort Sp. z o.o.
Warsaw University of Technology, Faculty of Architecture

MASTER'S STUDENTS
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Łukasz Danilczuk
Kseniia Fedorchuk, Eng. Arch.
Zofia Gancarczyk
Aleksander Janowski
Szymon Kassowski, Eng. Arch.
Klaudia Kochanowska
Julia Koperska, Eng. Arch.
Maja Kordalska, Eng. Arch.
Barbara Krawiecka, Eng. Arch.
Szymon Kucharczyk
Ewa Maniak
Piotr Marczak
Maciej Oberzig
Maria Olko
Mateusz Robak, Eng. Arch.
Karolina Rorat
Katarzyna Sikora
Magdalena Smolińska, Eng. Arch.
Aleksandra Sułecka, Eng. Arch.
Gabriela Szewczak, Eng. Arch.
Natalia Szewczyk, Eng. Arch.
Julia Świstuniuk
Jakub Wiązowski
Maria Wolska
Dominik Zagajewski
Jan Zwierzchowski, Eng. Arch.
Zofia Zwijacz

The urban and architectural design workshop was conducted during the winter semester of the 2025/26 academic year. The primary objective of the workshop was to develop an urban and architectural concept for a new hotel and recreational complex in Sztynort, together with a comprehensive masterplan for the waterfront and marina area. Sztynort is a small settlement located in the northern part of the Masurian Lake District, home to one of the largest marinas on the Great Masurian Lakes route. At the same time, the site possesses a rich historical legacy reflected in numerous heritage structures, including the palace complex — regarded as one of the architectural gems of the Warmia and Masuria region - the historic park, and the remnants of the former Lehndorff family estate. The students were therefore confronted with a highly complex design challenge: how to develop such an exceptional place without compromising its identity.

Ten student teams participated in the workshop, preparing design proposals based on site visits, meetings with landowners, materials provided by the investor, and consultations with academic supervisors.

The accompanying objectives of the workshop included:

– selecting the most compelling urban, architectural, and functional concept,
– creating a realistic foundation for the future implementation of the investment,
– supporting students through competition awards and opportunities for professional development and visibility.

The projects were developed across multiple scales, ranging from urban and landscape analyses to architectural solutions for a hotel accommodating 150 rooms together with associated functions. The workshop became a unique opportunity to confront students' design ideas with real investor expectations and professional practice.

The workshop demonstrated both the students' strong engagement and the diversity of design approaches, from landscape-oriented visions to precise architectural solutions. This experience of collaborative design practice and dialogue with place became an important component of the educational methodology of the A3 specialization and of the Faculty of Architecture at the Warsaw University of Technology as a space for inclusive dialogue.

The workshop concluded with an architectural competition. The evaluation of the projects and the selection of the best design proposal were carried out by the competition jury composed of:

- Chair – Przemo Łukasik (Medusa Group)

- Rapporteur – Małgorzata Nowak-Pieńkowska (Faculty of Architecture, WUT)

- Maciej Miłobędzki (JEMS Architekci, Vice-Dean for Development, Faculty of Architecture, WUT)

- Krystyna Solarek (Professor at the Faculty of Architecture, WUT)

- Karolina Tulkowska-Słyk (Vice-Dean for Academic Affairs, Professor at the Faculty of Architecture, WUT)

- Piotr Bujnowski (Bujnowski Architekci)

- Justyna Dziedziejko (Top_Scape)

- Marcin Maraszek (Archigrest)

- Marek Makowski (King Cross Development)

- Mikołaj Nowakowski (Chiliart)

- Marcin Trybus (King Cross Development)

The workshop was organized through a collaboration between the Faculty of Architecture at the Warsaw University of Technology and the company King Cross Development.

ARCHITECTURAL WORKSHOP IN SZTYNORT

An Architectural and Urban Planning Concept for a New Hotel and Recreation Complex in Sztynort

ARCHITECTURAL DESIGN TUTORS:
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CONSULTATIONS
Kinga Zinowiec-Cieplik, PhD Eng. in Landscape Architecture
Mariusz Wrona, MSc Eng. Arch. - Construction
Alicja Szmelter, PhD Eng. Arch. - History of Architecture

MASTER'S STUDENTS
Zofia Całka, Eng. Arch.
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Master's degree (second-cycle studies)
sem. 3/4 WUT

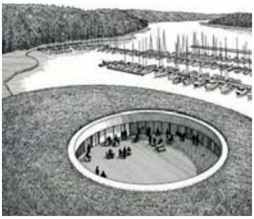
THE ORGANIZERS OF THE WORKSHOP
King Cross Developments Sp. z o.o.
Nowy Sztynort Sp. z o.o.
Warsaw University of Technology, Faculty of Architecture

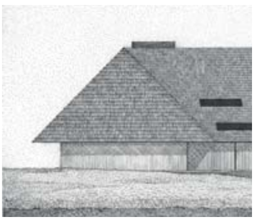
1st place

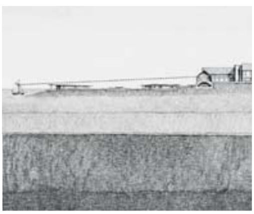
NORT

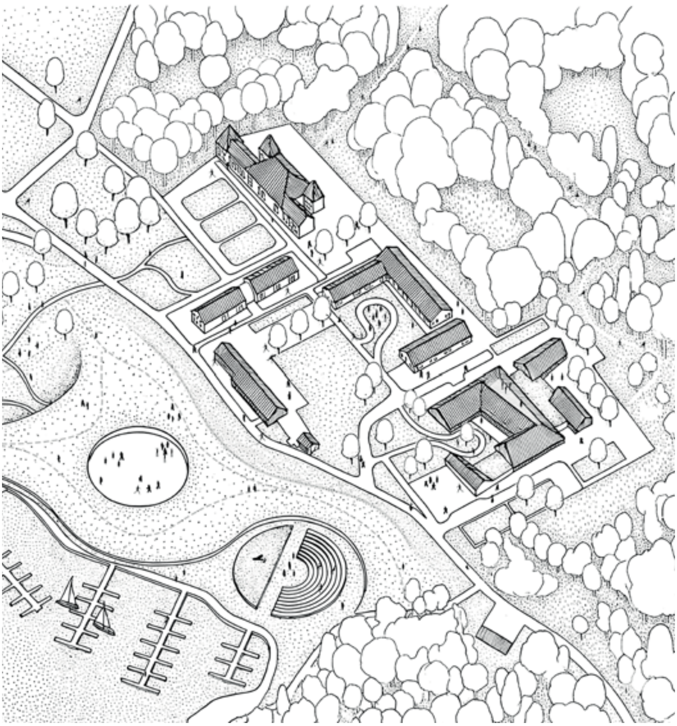
The aim of the project is to create a building complex that develops and organizes the existing spatial structure of Sztynort while respecting its historical and landscape character. The new architecture is conceived as a complementary addition to the existing layout, subordinated to the site's cultural and spatial values. A key design strategy is the partial embedding of the buildings into the terrain, reducing their visual impact and preserving view corridors towards the historic complex, maintained as the compositional dominant. The functional layout is divided into two zones: a hotel area referencing traditional farm architecture and a publicly accessible waterfront marina integrated with the surrounding landscape.

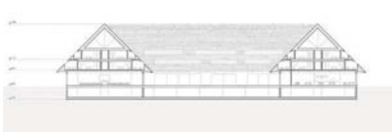
The north-eastern part of the site is occupied by the hotel complex, whose composition refers to existing farmstead structures. A traditional arrangement of buildings organized around courtyards and connected by visual axes has been preserved, with scale and rhythm adapted to existing spatial relationships. The waterfront zone has been shaped in a distinct manner, subordinated to the landscape and open views. Marina, gastronomic, and recreational facilities are partially embedded in the terrain and covered with green roofs. Their soft geometric forms resemble natural landforms, minimizing the visual presence of architecture and maintaining openness towards the lake. Interior spaces are organized around a central courtyard serving as a semi-public meeting area. The hotel is designed as a composition of interconnected volumes divided into functional wings, allowing maximum use of the site's scenic qualities while maintaining an intimate scale. The investment is planned in stages: the southern part, including the primary accommodation base and key public functions, will be implemented first, followed by the northern section as an extension. The complex includes 148 rooms of 24 m², each with a private loggia. The ground floor accommodates a wellness and spa area with a swimming pool and saunas opened towards the surrounding forest, complemented by a lakeside restaurant and conference facilities supporting year-round use.

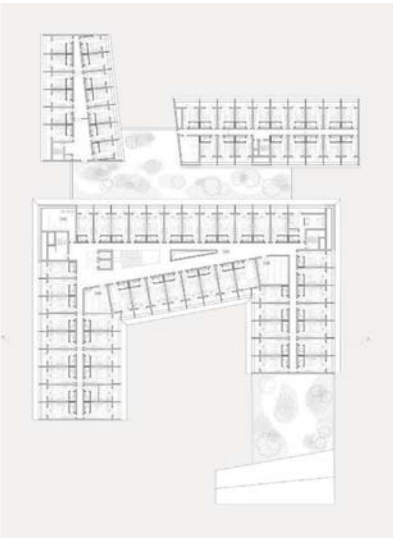












ARCHITECTURAL WORKSHOP IN SZTYNORT

An Architectural and Urban Planning Concept for a New Hotel and Recreation Complex in Sztynort

ARCHITECTURAL DESIGN TUTORS:
Anna Maria Wierzbicka, DSc PhD Eng. Arch., Assoc. Prof.
Jakub Pieńkowski, MSc Eng. Arch.
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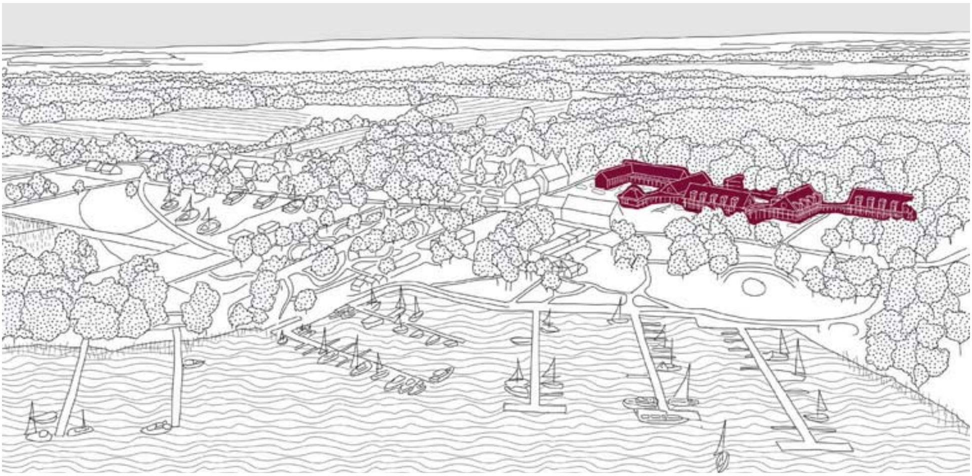
sem. 3/4 WUT

THE ORGANIZERS OF THE WORKSHOP
King Cross Developments Sp. z o.o.
Nowy Sztynort Sp. z o.o.
Warsaw University of Technology, Faculty of Architecture

2nd place
Audience Awards

ZAKORZENIENIE

The project's concept is to use reeds as a clear organizing principle for both the urban structure and architectural solutions. Reeds, naturally present in the Sztynort landscape, have become a reference point for establishing a spatial hierarchy, organizing pedestrian routes, and defining the relationships between buildings, greenery, and the forest. The project is based on ecological principles and regeneration, shaping diverse microclimates and spaces conducive to education, recreation, and contact with nature. The entire concept emphasizes the role of natural materials, greenery, and landscape heritage as the fundamental carriers of the place's identity. Adopting reeds as a structural inspiration allowed for a departure from schematic functional divisions in favor of a more organic, landscape-oriented approach to shaping space. The design does not impose a uniform scale or a single mode of use, but rather enables fluid transitions between public, semi-public, and private spaces. This approach strengthens the user's relationship with the surroundings and fosters a mindful experience of the lake, reeds, and forest. Trzcinort functions as a cohesive spatial ecosystem in which architecture serves as a framework for natural and social processes.

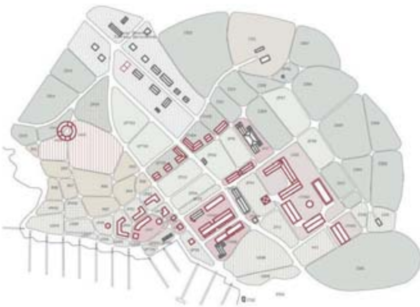


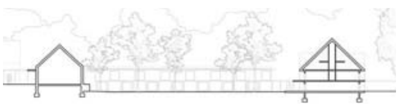
The roof of the former stable as a point of departure for the colour palette and the character of the new development.

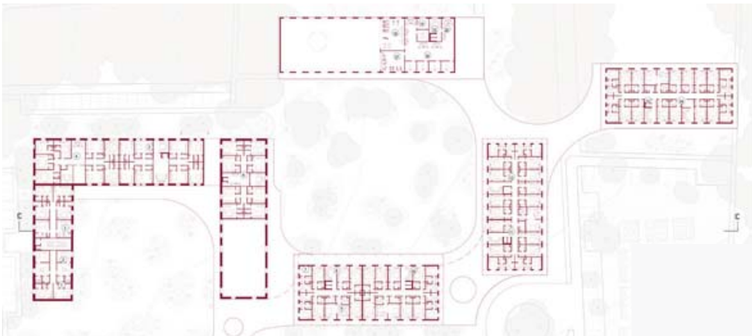
The reuse of bricks from the demolished wing of the stable, incorporating architectural spolia.

The integration of the historic fabric with the new structure through an organic roof form.

Courtyards of varying character.







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THE ORGANIZERS OF THE WORKSHOP
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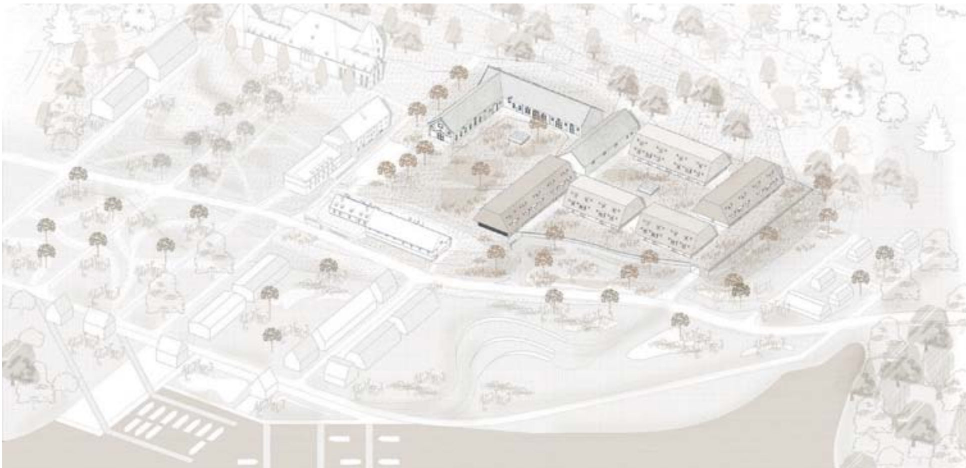
3rd place

KŁĄCZE TRZCINY

The project draws inspiration from the morphology of a semi-arid cactus and its root system, using a biomimetic approach to organize circulation and the underground level. Wide underground corridors distribute movement from the vertical circulation cores while separating technical and service functions from the above-ground public areas. This structure enables intuitive access to wellness facilities, lounges, and parking. The urban composition references the typology of a traditional manor farm. The historic stable acts as the dominant element, while the new volumes reconstruct the spatial logic of a central courtyard. The arrangement creates intimate buffer spaces and establishes a dialogue between historic fabric and contemporary architecture.

The spatial concept is based on the diversification of functional levels, relocating the main circulation and technical infrastructure underground in order to prioritize pedestrian areas and greenery at ground level. Between the proposed volumes, a sequence of intimate courtyards was introduced, inspired by traditional farm layouts. Hidden rooflights within these courtyards provide natural daylight to the pool and gym below. The terrain was reshaped to improve accessibility and spatial quality. Reprofiling the embankment in front of the stable enabled the creation of a representative entrance level with a drop-off zone leading to a two-storey lobby. Full-height glazing of the restaurant façade visually connects the interior with the lowered entrance plaza.

The geometry of the new buildings derives directly from the proportions of the historic stable. The hotel rooms follow a strict modular system, while larger apartments are located in the attic level and within the adapted stable structure. The roof was designed as a monolithic form, articulated through deep recessed loggias rather than traditional dormers, preserving the integrity of the silhouette while ensuring privacy and daylight access. The programme is divided into two zones: the historic stable accommodates the 5-star hotel with apartments, private wellness facilities, and dedicated services, while the newly designed section operates as a 4-star hotel. Both parts are connected through the underground level containing shared recreational facilities, including the pool, gym, SPA & Wellness area, and game rooms.

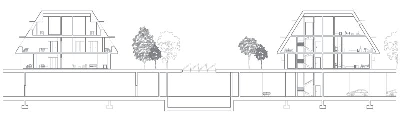


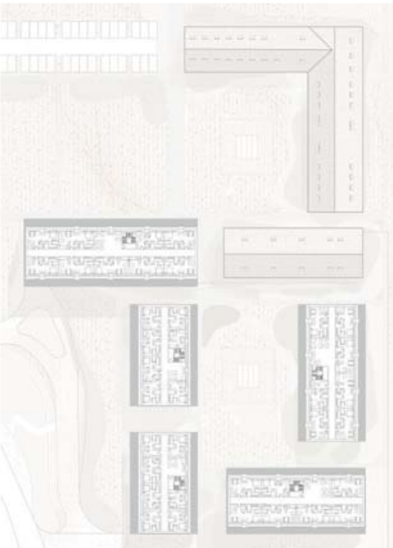
rhizome

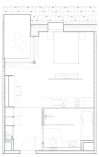
the broadleaf cattail concept

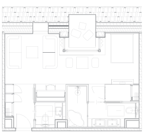
urban design concept

urban design concept









ARCHITECTURAL WORKSHOP IN SZTYNORT

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Warsaw University of Technology, Faculty of Architecture

1st Honorable Mention

MIĘDZY NATURĄ A DZIEDZICTWEM

The proposed hotel complex is designed as a contemporary response to the exceptional landscape and cultural identity of the Masurian region. Located on the shore of Lake Sztynorckie, the development establishes a new waterfront landmark while remaining closely connected to the existing marina and local urban fabric.

The architectural concept is based on the creation of a high-standard leisure destination that naturally extends the character of the port area. At the centre of the composition lies a representative colonnaded courtyard, forming a transitional space between the vibrant marina and the quieter hotel interiors. The courtyard opens visually toward the lake, creating a unique recreational setting for sailors, tourists, and guests seeking relaxation.

The complex comprises 150 accommodation units of varying standards, ranging from standard rooms to premium suites and luxury apartments with panoramic waterfront views. The functional layout has been developed to combine leisure functions with business infrastructure, including a contemporary conference facility adaptable to conferences, cultural events, and exhibitions.

The gastronomic zone, centred around a representative waterfront restaurant, serves as the social heart of the complex and a meeting place for the sailing community. Complementing the program is an extensive SPA and swimming pool area designed to ensure year-round attractiveness and comfort.

Through its restrained architectural expression, natural materials, and carefully articulated proportions, the project introduces a contemporary interpretation of luxury hospitality architecture while respecting the identity and spatial character of Sztynort.

ARCHITECTURAL WORKSHOP IN SZTYNORT

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THE ORGANIZERS OF THE WORKSHOP
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Nowy Sztynort Sp. z o.o.
Warsaw University of Technology, Faculty of Architecture

2nd Honorable Mention

SZTYNORT DO KWADRATU

The central idea of the project is to reinforce and reorganize the existing spatial axes of Sztynort through a coherent quadrangular composition that integrates the key elements of the settlement into a unified urban structure. The proposed hotel complex develops the concept of the courtyard block as a contemporary interpretation of traditional spatial organization, establishing clear relationships between architecture, landscape, and public space.

The urban layout is based on a geometrically ordered plan that ensures continuity of pedestrian circulation and visual connections throughout the site. Four interpenetrating building volumes are linked on the upper levels, accommodating hotel, gastronomic, SPA, and recreational functions within a spatially open and permeable structure. The arrangement creates a sequence of interconnected public and semi-public spaces that strengthen the relationship between the complex and the surrounding waterfront landscape.

A key objective of the proposal is the restoration and exposure of the historic palace–lake axis, currently diminished within the spatial structure of Sztynort. This axis becomes the principal compositional element of the development, organizing the entire urban arrangement while emphasizing the historical significance of the site. The project integrates the most important components of the local environment — the forest, the village, the historical manor context, and the lakefront — into a cohesive spatial system that respects the cultural identity and natural character of the region.

The architecture combines restrained contemporary expression with references to local building traditions, creating a balanced relationship between modern hospitality functions and the historical landscape of the Masurian waterfront.

ARCHITECTURAL WORKSHOP IN SZTYNORT

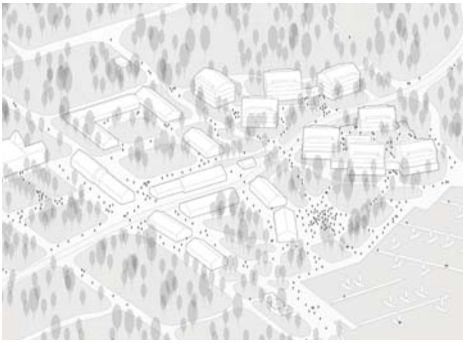
An Architectural and Urban Planning Concept for a New Hotel and Recreation Complex in Sztynort | WIDOKI | Zofia Garncarczyk, Klaudia Kochanowska, Julia Świstuniuk

An Architectural and Urban Planning Concept for a New Hotel and Recreation Complex in Sztynort | MIĘDZY WODA A ZIEMIĄ | Piotr Marczak, Szymon Kucharczyk, Dominik Zagajewski

ARCHITECTURAL WORKSHOP IN SZTYNORT

An Architectural and Urban Planning Concept for a New Hotel and Recreation Complex in Sztynort | PUNKT SZTYNORT | Szymon Kassowski - Eng. Arch., Jakub Wiązowski

An Architectural and Urban Planning Concept for a New Hotel and Recreation Complex in Sztynort | POMIĘDZY | Aleksander Janowski, Mateusz Robak - Eng. Arch.



WE STAND FOR THE EXPERIMENT PAVILION 2026

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Master's degree sem. 3/11 WUT

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Warsaw University of Technology
Faculty of Architecture WUT
The Faculty of Geodesy and Cartography WUT
New European Bauhaus

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Gembal Agata
Głomski Bartłomiej, Eng. Arch.
Grabowski Anna
Gródek Natasza
Jaszczak Zuzanna
Kowalczyk Zuzanna
Kuć Weronika, Eng. Arch.
Kurzela Julia
Mencwal Marcin, Eng. Arch.
Mróz Gabriela
Piróg Olga, Eng. Arch.
Robak Julia
Różewska Natalia, Eng. Arch.
Rutecka Maria, Eng. Arch.
Rybkowska Natalia
Ryś Zuzanna
Sawicka Katarzyna
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The fourth edition of the Pavilion of Experiment confronted students of the Faculty of Architecture with one of the most challenging questions related to temporary architecture: is a pavilion primarily an experiment – a laboratory exploring the boundaries of form, material, and metabolism – or is it rather an element of public space that simultaneously performs a socially engaged function?

The task required not only design skills but also a conscious definition of an architectural attitude: should the structure function as an autonomous experiment, or rather as a catalyst for social interaction within public space?

Already at the initial stage, an exceptionally large number of scenographic and parametric concepts emerged. Students intensively explored adaptation to metabolic changes – responsiveness to variable weather conditions, daily cycles, and seasonal transformations. Most proposals referenced lightweight timber architecture and temporary canopies: demountable frames, membranes, movable elements, shading systems, and natural ventilation strategies. The forms were open, modular, and designed for transformation. The pavilion was conceived not as a static object, but as a living organism responding to its surroundings – testing the limits of ephemeral architecture and its capacity for self-organisation.

The entire project was based on a clear and compelling narrative: the act of constructing the pavilion within the space of the department plays an important and meaningful role. It becomes a place where educational activities intertwine with collaborative work – both design-oriented and social. The pavilion functions as a physical and symbolic bridge between theory and practice, between individual and collective work, and between architecture as a discipline and architecture as a socially engaged activity.

Two principal thematic axes clearly emerged in the students' work, complementing and overlapping with one another.

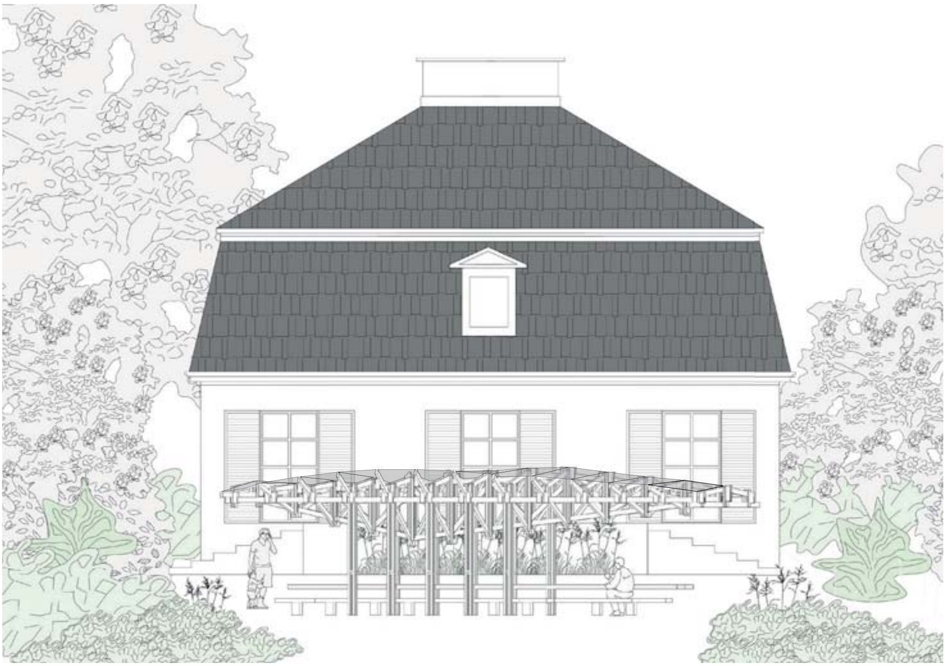
The first axis – "Experiment in Space" – focused on technical, material, and performative aspects. The pavilion was treated as a field laboratory. Students tested structural solutions (timber profiles, joints, anchoring systems) and experimented with materials such as ETFE membranes, technical fabrics, composite timber, and recycled elements. Particular attention was devoted to the building's metabolism: the way it "breathes," responds to rain, wind, and temperature, and transforms throughout the day and night cycle. Projects within this category demonstrated a high level of technological and parametric awareness – ranging from simple movable mechanisms to advanced passive cooling systems and rainwater retention strategies.

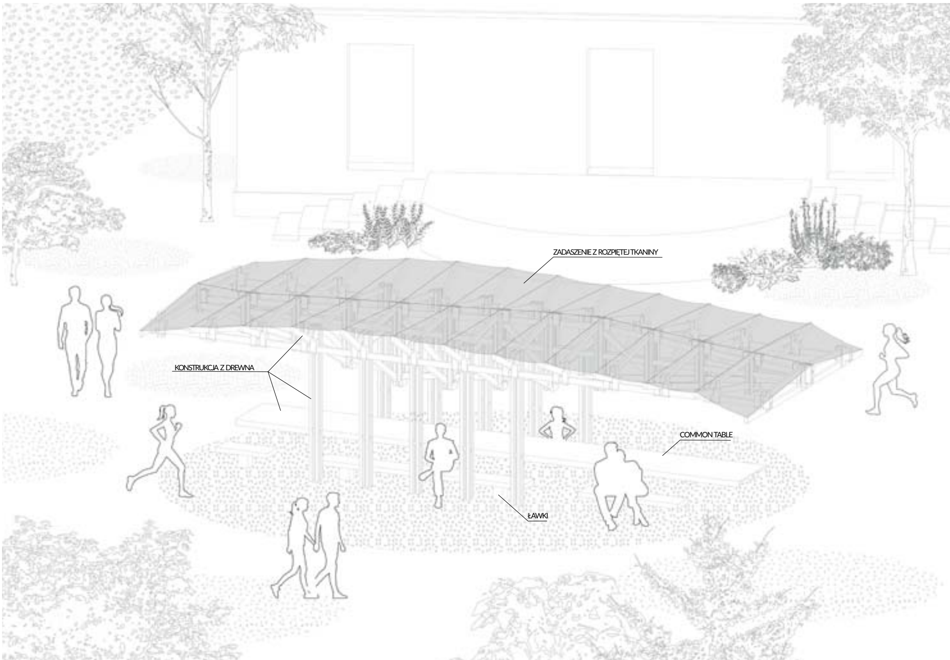
The second axis – "Space of Experiment" (social engagement) – emphasised the social and integrative dimension of architecture. The pavilion ceased to function merely as an architectural object and instead became a stage, a table, and a platform for collective presence. The recurring motif of the communal table proved especially significant – a place for sitting, cooking, conversation, workshops, and informal meetings. Students searched for solutions that would not only occupy space, but actively animate it – attracting passers-by, students from different years, faculty staff, and local residents. Social engagement was understood as openness, hospitality, and the creation of temporary communities. The pavilion was intended to provoke encounters between people who would not normally meet in everyday academic life.

The most compelling projects did not choose one thematic axis over the other, but instead demonstrated their deep interconnection. The best example was the winning proposal, "Table" – a lightweight timber structure with a movable canopy and multifunctional tabletop, simultaneously testing innovative material and structural solutions while serving as a central, inclusive place of integration for the department community.

The fourth edition of the Pavilion of Experiment demonstrated that temporary architecture can function simultaneously as a research laboratory and as a space for building inclusive communities. It confirmed the fluid boundary between technical experimentation and social action, showing that the true strength of architectural design lies in their conscious and thoughtful integration. The pavilion is therefore neither merely an object nor simply an event – it is a living, inclusive spatial experiment that each year redefines what architecture can become within academic and social contexts.

AMW




ZADASZENIE Z ROZPIĘTEJ TKANINY
KONSTRUKCJA Z DREWNA
COMMON TABLE
ŁAWKI

WE STAND FOR THE COMMUNITY PAVILION 2025

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PATRONAGE
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Faculty of Architecture WUT
The Faculty of Geodesy and Cartography WUT
New European Bauhaus
Polish Presidency of the Council of the EU in 2025

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Oberzig Maciej
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Świstuniuk Julia
Wiązowski Jakub
Wolska Maria
Zagajewski Dominik
Zioło Alicja, Eng. Arch.
Zwierzchowski Jan, Eng. Arch.
Zwijacz Zofia

In 2025, as part of the “Stawiamy_” initiative, the third pavilion – the Community Pavilion – was built at the Faculty of Architecture of the Warsaw University of Technology. It was designed within the A3 Technology and Structural Architecture specialization under the supervision of Prof. Anna Maria Wierzbicka. Like the previous ones, the structure was erected at the Royal Łazienki Museum in Warsaw and was closely linked to the international conference “Architecture of Challenges. The New European Bauhaus – Building Community.”

Right at the start of the project, students and faculty posed a key question: what exactly is community, and how can it be expressed through the language of architecture? Discussions centered on warmth, integration, harmony, synergy, belonging, diversity, and closeness. After many hours of debate and a democratic vote, the theme of “symbiosis” was chosen – an idea that most fully captured the coexistence of people, space, and nature.

The pavilion’s symbolism was multi-layered and highly expressive. Its final form referenced the real circus – a circular, egalitarian space where everyone is equal. The circle symbolized eternal unity and wholeness. Three large planters were placed at the center of the structure, representing the balance of three spheres: humanity, community, and nature. The twelfth vertical support expressed the idea of completion and unity – the twelve months of the year, the twelve apostles, the twelve signs of the zodiac. The self-supporting wooden structure rose toward the sky in the form of a vertical axis connecting earth and heaven – a gesture symbolizing the union of human existence with the forces of nature and the spiritual. The planters, inspired by the work of Alina Scholtz, are filled with water in the summer and transformed into living microecosystems, becoming a tangible lesson in the fragility, interdependence, and beauty of natural cycles.

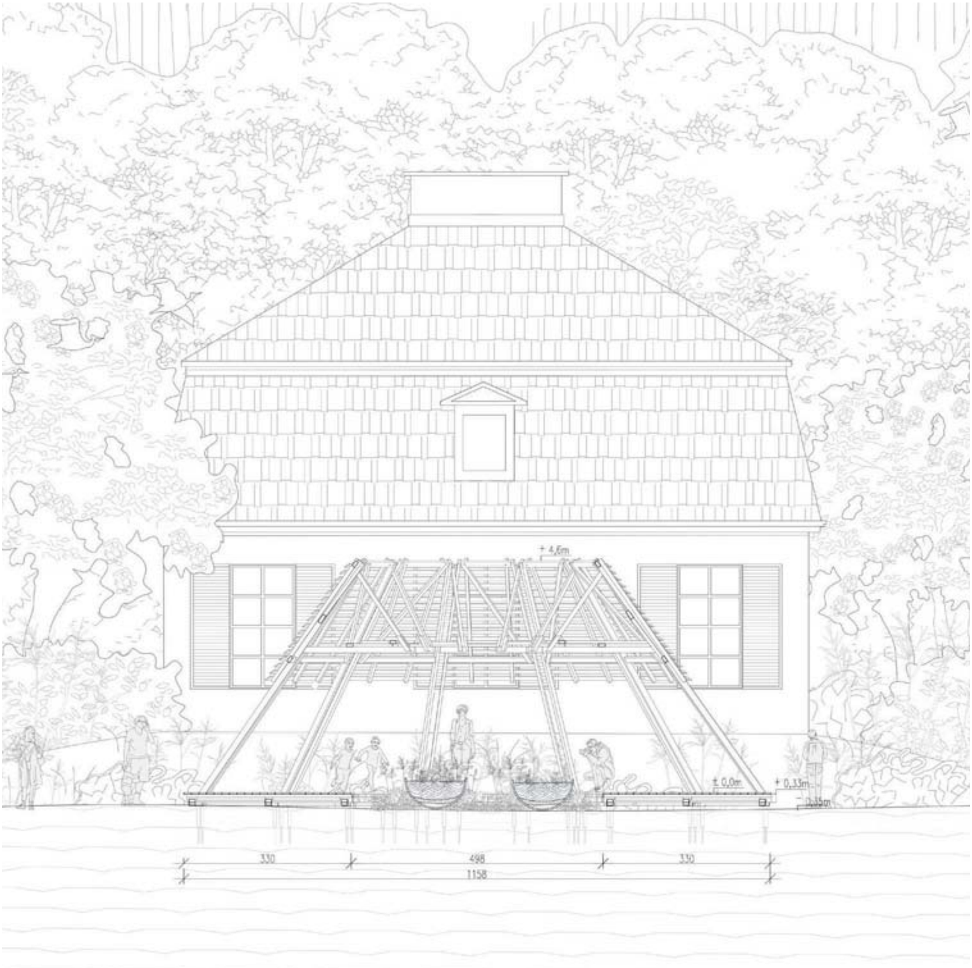
In designing the pavilion, the team emphasized the relationship between people and the natural world. The main goal was to create a space conducive to reflection, authentic social interaction, and a deep, respectful connection with the living environment. The pavilion also served an important educational role – it encouraged visitors to contemplate the rhythms of nature – the cycle of growth, decay, and regeneration – and to reflect on the transience of architecture.

Its location in the Royal Łazienki Garden further reinforced this message. Set among centuries-old trees and classical architecture, the pavilion did not clash with its surroundings but engaged in a respectful dialogue with them. It became a spatial metaphor for continuity – a bridge between generations, eras, and different systems of knowledge. This subtle yet deliberate architectural gesture served as a reminder that true progress can only spring from a deep understanding of history and nature.

The pavilion’s realization was made possible thanks to the support of investors and donors who saw this project as an investment in a more sustainable and thoughtful future. Parallel to the main project, a workshop edition took place, in which students divided into 15 teams and experimented with form and material, creating their own variations of pavilions focused on organic structures and environmental harmony.

Today, both the main Community Pavilion and the workshop pavilions constitute the tangible legacy of the 2025 edition. They serve as a powerful reminder that architecture is not merely the art of building, but above all a reflection on values – an expression of how we choose to live with one another, with history, and with the Earth. Their presence in the park remains an invitation to pause, observe, and listen to nature, to time, and to one another.

AMW


+ 4,6m
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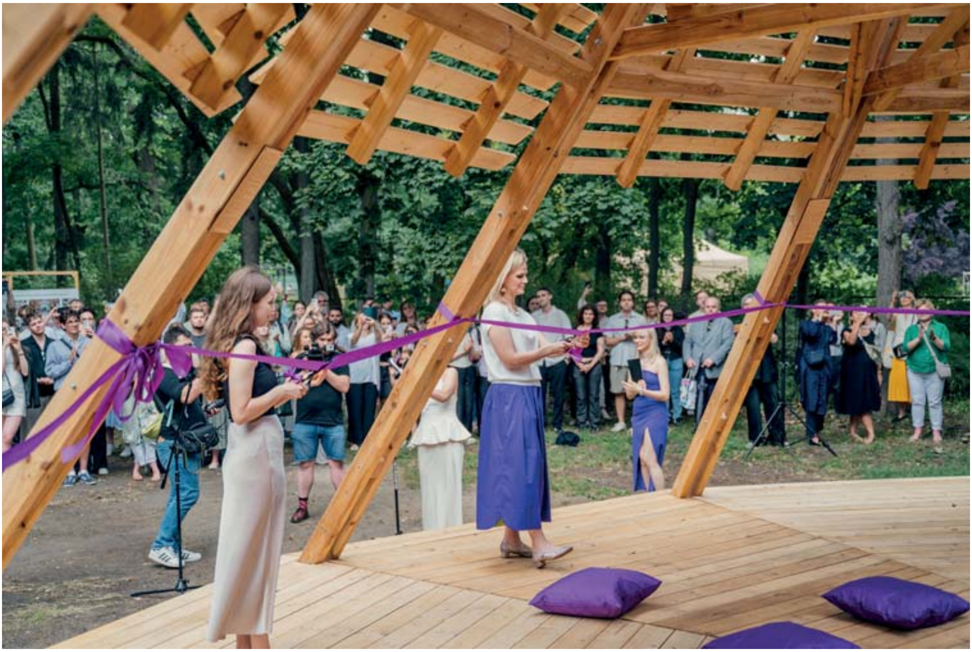
Photography by: Artur Brozowski



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WE STAND FOR THE FUTURE PAVILION 2024

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PATRONAGE
Warsaw University of Technology
Faculty of Architecture WUT
The Faculty of Geodesy and Cartography WUT
University of Warsaw
Instytut Łukasiewicza
New European Bauhaus

STUDENTS
Alicja Bakalarska, Eng. Arch.
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Stanisław Dawidziuk, Eng. Arch.
Maria Hofman, Eng. Arch.
Zofia Jemioło, Eng. Arch.
Anita Karczmarczyk, Eng. Arch.
Konrad Kmiecik, Eng. Arch.
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Marta Zawadka, Eng. Arch.

In 2024, as part of the “stawiamy_” initiative, a second pavilion was created – the Pavilion of the Future. Its central theme was a profound reflection on the future and the responsibility we bear as the current generation toward the planet we will pass on to our children. Students and faculty asked themselves fundamental questions: Are we, as inhabitants of Earth, truly responsible for it? Shouldn’t we leave it in better condition than we found it?

The main theme of this edition was the search for materials of the future – biodegradable, organic, and fully respectful of natural resources. The most important and groundbreaking symbol of this idea was mycelium – an innovative material that the students called a “mushroom brick.” Mycelium is grown on organic waste such as sawdust, straw, or dead construction wood. Under laboratory conditions, it grows into a dense, fibrous structure, and after drying and heat treatment, it becomes a hard, light, and durable building material. It has a zero carbon footprint, is fully biodegradable, and at the end of its life cycle can return to the soil as natural fertilizer. Additionally, it has excellent thermal and acoustic insulation properties, is resistant to mold and fire, and is extremely lightweight compared to traditional materials.

Thanks to close collaboration with researchers from the Łukasiewicz Center, particularly Prof. Katarzyna Misiewicz, students were able to work with this pioneering material at the experimental stage. The process of manufacturing elements from mycelium – from substrate preparation, through controlled mycelium growth, to drying and assembly – became one of the most important educational experiences of the entire program.

The 2023/2024 summer semester in the A3 Technology and Structures Architecture specialization was held under the theme “materials of the future.” Students went through the entire design and construction cycle – from initial concepts and laboratory research through prototyping to the full realization of the pavilion and the ribbon-cutting ceremony. Using the narrative methodology of Prof. Anna Maria Wierzbicka, they treated the pavilion as a story of harmony between humans and nature and of responsibility to future generations.

The most interesting design was realized – an openwork, organic structure built from four identical modules. Its geometric shape directly references the structure of mushroom fruiting bodies. The pavilion does not dominate its surroundings but subtly and organically enriches the landscape of Łazienki Park, creating an intimate place for rest and contemplation.

In the same year, in collaboration with XYstudio Architectural Practice, the pavilion was relocated to the House of Our Lady of Mercy on Foliałowa Street in Warsaw. There, it was adapted to the needs of users with special needs – older people, people experiencing homelessness, and people with disabilities. This relocation demonstrated that the pavilions’ architecture can dynamically change and adapt to real social needs, gaining a new, deeply humanistic significance.

Today, the Pavilion of the Future embodies a modern vision of architecture – one that harmoniously blends with the city’s historic fabric while boldly pointing the way toward sustainable and responsible development. As the second pavilion in the “stawiamy_” series, it confirms that architectural education can effectively address the most critical challenges of our time.

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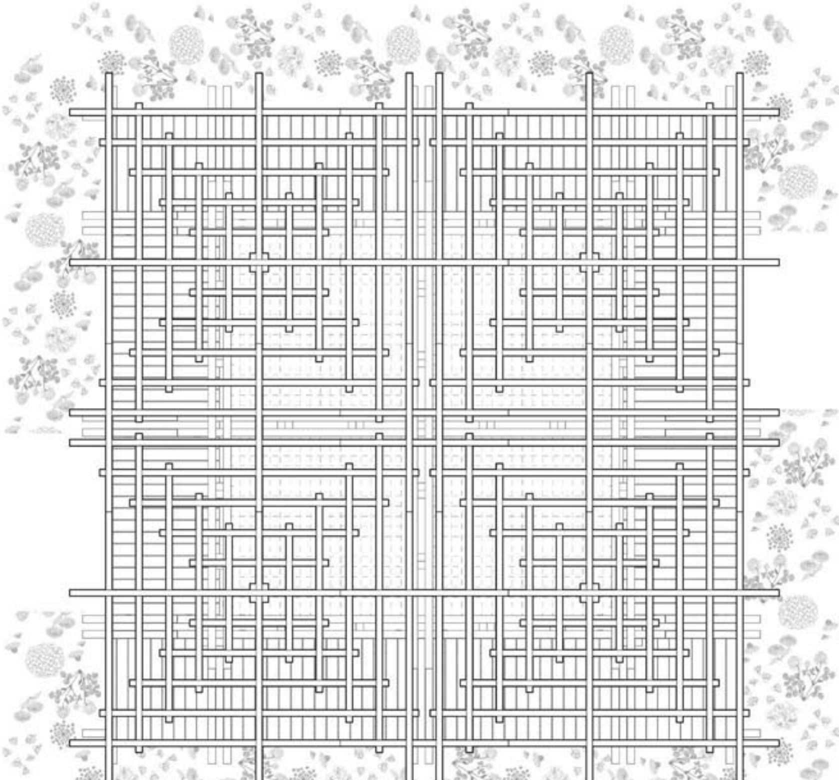
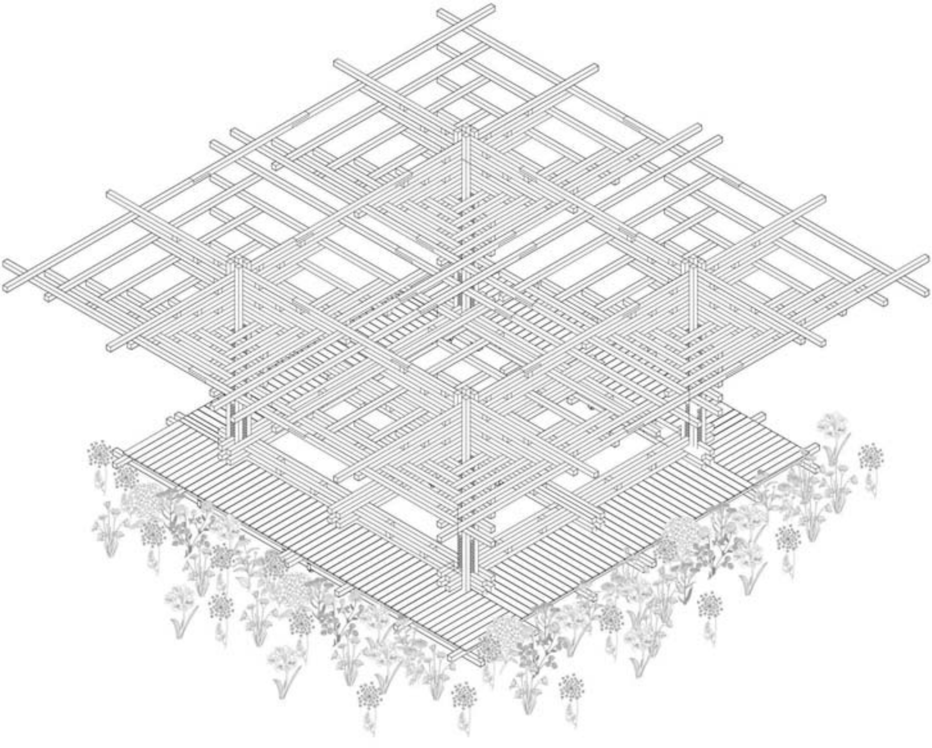
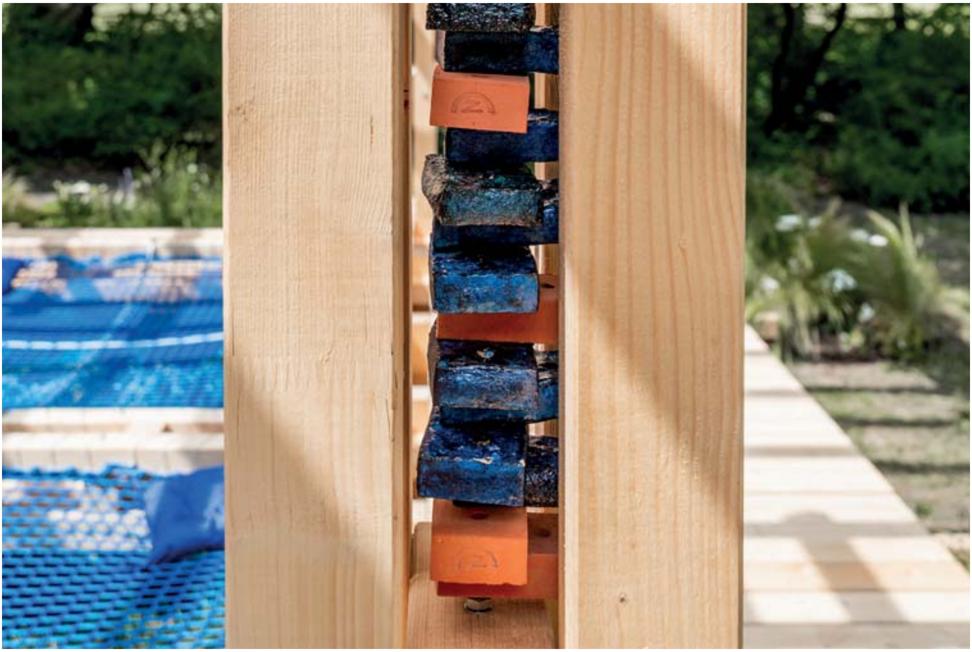




Photography by: Artur Brozowski



Photography by: Artur Brozowski

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WE STAND FOR THE FREEDOM PAVILION 2023

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Kinga Zinowiec-Cieplik, PhD Eng. in Landscape Architecture
Maciej Kaufman, MSc Eng. Arch.
Szymon Kalata, MSc Eng. Arch.

CONSULTATIONS
Ewelina Gawell - PhD Eng. Arch. - Construction
Mariusz Wrona - MSc Eng. Arch. - Construction
Paweł Trębacz - PhD Eng. Arch. - Urban Design

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Master's degree sem. 3/11 WUT

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Natalia Trzaskowska, Eng. Arch.
Maria Wito, Eng. Arch.

In 2023, an initiative by students and faculty of the Faculty of Architecture at the Warsaw University of Technology gave rise to the idea of the Pavilion of Freedom – the first pavilion realized as part of the “stawiamy_” initiative. It was the true Mother Pavilion of the entire series, setting the direction, methodology, and spirit for all subsequent projects. The main idea behind the pavilion was to express the universal values of peace, solidarity, and freedom in the face of war and the Russian invasion of Ukraine. Prof. Anna Maria Wierzbicka’s narrative methodology became the key design tool. The students treated the pavilion as a multi-layered spatial narrative in which every detail contributed to a coherent symbolic story. Strong, ambiguous gestures were clearly visible in the project’s symbolism: the circle symbolized eternity, unity, and equality; the twelve identical trusses forming a funnel-shaped roof served as a metaphor for the community of the full year and the solidarity of twelve months of struggle; the swings inside were a symbol of freedom of movement and childlike carefreeness in the face of tragedy; the white fabric signified purity and hope; sunflowers and millet were a direct reference to the colors of the Ukrainian flag as well as the Ukrainian land and traditions.

This circular wooden structure, a symbol of peace and solidarity with Ukraine, was created as part of the A3 specialization in Architecture of Technology and Structures of Context and Meaning. The installation was open to the public until the end of December 2023 and was located in Łazienki Park in Warsaw. The pavilion’s origins were inextricably linked to the annual “stawiamy_” initiative led by Prof. Anna Maria Wierzbicka and the international academic conference “Architecture of Challenges – Rebuilding Ukraine.” The pavilion became a space of hope and unity, surrounding a symbolic garden of hope within the historic Royal Łazienki Gardens. During the 2022/2023 academic year, students, divided into 10 teams, spent weeks brainstorming, designing, and prototyping. Each team of ten people created its own unique vision of the pavilion, striving to capture the essence of freedom. Using a narrative methodology, the creators delved into various aspects of the project – place, time, narrative, ideas, and references – to create a coherent, multidimensional story. This marked the debut of this methodology in the implementation phase, combining theory with tangible reality. Significantly, the authors themselves erected the pavilion in Łazienki Park – a testament to their full commitment and determination.

The structure consisted of twelve identical trusses forming a funnel-like roof. The interior was adorned with white fabric strips serving as benches or swings. Sunflowers and millet adorned the space, and the interplay of light and materials created a reflective atmosphere inviting contemplation. Through its circular structure and thoughtful form, the pavilion reminds us that unity can triumph over adversity.

Today, in 2024, the pavilion stands along the Vistula River, on the boulevards. Awarded for the best architectural design in the landscape category in the U.S., it continues to inspire viewers worldwide. As the first and flagship pavilion of the entire “stawiamy_” initiative, it has remained not only a temporary structure but, above all, a manifesto of solidarity, hope, and faith in architecture’s power as a tool for building peace and community in times of crisis.

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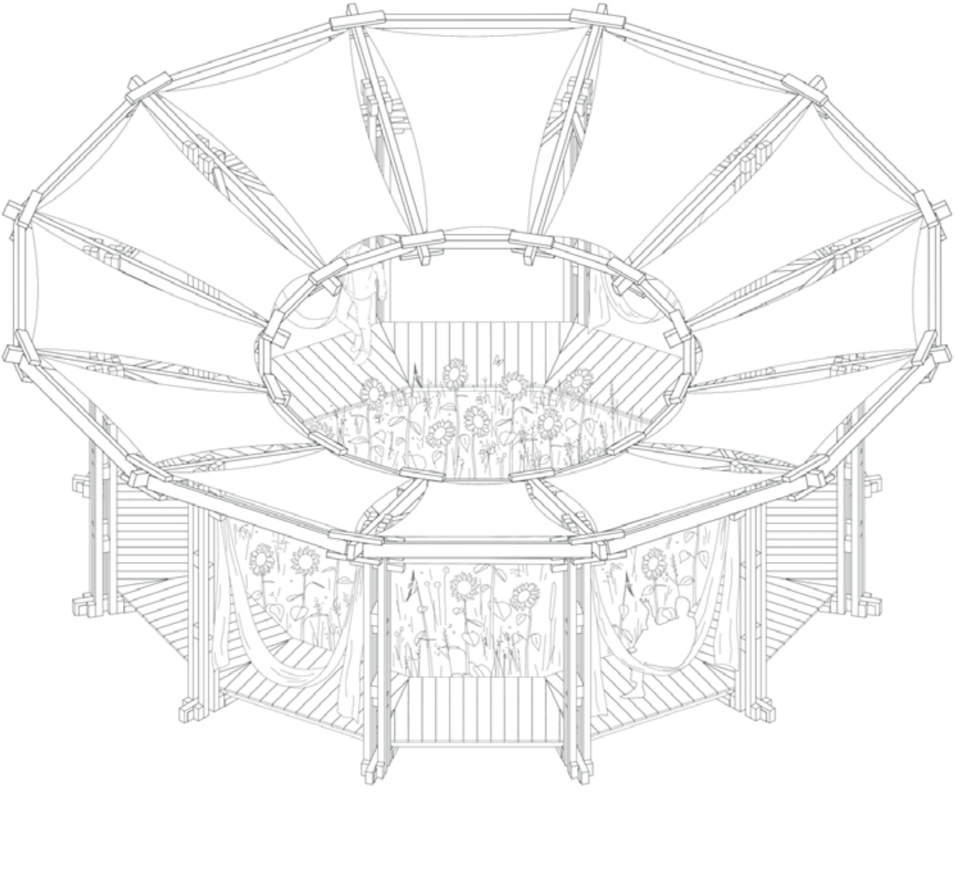

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Photography by: Krzysztof Koszewski

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INTERNATIONAL ARCHITECTURAL WORKSHOPS IN VALENCIA – CONSTRUCTION OF WOODEN PAVILIONS WITH PW STUDENTS

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Szymon Kalata, MSc Eng. Arch.
Kinga Zinowiec-Cieplik, PhD Eng. in Landscape Architecture

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Marta Zawadka, Eng. Arch.

In February 2026, the campus of Universitat Politècnica de València (UPV) hosted the second edition of international architectural workshops devoted to building wooden pavilions. The event brought together students from the Faculty of Architecture at Warsaw University of Technology (PW), specialising in A3 – Architecture, Technology and Structures. Organised within the ENHANCE consortium and the New European Bauhaus initiative, the workshops promoted beauty, sustainability and inclusiveness in contemporary European architecture.
The Polish group was led by Dr. hab. inż. arch. Anna Maria Wierzbicka, university professor, head of the A3 specialisation and the educational-research programme "Stawiamy_", with support from Dr. inż. arch. Anita Orchowska. The workshops were conducted by Prof. Begoña Serrano Lanzarote from UPV, an expert in sustainable architecture and timber construction. This cooperation allowed students to learn from two architectural traditions: the Polish one, rooted in technology and materials, and the Spanish one, shaped by Mediterranean context and innovation. The first edition, held in February 2025, included participants such as Stanisław Dawidziuk, Michał Szczepanek, Mikołaj Szafrański and Julia Stefanowa. In 2026, Piotr Marczak returned, joined by Ewa Maniak, Hanna Lechowska and Alicja Zioło. The Polish students worked in an international team with participants from Spain, Japan and Equatorial Guinea, which enriched the creative and cultural exchange.
The main task was to design and build a full-scale educational pavilion made entirely of wood. In the first edition, the pavilion was inspired by traditional Valencian fruit and vegetable sheds, using a light timber frame and perforated cladding for natural ventilation and passive cooling. In 2026, the team created a prototype pavilion for outdoor work, meetings and teaching activities. Its wooden structure reflected the New European Bauhaus values: low carbon footprint, material reusability, integration with nature and human-centred aesthetics.
The process was intensive and multi-stage. Students began with concepts and structural analyses, then developed construction details such as connectors, nodes and fastening systems, before carrying out the physical build. Lectures on wood properties, joining techniques and ecological design alternated with hands-on work.
The "Stawiamy_" programme at the Faculty of Architecture of PW has long promoted this kind of project-based research education. In Warsaw, students create pavilions and installations later exhibited in public spaces, including the Freedom Pavilion and the Community Pavilion in the Royal Łazienki Park. The Valencia workshops became an international extension of this approach. Students gained technical experience as well as soft skills: multicultural teamwork, communication in English and Spanish, and greater ecological awareness.
Prof. Anna Maria Wierzbicka emphasises that the workshops are not only about building a pavilion, but also about learning by doing, strengthening cooperation between PW and UPV, and preparing students for low-carbon, circular and climate-sensitive architecture. The result is both a completed pavilion serving the Valencia campus and a set of new competencies for young architects, including practical skills, interdisciplinarity and international experience.
The workshops also fit the broader strategy of the Faculty of Architecture at Warsaw University of Technology, which develops exchanges, joint research and practical realisations beyond traditional lecture halls. Wood, as a renewable, carbon-sequestering material with strong formal and technical potential, is becoming central to this education. In the context of the climate crisis, such workshops carry both educational and symbolic importance, showing that young architects are ready to change the way Europe is built.
The organisers have announced the continuation of the cooperation, with future editions expected to bring new and more advanced timber realisations, possibly on a larger scale or using innovative wood-processing technologies. For PW students, the programme offers a valuable opportunity to build international contacts for their future professional and academic careers.

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ESCUELA DE
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WE STAND FOR THE FUTURE PAVILION 2024

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Marta Zawadka, Eng. Arch.

In 2024, a meaningful social and architectural project took shape through the close partnership between XYstudio Architectural Practice and Professor Anna Maria Wierzbicka of the Faculty of Architecture at Warsaw University of Technology.

The initiative focused on relocating the "Pavilion of the Future" from the historic grounds of the Royal Łazienki Museum to the garden of the House of Our Lady of the Heart. This care facility, run by the "Homes of the Bread of Life Community" Foundation and led by Sister Małgorzata Chmielewska, is located at 10 Foliałowa Street in Warsaw. The move was driven by a clear goal: to create a friendly, safe, and accessible outdoor space for the center's residents – primarily elderly people, individuals experiencing homelessness, and persons with disabilities – who had previously lacked any proper recreational area or greenery during the spring and summer months. Generous support from Unibep S.A., under President Leszek Marek Gołąbiecki, enabled the planting of low-maintenance trees. The demanding task of transporting and installing the pavilion was expertly managed by Unihouse, led by President Marcin Gołębiewski, whose personal commitment and organizational expertise were instrumental to the project's success. At the heart of the initiative were participatory workshops guided by Professor Anna Maria Wierzbicka. They united the residents of the House of Our Lady of the Heart with architects Dorota Sibińska, Filip Domaszczyński, Paweł Trębacz, Renata Jóźwik, Magdalena Duda, and Małgorzata Naumann, as well as students from the A3 studio at the Faculty of Architecture of WUT: Natalia Małolepsza, Zofia Jemioło-Cupriak, Patrycja Kuczyńska, Szymon Panek, Grzegorz Staroń, Bartłomiej Urbanowski, Grzegorz Środa, Julia Stepanow, and Konrad Kmiecik.

Together, participants examined the pavilion's existing layout, developed fresh ideas, and carried out practical adaptations to meet daily user needs. A 1:20 scale model helped everyone visualize solutions related to accessibility, shading, relaxation zones, small architectural elements, and smoke-free areas. Particular care was taken to accommodate people with reduced mobility: entrances were lowered, barriers removed, and surfaces adjusted for easy wheelchair access. Far more than a technical relocation, the project held powerful symbolic value. It represented a genuine act of care for those often pushed to the margins of society and demonstrated how thoughtfully designed space can foster community, safety, and human dignity.

XYstudio, led by Dorota Sibińska, Filip Domaszczyński, and Marta Nowosielska, has built its reputation on creating architecture for excluded and vulnerable groups. Their earlier projects – kindergartens, care homes, and a shelter for people experiencing homelessness in Jankowice – have earned numerous awards and brought about real, lasting change. For the studio, architecture is not only about structure, but about how space influences everyday behaviors and human connections. Anna Maria Wierzbicka has long emphasized the educational power of architecture and the essential role of genuine social participation in the design process. Her approach shows that architecture must never be separated from real-life social challenges, and that including voices typically left unheard is both an act of justice and a way to create more responsible, inclusive environments.

This relocation perfectly embodies the spirit of the New European Bauhaus, uniting sustainability, inclusivity, and beauty in the built environment. By moving from the prestigious royal gardens to a place serving those often invisible in public life, the Pavilion of the Future gained a deeper, more authentic identity – becoming a living symbol of solidarity, empathy, and tangible care.

AMW

Ladies and Gentlemen,

Experimentation has always constituted one of the foundations of architecture, urban planning, and design education. It is through the courage to transcend established patterns that it becomes possible to seek new spatial forms, new social relationships, and new ways of understanding the city as a dynamic living environment. Contemporary challenges – technological, environmental, and social – particularly require an open and critical approach to design today.

The city may be perceived as a laboratory of constant transformation, where experimentation becomes a tool for investigating the quality of space and the ways in which inhabitants coexist. Both design practice and scholarly reflection make it possible to test new models of architecture and urbanism, while also enabling a deeper understanding of the impact of space on human beings. In this context, interdisciplinarity is of particular importance, as it allows knowledge from various fields of science and practice to be integrated.

This year's conference provides an opportunity to exchange experiences, present innovative ideas, and jointly reflect on the role of experimentation in shaping the future of the built environment. The gathering of representatives from diverse academic and professional communities demonstrates that experimentation is not merely a method of formal exploration, but above all a way of responsibly thinking about space, society, and quality of life.

***

Therefore, the idea of experimentation, to which we have dedicated the fourth edition of our Architecture of Challenges conference series, is both a necessity and a response to the challenges of our times. I would like to thank the academic community of the Faculty of Architecture at the Warsaw University of Technology, and in particular the Dean, Krzysztof Koszewski, DSc Eng. Arch., Professor of the Warsaw University of Technology, for the organizational support provided for this conference.

The fourth edition of the conference, which has already become an important and recognizable academic undertaking, marks a moment when the emerging relationships, partnerships, and community of individuals committed to the idea of experimentation are clearly visible. I would especially like to thank the co-organizers of the event, who once again demonstrated kindness, trust, and openness to collaborative activities.

My sincere gratitude goes to Dr. Marianna Otmianowska, Director of the Royal Łazienki Museum in Warsaw, for once again welcoming the conference participants to the unique setting of the Palace on the Isle, and for granting permission for the realization of the "Pavilion of Experiment," prepared within the A3 Specialization in Technology and Structure – Context and Meaning, conducted at the Chair of Architectural and Urban Design of the Faculty of Architecture, Warsaw University of Technology, under my supervision. I also extend my gratitude to the Rector of the Academy of Fine Arts in Warsaw, Professor Błażej Ostoja Lniski, for providing the University's premises and assuming the role of host for the second day of the conference.

I would also like to thank Professor Bolesław Stelmach, DSc Eng. Arch., Director of the National Institute of Architecture and Urban Planning, for his organizational support and commitment to the realization of the conference. Thanks to the cooperation of institutions representing diverse communities, it is possible to create a space for dialogue, exchange of ideas, and presentation of experimental activities in architecture, urban planning, and education.

I would like to thank the conference partners: Kyiv National University of Construction and Architecture (KNUCA), Zaporizhzhia Polytechnic National University, Lviv Polytechnic National University, the National Academy of Fine Arts and Architecture of Ukraine, the

National Union of Architects of Ukraine, the Association of Polish Architects (SARP), and the Warsaw Branch of SARP.

The conference has been granted the Honorary Patronage of the Rector of the Warsaw University of Technology, Professor Krzysztof Zaremba. This constitutes a great honor and important support for the event in this exceptional year marking the 200th anniversary of the Warsaw University of Technology. I would also like to express my profound gratitude that the Honorary Patronage continues to be held by the Ministers of Culture and National Heritage, as well as Science and Higher Education. Thank you once again for this distinction.

My heartfelt thanks go to all Members of the Scientific Committee, who once again agreed to oversee the substantive dimension of the conference – dedicating their time, sharing their knowledge, and contributing valuable experience. The preparation of this conference and publication required an enormous amount of work by many individuals. Its organization would not have been possible without the invaluable efforts of the members of the Organizing Committee. Special thanks are due to Magdalena Duda, MSc Eng. Arch., who served this year as Secretary, and I sincerely hope for the continuation of this cooperation in the future. I would also like to thank once again Professor Konrad Kucza-Kuczyński for creating another outstanding original conference poster. I extend my gratitude to all those involved in the organizational work.

I would also like to thank the architects co-leading with me the A3 specialization project in Architecture of Technology and Structure – Context and Meaning: Maciej Kaufman, Szymon Kalata, also PhD Eng. in Landscape Architect Kinga Zinowiec-Cieplik, and Mariusz Wrona, MSc Eng., as well as the students of the Faculty of Architecture who, within the chosen specialization, became involved in the design and construction of the "Pavilion of Experiment."

The organization of this undertaking was financially and professionally supported through invaluable industry contributions by Arche – the main sponsor, Warsaw University of Technology, Hilti Polska, Ramirent Polska, Schréder Polska, Urban Jungle, and Grochowski Construction, for which I am deeply grateful. I would also like to thank the media patrons, who once again this year were Architektura & Biznes and Architektura-Murator, thanks to whom the conference received wide and professional publicity. Special thanks go to Professor Magdalena Jagiełło-Kowalczyk, Editor-in-Chief of the Journal Housing Environment, for granting the conference Publishing Patronage.

I would also like to thank the students who collaborated in the construction of the Pavilion and the volunteers assisting in the operation of the conference. The specialization group included undergraduate students: Weronika Boczar, Krystian Brynda, Lena Firlińska, Agata Gembal, Anna Grabowski, Natasza Gródek, Zuzanna Jaszczak, Zuzanna Kowalczyk, Julia Kurzela, Gabriela Mróz, Julia Robak, Natalia Rybkowska, Zuzanna Ryś, Katarzyna Sawicka, Karolina Wojak, as well as graduate architecture students: Rafał Chmielewski, Alicja Dyśko, Bartłomiej Głomski, Weronika Kuć, Marcin Mencwal, Olga Piróg, Natalia Różewska, Maria Rutecka, Zofia Staszel, Jędrzej Szepietowski, Aleksandra Szurek, Marta Wołoszyn, Franciszek Woźnica, and Zofia Zimowska.

Finally, I extend my warmest thanks to all Participants of this year's conference who share a concern for architecture, community, and the success of every meaningful experiment. Thank you for your presence, participation, and for all actions contributing to the building of our conference community.

Anna Maria Wierzbicka, DSc. Eng. Arch.,
Professor at the Warsaw University of Technology

Anna Maria Wierzbicka

Architect, teacher and professor, D.Sc. (habilitation) in Architecture, Professor at the Faculty of Architecture, Warsaw University of Technology. Head of the Department of Architectural and Urban Design and the Studio of Sacred and Monumental Architecture.

Her scientific and teaching activities focus on architecture as a carrier of meaning, spatial narrative, and the ethical dimensions of design. She specialises in sacred and monumental architecture, the adaptation of religious heritage buildings, and experimental methods of teaching architecture through actual construction. She is the author of monographs and numerous scientific publications on symbolism in architecture, contemporary churches, and the role of narrative in design.

An active advocate for equal opportunities for women in architecture and engineering fields – she supports initiatives promoting gender balance in technical professions, mentoring, and the visibility of women in public space and academia. Author and co-author of many architectural projects, including sacred buildings, adaptations of historic structures, and experimental pavilions constructed as part of her educational programme. Initiator and coordinator of the stawiamy_ project – a unique didactic programme in which students build full-scale pavilions in public spaces (including the Royal Łazienki Park in Warsaw).

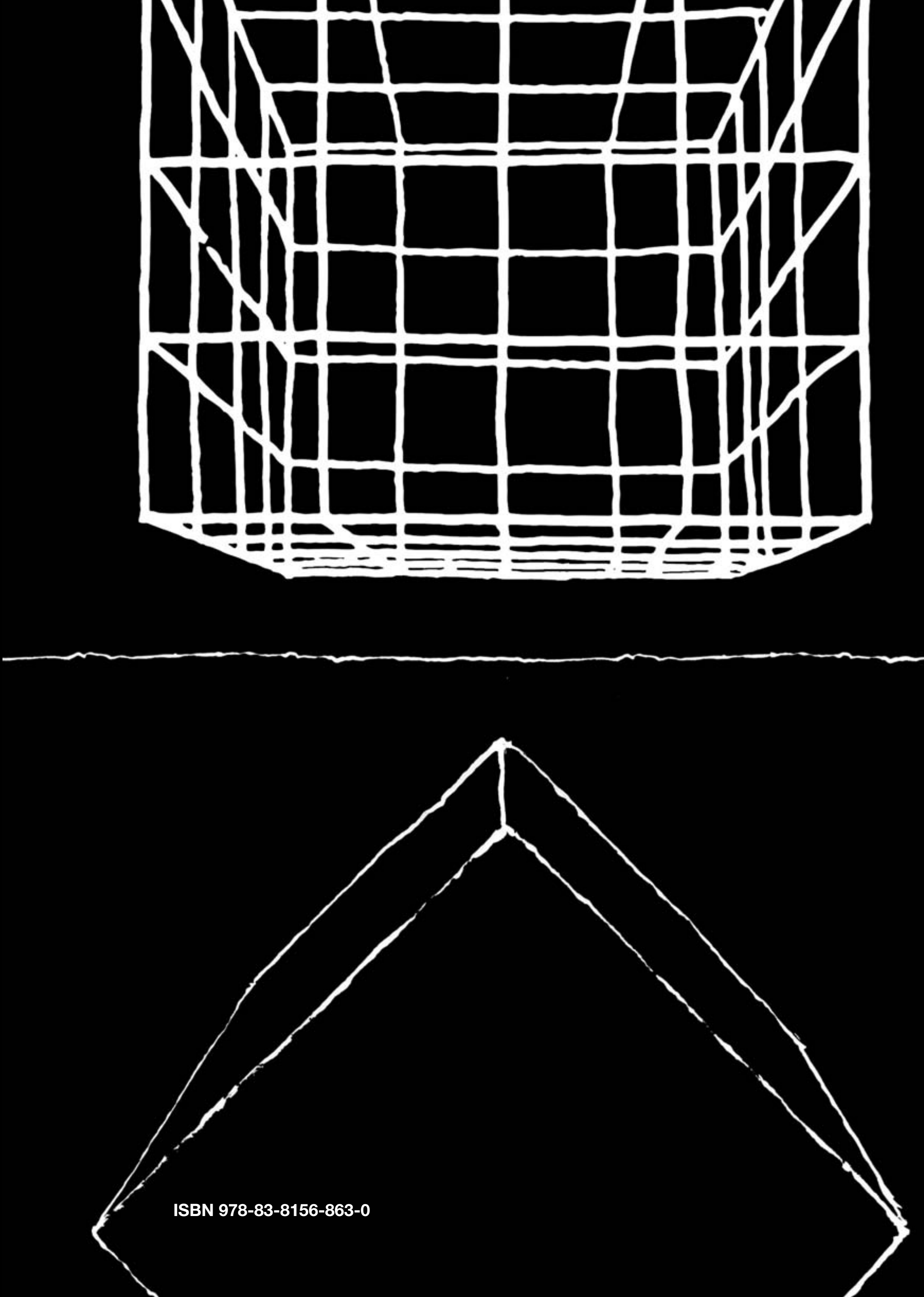
Co-founder and editor of the "Architecture of Challenges" series – already more than 4 international conferences and collective publications in which she develops the concept of architecture as a response to contemporary civilisational, climate, and social crises. Participant in international conferences, including those of the AIA, and supervisor of numerous diploma theses that combine theory with architectural practice.

In the monograph Architecture of Challenges. Experiment in Space – Space of Experiment she explores experiment as the essence of contemporary architectural practice – not only as a design tool, but as a fundamental way of thinking about space as a laboratory of meanings and shared experiences.

She lives and works in Warsaw.

Experiment is the sole judge of scientific truth.

Richard Feynman



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